

Space–Time Algebra

A compact development of paravectors and their physical applications

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Abstract

A paravector algebra is formed from a complex scalar and three-vector, with dot and cross products combined. Quaternions, rotations, Lorentz boosts, projections, spectral functions, spacetime geometry, relativistic dynamics, electromagnetism, gauge coupling, Klein–Gordon and Dirac equations, Schrödinger and Pauli limits, and spin amplitudes and probabilities are expressed in this algebra. These results are connected through shared definitions and short derivations: proper time, four-momentum, and the mass shell are obtained from spacetime determinants; Klein–Gordon is obtained from energy and momentum operators, and Dirac from first-order factorization. Spin one-half is determined by the rotation structure of paravector spinors; its magnetic coupling and measurement probabilities are obtained from the same products and projectors.

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1 Algebra and paravectors

1.1 Notation and physical conventions

Units and type. Natural units $\hbar = c = 1$, rationalized electromagnetic units, and signature $(+, -, -, -)$ apply except in boxes labeled SI units, where symbols carry the stated physical units. Scalars and paravectors use ordinary math lettering; vectors and matrices use upright bold Latin letters or bold Greek glyphs. Domains distinguish scalars from paravectors; letter case carries no type convention. For example,

$$c \in \mathbb{C} \quad (\text{scalar}), \quad \mathbf{Z} \in \mathbb{C}^3 \quad (\text{vector}), \quad Z \in \mathbb{C} \oplus \mathbb{C}^3 \quad (\text{paravector}).$$

Vector derivatives $\partial_{\mathbf{r}}$ have a bold derivative and subscript; scalar derivatives ∂_t remain ordinary. Likewise, the scalar trip distance ℓ differs from position $\mathbf{r}$, and paravector momentum P from its vector part $\mathbf{P}$. Terminology is compared with the literature in Appendix C.

Paravectors and sigmas. The algebra uses coefficient pairs $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma} \leftrightarrow (c, \mathbf{Z}) \in \mathbb{C} \oplus \mathbb{C}^3$. Real paravectors lie in $\mathbb{R} \oplus \mathbb{R}^3$; mixed sums specify each coefficient space separately. Direct sums specify coefficient values. Each σ_k is a distinct symbolic object whose products follow the order-dependent rules given below. The bold symbol $\boldsymbol{\sigma}$ is the three-vector of these objects. A numerical $\mathbf{X} \in \mathbb{R}^3$ and its algebra image $\mathbf{X} \cdot \boldsymbol{\sigma}$ are distinct types. Spatial indices range over $\{1, 2, 3\}$ unless stated otherwise.

Vectors, matrices, and constants. For positive integers m, n , numerical columns lie in $\mathbb{R}^n$ or $\mathbb{C}^n$, rows in $\mathbb{R}^{1 \times n}$ or $\mathbb{C}^{1 \times n}$, and matrices in $\mathbb{R}^{m \times n}$ or $\mathbb{C}^{m \times n}$; their entries are scalars. A paravector column instead has paravector entries. Identity dimensions are stated locally; scalar multiples of identities may be written as scalars in block formulas. The constants $i \in \mathbb{C}$, $e, \pi \in \mathbb{R}$, and 1 retain ordinary type. The algebraic coefficient $c \in \mathbb{C}$ differs from the SI speed of light. SI units change physical dimensions, not real or complex value spaces; $c, \hbar, h, \epsilon_0, \mu_0 \in \mathbb{R}_{>0}$ in the SI boxes.

Domains and derivatives. Domains persist within a section unless specialized or redefined. Primes, initial-value subscripts, increments, and real-parameter derivatives inherit the declared value space; differentiation need not preserve constraints such as unit norm. An indexed numerical vector component is scalar; an indexed paravector-tuple entry remains a paravector. For fields, membership specifies the value at each real spacetime point, while function notation gives the independent variables. Required regularity is assumed. Local differential operators act on smooth fields on the stated open coordinate region; scalar operators act componentwise on vector, matrix, and paravector fields. Outputs have the declared value spaces, but operators themselves are not numerical values in those spaces.

1.2 Sigmas and ordered multiplication

The sigmas σ_n , $n = 1, 2, 3$, are defined by the multiplication rules below. The general complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ is expressed in terms of these sigmas (1, pp. 4–5):

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\ \boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i} \boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z &= (a + b\mathbf{i}) + (\mathbf{X} + \mathbf{Y}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b &\in \mathbb{R}, \quad \mathbf{X}, \mathbf{Y} \in \mathbb{R}^3. \end{aligned}$$

Ordinary three-vector operations are defined by

$$\begin{aligned} \mathbf{X} \cdot \mathbf{Y} &= X_1 Y_1 + X_2 Y_2 + X_3 Y_3, \\ \mathbf{X} \times \mathbf{Y} &= \{X_2 Y_3 - X_3 Y_2, X_3 Y_1 - X_1 Y_3, X_1 Y_2 - X_2 Y_1\}, \\ \mathbf{X}^2 &:= \mathbf{X} \cdot \mathbf{X}. \end{aligned}$$

Thus the square of a vector always denotes its self dot product. For complex vectors, the same formulas are used for dot and cross products, with neither factor conjugated. The elementary Cartesian vectors $\mathbf{e}_k \in \mathbb{R}^3$ used below are

$$\mathbf{e}_1 = \{1, 0, 0\}, \quad \mathbf{e}_2 = \{0, 1, 0\}, \quad \mathbf{e}_3 = \{0, 0, 1\}.$$

Thus $\mathbf{e}_k \cdot \boldsymbol{\sigma} = \sigma_k$; the scalar basis element is 1. Unindexed e denotes only Euler's number.

A basis over $\mathbb{R}$ is provided by the eight elements

$$1, \quad \sigma_1, \sigma_2, \sigma_3, \quad \mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3, \quad \mathbf{i}.$$

By anticommutation, $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$. Two sign changes are introduced when σ_n is moved through the other two sigmas, so

$$\mathbf{i}\sigma_n = \sigma_n \mathbf{i}.$$

Thus $\mathbf{i}$ commutes with every sigma and hence with every paravector.

For $X, Y \in \mathbb{C} \oplus \mathbb{C}^3$, the commutator is

$$\begin{aligned} [X, Y] &:= XY - YX, \\ [Y, X] &= -[X, Y]. \end{aligned}$$

Thus $[X, Y] = 0$ exactly when X and Y commute. For $n, m, k = 1, 2, 3$, the Kronecker delta is defined by

$$\delta_{nm} \in \{0, 1\}, \quad \delta_{nm} := 1 \quad \text{if } n = m, \\ 0 \quad \text{if } n \neq m,$$

and the Levi-Civita symbol is defined by

$$\begin{aligned} \epsilon_{nmk} &\in \{-1, 0, 1\}, \quad \epsilon_{nmk} := +1 \quad \text{if } (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ &-1 \quad \text{if } (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ &0 \quad \text{if two or more indices are equal.} \end{aligned}$$

For $n, m = 1, 2, 3$, the ordered multiplication rule and commutator are (1, Eq. (3)) (1, Eq. (10))

$$\begin{aligned}\sigma_n \sigma_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k, \\ [\sigma_n, \sigma_m] &= 2 \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k.\end{aligned}\tag{1.1}$$

1.3 Hermitian and sigma conjugation

Hermitian conjugation is defined by (1.2), and *sigma conjugation* is defined by (1.3) (2, Eq. (3)):

$$(\cdot)^{\text{H}} := \text{the multiplication order of the sigmas is reversed},\tag{1.2}$$

$$(\cdot)^* := \text{every } \sigma_n \text{ is replaced by } -\sigma_n \text{ with product order preserved}.\tag{1.3}$$

Thus, for two sigmas,

$$\begin{aligned}(\sigma_m \sigma_n)^{\text{H}} &= \sigma_n \sigma_m, \\ (\sigma_m \sigma_n)^* &= (-\sigma_m)(-\sigma_n).\end{aligned}$$

Both operations are applied term by term to sums, with real coefficients left unchanged. Since $i = \sigma_1 \sigma_2 \sigma_3$, under sigma conjugation each sigma is negated and multiplication order is preserved:

$$i^* = (-\sigma_1)(-\sigma_2)(-\sigma_3) = -\sigma_1 \sigma_2 \sigma_3 = -i.$$

Complex coefficients are therefore conjugated, and the sign of σ is reversed. The resulting signs on the basis elements are

element	1	σ_n	$i\sigma_n$	i
$(\cdot)^*$	+	-	+	-
$(\cdot)^{\text{H}}$	+	+	-	-
$(\cdot)^{*{\text{H}}}$	+	-	-	+

For $i\sigma_n$, the signs are obtained directly:

$$\begin{aligned}(i\sigma_n)^* &= i^* \sigma_n^* = i\sigma_n, \\ (i\sigma_n)^{\text{H}} &= \sigma_n^{\text{H}} i^{\text{H}} = -i\sigma_n.\end{aligned}$$

For $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$, the definitions give the product laws (1, p. 4) (2, p. 2)

$$(ZZ')^{\text{H}} = Z'^{\text{H}} Z^{\text{H}},\tag{1.4}$$

$$(ZZ')^* = Z^* Z'^*.\tag{1.5}$$

The operations commute with each other: $Z^{*\text{H}} = Z^{\text{H}*}$. Appendix A gives both operations in the Pauli-matrix representation, where $^{\text{H}}$ is the usual matrix Hermitian conjugation.

1.4 Complex paravectors and their components

$a, b \in \mathbb{R}$ and $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^3$ are assumed. The two real paravectors $X, Y \in \mathbb{R} \oplus \mathbb{R}^3$ are combined into the general complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$:

$$\begin{aligned} X &= a + \mathbf{X} \cdot \boldsymbol{\sigma}, \\ Y &= b + \mathbf{Y} \cdot \boldsymbol{\sigma}, \\ Z &= X + iY \\ &= (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma} \\ &= c + \mathbf{Z} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here $c = a + bi \in \mathbb{C}$ and $\mathbf{Z} = \mathbf{X} + i\mathbf{Y} \in \mathbb{C}^3$ are its complex scalar and vector coefficients.

Thus $\text{sc}(Z), \det(Z), \text{tr}(Z)$ lie in $\mathbb{C}$, $\text{vec}(Z)$ in $\mathbb{C}^3$, and $\text{re}(Z), \text{im}(Z)$ in $\mathbb{R} \oplus \mathbb{R}^3$. Conjugates, adjugates, and existing inverses take values in $\mathbb{C} \oplus \mathbb{C}^3$. The complex scalar and vector extractions are defined by

$$\begin{aligned} \text{sc}(Z) &:= c = a + ib, \\ \text{vec}(Z) &:= \mathbf{Z} = \mathbf{X} + i\mathbf{Y}. \end{aligned}$$

This notation is used throughout unless a real specialization is stated.

1.5 Conjugates, determinant, and inverse

On the coefficients, both conjugations are expressed by $c^* = c^H = a - bi$ and $\mathbf{Z}^* = \mathbf{Z}^H = \mathbf{X} - i\mathbf{Y}$; on the paravector, the two conjugations differ in the vector sign (2, Eq. (45)):

$$Z^* = c^* - \mathbf{Z}^* \cdot \boldsymbol{\sigma} = (a - bi) + (-\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}, \quad (1.6)$$

$$Z^H = c^* + \mathbf{Z}^* \cdot \boldsymbol{\sigma} = (a - bi) + (\mathbf{X} - i\mathbf{Y}) \cdot \boldsymbol{\sigma}. \quad (1.7)$$

Their composition defines the adjugate (2, Eq. (39))

$$\text{adj}(Z) := Z^{*H} = c - \mathbf{Z} \cdot \boldsymbol{\sigma}. \quad (1.8)$$

The trace is twice the scalar part:

$$\text{tr}(Z) := Z + \text{adj}(Z) = 2\text{sc}(Z) = 2c. \quad (1.9)$$

The determinant is the complex scalar (3, p. 104)

$$\det(Z) := \text{adj}(Z) Z = Z \text{adj}(Z) = c^2 - \mathbf{Z}^2. \quad (1.10)$$

For $Z' \in \mathbb{C} \oplus \mathbb{C}^3$, adjugation reverses product order, giving determinant multiplicativity:

$$\begin{aligned} \text{adj}(ZZ') &= \text{adj}(Z') \text{adj}(Z), \\ \det(ZZ') &= \text{adj}(Z')(\text{adj}(Z) Z)Z' = \det(Z) \det(Z'). \end{aligned}$$

Since the determinants are scalars, the determinant of any product is unchanged by reordering its factors. For a scalar $z \in \mathbb{C}$,

$$\begin{aligned} \text{adj}(zZ) &= z \text{adj}(Z), \\ \det(zZ) &= z^2 \det(Z). \end{aligned} \quad (1.11)$$

When $\det(Z) \neq 0$, the inverse is (2, Eq. (47))

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)} = \frac{c - \mathbf{Z} \cdot \boldsymbol{\sigma}}{c^2 - \mathbf{Z}^2}. \quad (1.12)$$

For $Z' = c' + \mathbf{Z}' \cdot \boldsymbol{\sigma}$, with $c' \in \mathbb{C}$, $\mathbf{Z}' \in \mathbb{C}^3$, division is order-sensitive. An unambiguous quotient exists when Z is invertible and the factors commute:

$$\frac{Z'}{Z} := Z' Z^{-1} = Z^{-1} Z' \quad \text{if } \mathbf{Z} \times \mathbf{Z}' = 0.$$

Differentiating $Z^{-1}Z = 1$ gives the inverse rule

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}.$$

For $Z, Z', Z'' \in \mathbb{C} \oplus \mathbb{C}^3$, scalar part obeys the cyclic laws

$$\begin{aligned} \text{sc}(ZZ') &= \text{sc}(Z'Z), \\ \text{sc}(ZZ'Z'') &= \text{sc}(Z'Z''Z) = \text{sc}(Z''ZZ'). \end{aligned} \quad (1.13)$$

Multiplying these identities by 2 gives trace cyclicity. Their similarity invariants are derived in Section 5.

1.6 Powers, inverses, and combined superscripts

Let $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$ and let n be a positive integer. The product laws give, by induction,

$$\boxed{\begin{aligned} (Z^n)^{\mathbf{H}} &= (Z^{\mathbf{H}})^n =: Z^{n\mathbf{H}}, \\ (Z^n)^* &= (Z^*)^n =: Z^{n*}, \\ \text{adj}(Z^n) &= \text{adj}(Z)^n. \end{aligned}} \quad (1.14)$$

The case $n = 1$ is immediate. For the inductive step, for example,

$$(Z^{n+1})^{\mathbf{H}} = (Z^n Z)^{\mathbf{H}} = Z^{\mathbf{H}} (Z^n)^{\mathbf{H}} = (Z^{\mathbf{H}})^{n+1}.$$

The other two operations follow in the same way: reversing the order of identical factors has no effect. These identities also allow singular Z .

For the exponent -1 , assume $\det(Z) \neq 0$. Applying the product laws to both $ZZ^{-1} = 1$ and $Z^{-1}Z = 1$ identifies the transformed two-sided inverses:

$$\boxed{\begin{aligned} (Z^{-1})^{\mathbf{H}} &= (Z^{\mathbf{H}})^{-1} =: Z^{-\mathbf{H}}, \\ (Z^{-1})^* &= (Z^*)^{-1} =: Z^{-*}, \\ \text{adj}(Z^{-1}) &= \text{adj}(Z)^{-1}. \end{aligned}} \quad (1.15)$$

Combined superscripts denote successive operations; the minus in $-\mathbf{H}$ or $-*$ denotes inversion. General real powers are introduced in Section 4.6.

1.7 Real and imaginary extractions

The real and imaginary extractions are defined using Hermitian conjugation (1, Eq. (9)):

$$\text{re}(Z) := \frac{1}{2}(Z + Z^H) = X, \quad (1.16)$$

$$\text{im}(Z) := \frac{Z - Z^H}{2i} = Y. \quad (1.17)$$

Adjugation evaluates the scalar and vector extractions explicitly (1, Eq. (14)):

$$\begin{aligned} \text{sc}(Z) &= \frac{1}{2}(Z + \text{adj}(Z)) = c, \\ \text{vec}(Z) \cdot \boldsymbol{\sigma} &= \frac{1}{2}(Z - \text{adj}(Z)) = \mathbf{Z} \cdot \boldsymbol{\sigma}. \end{aligned}$$

The real- and imaginary-part functions also act on complex coefficients. Composing extractions recovers all four real parts:

$$\begin{aligned} a &= \text{re}(c) = \text{sc}(\text{re}(Z)), \\ b &= \text{im}(c) = \text{sc}(\text{im}(Z)), \\ \mathbf{X} &= \text{re}(\mathbf{Z}) = \text{vec}(\text{re}(Z)), \\ \mathbf{Y} &= \text{im}(\mathbf{Z}) = \text{vec}(\text{im}(Z)). \end{aligned}$$

Appendix A reproduces these intrinsic operations with Pauli matrices.

2 Products and normalizations

Throughout this section $Z \in \mathbb{C} \oplus \mathbb{C}^3$, with scalar coefficient $c \in \mathbb{C}$ and vector coefficient $\mathbf{Z} \in \mathbb{C}^3$; real specializations are stated where used. Scalar dot products and determinants lie in $\mathbb{C}$, vector cross products in $\mathbb{C}^3$, and norms in $\mathbb{R}_{\geq 0}$.

2.1 Invertibility and normalization

Notation: $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$, with $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$. For $\mathbf{Z} \in \mathbb{C}^3$, the bilinear square and squared norm are distinguished:

$$\begin{aligned} \mathbf{Z}^2 &:= \mathbf{Z} \cdot \mathbf{Z}, \\ \|\mathbf{Z}\|^2 &:= \mathbf{Z} \cdot \mathbf{Z}^*. \end{aligned}$$

For $c \in \mathbb{C}$, $\|c\|^2 := cc^*$ is defined. The squared paravector norm is (2, Eq. (46))

$$\|Z\|^2 := \|c\|^2 + \|\mathbf{Z}\|^2 = cc^* + \mathbf{Z} \cdot \mathbf{Z}^*. \quad (2.1)$$

All norms take values in $\mathbb{R}_{\geq 0}$ and denote nonnegative real square roots. Here $\mathbf{u} \in \mathbb{C}^3$ and $Z \in \mathbb{C} \oplus \mathbb{C}^3$. Each name below specifies a condition; a letter alone implies none.

- **Vector of square one.** $\mathbf{u}$ satisfies

$$\mathbf{u}^2 = \mathbf{u} \cdot \mathbf{u} = 1. \quad (2.2)$$

For $\mathbf{Z}^2 \neq 0$, $\mathbf{u} = \mathbf{Z}/\sqrt{\mathbf{Z}^2}$ is defined with either square root. Its norm need not be one. A nonzero vector with $\mathbf{Z}^2 = 0$ cannot be normalized this way.

- **Vector of unit norm.** $\mathbf{u}$ satisfies

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u}^* = 1.$$

For any $\mathbf{Z} \neq 0$, $\mathbf{u} = \mathbf{Z}/\|\mathbf{Z}\|$ is defined; then $\mathbf{u}^2 = \mathbf{Z}^2/\|\mathbf{Z}\|^2$ need not be one. For real vectors the two conditions coincide: a *real unit vector* has $\mathbf{u}^2 = \|\mathbf{u}\|^2 = 1$. For complex vectors, specify which holds.

- **Invertible paravector.** Z has an inverse on both sides:

$$ZZ^{-1} = Z^{-1}Z = 1. \quad (2.3)$$

Equivalently, $\det(Z) \neq 0$. Neither unit norm nor determinant one is required.

- **Unitary paravector.** Its Hermitian conjugate is its inverse:

$$Z^H Z = 1. \quad (2.4)$$

Then $Z^H = Z^{-1}$ and $ZZ^H = 1$.

- **Paravector of determinant one.** Z satisfies

$$\det(Z) = 1. \quad (2.5)$$

It is invertible with $Z^{-1} = \text{adj}(Z)$; it need not be unitary. For real Z , $\text{adj}(Z) = Z^*$, so this condition is also $Z^*Z = 1$. This equivalence need not hold for complex Z .

- **Paravector of unit norm.** Z satisfies

$$\|Z\|^2 = 1. \quad (2.6)$$

For $Z \neq 0$, $Z' = Z/\|Z\|$ has unit norm. It need not have an inverse: $(1 + \mathbf{u} \cdot \boldsymbol{\sigma})/\sqrt{2}$ has norm one and determinant zero for any real unit vector $\mathbf{u}$. Unitarity implies unit norm by (2.14).

- **Self-inverse paravector.** Z is its own inverse:

$$Z^2 = 1. \quad (2.7)$$

Then $Z^{-1} = Z$; it is unitary exactly when $Z^H = Z$. Its parts obey $c^2 + \mathbf{Z}^2 = 1$ and $2c\mathbf{Z} = 0$: thus $Z = \pm 1$ or $Z = \mathbf{u} \cdot \boldsymbol{\sigma}$ with $\mathbf{u}^2 = 1$.

The physical roles of the paravector conditions are compared in Section 5.

2.2 Products and quadratic forms

For $\mathbf{X}, \mathbf{Y} \in \mathbb{C}^3$, multiplication combines the dot and cross products; both products are complex bilinear, with no conjugation. The sigma rule (1.1) gives the dot-cross identity (2, Eq. (38))

$$\begin{aligned} (\mathbf{X} \cdot \boldsymbol{\sigma})(\mathbf{Y} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 X_n Y_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 \sum_{m=1}^3 X_n Y_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 X_n Y_m \epsilon_{nmk} i \sigma_k \\ &= \mathbf{X} \cdot \mathbf{Y} + i(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (2.8)$$

For real $\mathbf{X}, \mathbf{Y}$, the product lies in $\mathbb{R} \oplus i\mathbb{R}^3$. Adding the product with $\mathbf{X}$ and $\mathbf{Y}$ interchanged gives

$$2(\mathbf{X} \cdot \mathbf{Y}) = (\mathbf{X} \cdot \boldsymbol{\sigma})(\mathbf{Y} \cdot \boldsymbol{\sigma}) + (\mathbf{Y} \cdot \boldsymbol{\sigma})(\mathbf{X} \cdot \boldsymbol{\sigma}).$$

Consequently, for every nonnegative integer n ,

$$\begin{aligned} (\mathbf{X} \cdot \boldsymbol{\sigma})^2 &= \mathbf{X}^2, \\ (\mathbf{X} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{X}^2)^n, \\ (\mathbf{X} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{X}^2)^n \mathbf{X} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Also,

$$(\mathbf{X} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{X} - i\mathbf{X} \times \boldsymbol{\sigma}.$$

Subtracting the two orders gives

$$[\mathbf{X} \cdot \boldsymbol{\sigma}, \mathbf{Y} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}.$$

Hence

$$\begin{aligned} [\mathbf{X} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] &= -2i\mathbf{X} \times \boldsymbol{\sigma}, \\ [\mathbf{X} \cdot i\boldsymbol{\sigma}, \boldsymbol{\sigma}] &= 2\mathbf{X} \times \boldsymbol{\sigma}, \\ [\mathbf{X} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{Y} &= 2i(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

2.3 Real products

For real $X = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$, with $a, b \in \mathbb{R}$ and $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^3$, (2.8) gives

$$\begin{aligned} XY &= (ab + \mathbf{X} \cdot \mathbf{Y}) + (a\mathbf{Y} + b\mathbf{X} + i\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \\ [X, Y] &= 2i(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \\ X^2 &= (a^2 + \mathbf{X}^2) + 2a\mathbf{X} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Thus $XY \in \mathbb{R} \oplus \mathbb{C}^3$, $[X, Y] \in \{0\} \oplus i\mathbb{R}^3$, and $X^2 \in \mathbb{R} \oplus \mathbb{R}^3$. Real paravectors are not closed under multiplication. The real quadratic identity is (3.1) with $c = a$. If $\mathbf{X} \times \mathbf{Y} = 0$, then $XY = YX$; when $\det(X) \neq 0$, also $X^{-1}Y = YX^{-1}$.

2.4 Complex products

The decompositions $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ and $Z' = c' + \mathbf{Z}' \cdot \boldsymbol{\sigma}$ are used, where $c, c' \in \mathbb{C}$ and $\mathbf{Z}, \mathbf{Z}' \in \mathbb{C}^3$. Complex bilinearity extends (2.8) to the general product

$$ZZ' = (cc' + \mathbf{Z} \cdot \mathbf{Z}') + (c\mathbf{Z}' + c'\mathbf{Z} + i\mathbf{Z} \times \mathbf{Z}') \cdot \boldsymbol{\sigma}. \quad (2.9)$$

The result $ZZ' \in \mathbb{C} \oplus \mathbb{C}^3$ has a complex scalar and complex vector part. Thus

$$\begin{aligned} \text{sc}(ZZ') &= cc' + \mathbf{Z} \cdot \mathbf{Z}', \\ \text{vec}(ZZ') &= c\mathbf{Z}' + c'\mathbf{Z} + i\mathbf{Z} \times \mathbf{Z}'. \end{aligned}$$

Equivalently, $Z = X + iY$ and $Z' = X' + iY'$ are used with all four factors real. Their products need not be real, but distributivity gives

$$ZZ' = (XX' - YY') + i(XY' + YX').$$

The decomposition $Z = (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}$ is used with real $a, b, \mathbf{X}, \mathbf{Y}$, and primes are used for the second paravector's real parts. The real and imaginary scalar and vector parts of ZZ' are

$$\begin{aligned} \text{real scalar: } & aa' - bb' + \mathbf{X} \cdot \mathbf{X}' - \mathbf{Y} \cdot \mathbf{Y}', \\ \text{imaginary scalar: } & ab' + ba' + \mathbf{X} \cdot \mathbf{Y}' + \mathbf{Y} \cdot \mathbf{X}', \\ \text{real vector: } & a\mathbf{X}' - b\mathbf{Y}' + a'\mathbf{X} - b'\mathbf{Y} - \mathbf{X} \times \mathbf{Y}' - \mathbf{Y} \times \mathbf{X}', \\ \text{imaginary vector: } & a\mathbf{Y}' + b\mathbf{X}' + a'\mathbf{Y} + b'\mathbf{X} + \mathbf{X} \times \mathbf{X}' - \mathbf{Y} \times \mathbf{Y}'. \end{aligned}$$

Specializing (2.9) gives the quadratic products:

$$\begin{aligned} Z^2 &= (a^2 - b^2 + \mathbf{X}^2 - \mathbf{Y}^2) + 2i(ab + \mathbf{X} \cdot \mathbf{Y}) \\ &\quad + 2(a\mathbf{X} - b\mathbf{Y}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{Y} + b\mathbf{X}) \cdot \boldsymbol{\sigma}, \end{aligned} \tag{2.10}$$

$$\begin{aligned} ZZ^* &= (a^2 + b^2 - \mathbf{X}^2 - \mathbf{Y}^2) - 2(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{Y} - b\mathbf{X}) \cdot \boldsymbol{\sigma}, \\ Z^*Z &= (a^2 + b^2 - \mathbf{X}^2 - \mathbf{Y}^2) + 2(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{Y} - b\mathbf{X}) \cdot \boldsymbol{\sigma}, \end{aligned} \tag{2.11}$$

$$ZZ^H = a^2 + b^2 + \mathbf{X}^2 + \mathbf{Y}^2 + 2(a\mathbf{X} + b\mathbf{Y} + \mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \tag{2.12}$$

$$Z^H Z = a^2 + b^2 + \mathbf{X}^2 + \mathbf{Y}^2 + 2(a\mathbf{X} + b\mathbf{Y} - \mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \tag{2.13}$$

$$ZZ^{*H} = \det(Z) = a^2 - b^2 - \mathbf{X}^2 + \mathbf{Y}^2 + 2i(ab - \mathbf{X} \cdot \mathbf{Y}).$$

Equations (2.10) and (2.11) have values in $\mathbb{R} \oplus \mathbb{C}^3$ and show $ZZ^* = Z^*Z$ exactly when $\mathbf{X} \times \mathbf{Y} = 0$. Equations (2.12) and (2.13) take values in $\mathbb{R} \oplus \mathbb{R}^3$ and give the same condition for $ZZ^H = Z^H Z$.

2.5 Norm and determinant

For the complex coefficients $c = a + bi$ and $\mathbf{Z} = \mathbf{X} + i\mathbf{Y}$, the squared norms defined above are

$$\begin{aligned} \|c\|^2 &= a^2 + b^2, \\ \|\mathbf{Z}\|^2 &= \mathbf{X}^2 + \mathbf{Y}^2. \end{aligned}$$

For the squared norm (2.1), the quadratic products and cyclicity give the Hermitian form (2, Eq. (46))

$$\begin{aligned} \|Z\|^2 &= \text{sc}(Z^H Z) = \frac{1}{2} \text{tr}(Z^H Z) = \text{sc}(ZZ^H) \\ &= \text{sc}(Z^* Z^{*H}) = \text{sc}(Z^{*H} Z^*). \end{aligned} \tag{2.14}$$

For every complex $Z \in \mathbb{C} \oplus \mathbb{C}^3$, conjugating its coefficients gives

$$\det(Z^H) = (c^*)^2 - (\mathbf{Z}^*)^2 = \det(Z)^*.$$

Determinant multiplicativity therefore yields

$$\begin{aligned} \det(ZZ^H) &= \det(Z^H Z) \\ &= \det(Z) \det(Z)^* = \|\det(Z)\|^2 \geq 0. \end{aligned} \tag{2.15}$$

This also holds when Z is singular.

For complex coefficients, c^2 and $\mathbf{Z}^2$ generally differ from these squared norms. Also $\det(Z)$ may be complex. For a complex scalar $z = re^{i\theta}$, with real $r \geq 0$ and θ , scalar scaling gives

$$\begin{aligned} \det(zZ) &= r^2 e^{2i\theta} \det(Z), \\ \|zZ\|^2 &= \|z\|^2 \|Z\|^2 = r^2 \|Z\|^2. \end{aligned} \tag{2.16}$$

The determinant retains the scalar phase by (1.11); the norm loses it by (2.16).

3 Projectors and spectral functions

The general element is $Z \in \mathbb{C} \oplus \mathbb{C}^3$, with $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$. Eigenvalues and spectral parameters are complex scalars. A scalar analytic function $f : D \rightarrow \mathbb{C}$, with $D \subseteq \mathbb{C}$ open and containing the relevant eigenvalues, induces a paravector value $f(Z)$. Real and complex projector domains are distinguished below.

3.1 Quadratic identity and eigenvalues

Notation: $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$, with $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$. Since $Z - c = \mathbf{Z} \cdot \boldsymbol{\sigma}$ and $(\mathbf{Z} \cdot \boldsymbol{\sigma})^2 = \mathbf{Z}^2$,

$$\begin{aligned} (Z - c)^2 &= \mathbf{Z}^2, \\ Z^2 - 2cZ + (c^2 - \mathbf{Z}^2) &= 0. \end{aligned}$$

Using $\text{tr}(Z) = 2c$ and $\det(Z) = c^2 - \mathbf{Z}^2$ gives the trace-determinant identity

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (3.1)$$

An eigenvalue is a scalar $\lambda \in \mathbb{C}$ for which $Z - \lambda$ has no inverse. Thus $\det(Z - \lambda) = \lambda^2 - 2c\lambda + \det(Z) = 0$, the scalar quadratic corresponding to (3.1). Its roots are

$$\lambda_{\pm} = \frac{\text{tr}(Z) \pm \sqrt{\text{tr}(Z)^2 - 4\det(Z)}}{2} = c \pm \sqrt{\mathbf{Z}^2}.$$

Because $\mathbf{Z}$ is generally complex, $\mathbf{Z}^2 = \mathbf{Z} \cdot \mathbf{Z}$ and its square root may also be complex. The quadratic identity therefore factors as

$$(Z - \lambda_-)(Z - \lambda_+) = 0.$$

3.2 Projection operators

For a [real unit vector](#) $\mathbf{u}$, the paravector $\mathbf{u} \cdot \boldsymbol{\sigma} \in \{0\} \oplus \mathbb{R}^3$ is Hermitian and [self-inverse](#). Its two orthogonal projectors are defined by [\(3, p. 121\)](#)

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.2)$$

Here $\Pi_{\pm} \in \mathbb{R} \oplus \mathbb{R}^3$ with scalar part 1/2. Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$,

$$\begin{aligned} \Pi_{\pm}^2 &= \Pi_{\pm}, \\ \Pi_+ \Pi_- &= 0, \\ \Pi_+ + \Pi_- &= 1, \\ \Pi_{\pm}^{\text{H}} &= \Pi_{\pm}, \\ \Pi_{\pm}^* &= \Pi_{\mp}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} &= \pm \Pi_{\pm}. \end{aligned}$$

Thus $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. For a second [real unit vector](#) $\mathbf{u}'$,

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ [\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

The projections $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$ are defined in $\mathbb{R}^3$. Then

$$\begin{aligned}\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\pm} &= \pm(\mathbf{u} \cdot \mathbf{u}')\Pi_{\pm}, \\ \Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\mp} &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}.\end{aligned}$$

For $\Pi'_{\pm} = (1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})/2 \in \mathbb{R} \oplus \mathbb{R}^3$, all four products lie in $\mathbb{R} \oplus \mathbb{C}^3$:

$$\begin{aligned}\Pi_{\pm}\Pi'_{\pm} &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \pm \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ \Pi_{\pm}\Pi'_{\mp} &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \mp \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

3.3 Spectral rule

The decomposition $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is used, with $c \in \mathbb{C}$, $\mathbf{Z} \in \mathbb{C}^3$, and $\mathbf{Z}^2 \neq 0$. The complex scalars $z, \lambda_{\pm}$, complex vector $\mathbf{u}$, and projectors $\Pi_{\pm} \in \mathbb{R} \oplus \mathbb{C}^3$ are defined by

$$\begin{aligned}z &:= +\sqrt{\mathbf{Z}^2}, \\ \mathbf{u} &:= \mathbf{Z}/z, & Z &= c + z\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \lambda_{\pm} &:= c \pm z, & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).\end{aligned}$$

The root is the principal scalar root. Thus $\mathbf{u}$ is a [vector of square one](#); $\mathbf{u} \cdot \boldsymbol{\sigma}$ is [self-inverse](#). By squaring,

$$\begin{aligned}\Pi_{\pm}^2 &= \frac{1}{4}(1 \pm 2\mathbf{u} \cdot \boldsymbol{\sigma} + 1) = \Pi_{\pm}, \\ \Pi_+\Pi_- &= \frac{1}{4}(1 - (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = 0, \\ \Pi_+ + \Pi_- &= 1, \\ Z &= c(\Pi_+ + \Pi_-) + z(\Pi_+ - \Pi_-) = \lambda_+\Pi_+ + \lambda_-\Pi_-.\end{aligned}$$

Here $z, \lambda_{\pm} \in \mathbb{C}$, $\mathbf{u} \in \mathbb{C}^3$, and $Z, f(Z) \in \mathbb{C} \oplus \mathbb{C}^3$ for the analytic functions below. These projectors are Hermitian when $\mathbf{u}$ is real; in general they need not be. The same formulas apply directly to any given $Z = c + z\mathbf{u} \cdot \boldsymbol{\sigma}$, with $c, z \in \mathbb{C}$ and $\mathbf{u}^2 = 1$. If $z = 0$, $Z = c$ is scalar and $f(Z) = f(c)$. The complementary-projector identities give, for every nonnegative integer n ,

$$Z^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_- = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n)\mathbf{u} \cdot \boldsymbol{\sigma}.$$

For a scalar power series $f(z) = \sum_{n=0}^{\infty} c_n z^n$ converging at both eigenvalues, with $c_n \in \mathbb{C}$, inserting these powers into $f(Z) = \sum_n c_n Z^n$ gives the spectral rule [\(4, p. 3\)](#)

$$f(Z) = f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_-. \quad (3.3)$$

The same rule defines chosen analytic branches at the two eigenvalues. Only the complementary projector identities are used in the proof, so the result is valid for any decomposition with these identities.

Adjugation exchanges the eigenvalues, so

$$\begin{aligned}\text{adj}(Z) &= \lambda_-\Pi_+ + \lambda_+\Pi_-, \\ \det(Z) &= \lambda_+\lambda_-, \\ Z^{-1} &= \frac{\text{adj}(Z)}{\det(Z)} = \lambda_+^{-1}\Pi_+ + \lambda_-^{-1}\Pi_-.\end{aligned}$$

The inverse requires both eigenvalues to be nonzero. Independently choosing the signs of their principal square roots gives

$$\sqrt{Z} = \pm\sqrt{\lambda_+}\Pi_+ \pm \sqrt{\lambda_-}\Pi_-.$$

There are four distinct roots when both eigenvalues are nonzero, and two when one vanishes. Changing z to $-z$ exchanges the eigenvalue–projector pairs and leaves the function unchanged when its branch choices are exchanged with them.

3.4 Square roots and repeated eigenvalues

3.4.1 Real paravectors and direct square roots

For real $X = a + \mathbf{X} \cdot \boldsymbol{\sigma}$, with $a \in \mathbb{R}$, $\mathbf{X} \in \mathbb{R}^3$, and $\mathbf{X} \neq 0$, the spectral quantities above reduce to

$$z = \|\mathbf{X}\|, \quad \mathbf{u} = \mathbf{X}/\|\mathbf{X}\|, \quad \lambda_{\pm} = a \pm \|\mathbf{X}\|.$$

Here $X, \Pi_{\pm} \in \mathbb{R} \oplus \mathbb{R}^3$, $z, \lambda_{\pm} \in \mathbb{R}$, and $\mathbf{u} \in \mathbb{R}^3$. The projectors are now Hermitian; complex-valued functions or square roots may still give $f(X) \in \mathbb{C} \oplus \mathbb{C}^3$. In particular, for a [real unit vector](#) $\mathbf{u}$,

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\Pi_+ + f(-1)\Pi_- \\ &= \tfrac{1}{2}(f(1) + f(-1)) + \tfrac{1}{2}(f(1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm\Pi_+ \pm i\Pi_-. \end{aligned}$$

The two root signs are independent. For a fixed analytic branch, $f(X) \rightarrow f(a)$ as $\mathbf{X} \rightarrow 0$; independent root choices require the scalar case below.

A direct component derivation also determines all *real* roots. $Y = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$ is taken to be real, with $b \in \mathbb{R}$, $\mathbf{Y} \in \mathbb{R}^3$, and real roots $X = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ are sought such that $X^2 = Y$. By equating parts,

$$\begin{aligned} a^2 + \mathbf{X}^2 &= b, \\ 2a\mathbf{X} &= \mathbf{Y}. \end{aligned}$$

If $\mathbf{Y} \neq 0$, then $a \neq 0$ and $\mathbf{X} = \mathbf{Y}/(2a)$. By substitution,

$$\begin{aligned} a^4 - ba^2 + \tfrac{1}{4}\mathbf{Y}^2 &= 0, \\ a^2 &= \tfrac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{Y}^2} \right). \end{aligned} \tag{3.4}$$

For each nonnegative a^2 in (3.4), both scalar roots a are included, with $\mathbf{X} = \mathbf{Y}/(2a)$; then $\mathbf{X}^2 = b - a^2$. There are four real roots when $b > \|\mathbf{Y}\|$, two when $b = \|\mathbf{Y}\| > 0$, and none when $b < \|\mathbf{Y}\|$.

If $\mathbf{Y} = 0$, the equations instead give the scalar roots $X = \pm\sqrt{b}$ and the pure-vector roots $X = \mathbf{X} \cdot \boldsymbol{\sigma}$ with $\mathbf{X}^2 = b$ for $b > 0$. For $b = 0$ only $X = 0$ is real; for $b < 0$ there is no real root.

3.4.2 Coalescing eigenvalues

For complex $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ with $\mathbf{Z}^2 = 0$, $(\mathbf{Z} \cdot \boldsymbol{\sigma})^2 = 0$. A scalar function analytic at c has the terminating Taylor expansion (4, p. 4)

$$f(Z) = f(c) + f'(c)\mathbf{Z} \cdot \boldsymbol{\sigma}. \tag{3.5}$$

For $c \neq 0$, the chosen square root and logarithm are

$$\sqrt{Z} = r + \frac{\mathbf{Z} \cdot \boldsymbol{\sigma}}{2r},$$

$$\log(Z) = \log(c) + \frac{\mathbf{Z} \cdot \boldsymbol{\sigma}}{c}.$$

Here $r \in \mathbb{C}$ is a chosen scalar root of c ($r^2 = c$); a consistent logarithm branch is used. A nonzero $\mathbf{Z} \cdot \boldsymbol{\sigma}$ with $c = 0$ has no square root: such a root would have only zero eigenvalues, and its quadratic identity would force its square to vanish. It has no logarithm because it is not invertible. If $\mathbf{Z} \cdot \boldsymbol{\sigma} = 0$, a fixed analytic branch gives $f(c)$; non-scalar square roots of a scalar must instead be found from the component equations $a^2 + \mathbf{X}^2 = c$, $2a\mathbf{X} = 0$, now allowing $a \in \mathbb{C}$ and $\mathbf{X} \in \mathbb{C}^3$. Independent branch choices need not remain finite as distinct eigenvalues coalesce.

4 Exponentials and elementary functions

The complex-function formulas use $Z \in \mathbb{C} \oplus \mathbb{C}^3$, $c, z, \lambda_{\pm} \in \mathbb{C}$, $\mathbf{Z}, \mathbf{u} \in \mathbb{C}^3$, and $\Pi_{\pm} \in \mathbb{R} \oplus \mathbb{C}^3$. In the real specializations, $a, b, \theta \in \mathbb{R}$ and $\mathbf{X}, \mathbf{Y}, \mathbf{u} \in \mathbb{R}^3$. Exponentials, logarithms, and roots are algebra values in $\mathbb{C} \oplus \mathbb{C}^3$; branch conditions are stated with each formula.

Explicit functions and the factors used for rotations and boosts are obtained from the spectral rule. Algebraic functions are connected to geometric transformations through composition and logarithms.

4.1 Exponential: projection and series derivations

For complex $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ with $\mathbf{Z}^2 \neq 0$, the following notation is retained: $z = +\sqrt{\mathbf{Z}^2}$, the **vector of square one** $\mathbf{u} = \mathbf{Z}/z$, and $\lambda_{\pm} = c \pm z$. The spectral rule (3.3) gives

$$\begin{aligned} \exp(Z) &= \frac{1}{2}e^{c+z}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) + \frac{1}{2}e^{c-z}(1 - \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^c \left(\frac{e^z + e^{-z}}{2} + \frac{e^z - e^{-z}}{2} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Independently, c commutes with the vector part and $(z\mathbf{u} \cdot \boldsymbol{\sigma})^{2n} = z^{2n}$, $(z\mathbf{u} \cdot \boldsymbol{\sigma})^{2n+1} = z^{2n+1}\mathbf{u} \cdot \boldsymbol{\sigma}$. Separating the defining series therefore gives

$$\begin{aligned} \exp(Z) &= e^c \sum_{n=0}^{\infty} \frac{(z\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= e^c \left(\sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Both derivations give (2, Eq. (52))

$$\exp(Z) = e^c (\cosh(z) + \sinh(z) \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (4.1)$$

This is independent of the choice z or $-z$: $\cosh(z)$ is even in z , while both $\sinh(z)$ and $\mathbf{u}$ change sign. When $\mathbf{Z}^2 = 0$, (3.5) instead gives

$$\exp(c + \mathbf{Z} \cdot \boldsymbol{\sigma}) = e^c (1 + \mathbf{Z} \cdot \boldsymbol{\sigma}).$$

The defining series also gives

$$\begin{aligned}\exp(Z^*) &= \exp(Z)^*, \\ \exp(Z^H) &= \exp(Z)^H.\end{aligned}$$

For the real components $a, b, \mathbf{X}, \mathbf{Y}$ of $Z = (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}$, the following form is used:

$$z = +\sqrt{\mathbf{X}^2 - \mathbf{Y}^2 + 2i\mathbf{X} \cdot \mathbf{Y}}.$$

Substituting $\mathbf{u} = (\mathbf{X} + i\mathbf{Y})/z$ into (4.1) gives the component form

$$\exp(Z) = e^a (\cos(b) + i \sin(b)) \left(\cosh(z) + \frac{\sinh(z)}{z} (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma} \right).$$

Its real and imaginary specializations are

$$\exp(a + \mathbf{X} \cdot \boldsymbol{\sigma}) = e^a \left(\cosh(\|\mathbf{X}\|) + \frac{\sinh(\|\mathbf{X}\|)}{\|\mathbf{X}\|} \mathbf{X} \cdot \boldsymbol{\sigma} \right), \quad (4.2)$$

$$\exp(i(b + \mathbf{Y} \cdot \boldsymbol{\sigma})) = (\cos(b) + i \sin(b)) \left(\cos(\|\mathbf{Y}\|) + i \frac{\sin(\|\mathbf{Y}\|)}{\|\mathbf{Y}\|} \mathbf{Y} \cdot \boldsymbol{\sigma} \right). \quad (4.3)$$

Setting $a = 0$ in (4.2) or $b = 0$ in (4.3) gives the pure-vector exponentials. Every quotient uses its analytic limit at a zero argument.

4.2 Rotation and boost factors

For a general real exponent $a + \mathbf{X} \cdot \boldsymbol{\sigma}$, where $a \in \mathbb{R}$ and $\mathbf{X} \in \mathbb{R}^3$, (4.2) gives

$$\det(\exp(a + \mathbf{X} \cdot \boldsymbol{\sigma})) = e^{2a}. \quad (4.4)$$

Thus a real exponential has determinant one exactly when its scalar exponent is zero. Its value still has a nonzero scalar part, supplied by the hyperbolic cosine.

A real parameter $\theta \in \mathbb{R}$ is used, and $\mathbf{u} \in \mathbb{R}^3$ is taken to be a **real unit vector**, $\mathbf{u}^2 = 1$. The two special exponentials just derived define the complex rotation factor $R \in \mathbb{R} \oplus i\mathbb{R}^3$ and the real boost factor $L \in \mathbb{R} \oplus \mathbb{R}^3$ (1, Eq. (6)) (5, p. 23):

$$R(\theta, \mathbf{u}) := \exp\left(-i\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cos\left(\frac{\theta}{2}\right) - i \sin\left(\frac{\theta}{2}\right) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (4.5)$$

$$L(\theta, \mathbf{u}) := \exp\left(-\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cosh\left(\frac{\theta}{2}\right) - \sinh\left(\frac{\theta}{2}\right) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (4.6)$$

In (4.5) and (4.6), the parameter is a rotation angle for R and a rapidity for L ; both are real dimensionless numbers. Here these names specify algebraic objects. Their actions on a general paravector are derived in Section 5; their physical applications follow in Sections 7 and 18.

Because $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, reversing the parameter gives the inverses

$$\begin{aligned}R^{-1} &= R(-\theta, \mathbf{u}) = \cos\left(\frac{\theta}{2}\right) + i \sin\left(\frac{\theta}{2}\right) \mathbf{u} \cdot \boldsymbol{\sigma}, \\ L^{-1} &= L(-\theta, \mathbf{u}) = \cosh\left(\frac{\theta}{2}\right) + \sinh\left(\frac{\theta}{2}\right) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned} \quad (4.7)$$

Sigma conjugation changes both $\mathbf{i}$ and $\boldsymbol{\sigma}$ in sign; Hermitian conjugation changes $\mathbf{i}$ but fixes a real vector. Hence

$$\begin{aligned} R^* &= R, \\ R^H &= R^{-1}, \end{aligned} \tag{4.8}$$

$$\begin{aligned} L^* &= L^{-1}, \\ L^H &= L. \end{aligned} \tag{4.9}$$

Equation (4.8) makes R **unitary**; (4.9) makes L Hermitian, so $L^H L = L^2$. Their determinants and squared norms follow from the circular and hyperbolic identities:

$$\det(R) = \cos^2(\theta/2) + \sin^2(\theta/2) = 1, \tag{4.10}$$

$$\det(L) = \cosh^2(\theta/2) - \sinh^2(\theta/2) = 1, \tag{4.11}$$

$$\|R\|^2 = 1, \tag{4.12}$$

$$\|L\|^2 = \cosh(\theta). \tag{4.13}$$

Equations (4.10) and (4.11) make both **paravectors of determinant one**, so their adjugates equal their inverses. Equations (4.12) and (4.13) distinguish their squared norms; only R is unitary for nonzero θ . The half-angle gives the sign and period:

$$R(\theta + 2\pi, \mathbf{u}) = -R(\theta, \mathbf{u}), \tag{4.14}$$

$$R(\theta + 4\pi, \mathbf{u}) = +R(\theta, \mathbf{u}).$$

Gamma form. Notation: $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.6). Its real velocity parameter and Lorentz factor are defined by

$$\beta := \tanh(\theta),$$

$$\gamma := \cosh(\theta) = \frac{1}{\sqrt{1 - \beta^2}}.$$

Here $-1 < \beta < 1$, $\gamma \geq 1$, and $\gamma\beta = \sinh(\theta)$. By the half-angle identities,

$$\begin{aligned} \cosh\left(\frac{\theta}{2}\right) &= \sqrt{\frac{\gamma + 1}{2}}, \\ \sinh\left(\frac{\theta}{2}\right) &= \frac{\gamma\beta}{\sqrt{2(\gamma + 1)}}. \end{aligned}$$

Hence the boost factor itself is

$$L = \sqrt{\frac{\gamma + 1}{2}} \left(1 - \frac{\gamma\beta}{\gamma + 1} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \tag{4.15}$$

Squaring doubles the exponent, so the full-rapidity paravector is

$$L^2 = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma}). \tag{4.16}$$

Equation (4.16) has prefactor γ ; L has the half-angle prefactor in (4.15). Equivalently, for any real vector $\mathbf{X} \neq 0$, $\mathbf{v} = \tanh(\|\mathbf{X}\|) \mathbf{X}/\|\mathbf{X}\|$ and $\gamma = \cosh(\|\mathbf{X}\|)$ are defined. Then

$$\exp(\mathbf{X} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).$$

The zero-vector case follows by continuity. This last real paravector has **determinant one** and represents four-velocity in Section 6.

4.3 Cayley transform

$Y = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$ is taken to be real, with $b \in \mathbb{R}$, $\mathbf{Y} \in \mathbb{R}^3$, and $0 < \|\mathbf{Y}\| < b$. The [real unit vector](#) $\mathbf{u} = \mathbf{Y}/\|\mathbf{Y}\|$ is used, with $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. Its sigma conjugate $Y^* = b - \mathbf{Y} \cdot \boldsymbol{\sigma} = \text{adj}(Y)$ commutes with Y , so

$$\frac{Y}{Y^*} = \frac{Y^2}{\det(Y)} = \frac{b^2 + \mathbf{Y}^2 + 2b\mathbf{Y} \cdot \boldsymbol{\sigma}}{b^2 - \mathbf{Y}^2}.$$

The real parameter $\theta = 2 \tanh^{-1}(\|\mathbf{Y}\|/b)$ is defined. By the double-angle identities,

$$\begin{aligned} \cosh(\theta) &= \frac{b^2 + \mathbf{Y}^2}{b^2 - \mathbf{Y}^2}, \\ \sinh(\theta) &= \frac{2b\|\mathbf{Y}\|}{b^2 - \mathbf{Y}^2}. \end{aligned}$$

Thus the quotient is the full-rapidity exponential

$$\begin{aligned} \frac{b + \mathbf{Y} \cdot \boldsymbol{\sigma}}{b - \mathbf{Y} \cdot \boldsymbol{\sigma}} &= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \Pi_+ + e^{-\theta} \Pi_-. \end{aligned} \tag{4.17}$$

Equation (4.17) is $L(\theta, \mathbf{u})^{-2}$ from (4.16). Its real exponent is written using $\mathbf{X} = \theta \mathbf{u}$, while the Cayley vector is determined by

$$\mathbf{Y} = b \tanh\left(\frac{\|\mathbf{X}\|}{2}\right) \frac{\mathbf{X}}{\|\mathbf{X}\|}.$$

Thus $\|\mathbf{Y}\|/b = \tanh(\theta/2)$, the half-rapidity parameter. The corresponding real Lorentz factor and velocity vector are

$$\begin{aligned} \gamma &= \frac{b^2 + \mathbf{Y}^2}{b^2 - \mathbf{Y}^2}, \\ \mathbf{v} &= \frac{2b\mathbf{Y}}{b^2 + \mathbf{Y}^2}. \end{aligned}$$

They give the equivalent forms

$$\exp(\mathbf{X} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{b + \mathbf{Y} \cdot \boldsymbol{\sigma}}{b - \mathbf{Y} \cdot \boldsymbol{\sigma}} = \frac{Y}{Y^*}.$$

Choosing $b = 1$ normalizes the scalar part of Y to one; $\mathbf{Y} = 0$ follows by continuity.

4.4 Composing the factors

Notation: $L(\theta, \mathbf{u}) = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $R(\theta, \mathbf{u}) = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.6) and (4.5). For a fixed [real unit direction](#) $\mathbf{u}$ and real parameters θ, θ' , the parameters can be added because the exponents commute:

$$L(\theta, \mathbf{u})L(\theta', \mathbf{u}) = L(\theta + \theta', \mathbf{u}), \tag{4.18}$$

$$\begin{aligned} R(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta + \theta', \mathbf{u}), \\ L(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta', \mathbf{u})L(\theta, \mathbf{u}). \end{aligned}$$

For a common axis, the following complex parameter is defined: $z = -(\theta + i\theta')/2 \in \mathbb{C}$. Its exponential is

$$\exp(z\mathbf{u} \cdot \boldsymbol{\sigma}) = L(\theta, \mathbf{u})R(\theta', \mathbf{u}) = \cosh(z) + \sinh(z) \mathbf{u} \cdot \boldsymbol{\sigma}.$$

By expansion of the complex scalar coefficients,

$$\begin{aligned}\cosh(z) &= +\cosh\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta'}{2}\right) + i\sinh\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta'}{2}\right), \\ \sinh(z) &= -\sinh\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta'}{2}\right) - i\cosh\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta'}{2}\right).\end{aligned}$$

This factorization is valid when the real and imaginary vector parts of the exponent are parallel. For different [real unit directions](#) $\mathbf{u}, \mathbf{v}$, the vector product instead gives

$$\left[-\frac{1}{2}\mathbf{u} \cdot \boldsymbol{\sigma}, -\frac{1}{2}\mathbf{v} \cdot \boldsymbol{\sigma}\right] = \frac{i}{2}(\mathbf{u} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (4.19)$$

This commutator has the imaginary-vector form used in the exponent of a rotation; boosts along nonparallel directions cannot be interchanged.

4.5 Logarithm

For complex $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ with $\mathbf{Z}^2 \neq 0$, the following notation is retained: $z = +\sqrt{\mathbf{Z}^2}$, the [vector of square one](#) $\mathbf{u} = \mathbf{Z}/z$, $\lambda_{\pm} = c \pm z$, and $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. Both eigenvalues are assumed to be nonzero. For a nonzero complex scalar λ and any integer n , the scalar branches are

$$\log(\lambda) = \log(\|\lambda\|) + i(\arg(\lambda) + 2n\pi).$$

One branch is chosen for each eigenvalue. By the spectral rule,

$$\begin{aligned}\log(Z) &= \log(\lambda_+)\Pi_+ + \log(\lambda_-)\Pi_- \\ &= \frac{1}{2}(\log(\lambda_+) + \log(\lambda_-)) + \frac{1}{2}(\log(\lambda_+) - \log(\lambda_-))\mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned} \quad (4.20)$$

Exchanging z and $-z$ exchanges both eigenvalue–projector pairs and preserves the same chosen logarithm. Independent scalar branches give the different logarithms.

For a [real unit vector](#) $\mathbf{u}$, (4.20) uses eigenvalues 1 and -1 , giving, for integers n, m ,

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2n\pi i\Pi_+ + (2m+1)\pi i\Pi_-.$$

For real $Y = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$, with $b \in \mathbb{R}$, $\mathbf{Y} \in \mathbb{R}^3$, and $b > \|\mathbf{Y}\|$, both eigenvalues are positive. Combining their real logarithms yields

$$\log(Y) = \frac{1}{2}\log(b^2 - \mathbf{Y}^2) + \tanh^{-1}\left(\frac{\|\mathbf{Y}\|}{b}\right)\frac{\mathbf{Y}}{\|\mathbf{Y}\|} \cdot \boldsymbol{\sigma}.$$

The vector term vanishes at $\mathbf{Y} = 0$; the general coalescing case was treated above.

4.6 Real powers

For invertible $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$, choose an analytic logarithm branch at each distinct eigenvalue. Here $a \in \mathbb{R}$ denotes the exponent, and

$$\boxed{Z^a := \exp(a \log(Z))}. \quad (4.21)$$

For distinct eigenvalues, the spectral rule gives

$$Z^a = \lambda_+^a \Pi_+ + \lambda_-^a \Pi_-, \quad \lambda_{\pm}^a := \exp(a \log(\lambda_{\pm})).$$

For coalescing eigenvalues use (3.5). Integer powers agree with the algebraic powers; noninteger powers can depend on the logarithm branches.

For the following identities, use principal logarithms and assume neither eigenvalue lies on the nonpositive real axis (4, Sec. 4, Fact 1). The integer-power identities pass through the exponential series; uniqueness of the principal logarithm then gives

$$\log(Z^H) = \log(Z)^H, \quad \log(Z^*) = \log(Z)^*, \quad \log(\text{adj}(Z)) = \text{adj}(\log(Z)).$$

Since a is real, conjugation leaves it unchanged. For example,

$$(Z^a)^H = \exp(a \log(Z))^H = \exp(a \log(Z^H)) = (Z^H)^a.$$

Thus the combined-superscript convention extends to real powers:

$$\boxed{\begin{aligned} (Z^a)^H &= (Z^H)^a =: Z^{aH}, \\ (Z^a)^* &= (Z^*)^a =: Z^{a*}, \\ \text{adj}(Z^a) &= \text{adj}(Z)^a. \end{aligned}} \quad (4.22)$$

Using the same logarithm for opposite exponents gives $Z^a Z^{-a} = \exp(a \log(Z)) \exp(-a \log(Z)) = 1$, hence

$$Z^{-a} = (Z^a)^{-1}, \quad Z^{-aH} = (Z^{aH})^{-1}, \quad Z^{-a*} = (Z^{a*})^{-1}, \quad \text{adj}(Z^{-a}) = \text{adj}(Z^a)^{-1}.$$

Other logarithm branches obey the same rules when their transformed logarithms satisfy the three compatibility identities above. Integer powers require no branch restriction; negative integer powers require only invertibility.

Complex exponents. For $w \in \mathbb{C}$, define

$$Z^w := \exp(w \log(Z)).$$

With the same compatible logarithm branches, Hermitian and sigma conjugation also conjugate the exponent, whereas adjugation leaves it unchanged:

$$\boxed{\begin{aligned} (Z^w)^H &= (Z^H)^{w*}, \\ (Z^w)^* &= (Z^*)^{w*}, \\ \text{adj}(Z^w) &= \text{adj}(Z)^w. \end{aligned}} \quad (4.23)$$

For example,

$$(Z^w)^H = \exp(w^* \log(Z)^H) = \exp(w^* \log(Z^H)).$$

Adjugation retains w because it is complex-linear. Principal logarithms satisfy the compatibility conditions under the eigenvalue restriction above. Using the same logarithm also gives $Z^{-w} = (Z^w)^{-1}$. For complex exponents, parentheses are retained in $(Z^w)^*$ to distinguish conjugation of the whole power from $Z^{(w^*)}$, a power with the conjugated exponent.

5 Transformations and invariants

Here $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is a complex paravector: $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$, hence $Z \in \mathbb{C} \oplus \mathbb{C}^3$. Transformation factors $T, T' \in \mathbb{C} \oplus \mathbb{C}^3$ are constant. Physical rotation and boost factors have domains $R \in \mathbb{R} \oplus i\mathbb{R}^3$ and $L \in \mathbb{R} \oplus \mathbb{R}^3$; their angles or rapidities are $\theta \in \mathbb{R}$ and axes $\mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$. A distinction is made between the scalar part $\text{sc}(Z) = c = \frac{1}{2} \text{tr}(Z)$ from (1.9), the complex determinant $c^2 - \mathbf{Z}^2$ from (1.10), and the squared norm $\text{sc}(Z^H Z)$ from (2.14). Scalar part and trace carry exactly the same information; the scalar part is used below.

5.1 Similarity, determinant, and norm

For a complex invertible paravector T , similarity means $Z' = T Z T^{-1}$. By scalar-part cyclicity,

$$\text{sc}(Z') = \text{sc}(T Z T^{-1}) = \text{sc}(Z T^{-1} T) = \text{sc}(Z).$$

Thus trace is also preserved. By multiplicativity,

$$\det(Z') = \det(T) \det(Z) \det(T^{-1}) = \det(Z). \quad (5.1)$$

The opposite similarity $Z' = T^{-1} Z T$ has the same invariants. For independent complex factors T, T' , the law $Z' = T' Z T$ instead gives

$$\det(Z') = \det(T') \det(T) \det(Z).$$

This preserves the determinant for every Z exactly when $\det(T') \det(T) = 1$. This alone does not preserve scalar part, norm, or the real paravector subspace.

If the factor T is unitary, then $T^H = T^{-1}$ by (2.4). For $Z' = T Z T^H$, cancellation and scalar-part cyclicity give

$$\begin{aligned} \|Z'\|^2 &= \text{sc}(T Z^H T^H T Z T^H) \\ &= \text{sc}(Z^H Z T^H T) = \|Z\|^2. \end{aligned}$$

The norm is also preserved by $Z' = T Z$ and $Z' = Z T$. The congruence additionally preserves scalar part and determinant; a one-sided unitary product need not.

5.1.1 Determinant one is not Hermitian unitarity

Invertibility, determinant one, and Hermitian unitarity are distinct conditions. Invertibility means that a two-sided inverse exists, as in (2.3); determinant one means $\det(Z) = 1$; and Hermitian unitarity means $Z^H Z = 1$, as in (2.4). The phrase *unit determinant* means only that the determinant equals one. It does not mean unit norm or Hermitian unitarity.

Determinant one implies invertibility, but does not imply Hermitian unitarity. For comparison, a real paravector $X = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ with $a \in \mathbb{R}$, $\mathbf{X} \in \mathbb{R}^3$, and $a^2 - \mathbf{X}^2 = 1$ is considered. Then

$$X^* X = \det(X) = 1, \quad X^{-1} = X^* = a - \mathbf{X} \cdot \boldsymbol{\sigma}.$$

Thus determinant one implies invertibility. Hermitian conjugation fixes X , whereas sigma conjugation reverses its vector part, so

$$\begin{aligned} X^H X &= X^2 = a^2 + \mathbf{X}^2 + 2a\mathbf{X} \cdot \boldsymbol{\sigma}, \\ \|X\|^2 &= a^2 + \mathbf{X}^2 = 1 + 2\mathbf{X}^2. \end{aligned}$$

Both equal 1 only when $\mathbf{X} = 0$, in which case $X = \pm 1$. Choosing any nonzero real $\mathbf{X}$ and $a = \sqrt{1 + \mathbf{X}^2}$ therefore gives an invertible paravector of determinant one that is not unitary. These properties follow from the defining equations, not from the letter used for the paravector.

5.2 Real paravectors and the Lorentz metric

$X = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ is taken to be real, with $a \in \mathbb{R}$ and $\mathbf{X} \in \mathbb{R}^3$. The determinant and squared norm give different quadratic forms:

$$\begin{aligned}\det(X) &= a^2 - \mathbf{X}^2, \\ \|X\|^2 &= a^2 + \mathbf{X}^2.\end{aligned}$$

The first is the Lorentz quadratic form of signature $(+, -, -, -)$; the second is the positive sum of squared coefficients. A real-linear isometry of the first form must preserve the real subspace and its determinant. For a constant complex [invertible paravector](#) T with $\det(T) = 1$, the following Hermitian congruence is used ([1, Eq. \(17\)](#))

$$\boxed{X' = TXT^{\mathbf{H}}.} \tag{5.2}$$

Hermitian conjugation and determinant multiplication give

$$\begin{aligned}X'^{\mathbf{H}} &= TX^{\mathbf{H}}T^{\mathbf{H}} = X', \\ \det(X') &= \det(T) \det(X) \det(T^{\mathbf{H}}) = \det(X).\end{aligned}$$

For a second real paravector $Y = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$, with $b \in \mathbb{R}$ and $\mathbf{Y} \in \mathbb{R}^3$, the associated metric pairing is ([1, Eq. \(15\)](#))

$$\text{sc}(X^*Y) = ab - \mathbf{X} \cdot \mathbf{Y} = \frac{1}{2}(\det(X + Y) - \det(X) - \det(Y)). \tag{5.3}$$

Linearity and determinant invariance therefore give $\text{sc}(X'^*Y') = \text{sc}(X^*Y)$: the full metric is preserved. These maps preserve the positive cone as well. For $a \geq \|\mathbf{X}\|$, the eigenvalues $a \pm \|\mathbf{X}\|$ are nonnegative, so X has a real square root and

$$\text{sc}(X') = \|\sqrt{X} T^{\mathbf{H}}\|^2 \geq 0.$$

Together with determinant invariance, this keeps X' in the cone $a \geq \|\mathbf{X}\|$. These are the proper Lorentz transformations preserving that cone; T and $-T$ give the same map.

Why the scalar exponent vanishes. More generally, the congruence multiplies $\det(X)$ by $\|\det(T)\|^2$. For real $T = \exp(a + \mathbf{X} \cdot \boldsymbol{\sigma})$, ([4.4](#)) therefore gives

$$\det(TXT^{\mathbf{H}}) = e^{4a} \det(X).$$

A nonzero real scalar exponent dilates the metric; it is excluded by determinant invariance. An imaginary scalar exponent contributes a phase e^{ib} , $b \in \mathbb{R}$, which cancels between T and $T^{\mathbf{H}}$. Requiring $\det(T) = 1$ removes this continuous phase freedom. Thus a pure boost needs only a real vector exponent.

Why the squared norm is different. For $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real θ and real unit $\mathbf{u}$, applying (5.2) to $X = 1$ gives

$$\begin{aligned} X' &= L^2 = \cosh(\theta) - \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \det(X') &= 1, \\ \text{sc}(X') &= \cosh(\theta), \\ \|X'\|^2 &= \cosh(2\theta). \end{aligned}$$

A nonzero boost preserves the determinant while changing the scalar part and squared norm. In fact, a determinant-one congruence preserves the squared norm for every real X exactly when T is [unitary](#). Necessity follows by taking $X = 1$: the real, positive TT^H can have both determinant and norm one only if it is 1, which makes T unitary. Sufficiency was proved above. Normalizing L to $\|L\| = 1$ would change its determinant and lose the Lorentz metric.

Every factor separates into a boost and a rotation. For any complex T , the Hermitian product rule gives

$$(TT^H)^H = (T^H)^H T^H = TT^H.$$

By (1.7), a paravector equals its Hermitian conjugate exactly when its scalar and vector coefficients are real. Thus $TT^H \in \mathbb{R} \oplus \mathbb{R}^3$, without any determinant constraint. For a determinant-one complex T , i.e., $\det(T) = 1$, this real paravector also has determinant one and positive scalar part $\|T\|^2$. Both eigenvalues are therefore positive. Using the positive real square root,

$$L = \sqrt{TT^H}, \quad R = L^{-1}T, \quad T = LR.$$

The factors satisfy

$$\det(L) = \det(R) = 1, \quad L^H = L, \quad R^H = R^{-1},$$

with $L \in \mathbb{R} \oplus \mathbb{R}^3$ and $R \in \mathbb{R} \oplus i\mathbb{R}^3$. The real logarithm of L has zero scalar part by (4.4), so L is a real vector exponential. For R , the identity $R^H = R^{-1} = \text{adj}(R)$ makes its scalar coefficient real and its vector coefficient imaginary; determinant one then gives $R = \cos(\theta/2) - i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}$ for some real θ and real unit $\mathbf{u}$. Consequently,

$$X' = T X T^H = L R X R^H L, \quad \det(X') = \det(X).$$

The rotation is followed by the boost. Their component actions follow next.

5.3 Rotation action and Rodrigues' formula

Notation: the unitary factor $R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.5), with real angle θ and [real unit axis](#) $\mathbf{u}$. For the complex paravector $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$, the following Hermitian congruence is used (1, Eq. (5))

$$\boxed{Z' = R Z R^H}. \tag{5.4}$$

Since $R^H = R^{-1}$ by (4.8), this keeps $c' = c$. The vector product (2.8) gives

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{Z} \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2(\mathbf{u} \cdot \mathbf{Z})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{Z} \cdot \boldsymbol{\sigma}, \\ [\mathbf{Z} \cdot \boldsymbol{\sigma}, \mathbf{u} \cdot \boldsymbol{\sigma}] &= -2i(\mathbf{u} \times \mathbf{Z}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Inserting the sine and cosine forms of R and R^H therefore yields

$$\begin{aligned}\mathbf{Z}' \cdot \boldsymbol{\sigma} = & \cos^2\left(\frac{\theta}{2}\right) \mathbf{Z} \cdot \boldsymbol{\sigma} + 2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) (\mathbf{u} \times \mathbf{Z}) \cdot \boldsymbol{\sigma} \\ & + \sin^2\left(\frac{\theta}{2}\right) (2(\mathbf{u} \cdot \mathbf{Z})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{Z} \cdot \boldsymbol{\sigma}).\end{aligned}$$

Rodrigues' formula is obtained by collecting terms (5, p. 14):

$$\boxed{\mathbf{Z}' = \mathbf{Z} \cos(\theta) + (\mathbf{u} \times \mathbf{Z}) \sin(\theta) + (\mathbf{u} \cdot \mathbf{Z})(1 - \cos(\theta))\mathbf{u}.} \quad (5.5)$$

This is the active right-handed rotation through $+\theta$ about $\mathbf{u}$. Rotating the coordinate axes through $+\theta$ instead uses the inverse map. It preserves the scalar part, determinant, norm, and the real paravector subspace. For complex $\mathbf{Z}$, it rotates its real and imaginary parts by the same real rotation. The sign change in (4.14) affects both factors in (5.4), so Z' has period 2π .

5.4 Quaternions as paravectors

The quaternion multiplication rules are satisfied by $(-i\sigma_k)$:

$$(-i\sigma_1)^2 = (-i\sigma_2)^2 = (-i\sigma_3)^2 = (-i\sigma_1)(-i\sigma_2)(-i\sigma_3) = -1,$$

$$(-i\sigma_1)(-i\sigma_2) = (-i\sigma_3),$$

$$(-i\sigma_2)(-i\sigma_3) = (-i\sigma_1),$$

$$(-i\sigma_3)(-i\sigma_1) = (-i\sigma_2).$$

The sign is reversed when two distinct factors are exchanged. Here i remains the commuting complex scalar.

For $a \in \mathbb{R}$ and $\mathbf{Q} \in \mathbb{R}^3$, the quaternion representation is (3, p. 92)

$$\boxed{Q := a + \mathbf{Q} \cdot (-i\boldsymbol{\sigma}) = a - i\mathbf{Q} \cdot \boldsymbol{\sigma}.} \quad (5.6)$$

Here $\mathbf{Q} \cdot (-i\boldsymbol{\sigma}) = Q_1(-i\sigma_1) + Q_2(-i\sigma_2) + Q_3(-i\sigma_3)$. In the decomposition $Z = (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}$, $b = 0$, $\mathbf{X} = 0$, and $\mathbf{Y} = -\mathbf{Q}$ are selected. Thus $Q, Q' \in \mathbb{R} \oplus i\mathbb{R}^3$: a real scalar coefficient and purely imaginary vector coefficients. This four-real-component form is preserved under addition and multiplication. Indeed, for $Q' = a' + \mathbf{Q}' \cdot (-i\boldsymbol{\sigma})$, the paravector product becomes (6, p. 9)

$$QQ' = (aa' - \mathbf{Q} \cdot \mathbf{Q}') + (a\mathbf{Q}' + a'\mathbf{Q} + \mathbf{Q} \times \mathbf{Q}') \cdot (-i\boldsymbol{\sigma}). \quad (5.7)$$

Since $(-i\sigma_k)^H = -(-i\sigma_k)$ and $(-i\sigma_k)^* = (-i\sigma_k)$, quaternion conjugation is the existing Hermitian conjugation:

$$Q^H = a - \mathbf{Q} \cdot (-i\boldsymbol{\sigma}) = a + i\mathbf{Q} \cdot \boldsymbol{\sigma},$$

$$Q^* = Q, \quad \text{adj}(Q) = Q^H,$$

$$QQ^H = Q^H Q = a^2 + \mathbf{Q}^2 = \det(Q) = \|Q\|^2. \quad (5.8)$$

For nonzero Q , (5.8) gives $Q^{-1} = Q^H/(a^2 + \mathbf{Q}^2)$. Nonzero quaternions are normalized with

$$Q \leftarrow \frac{Q}{\sqrt{QQ^H}},$$

which, with $\|\mathbf{Q}\|^2 = \mathbf{Q}^2$ for real $\mathbf{Q}$, gives

$$QQ^H = a^2 + \|\mathbf{Q}\|^2 = 1.$$

Thus a unit quaternion has $Q^{-1} = Q^H$.

For rotation, $\mathbf{r} \in \mathbb{R}^3$ is represented by the pure quaternion $\mathbf{r} \cdot (-i\boldsymbol{\sigma}) = -i\mathbf{r} \cdot \boldsymbol{\sigma}$. For a unit quaternion $Q = a + \mathbf{Q} \cdot (-i\boldsymbol{\sigma})$, the quaternion sandwich is used:

$$\mathbf{r}' \cdot (-i\boldsymbol{\sigma}) = Q(\mathbf{r} \cdot (-i\boldsymbol{\sigma}))Q^H, \quad \mathbf{r}' \cdot \boldsymbol{\sigma} = Q(\mathbf{r} \cdot \boldsymbol{\sigma})Q^H. \quad (5.9)$$

The two forms agree because $-i$ commutes with every paravector. Applying (5.7) twice gives

$$\begin{aligned} Q(\mathbf{r} \cdot (-i\boldsymbol{\sigma})) &= -\mathbf{Q} \cdot \mathbf{r} + (a\mathbf{r} + \mathbf{Q} \times \mathbf{r}) \cdot (-i\boldsymbol{\sigma}), \\ Q(\mathbf{r} \cdot (-i\boldsymbol{\sigma}))Q^H &= ((a^2 - \mathbf{Q}^2)\mathbf{r} + 2(\mathbf{Q} \cdot \mathbf{r})\mathbf{Q} + 2a\mathbf{Q} \times \mathbf{r}) \cdot (-i\boldsymbol{\sigma}). \end{aligned}$$

The scalar part cancels. For a unit quaternion with $\mathbf{Q} \neq 0$, its real unit axis $\mathbf{u}$ and rotation angle θ satisfy

$$\mathbf{u} = \frac{\mathbf{Q}}{\|\mathbf{Q}\|}, \quad a = \cos\left(\frac{\theta}{2}\right), \quad \mathbf{Q} = \mathbf{u} \sin\left(\frac{\theta}{2}\right).$$

Thus the quaternion is the rotation factor

$$Q = \cos\left(\frac{\theta}{2}\right) - i\mathbf{u} \cdot \boldsymbol{\sigma} \sin\left(\frac{\theta}{2}\right) = R.$$

Then $a^2 - \mathbf{Q}^2 = \cos(\theta)$, $2a\mathbf{Q} = \mathbf{u} \sin(\theta)$, and $2(\mathbf{Q} \cdot \mathbf{r})\mathbf{Q} = (1 - \cos(\theta))(\mathbf{u} \cdot \mathbf{r})\mathbf{u}$. Consequently (5.9) recovers

$$\boxed{\mathbf{r}' = \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta) + (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u}.}$$

This is precisely Rodrigues' formula (5.5) for $\mathbf{Z} = \mathbf{r}$. Thus $R = Q$, and the quaternion sandwich is exactly $R(\mathbf{r} \cdot \boldsymbol{\sigma})R^H$ from (5.4). The reversed sandwich $Q^H(\mathbf{r} \cdot \boldsymbol{\sigma})Q$ gives the inverse rotation.

5.5 Boost action and its sign convention

Notation: $L = \exp(-\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.6); $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, $\beta = \tanh(\theta)$, and $\gamma = \cosh(\theta)$. The same congruence is used with $T = L$ and $L^H = L$ from (4.9) (5, Eq. (24)):

$$\boxed{Z' = LZL.} \quad (5.10)$$

The paravector $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$ may be complex; the congruence also preserves its real subspace. $\mathbf{Z} \in \mathbb{C}^3$ is resolved into parallel and perpendicular parts in $\mathbb{C}^3$:

$$\begin{aligned} \mathbf{Z}_{\parallel} &:= (\mathbf{u} \cdot \mathbf{Z})\mathbf{u}, \\ \mathbf{Z}_{\perp} &:= \mathbf{Z} - \mathbf{Z}_{\parallel}. \end{aligned}$$

The parallel term commutes with $\mathbf{u} \cdot \boldsymbol{\sigma}$; the perpendicular term anticommutes with it. Consequently,

$$\begin{aligned} L(c + \mathbf{Z}_{\parallel} \cdot \boldsymbol{\sigma})L &= (c + \mathbf{Z}_{\parallel} \cdot \boldsymbol{\sigma})L^2, \\ L(\mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma})L &= \mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Using $L^2 = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma})$ from (4.16), multiplication gives (7, Eq. (5.1))

$$\boxed{c' = \gamma(c - \beta \mathbf{u} \cdot \mathbf{Z}),} \quad (5.11)$$

$$\boxed{\mathbf{Z}' = \mathbf{Z}_\perp + \gamma(\mathbf{Z}_\parallel - \beta c \mathbf{u}).} \quad (5.12)$$

The equivalent longitudinal combinations expose the invariant directly:

$$c' \pm \mathbf{u} \cdot \mathbf{Z}' = e^{\mp \theta} (c \pm \mathbf{u} \cdot \mathbf{Z}). \quad (5.13)$$

Their product is $c^2 - (\mathbf{u} \cdot \mathbf{Z})^2$; the transverse part is unchanged. Thus (5.13) preserves $c^2 - \mathbf{Z}^2$ and, for real Z , the null cone $c^2 - \mathbf{Z}^2 = 0$.

The inverse map is $Z = L^{-1} Z' L^{-1}$, obtained by reversing θ . Generally $LZ \neq ZL$: the scalar and parallel parts commute with L , while a nonzero perpendicular part does not for a nonzero boost. For any transverse vector $\mathbf{Z}_\perp$ with $\mathbf{u} \cdot \mathbf{Z}_\perp = 0$,

$$L(\mathbf{Z}_\perp \cdot \boldsymbol{\sigma}) = (\mathbf{Z}_\perp \cdot \boldsymbol{\sigma}) L^{-1}, \quad L(\mathbf{Z}_\perp \cdot \boldsymbol{\sigma}) L = \mathbf{Z}_\perp \cdot \boldsymbol{\sigma}.$$

Thus the two-sided action leaves every transverse component unchanged. For real inputs, the right factor cancels the imaginary transverse term introduced by the left factor. A single left factor, even at full rapidity, cannot represent the boost of every real paravector.

Similarity action of the boost factor. For $Z' = LZL^{-1}$, the same commutation rules give $c' = c$ and

$$\mathbf{Z}' = \mathbf{Z}_\parallel + \gamma(\mathbf{Z}_\perp - i\beta \mathbf{u} \times \mathbf{Z}). \quad (5.14)$$

This preserves scalar part and determinant, but fixes $Z = 1$ and generally makes a real vector complex. It does not mix scalar and vector components as the congruence does in (5.11)–(5.12). In particular, similarity preserves the scalar-free subspace; a nonzero boost congruence gives a nonzero scalar part to $Z = z \mathbf{u} \cdot \boldsymbol{\sigma}$ for any nonzero $z \in \mathbb{C}$.

6 Space–time kinematics

Here $t, s, \theta \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, $\beta \in (-1, 1)$, and $\gamma \in \mathbb{R}_{\geq 1}$. Worldline values X , four-velocities U , and longitudinal rest directions N lie in $\mathbb{R} \oplus \mathbb{R}^3$. Along a path these are functions of a real time interval; $\mathbf{r}$ and $\mathbf{v}$ have codomain $\mathbb{R}^3$, while X and U have codomain $\mathbb{R} \oplus \mathbb{R}^3$. Constant frame factors retain $T \in \mathbb{C} \oplus \mathbb{C}^3$, $\det(T) = 1$, $R \in \mathbb{R} \oplus i\mathbb{R}^3$, and $L \in \mathbb{R} \oplus \mathbb{R}^3$.

In an inertial frame, an event is $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$. A massive particle is described by a future-directed time-like path $\mathbf{r}(t)$, with coordinate velocity $\mathbf{v} = \dot{\mathbf{r}}$ and $\mathbf{v}^2 < 1$. Coordinate time is denoted by t ; the carried clock's proper time is represented by s in the natural units used here. Dots denote total t derivatives of scalars, vectors, and paravectors, even when a path is parametrized by s .

6.1 Intervals, light cones, and proper time

The displacement and its Minkowski interval are (5, Eq. (7))

$$\begin{aligned} dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \\ \det(dX) &= dX \, dX^* = dt^2 - d\mathbf{r}^2. \end{aligned} \quad (6.1)$$

The interval is positive, zero, or negative for time-like, null, or space-like displacements. Along a massive particle's path it equals ds^2 ; a null path has no elapsed proper time. By (5.2), a constant complex invertible paravector T with $\det(T) = 1$ gives

$$dX' = T \, dX \, T^H, \quad \det(dX') = \det(dX).$$

For a rotation, $X' = R X R^H$ leaves $t' = t$ and acts on $\mathbf{r}$ by Rodrigues' formula (5.5). For real displacements, the product also gives

$$\begin{aligned} [dX, dX'] &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}, \\ (dX)^{-1} &= \frac{dX^*}{\det(dX)}, \quad \text{if } \det(dX) \neq 0. \end{aligned}$$

A null interval has no inverse.

Light cones. For a finite real event separation $\Delta X = \Delta t + \Delta\mathbf{r} \cdot \boldsymbol{\sigma}$, the eigenvalues are $\Delta t \pm \|\Delta\mathbf{r}\|$. Both are nonnegative precisely in the future causal cone:

$$\Delta t \geq \|\Delta\mathbf{r}\|. \quad (6.2)$$

The future null boundary has $\det(\Delta X) = 0$ and $\Delta t \geq 0$; for a real unit vector $\mathbf{u}$, its nonzero future rays are

$$\Delta X = \Delta t(1 + \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \Delta t > 0, \quad \mathbf{u}^2 = 1.$$

The Hermitian congruence (5.2) preserves nonnegativity of the eigenvalues and the determinant, hence the cone and its boundary. For a future time-like or null path,

$$\|\Delta\mathbf{r}\| = \left\| \int \mathbf{v} \, dt \right\| \leq \int \|\mathbf{v}\| \, dt \leq \Delta t.$$

Its endpoints therefore satisfy (6.2); a space-like separation cannot carry a causal signal (9, p. 118).

Coordinate time and proper time. The coordinate t is read from synchronized clocks of the chosen inertial frame. Along a future-directed time-like path, $dt > 0$ and the positive spacetime arc-length increment $ds = \sqrt{\det(dX)}$ are chosen. Proper time is commonly denoted by τ .

SI units: Proper time
Reverting to SI units, with s measured in metres and τ in seconds,
$ds = c \, d\tau.$

With $c = 1$, the same elapsed quantity is represented by s and τ ; only s is used for proper time below. This interpretation is fixed by the positive interval normalization, not by an arbitrary choice

of path parameter. Null paths have $ds = 0$ and no proper-time parametrization; space-like paths have no real proper-time increment (9, pp. 133–134).

Dividing by dt gives

$$\begin{aligned}\dot{s} &= \frac{ds}{dt} = \frac{\sqrt{dt^2 - d\mathbf{r}^2}}{dt} \\ &= \sqrt{\frac{dt^2 - d\mathbf{r}^2}{dt^2}} \\ &= \sqrt{1 - \left(\frac{d\mathbf{r}}{dt}\right)^2} \\ &= \sqrt{1 - \dot{\mathbf{r}}^2} = \sqrt{1 - \mathbf{v}^2}.\end{aligned}$$

The Lorentz factor and the derivative conversion are therefore (9, Eq. (7.18))

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} = \frac{dt}{ds}, \quad (6.3)$$

$$\frac{d}{ds} = \gamma \frac{d}{dt}. \quad (6.4)$$

The coordinate-time and proper-time rates coincide only at rest in the chosen frame. For an ideal clock this relation also holds during acceleration: its instantaneous rate depends on its velocity, and its total reading depends on the complete path. This is the clock hypothesis. Acceleration and jerk change that path and the coordinates assigned by a carried frame; they do not add an independent term to the local proper-time interval.

Elapsed time and clock comparison. Between coordinate times t_0 and t the clock records (9, Eq. (7.19))

$$s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} dt'. \quad (6.5)$$

Fixed, time-like separated events are denoted by $X_0 = X(s_0)$ and $X_1 = X(s_1)$, with X_1 to the future of X_0 . The decomposition $X_k = t_k + \mathbf{r}_k \cdot \boldsymbol{\sigma}$ is used for $k = 0, 1$. In their separation's rest frame, $\mathbf{r}_1 = \mathbf{r}_0$ and $t_1 - t_0 = \sqrt{\det(X_1 - X_0)}$. Since $\sqrt{1 - \mathbf{v}^2} \leq 1$, (6.5) gives (7, p. 122)

$$s_1 - s_0 \leq \sqrt{\det(X_1 - X_0)}. \quad (6.6)$$

Equality requires the straight inertial path. The determinant of a finite displacement therefore does not generally equal the squared elapsed proper time of an accelerated clock.

For a return trip lasting real coordinate time $T > 0$, velocities $+\beta\mathbf{u}$ and $-\beta\mathbf{u}$ are maintained for $T/2$ each, with **real unit vector** $\mathbf{u}$ and $0 < \beta < 1$. The ideal instantaneous reversal is the limit of a short smooth turn. By the traveling clock, the following interval is recorded:

$$s(T) - s(0) = T\sqrt{1 - \beta^2},$$

while T is recorded by the clock at rest. This is the twin-clock comparison (9, p. 121).

6.2 Four-velocity and worldlines

The real four-velocity is the proper-time tangent (5, Eq. (8)):

$$U := \frac{dX}{ds} = \frac{dt}{ds}(1 + \dot{\mathbf{r}} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.7)$$

Its normalization and conjugates are

$$U^*U = \det(U) = \frac{\det(dX)}{ds^2} = 1, \quad U^H = U, \quad U^{-1} = U^* = \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.8)$$

Determinant one is distinct from Hermitian unitarity:

$$U^H U = U^2 = 2\gamma^2 - 1 + 2\gamma^2 \mathbf{v} \cdot \boldsymbol{\sigma}, \quad \|U\|^2 = 2\gamma^2 - 1.$$

Both equal 1 only at rest. Thus four-velocity has determinant one in every inertial frame but is Hermitian unitary only in its rest frame. For a [real unit direction](#) $\mathbf{u}$, $\mathbf{v} = \beta \mathbf{u}$ and $\theta = \tanh^{-1}(\beta)$ are used. Equation (4.16) gives $U = L^{-2}$ for $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$: the direct boost $LU L = 1$ takes this tangent to rest.

Integrating four-velocity. Integrating $U = dX/ds$ between $s_0 < s_1$ gives

$$X(s_1) - X(s_0) = \int_{s_0}^{s_1} U(s) ds. \quad (6.9)$$

Its real components are

$$\begin{aligned} t(s_1) - t(s_0) &= \int_{s_0}^{s_1} \gamma(s) ds, \\ \mathbf{r}(s_1) - \mathbf{r}(s_0) &= \int_{s_0}^{s_1} \gamma(s) \mathbf{v}(s) ds. \end{aligned}$$

The path is determined by the velocity history; an endpoint boost alone is insufficient.

6.3 Fixed-axis motion and rapidity

For a fixed real unit vector $\mathbf{u}$, the signed velocity along the axis is $\beta = \tanh(\theta)$, with $\mathbf{v} = \beta \mathbf{u}$ and $\gamma = \cosh(\theta)$. With the projectors $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ from (3.2), the four-velocity and longitudinal rest direction have equivalent hyperbolic, exponential, and projected forms:

$$\begin{aligned} U &= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \Pi_+ + e^{-\theta} \Pi_-, \\ N &= \sinh(\theta) + \cosh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= (\mathbf{u} \cdot \boldsymbol{\sigma}) \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \Pi_+ - e^{-\theta} \Pi_-. \end{aligned}$$

Here $\det(U) = 1$, $\det(N) = -1$, and $\text{sc}(U^*N) = 0$. The pair is an orthonormal time and spatial basis in the plane of motion. Although rapidity is given as $\theta(s)$, its coordinate-time derivatives are those of the composition $\theta(s(t))$.

6.4 Constant velocity

Subscript 0 denotes the initial event $t = s = 0$. For $\mathbf{v}(t) = \mathbf{v}_0$, $\mathbf{v}_0^2 < 1$, and $\mathbf{r}(0) = \mathbf{r}_0$,

$$\gamma_0 = \frac{1}{\sqrt{1 - \mathbf{v}_0^2}}, \quad X(t) = t + (\mathbf{r}_0 + \mathbf{v}_0 t) \cdot \boldsymbol{\sigma}.$$

With $s(0) = 0$, the clock integral gives $s = t/\gamma_0$. Hence

$$X(s) = \mathbf{r}_0 \cdot \boldsymbol{\sigma} + sU_0, \quad U_0 = \gamma_0(1 + \mathbf{v}_0 \cdot \boldsymbol{\sigma}). \quad (6.10)$$

This straight time-like worldline has $\dot{\mathbf{v}} = 0$ and $\dot{U} = dU/ds = 0$. For any two events on this inertial path, $\Delta X = \Delta s U_0$ and $\det(U_0) = 1$ give

$$\det(\Delta X) = (\Delta s)^2 = (\Delta t)^2(1 - \mathbf{v}_0^2).$$

In the particle's rest frame, $\mathbf{v}_0 = 0$, so $U_0 = 1$ and $s = t$.

7 Inertial frames and particle dynamics

The coordinate and particle domains are $t, s, \theta \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, and $X, U, P \in \mathbb{R} \oplus \mathbb{R}^3$. For boosts, $\beta \in (-1, 1)$ and $\gamma \in \mathbb{R}_{\geq 1}$; $L \in \mathbb{R} \oplus \mathbb{R}^3$ and composite factors $T, T' \in \mathbb{C} \oplus \mathbb{C}^3$ have determinant one. Particle energy $E \in \mathbb{R}$, momentum $\mathbf{P} \in \mathbb{R}^3$, and invariant mass $m \in \mathbb{R}_{>0}$ are introduced below. Primed quantities have the same domains.

Constant Lorentz transformations are applied to coordinates, clocks, and particle quantities. The proper time s defined in Section 6.1 is invariant; t, t' are frame-dependent coordinate times. Dots continue to denote total t derivatives.

7.1 Coordinates, intervals, and light cones

Notation: $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real rapidity θ , **real unit vector** $\mathbf{u}$, $\beta = \tanh(\theta)$, and $\gamma = \cosh(\theta)$. Here γ is the factor for frame velocity $\beta \mathbf{u}$. The gamma form (4.15) and its square (4.16) are the same exponential at half and full rapidity. For real $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, coincident inertial origins are chosen at $t = t' = 0$. Equation (5.2) with $T = L$ gives $X' = LXL$: the primed frame moves at $+\beta \mathbf{u}$. Equations (5.11)–(5.12) become (7, Eq. (5.1))

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (7.1)$$

$$\mathbf{r}' = \mathbf{r}_\perp + \gamma(\mathbf{r}_\parallel - \beta t \mathbf{u}), \quad (7.2)$$

where $\mathbf{r}_\parallel = (\mathbf{u} \cdot \mathbf{r})\mathbf{u}$ and $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$. In particular, $\mathbf{r} = \beta t \mathbf{u}$ maps to $\mathbf{r}' = 0$, fixing the physical meaning of the sign. Reversing θ as in (4.7) gives $X = L^{-1}X'L^{-1}$.

Specializing (5.13) to the real event X gives (9, p. 154)

$$t' \pm \mathbf{u} \cdot \mathbf{r}' = e^{\mp \theta}(t \pm \mathbf{u} \cdot \mathbf{r}). \quad (7.3)$$

Multiplying the two signs in (7.3) and using $\mathbf{r}'_\perp = \mathbf{r}_\perp$ gives

$$t'^2 - \mathbf{r}'^2 = (t' + \mathbf{u} \cdot \mathbf{r}')(t' - \mathbf{u} \cdot \mathbf{r}') - \mathbf{r}'_\perp^2 = t^2 - \mathbf{r}^2.$$

Thus the exponential stretches one light-cone coordinate and compresses the other reciprocally. The same identity holds for event differences; in particular,

$$dt'^2 - d\mathbf{r}'^2 = \det(dX') = \det(dX) = dt^2 - d\mathbf{r}^2. \quad (7.4)$$

Equation (7.4) preserves the interval (6.1) and light cones in every direction. Future causal increments stay future because $dt' = \gamma(dt - \beta \mathbf{u} \cdot d\mathbf{r}) > 0$ when $dt > 0$ and $dt \geq \|d\mathbf{r}\|$ (9, p. 118).

Along a future-directed time-like path, $ds^2 = dt'^2 - d\mathbf{r}'^2$. With $\mathbf{v}' = d\mathbf{r}'/dt'$ and particle Lorentz factor $\gamma' = dt'/ds = (1 - \mathbf{v}'^2)^{-1/2}$, the squared four-velocity norm is

$$\begin{aligned} \left\| \frac{dX'}{ds} \right\|^2 &= \left(\frac{dt'}{ds} \right)^2 + \left(\frac{d\mathbf{r}'}{ds} \right)^2 \\ &= 2 \left(\frac{dt'}{ds} \right)^2 - 1 \\ &= 2\gamma'^2 - 1 = \frac{1 + \mathbf{v}'^2}{1 - \mathbf{v}'^2}. \end{aligned}$$

Likewise, in the unprimed frame,

$$\left\| \frac{dX}{ds} \right\|^2 = 2 \left(\frac{dt}{ds} \right)^2 - 1 = \frac{1 + \mathbf{v}^2}{1 - \mathbf{v}^2}.$$

Each norm equals 1 only at rest in its corresponding frame, whereas $\det(dX'/ds) = 1$ in every inertial frame. Proper time and causality are determined by the determinant; these Lorentz transformations would not be obtained by preserving the squared norm.

7.2 The same events in two frames

An event is one spacetime point; $(t, \mathbf{r})$ and $(t', \mathbf{r}')$ are two coordinate assignments to it. A displacement compares two neighboring events. Since the boost factor is constant, its action on X gives

$$\boxed{\begin{aligned} dt' &= \gamma(dt - \beta \mathbf{u} \cdot d\mathbf{r}), \\ d\mathbf{r}'_{\parallel} &= \gamma(d\mathbf{r}_{\parallel} - \beta \mathbf{u} dt), \\ d\mathbf{r}'_{\perp} &= d\mathbf{r}_{\perp}. \end{aligned}} \quad (7.5)$$

These equations also hold for finite separations $\Delta t, \Delta \mathbf{r}$. They compare the same pair of events, rather than different events chosen to have equal coordinate times. Clock readings and rod lengths require that choice to be specified.

For example, take $\beta = 3/5$, so $\gamma = 5/4$, and zero transverse separation. In any common natural unit, the pairs below are measured from the common origin:

Event pair	Δt	$\mathbf{u} \cdot \Delta \mathbf{r}$	$\Delta t'$	$\mathbf{u} \cdot \Delta \mathbf{r}'$	$\det(\Delta X)$
Unprimed clock at rest	1	0	5/4	-3/4	1
Primed clock at rest	5/4	3/4	1	0	1
Light along $+\mathbf{u}$	1	1	1/2	1/2	0
Simultaneous unprimed endpoints	0	1	-3/4	5/4	-1

Both clocks in the first two rows record unit proper time along their respective straight worldlines. The light ray stays null, and the last pair remains space-like even though its coordinate-time ordering changes. For an accelerated clock between two fixed events, its elapsed proper time is instead the path integral (6.5).

These are assigned coordinates, not the time at which a distant observer receives an image. Reception is a further event connected to emission by a future null displacement. This distinction remains necessary for an accelerating observer.

Figure 1 shows the four event pairs in the table. The same labels refer to the same events in both panels.

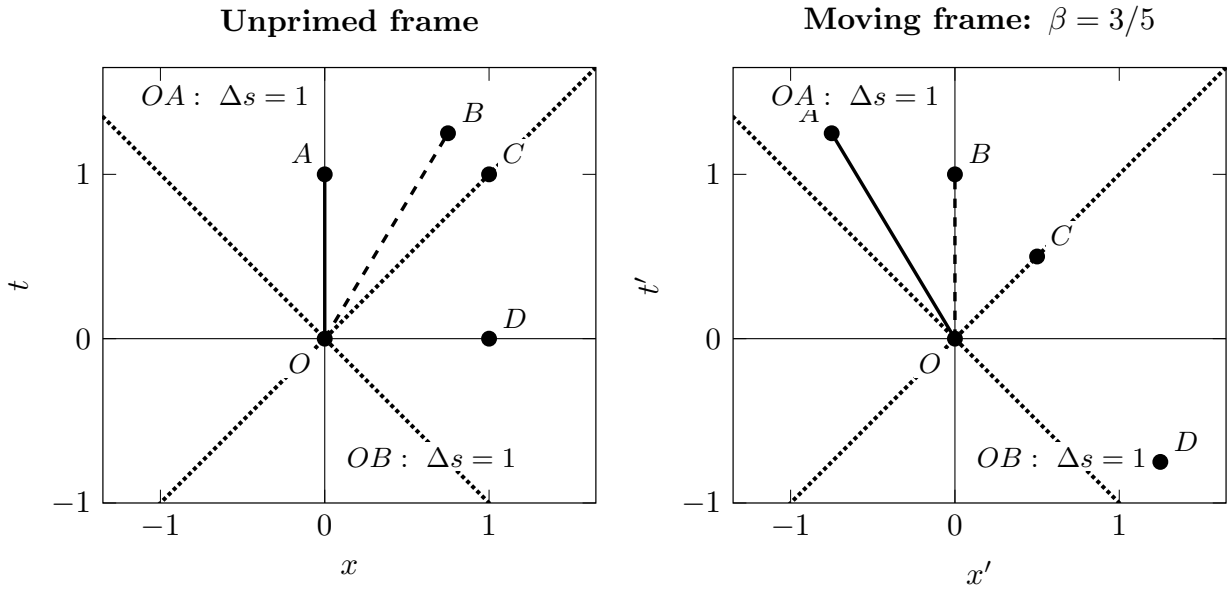


Figure 1: The same events in two inertial frames, with $x = \mathbf{u} \cdot \mathbf{r}$, $x' = \mathbf{u} \cdot \mathbf{r}'$, and $\beta = 3/5$. From the common event O , A lies on the unprimed rest clock (solid), B on the primed rest clock (dashed), C on the future light cone (dotted), and D at simultaneous unprimed separation. Both clocks accumulate $\Delta s = 1$, although their coordinate increments differ. The space-like pair OD is simultaneous only in the left panel. Equal spatial and temporal scales keep null lines at 45° .

SI units: The same event displacement

With t, t' in seconds, $\mathbf{r}, \mathbf{r}'$ in metres, and frame velocity $\beta c \mathbf{u}$, the final relations are

$$dt' = \gamma \left(dt - \frac{\beta}{c} \mathbf{u} \cdot d\mathbf{r} \right), \quad d\mathbf{r}'_{\parallel} = \gamma (d\mathbf{r}_{\parallel} - \beta c \mathbf{u} dt), \quad d\mathbf{r}'_{\perp} = d\mathbf{r}_{\perp},$$

$$ds^2 = c^2 d\tau^2 = c^2 dt^2 - d\mathbf{r}^2 = c^2 dt'^2 - d\mathbf{r}'^2.$$

The proper-time equality applies to time-like displacements; for light the interval is zero.

7.3 Spacetime increments and velocity

Notation: the real four-velocity $U = dX/ds$, an [invertible paravector](#) with $UU^{-1} = 1$, along a real world line $X(s)$. For constant L , differentiation gives

$$dX' = L dX L. \quad (7.6)$$

Because ds is invariant, $U' = LUL$. At a point where $U = L^{-2} = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma})$, this gives $U' = 1$ and $dX' = ds$: the boost takes the particle's instantaneous tangent to rest. For varying velocity, a different L is needed at each point; Section 11.4 supplies its derivative terms.

The particle velocity $\mathbf{v} = \dot{\mathbf{r}}$ is resolved along $\mathbf{u}$. Dividing the spatial increment by $dt' = \gamma(1 - \beta \mathbf{u} \cdot \mathbf{v}) dt$ gives (9, Eq. (7.22))

$$\mathbf{v}' = \frac{\mathbf{v}_\perp / \gamma + \mathbf{v}_\parallel - \beta \mathbf{u}}{1 - \beta \mathbf{u} \cdot \mathbf{v}}. \quad (7.7)$$

By interval invariance,

$$1 - \mathbf{v}'^2 = \frac{1 - \mathbf{v}^2}{\gamma^2(1 - \beta \mathbf{u} \cdot \mathbf{v})^2}. \quad (7.8)$$

Thus (7.7) preserves subluminal motion and light speed. Since L is constant, the same transformation applies to the integrated worldline in (6.9).

Rest, simultaneity, and the light-speed limit. For an unprimed clock at rest, $d\mathbf{r} = 0$, so $ds = dt$, $dt' = \gamma dt$, and $d\mathbf{r}' = -\gamma\beta \mathbf{u} dt$. For a primed clock at rest, $d\mathbf{r}' = 0$, the inverse calculation gives $ds = dt' = dt/\gamma$. Neither clock stops evolving in its own proper time. If $dt = 0$ and $d\mathbf{r} \neq 0$, the displacement is space-like; it is a simultaneous separation of events, not a massive observer's path. If both increments vanish, there is no displacement.

For massive motion, $ds/dt = \sqrt{1 - \mathbf{v}^2} \rightarrow 0$ as $\|\mathbf{v}\| \rightarrow 1$, while $dt/ds \rightarrow \infty$. This limiting comparison of clock rates does not define a frame that is “not moving through time.” At light speed $ds = 0$, so $U = dX/ds$ and a comoving massive clock cease to be defined. Likewise, a frame boost requires $|\beta| < 1$: its light-speed limit has unbounded rapidity and no finite invertible rest-frame transformation.

The null directions show exactly what remains regular. For light along $\pm \mathbf{u}$, substitute $d\mathbf{r} = \pm \mathbf{u} dt$ to obtain

$$dt' = e^{\mp\theta} dt > 0, \quad d\mathbf{r}' = \pm \mathbf{u} dt'.$$

Every finite boost preserves the light speed and future direction. Acceleration changes which boost the observer uses at successive events; it never turns a null tangent into a time-like one.

7.4 Clock rates and measured lengths

Time dilation and a transverse light clock. For a clock at rest in the primed frame, (7.1) uses $d\mathbf{r} = \beta \mathbf{u} dt$, hence (9, p. 119)

$$dt' = ds = \gamma(1 - \beta^2) dt = dt/\gamma. \quad (7.9)$$

Here $dt' = ds$; finite elapsed time follows from (6.5). To realize this clock, place two mirrors at primed rest separation $\ell > 0$ perpendicular to $\mathbf{u}$. A light pulse returns after $\Delta t' = 2\ell$. In the

unprimed frame each half-trip is a null displacement with transverse length ℓ and longitudinal displacement $\beta\Delta t/2$. Its determinant gives

$$(\Delta t/2)^2 - \ell^2 - (\beta\Delta t/2)^2 = 0, \quad \Delta t = 2\gamma\ell = \gamma\Delta t'.$$

Thus the clock relation follows from the light cone itself. For $\beta = 3/5$, a primed elapsed time of 1 corresponds to unprimed elapsed time $5/4$. The reciprocal clock statement compares a different worldline, as in the first two rows of Section 7.2.

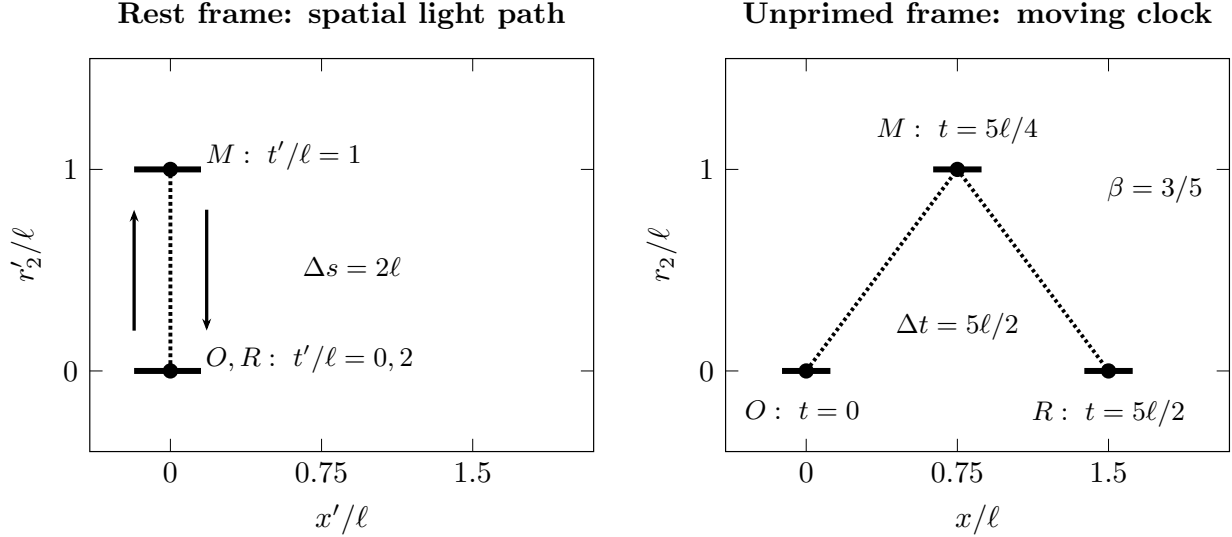


Figure 2: A transverse light clock with mirror separation ℓ and $\beta = 3/5$. These are spatial projections, with the time of each event written beside it: emission O , reflection M , and return R . Choose the first spatial axis along $\mathbf{u}$ and the second along the mirrors. In the clock's rest frame the pulse retraces its path; in the unprimed frame each dotted leg has length $5\ell/4$. Light speed is 1 in both frames, so the same round trip takes $\Delta t = 5\ell/2$ while the clock records $\Delta s = 2\ell$.

Length contraction and simultaneity. For a rod at rest in the unprimed frame, the fixed endpoint separation is denoted by $\Delta \mathbf{r} \in \mathbb{R}^3$. A simultaneous measurement in the primed frame requires $\Delta t' = 0$, so $\Delta t = \beta \mathbf{u} \cdot \Delta \mathbf{r}$. Substituting in (7.2) gives (9, p. 123)

$$\boxed{\Delta \mathbf{r}'_{\parallel} = \Delta \mathbf{r}_{\parallel}/\gamma, \quad \Delta \mathbf{r}'_{\perp} = \Delta \mathbf{r}_{\perp}.} \quad (7.10)$$

The transverse separation stays unchanged; thus a rod of any orientation has

$$\|\Delta \mathbf{r}'\|^2 = \|\Delta \mathbf{r}_{\perp}\|^2 + \|\Delta \mathbf{r}_{\parallel}\|^2/\gamma^2.$$

Longitudinal contraction (7.10) therefore compares endpoint events at equal primed times. Equal unprimed times instead give $\Delta t' = -\gamma\beta \mathbf{u} \cdot \Delta \mathbf{r}$; simultaneity is frame-dependent. The reciprocal statement follows by exchanging primes and reversing β . For a rod of rest length ℓ parallel to the motion, $\beta = 3/5$ therefore gives measured length $4\ell/5$. This is the separation of its endpoints measured simultaneously in the observing frame, rather than an optical image or a change of the rod's rest length.

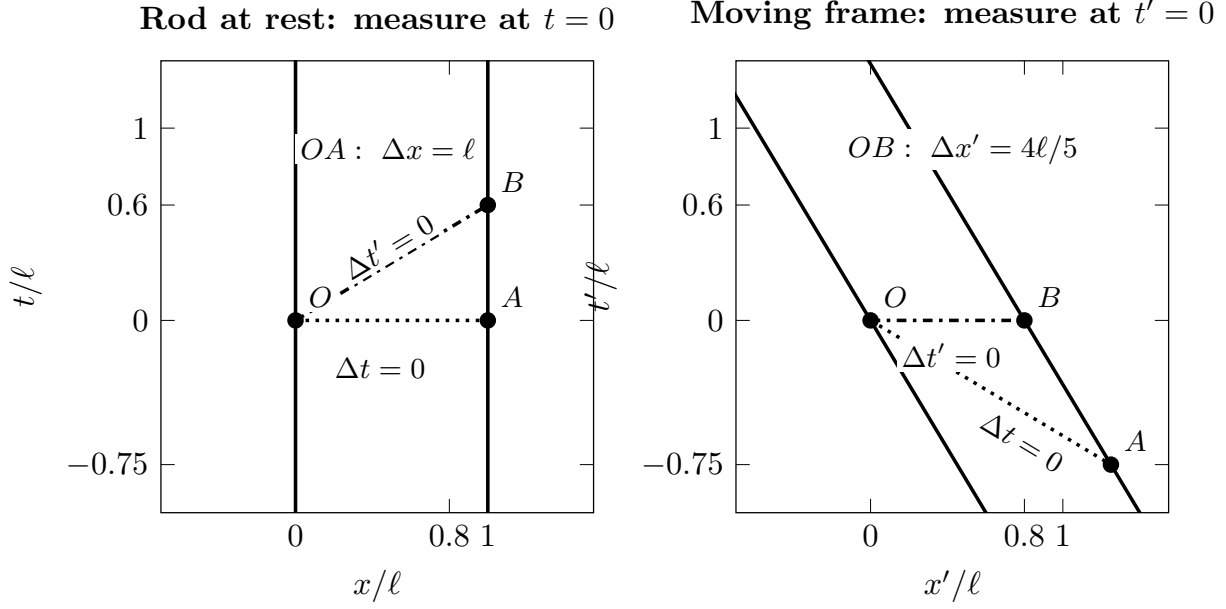


Figure 3: Length contraction selects simultaneous endpoint events. The solid lines are the two ends of a rod at unprimed rest, with $x = \mathbf{u} \cdot \mathbf{r}$ and rest length ℓ ; $\beta = 3/5$. The dotted segment OA measures its length at $t = 0$. The dash-dotted segment OB instead has $t' = 0$ and measures $\Delta x' = 4\ell/5$ in the moving frame. The event A has $t' = -3\ell/4$, so OA cannot be used as that frame's simultaneous length measurement. Labels identify the same three events in both panels.

SI units: Clock dilation and length contraction

For frame speed βc , $\gamma = (1 - \beta^2)^{-1/2}$:

$$\begin{aligned} d\tau &= dt' = \frac{dt}{\gamma} \quad \text{for a clock at primed rest,} \\ \Delta t' &= \frac{2\ell}{c}, \quad \Delta t = \frac{2\gamma\ell}{c} \quad \text{for the transverse light clock,} \\ \Delta t &= \frac{\beta}{c} \mathbf{u} \cdot \Delta \mathbf{r}, \quad \Delta \mathbf{r}'_{\parallel} = \frac{\Delta \mathbf{r}_{\parallel}}{\gamma}, \quad \Delta \mathbf{r}'_{\perp} = \Delta \mathbf{r}_{\perp} \quad (\Delta t' = 0). \end{aligned}$$

The last line measures a rod at unprimed rest using simultaneous primed endpoint events.

7.5 Composing frames

For constant determinant-one factors T, T' , applying the frame change T and then T' gives

$$Z'' = T'(TZT^H)T'^H = (T'T)Z(T'T)^H.$$

Thus the combined factor is $T'T$: the rightmost factor acts first.

Collinear boosts add rapidities by (4.18). For a second real rapidity θ' , the parameter $\beta' = \tanh(\theta')$

is defined, and $\gamma' = \cosh(\theta')$. Their combined parameters are

$$\begin{aligned}\beta'' &= \tanh(\theta + \theta') = \frac{\beta + \beta'}{1 + \beta\beta'}, \\ \gamma'' &= \gamma\gamma'(1 + \beta\beta').\end{aligned}$$

These are the collinear velocity and clock-factor composition laws. Noncollinear boosts instead have the rotation commutator (4.19), producing the Thomas–Wigner rotation.

7.6 Four-momentum and the mass shell

For constant invariant mass $m > 0$, the real four-momentum $P \in \mathbb{R} \oplus \mathbb{R}^3$ is defined by (9, Eq. (7.24))

$$P := mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}. \quad (7.11)$$

Energy and momentum are

$$\begin{aligned}E = \text{sc}(P) &= m \frac{dt}{ds} = \frac{m}{\sqrt{1 - \mathbf{v}^2}}, \\ \mathbf{P} = \text{vec}(P) &= E\mathbf{v}.\end{aligned}$$

Since $\det(U) = 1$, scalar scaling gives the mass shell (9, Eq. (7.30)):

$$E^2 - \mathbf{P}^2 = \det(P) = m^2, \quad E > 0. \quad (7.12)$$

SI units: Momentum

With dimensionless four-velocity U , mass m in kilograms, energy E in joules, and momentum $\mathbf{P}$ in kg m s^{-1} ,

$$\begin{aligned}P &= mcU = \frac{E}{c} + \mathbf{P} \cdot \boldsymbol{\sigma}, \\ \det(P) &= m^2 c^2, \quad E^2 = m^2 c^4 + c^2 \mathbf{P}^2.\end{aligned}$$

The scalar and vector coefficients of P now both have momentum units.

Returning to natural units, the inverse and speed follow directly:

$$P^{-1} = \frac{E - \mathbf{P} \cdot \boldsymbol{\sigma}}{m^2}, \quad \mathbf{v}^2 = \frac{\mathbf{P}^2}{E^2} = 1 - \frac{m^2}{E^2} < 1.$$

For fixed $m > 0$, approaching light speed requires unbounded energy. A massless particle has no proper-time four-velocity. Its momentum is defined directly by $P = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, with $E > 0$ and $E^2 = \mathbf{P}^2$.

For the real momentum $P = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, the same congruence gives (21, Eq. (17.12))

$$E' = \gamma(E - \beta \mathbf{u} \cdot \mathbf{P}), \quad (7.13)$$

$$\mathbf{P}' = \mathbf{P}_\perp + \gamma(\mathbf{P}_\parallel - \beta E \mathbf{u}). \quad (7.14)$$

Here the projections are along $\mathbf{u}$. The factor $\gamma = \cosh(\theta)$ belongs to the frame boost. Since $\mathbf{P} = E\mathbf{v}$, energy and coordinate-time increments acquire the same factor along a massive particle's path:

$$\frac{E'}{E} = \frac{dt'}{dt} = \gamma(1 - \beta \mathbf{u} \cdot \mathbf{v}). \quad (7.15)$$

This is also the time factor used to obtain the velocity transformation. Determinant invariance gives $E'^2 - \mathbf{P}'^2 = E^2 - \mathbf{P}^2 = m^2$ for a particle of rest mass $m > 0$. The energy increase is obtained directly from $P' = m$ in the rest frame by inversion of the boost:

$$P = L^{-1}mL^{-1} = m\gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma}).$$

Thus $E = m\gamma$ and $\mathbf{P} = m\gamma\beta\mathbf{u}$, while m remains unchanged (9, Eqs. (7.25), (7.28)).

Energy increase and the meaning of “mass dilation.” The same boost that compares clock rates gives

$$\boxed{\frac{E}{m} = \frac{dt}{ds} = \gamma, \quad E^2 - \mathbf{P}^2 = m^2.}$$

For $\beta = 3/5$, $E = 5m/4$ and $\mathbf{P} = 3m\mathbf{u}/4$; their difference of squares still gives m^2 . The older term *relativistic mass* denotes the energy E in natural units, or E/c^2 in SI. Calling its increase “mass dilation” must not be read as a change of the invariant m used here. The different longitudinal and transverse inertias follow in Section 8.2. For fixed $m > 0$, $E \rightarrow \infty$ as $|\beta| \rightarrow 1$; a massless particle is a separate null case, with no rest frame or proper-time four-velocity.

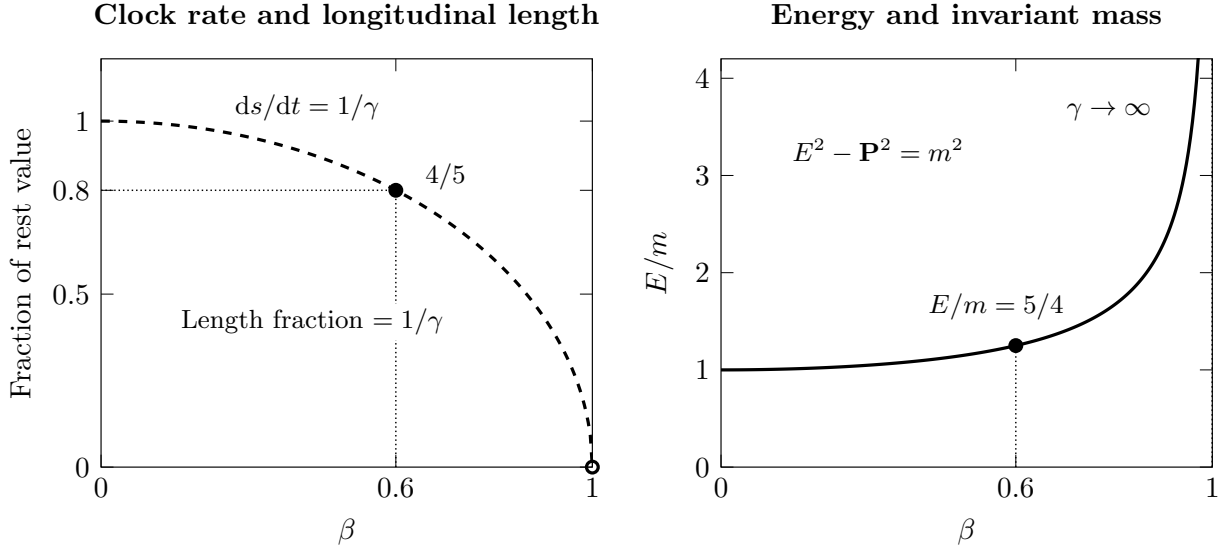


Figure 4: Rest and the light-speed limit. The dashed curve gives both a moving clock’s proper-time rate and a longitudinal rod’s measured length divided by its rest length. The solid curve gives a massive particle’s energy divided by its invariant mass. At $\beta = 3/5$ the first factor is $4/5$ and the second $5/4$. As $\beta \rightarrow 1$, the clock rate and length fraction tend to zero while the energy diverges. The open endpoint is a limit, not a massive observer at light speed; m remains invariant throughout.

SI units: Energy increase and invariant mass

With frame-relative particle speed βc and invariant mass m ,

$$E = \gamma mc^2, \quad \mathbf{P} = \gamma m\beta c\mathbf{u}, \quad \frac{E^2}{c^4} - \frac{\mathbf{P}^2}{c^2} = m^2.$$

Thus $E/c^2 = \gamma m$ is frame-dependent, while m is not.

7.7 Free-particle action

Notation: real $m > 0$, $\mathbf{v} = \dot{\mathbf{r}}$, particle factor $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, and $ds = dt/\gamma$. Here $L : \mathbb{R} \times \mathbb{R}^3 \times \{\mathbf{v} \in \mathbb{R}^3 : \mathbf{v}^2 < 1\} \rightarrow \mathbb{R}$ is the particle Lagrangian; the boost factor is $L \in \mathbb{R} \oplus \mathbb{R}^3$. $\partial_{\mathbf{v}}$ denotes its vector derivative with respect to the three independent velocity components, while $\partial_{\mathbf{r}}$ denotes the spatial gradient. The free action is the physical choice (7, Eqs. (5.38)–(5.39))

$$\int L dt = -m \int ds. \quad (7.16)$$

Using $ds = dt/\gamma$ gives the Lagrangian

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m/\gamma.$$

For endpoint-fixed real path variations $\delta \mathbf{r}(t)$, varying the Lagrangian and integrating by parts gives

$$\delta \int L dt = \int \left(\partial_{\mathbf{r}} L - \frac{d(\partial_{\mathbf{v}} L)}{dt} \right) \cdot \delta \mathbf{r} dt.$$

The Euler–Lagrange equation is obtained from stationarity for every variation. With Hamiltonian $H := \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \in \mathbb{R}$, the chain rule gives

$$\frac{d(\partial_{\mathbf{v}} L)}{dt} = \partial_{\mathbf{r}} L, \quad (7.17)$$

$$\dot{H} = \mathbf{v} \cdot \left(\frac{d(\partial_{\mathbf{v}} L)}{dt} - \partial_{\mathbf{r}} L \right) - \partial_t L = -\partial_t L. \quad (7.18)$$

By (7.17) and (7.18), spatial and temporal translation invariance conserve canonical momentum and energy. For the free action,

$$\begin{aligned} \partial_{\mathbf{v}} L &= m\gamma \mathbf{v} = \mathbf{P}, \\ H &= m\gamma \mathbf{v}^2 + m/\gamma = m\gamma = E. \end{aligned}$$

The notation $v = \|\mathbf{v}\| \ll 1$ is used. Here $O(v^n)$ denotes terms bounded by a constant times v^n as $v \rightarrow 0$. The binomial series gives

$$\begin{aligned} \sqrt{1 - v^2} &= 1 - \frac{1}{2}v^2 + O(v^4), \\ \gamma &= 1 + \frac{1}{2}v^2 + O(v^4), \\ L &= -m + \frac{1}{2}mv^2 + O(mv^4), \\ H &= m + \frac{1}{2}mv^2 + O(mv^4). \end{aligned}$$

By the mass shell, the exact kinetic energy above rest is

$$K := H - m = m(\gamma - 1) = \frac{\mathbf{P}^2}{E + m} \in \mathbb{R}_{\geq 0}.$$

This and both expansions recover $mv^2/2$ at low speed after removal of the constant rest term.

8 Acceleration

The real variables t, s, θ and real vectors $\mathbf{r}, \mathbf{v}, \mathbf{u} \in \mathbb{R}^3$ retain their kinematic meanings, with $\mathbf{v}^2 < 1$, $\mathbf{u}^2 = 1$, $\beta \in (-1, 1)$, and $\gamma \in \mathbb{R}_{\geq 1}$. X, U, N, P and their real-time derivatives lie in $\mathbb{R} \oplus \mathbb{R}^3$; $\dot{\mathbf{v}}, \dot{\mathbf{P}} \in \mathbb{R}^3$. Mass $m \in \mathbb{R}_{>0}$ and energy $E \in \mathbb{R}$ are scalars. Signed acceleration components are real; acceleration magnitudes are nonnegative real scalars.

Coordinate acceleration is $\dot{\mathbf{v}} = d\mathbf{v}/dt$; four-acceleration is dU/ds . Proper acceleration is measured by an accelerometer.

8.1 Four-acceleration and proper acceleration

Coordinate-time derivatives are first obtained:

$$\begin{aligned}\dot{\gamma} &= \gamma^3(\mathbf{v} \cdot \dot{\mathbf{v}}), \\ \dot{U} &= \dot{\gamma} + \frac{d(\gamma\mathbf{v})}{dt} \cdot \boldsymbol{\sigma}, \\ \frac{d(\gamma\mathbf{v})}{dt} &= \gamma\dot{\mathbf{v}} + \dot{\gamma}\mathbf{v}.\end{aligned}$$

For $\mathbf{v} \neq 0$, the coordinate acceleration is resolved into tangential (parallel to the velocity) and perpendicular components:

$$\dot{\mathbf{v}}_{\parallel} := \frac{(\mathbf{v} \cdot \dot{\mathbf{v}})\mathbf{v}}{\mathbf{v}^2}, \quad \dot{\mathbf{v}}_{\perp} := \dot{\mathbf{v}} - \dot{\mathbf{v}}_{\parallel}, \quad \dot{\mathbf{v}} = \dot{\mathbf{v}}_{\perp} + \dot{\mathbf{v}}_{\parallel}.$$

The projections of $\dot{\mathbf{v}}$ are indicated by these subscripts; at $\mathbf{v} = 0$, the unsplit formulas are used. Since $(\mathbf{v} \cdot \dot{\mathbf{v}})\mathbf{v} = \mathbf{v}^2\dot{\mathbf{v}}_{\parallel}$ and $1 + \gamma^2\mathbf{v}^2 = \gamma^2$, the spatial derivative becomes

$$\begin{aligned}\frac{d(\gamma\mathbf{v})}{dt} &= \gamma\dot{\mathbf{v}}_{\perp} + (\gamma + \gamma^3\mathbf{v}^2)\dot{\mathbf{v}}_{\parallel} \\ &= \gamma\dot{\mathbf{v}}_{\perp} + \gamma^3\dot{\mathbf{v}}_{\parallel}.\end{aligned}$$

Thus the full coordinate-time derivative of four-velocity is

$$\dot{U} = \gamma^3(\mathbf{v} \cdot \dot{\mathbf{v}}) + (\gamma\dot{\mathbf{v}}_{\perp} + \gamma^3\dot{\mathbf{v}}_{\parallel}) \cdot \boldsymbol{\sigma}. \quad (8.1)$$

Orthogonality and determinant. Differentiating $U^*U = 1$ gives

$$\dot{U}^*U + U^*\dot{U} = 0.$$

By taking scalar parts,

$$\text{sc}(U^*\dot{U}) = 0.$$

Thus U and $\dot{U}$ are Minkowski-orthogonal. Although $\dot{U}$ can have a scalar part, the product $U^*\dot{U}$ is scalar-free and belongs to $\{0\} \oplus \mathbb{C}^3$. Multiplication on the right by U^* also gives

$$\text{adj}(\dot{U}) = -U^*\dot{U}U^*.$$

By expansion of the commutator and the product,

$$\begin{aligned}[U, \dot{U}] &= 2i\gamma^2(\mathbf{v} \times \dot{\mathbf{v}}) \cdot \boldsymbol{\sigma}, \\ U^*\dot{U} &= \gamma^2(\dot{\mathbf{v}} - i\mathbf{v} \times \dot{\mathbf{v}}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The cross product depends only on acceleration perpendicular to $\mathbf{v}$. Since $\det(U^*) = 1$, multiplying by U^* preserves the determinant. The determinant of a scalar-free paravector is the negative of its square. Using $\dot{\mathbf{v}} \cdot (\mathbf{v} \times \dot{\mathbf{v}}) = 0$ therefore gives

$$\begin{aligned}\det(\dot{U}) &= -(U^* \dot{U})^2 \\ &= -\gamma^4 (\dot{\mathbf{v}}^2 - (\mathbf{v} \times \dot{\mathbf{v}})^2) \\ &= -\gamma^2 (\dot{\mathbf{v}}^2 + \gamma^2 (\mathbf{v} \cdot \dot{\mathbf{v}})^2).\end{aligned}$$

The last equality uses $(\mathbf{v} \times \dot{\mathbf{v}})^2 = \mathbf{v}^2 \dot{\mathbf{v}}^2 - (\mathbf{v} \cdot \dot{\mathbf{v}})^2$ and $1 - \mathbf{v}^2 = \gamma^{-2}$. Using the parallel and perpendicular components defined above, $\dot{\mathbf{v}}^2 = \dot{\mathbf{v}}_{\perp}^2 + \dot{\mathbf{v}}_{\parallel}^2$ and $(\mathbf{v} \cdot \dot{\mathbf{v}})^2 = \mathbf{v}^2 \dot{\mathbf{v}}_{\parallel}^2$ give

$$\det(\dot{U}) = -\gamma^2 \dot{\mathbf{v}}_{\perp}^2 - \gamma^4 \dot{\mathbf{v}}_{\parallel}^2.$$

Conversion to proper time. Multiplying $\dot{U}$ by γ gives (9, p. 141)

$$\frac{dU}{ds} = \gamma \dot{U} = \gamma^4 (\mathbf{v} \cdot \dot{\mathbf{v}}) + (\gamma^2 \dot{\mathbf{v}} + \gamma^4 (\mathbf{v} \cdot \dot{\mathbf{v}}) \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (8.2)$$

The determinant is quadratic, so

$$\begin{aligned}\det\left(\frac{dU}{ds}\right) &= \gamma^2 \det(\dot{U}) \\ &= -\gamma^4 \dot{\mathbf{v}}_{\perp}^2 - \gamma^6 \dot{\mathbf{v}}_{\parallel}^2.\end{aligned}$$

The same scaling gives $\text{sc}(U^* dU/ds) = 0$, so four-velocity and four-acceleration are Minkowski-orthogonal. Nonzero four-acceleration is space-like. Its negative determinant is the squared magnitude of the acceleration measured in the instantaneous rest frame, called the proper acceleration. At rest, $\gamma = 1$ and $dU/ds = \dot{\mathbf{v}} \cdot \boldsymbol{\sigma}$, so the determinant is $-\dot{\mathbf{v}}^2$ and the proper acceleration has magnitude $\|\dot{\mathbf{v}}\|$.

SI units: Proper acceleration

Here t, τ are measured in seconds, $s = c\tau$ and $\mathbf{r}$ in metres, and $\mathbf{v} = d\mathbf{r}/dt$ in metres per second. Dots denote derivatives with respect to this SI time. With $X = ct + \mathbf{r} \cdot \boldsymbol{\sigma}$,

$$\begin{aligned}\gamma &= \frac{1}{\sqrt{1 - \mathbf{v}^2/c^2}}, & U &= \frac{dX}{ds} = \gamma \left(1 + \frac{\mathbf{v}}{c} \cdot \boldsymbol{\sigma}\right), \\ c^2 \sqrt{-\det\left(\frac{dU}{ds}\right)} &= \sqrt{\gamma^6 \dot{\mathbf{v}}_{\parallel}^2 + \gamma^4 \dot{\mathbf{v}}_{\perp}^2}.\end{aligned}$$

Four-velocity remains dimensionless; both sides of the last equation give the accelerometer magnitude in m s^{-2} .

8.2 Force, work, and inertia

Here $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ is the particle Lorentz factor. For constant invariant mass $m > 0$, four-momentum is $P = mU$, so $\dot{P} = m\dot{U}$. Its scalar and spatial parts are $E = m\gamma$ and $\mathbf{P} = m\gamma\mathbf{v}$. The acceleration projections are parallel and perpendicular to $\mathbf{v}$, as in Section 8.1. Multiplying the spatial part of (8.1) by m therefore gives

$$\dot{\mathbf{P}} = m\gamma\dot{\mathbf{v}}_{\perp} + m\gamma^3\dot{\mathbf{v}}_{\parallel}. \quad (8.3)$$

Taking its dot product with $\mathbf{v}$ gives the work rate

$$\dot{E} = m\gamma^3\mathbf{v} \cdot \dot{\mathbf{v}} = \mathbf{v} \cdot \dot{\mathbf{P}}. \quad (8.4)$$

At rest the force is $m\dot{\mathbf{v}}$.

SI units: Force and energy
In the following SI formulas, t is in seconds, $\mathbf{v}$ in metres per second, and $\dot{\mathbf{P}}$ in newtons: $\frac{d\mathbf{P}}{dt} = m\gamma^3\dot{\mathbf{v}}_{\parallel} + m\gamma\dot{\mathbf{v}}_{\perp},$ $E = \gamma mc^2.$

In the natural units used elsewhere, invariant m is retained: the longitudinal and transverse inertia in (8.3) differ. For the Lorentz force, the corresponding acceleration components are derived in (12.22).

8.3 Differentiating a changing Lorentz frame

What the transformation guarantees. For constant $T \in \mathbb{C} \oplus \mathbb{C}^3$ with $\det(T) = 1$, $X' = TXT^H$ represents every Lorentz transformation that preserves spatial orientation and the future direction, with T and $-T$ representing the same transformation. These are changes of axes with a common spacetime origin. They preserve $\det(X)$ and, when applied to displacements, their time-like, null, or space-like character. An arbitrary event cannot be sent to an arbitrary other event: its interval from the origin is fixed. The speed limit constrains a physical trajectory's tangent, not the position of an event relative to a chosen origin.

A varying T supplies a sequence of local Lorentz frames. It does not by itself specify their moving origin or a simultaneity convention. In particular, differentiating TXT^H now includes derivatives of T ; the resulting coordinate differential is not simply $T dXT^H$. The moving-origin construction below supplies an actual noninertial coordinate chart and its interval.

Differentiate twice. Along a time-like curve use its proper time s , and let $T(s)$ be smooth. First distinguish the derivative of T from the derivative of its inverse. Differentiating $TT^{-1} = 1$ and then multiplying on the left by T^{-1} gives

$$\begin{aligned} (dT)T^{-1} + T d(T^{-1}) &= 0, \\ d(T^{-1}) &= -T^{-1}(dT)T^{-1}. \end{aligned}$$

The minus sign belongs to $d(T^{-1})$, the derivative of the inverse. Define the scalar-free frame generator as a *rate*:

$$\Omega := \frac{dT}{ds}T^{-1}, \quad \text{sc}(\Omega) = 0. \quad (8.5)$$

The zero scalar part follows from $d(\det(T))/ds = 2\text{sc}(\Omega) = 0$. Multiplying the definition of Ω on the right by T , and using the inverse-derivative identity separately, gives

$$\boxed{\frac{dT}{ds} = \Omega T, \quad \frac{d(T^{-1})}{ds} = -T^{-1}\Omega.}$$

Thus $(dT)T^{-1} = \Omega ds$, not Ω alone. Hermitian conjugation reverses the order of the factors, so

$$\Omega^H = T^{-H} \frac{dT^H}{ds}.$$

Now differentiate $X' = TXT^H$ directly:

$$\frac{dX'}{ds} = \frac{dT}{ds}XT^H + T \frac{dX}{ds}T^H + TX \frac{dT^H}{ds}.$$

There is no inverse in this product-rule expansion. To express the first and last terms using Ω and X' , insert identity factors without changing the multiplication order:

$$\begin{aligned} \frac{dT}{ds}XT^H &= \frac{dT}{ds} \underbrace{T^{-1}T}_1 XT^H = \Omega X', \\ TX \frac{dT^H}{ds} &= TX \underbrace{T^HT^{-H}}_1 \frac{dT^H}{ds} = X'\Omega^H. \end{aligned}$$

The inverse is therefore contained in Ω ; its adjacent T is contained in X' . The same factorization on the right gives $X'\Omega^H$. Hence

$$\frac{dX'}{ds} = T \frac{dX}{ds}T^H + \Omega X' + X'\Omega^H. \quad (8.6)$$

Equivalently, in differential form,

$$dX' = T(dX)T^H + (\Omega X' + X'\Omega^H) ds.$$

Applying the product rule again, without interchanging factors, gives

$$\begin{aligned} \frac{d^2X'}{ds^2} &= T \frac{d^2X}{ds^2}T^H + 2 \frac{dT}{ds} \frac{dX}{ds}T^H + 2T \frac{dX}{ds} \frac{dT^H}{ds} \\ &\quad + \frac{d^2T}{ds^2}XT^H + 2 \frac{dT}{ds}X \frac{dT^H}{ds} + TX \frac{d^2T^H}{ds^2}. \end{aligned}$$

Since $d^2T/ds^2 = (d\Omega/ds + \Omega^2)T$, substitution of (8.6) yields

$$\boxed{\begin{aligned} \frac{d^2X'}{ds^2} &= T \frac{d^2X}{ds^2} T^H + 2\Omega \frac{dX'}{ds} + 2 \frac{dX'}{ds} \Omega^H \\ &\quad + \left(\frac{d\Omega}{ds} - \Omega^2 \right) X' + X' \left(\frac{d\Omega^H}{ds} - \Omega^{2H} \right) - 2\Omega X' \Omega^H. \end{aligned}} \quad (8.7)$$

The terms linear in the first derivative, linear in the generator's derivative, and quadratic in the generator are the algebraic sources of the Coriolis, Euler, and centrifugal terms. The Hermitian part of Ω also includes changing boosts. To identify measured forces and coordinate accelerations, the origin and time parameter must first be specified. These product-rule identities hold for any common parameter; in the observer chart below that parameter is t' . Dots retain their earlier meaning of inertial t derivatives, so s and t' derivatives are written explicitly.

8.4 Event coordinates with changing motion and axes

Choose an arbitrary smooth future-directed time-like reference worldline $X(s)$ in flat spacetime. Choose a carried frame $T(s)$ with $TUT^H = 1$ and use $t' = s$ on the origin. The frame determines the spatial axes; the instantaneous rest planes determine simultaneity. The event map is

$$t + \mathbf{r} \cdot \boldsymbol{\sigma} = X(t') + T(t')^{-1}(\mathbf{r}' \cdot \boldsymbol{\sigma})T(t')^{-H}. \quad (8.8)$$

Again the worldline value on the right is the origin's event at t' , whereas the left side is the event being assigned coordinates. This construction is restricted to a one-to-one neighborhood of the worldline. Conversely, any smooth finite $T(t')$ with determinant one determines a future unit tangent

$$U = T^{-1}T^{-H}, \quad \det(U) = 1.$$

Integrating this tangent fixes the origin once its initial event is chosen. Thus arbitrary smooth time-like observer histories and choices of spatial axes are covered locally. No light-speed rest frame is included, and neither a global chart nor a new gravitational field follows from $\det(T) = 1$.

Differentiate using the inverse rule and $\Omega = (dT/dt')T^{-1}$. The moving origin contributes 1 in the local frame, while differentiating the two inverse factors gives

$$\boxed{T dXT^H = (1 - \Omega(\mathbf{r}' \cdot \boldsymbol{\sigma}) - (\mathbf{r}' \cdot \boldsymbol{\sigma})\Omega^H) dt' + d\mathbf{r}' \cdot \boldsymbol{\sigma}.} \quad (8.9)$$

Using the sigma product, its scalar and vector parts are

$$\begin{aligned} \text{sc}(T dXT^H) &= (1 - 2 \text{vec}(\text{re}(\Omega)) \cdot \mathbf{r}') dt', \\ \text{vec}(T dXT^H) &= d\mathbf{r}' + 2 \text{vec}(\text{im}(\Omega)) \times \mathbf{r}' dt'. \end{aligned}$$

These are orthonormal components in the carried axes; the coordinate differentials dt' , $d\mathbf{r}'$ alone are generally different. Determinant invariance now gives the interval without introducing a new metric rule:

$$\boxed{ds^2 = (1 - 2 \text{vec}(\text{re}(\Omega)) \cdot \mathbf{r}')^2 dt'^2 - \|d\mathbf{r}' + 2 \text{vec}(\text{im}(\Omega)) \times \mathbf{r}' dt'\|^2.} \quad (8.10)$$

The Hermitian part of Ω supplies the acceleration-dependent clock factor; its imaginary part supplies the spatial motion of rotating axes. A physical grid clock at fixed $\mathbf{r}'$ requires a positive future

time factor and a positive interval there. Thus a rotating grid also has a boundary where its fixed-coordinate worldlines would become null.

To name the two physical parts, let $\mathbf{a}, \boldsymbol{\omega} \in \mathbb{R}^3$ be the origin's proper-acceleration components and the carried axes' angular velocity relative to torque-free gyroscopes:

$$\boxed{\mathbf{a} = -2 \operatorname{vec}(\operatorname{re}(\Omega)), \quad \boldsymbol{\omega} = 2 \operatorname{vec}(\operatorname{im}(\Omega)), \quad \Omega = \frac{1}{2}(-\mathbf{a} + i\boldsymbol{\omega}) \cdot \boldsymbol{\sigma}.} \quad (8.11)$$

Here $\Omega = (dT/dt')T^{-1}$. Differentiating the rest condition $TUT^H = 1$ verifies the acceleration sign:

$$T \frac{dU}{dt'} T^H = -(\Omega + \Omega^H) = \mathbf{a} \cdot \boldsymbol{\sigma}.$$

For a rotation alone, the commutator with $i\boldsymbol{\omega} \cdot \boldsymbol{\sigma}/2$ changes vector components by $-\boldsymbol{\omega} \times \mathbf{r}'$; positive $\boldsymbol{\omega}$ therefore describes the rotation of the axes, rather than an active rotation of the vector. In these components the interval and the conditions for physical fixed grid clocks are

$$ds^2 = (1 + \mathbf{a} \cdot \mathbf{r}')^2 dt'^2 - \|\mathbf{dr}' + \boldsymbol{\omega} \times \mathbf{r}' dt'\|^2, \quad (8.12)$$

$$1 + \mathbf{a} \cdot \mathbf{r}' > 0, \quad \|\boldsymbol{\omega} \times \mathbf{r}'\| < 1 + \mathbf{a} \cdot \mathbf{r}'. \quad (8.13)$$

The first condition also keeps the local coordinate Jacobian nonsingular; global one-to-one behavior must still be checked. On a rotating grid, the carried orthonormal time direction need not be the rest direction of a distant fixed grid clock.

For axes carried by torque-free gyroscopes, $\Omega^H = \Omega$, the spatial rotation term vanishes. Equation (11.6) identifies $-2 \operatorname{vec}(\Omega)$ with the measured proper-acceleration vector; (11.7) identifies its proper-time derivative with proper jerk. The fixed-axis value $\Omega = -\frac{1}{2}(d\theta/dt')\mathbf{u} \cdot \boldsymbol{\sigma}$ recovers (8.25). This is the general nonrotating Fermi–Walker construction (23, Eqs. (6)–(9)).

At $\mathbf{r}' = 0$, every carried frame has the local interval $ds^2 = dt'^2 - d\mathbf{r}'^2$, regardless of its acceleration, rotation, or jerk. Away from the origin, light obeys

$$\left\| \frac{d\mathbf{r}'}{dt'} + 2 \operatorname{vec}(\operatorname{im}(\Omega)) \times \mathbf{r}' \right\| = 1 - 2 \operatorname{vec}(\operatorname{re}(\Omega)) \cdot \mathbf{r}'.$$

The apparent tilt and width of the coordinate cone can change, while its orthonormal components still satisfy the ordinary null condition. These coordinates describe flat spacetime because they came from an explicit coordinate map. Section 19 distinguishes this case from assigning a metric with actual curvature.

For a purely rotating frame with an inertial origin, $\operatorname{re}(\Omega) = 0$. At a chosen event, align the second spatial axis with the grid motion and write $2 \operatorname{vec}(\operatorname{im}(\Omega)) \times \mathbf{r}' = (0, \beta, 0)$, where $0 \leq \beta < 1$ is that grid point's inertial speed. In the first two spatial directions the preceding null condition becomes

$$\left(\frac{dr'_1}{dt'} \right)^2 + \left(\frac{dr'_2}{dt'} + \beta \right)^2 = 1, \quad \frac{ds}{dt'} = \sqrt{1 - \beta^2} \quad (\mathbf{r}' \text{ fixed}).$$

Figure 5 displays this shifted circle of coordinate light velocities and the corresponding clock rate.

SI units: A general carried frame
Use $s = c\tau$ for proper length and $\Omega = (dT/ds)T^{-1}$ in inverse metres. Chart time t' is the

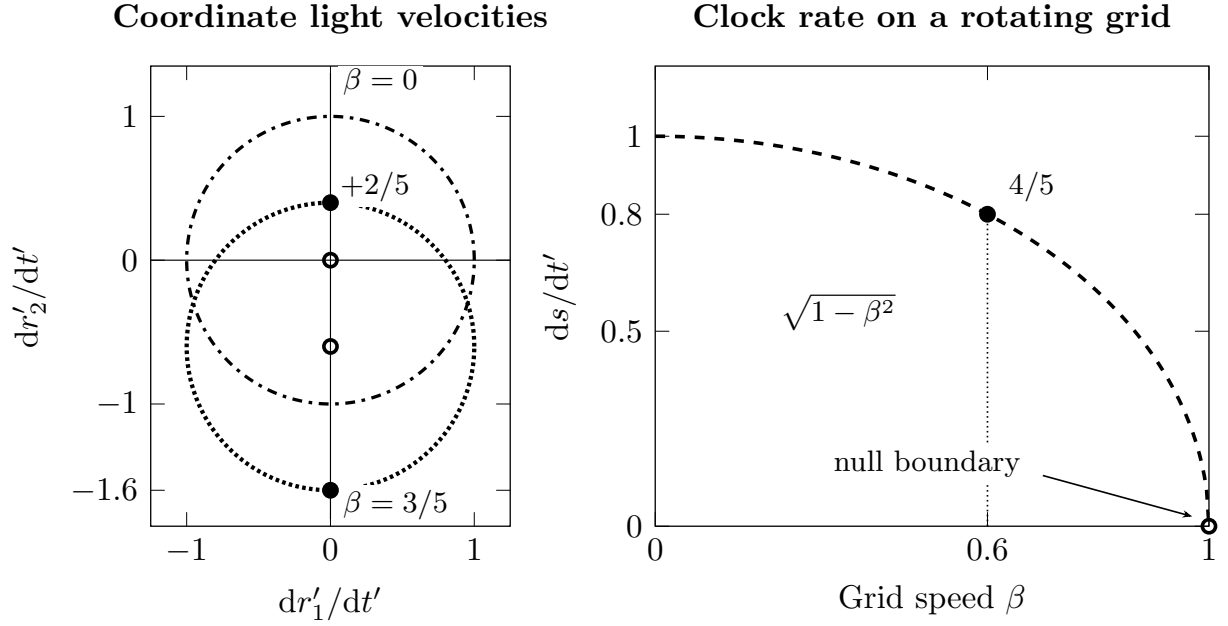


Figure 5: Rotating coordinates with an inertial origin. Left: cross sections of the local coordinate light cone, expressed as velocities in the rotation plane. The dash-dotted circle belongs to the origin ($\beta = 0$); the dotted circle to a grid point moving with $\beta = 3/5$ along the second local axis. The tangential coordinate light velocities there are $+2/5$ and $-8/5$; their different values reflect the rotating axes. Orthonormal measurements still give light speed 1. Right: a clock fixed to that grid point runs at $4/5$ of the origin's rate. The open endpoint $\beta = 1$ marks a null grid worldline, beyond the region occupied by physical rotating-grid clocks.

SI units: A general carried frame (continued)

origin's proper time in seconds. For another path,

$$c^2 d\tau^2 = (1 - 2 \text{vec}(\text{re}(\Omega)) \cdot \mathbf{r}')^2 c^2 dt'^2 - \|\mathbf{dr}' + 2c \text{vec}(\text{im}(\Omega)) \times \mathbf{r}' dt'\|^2.$$

The origin's proper acceleration has components $-2c^2 \text{vec}(\text{re}(\Omega))$. Setting the interval to zero gives the coordinate light cones; local orthonormal measurements give speed c . For the rotating example, β is the grid speed divided by c ; the fixed clock has $d\tau/dt' = \sqrt{1 - \beta^2}$ and the two tangential coordinate light velocities are $c(\pm 1 - \beta)$.

8.5 Inertial forces in the moving frame

A compact momentum law. Let $P = mU$ now refer to a test particle of constant mass $m > 0$, with its own proper time s . The observer's clock labels the chart time t' ; these times agree along the reference origin, but not generally along the test particle. The particle's momentum components in the carried orthonormal axes are $P' = TPT^H = E' + \mathbf{P}' \cdot \boldsymbol{\sigma}$. Differentiating this congruence gives the full force law in these axes:

$$\frac{dP'}{dt'} = \frac{ds}{dt'} T \frac{dP}{ds} T^H + \Omega P' + P' \Omega^H. \quad (8.14)$$

The first term is the physical four-force expressed in the carried axes and parametrized by t' . The remaining two terms arise entirely from changing those axes. Using (2.8), they expand as

$$\Omega P' + P' \Omega^H = -\mathbf{a} \cdot \mathbf{P}' - (E' \mathbf{a} + \boldsymbol{\omega} \times \mathbf{P}') \cdot \boldsymbol{\sigma}.$$

For a free particle, $dP/ds = 0$, so

$$\frac{dE'}{dt'} = -\mathbf{a} \cdot \mathbf{P}', \quad \frac{d\mathbf{P}'}{dt'} = -E' \mathbf{a} - \boldsymbol{\omega} \times \mathbf{P}'. \quad (8.15)$$

The acceleration of the frame changes the measured energy and momentum; rotation changes their spatial components. Direct substitution gives $d(E'^2 - \mathbf{P}'^2)/dt' = 0$, so the apparent force does not change the invariant mass. These momentum equations are particularly compact. The familiar factor of two in the Coriolis acceleration appears when momentum is related to the particle's changing position coordinates.

Differentiate the event map a second time. Write $\mathbf{v}' := d\mathbf{r}'/dt'$ for the test particle's *coordinate* velocity in this noninertial chart. It is not generally the velocity measured by a local observer. Equation (8.9) becomes, along the particle's path,

$$T \frac{dX}{dt'} T^H = 1 + \mathbf{a} \cdot \mathbf{r}' + (\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}') \cdot \boldsymbol{\sigma}. \quad (8.16)$$

Here X on the left is the particle's inertial event, rather than the reference origin's worldline value in (8.8). Taking its determinant fixes the clock conversion and hence the momentum:

$$\frac{ds}{dt'} = \sqrt{(1 + \mathbf{a} \cdot \mathbf{r}')^2 - \|\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}'\|^2}, \quad (8.17)$$

$$P' = m \frac{1 + \mathbf{a} \cdot \mathbf{r}' + (\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}') \cdot \boldsymbol{\sigma}}{ds/dt'}. \quad (8.18)$$

Positive radicand is the massive particle's speed restriction; zero gives the coordinate light cone. In particular, $\|\mathbf{v}'\| < 1$ is not the speed restriction away from the origin.

Differentiate (8.16), subtracting the two frame terms in (8.6). The scalar and vector parts of the transformed second derivative are

$$\text{sc}\left(T \frac{d^2 X}{dt'^2} T^H\right) = \frac{d\mathbf{a}}{dt'} \cdot \mathbf{r}' + 2\mathbf{a} \cdot \mathbf{v}' + \mathbf{a} \cdot (\boldsymbol{\omega} \times \mathbf{r}'), \quad (8.19)$$

$$\begin{aligned} \text{vec}\left(T \frac{d^2 X}{dt'^2} T^H\right) &= \frac{d\mathbf{v}'}{dt'} + 2\boldsymbol{\omega} \times \mathbf{v}' + \frac{d\boldsymbol{\omega}}{dt'} \times \mathbf{r}' \\ &\quad + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') + (1 + \mathbf{a} \cdot \mathbf{r}')\mathbf{a}. \end{aligned} \quad (8.20)$$

One copy of $\boldsymbol{\omega} \times \mathbf{v}'$ comes from differentiating the position-dependent term in (8.16); the other comes from differentiating the carried axes. This gives the Coriolis factor of two directly from the product rule.

For a free massive particle, $d^2 X/ds^2 = 0$. Chart time t' is generally not an affine parameter along that particle, so the second derivative with respect to t' need not vanish. The chain rule gives

$$\frac{d^2 X}{dt'^2} = \frac{d^2 s/dt'^2}{ds/dt'} \frac{dX}{dt'}.$$

Its transformed scalar part determines the common proportionality factor. Substituting it into the vector part yields the exact free coordinate acceleration:

$$\begin{aligned} \frac{d\mathbf{v}'}{dt'} &= -(1 + \mathbf{a} \cdot \mathbf{r}')\mathbf{a} - 2\boldsymbol{\omega} \times \mathbf{v}' - \frac{d\boldsymbol{\omega}}{dt'} \times \mathbf{r}' \\ &\quad - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \\ &\quad + \frac{\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}'}{1 + \mathbf{a} \cdot \mathbf{r}'} \left(\frac{d\mathbf{a}}{dt'} \cdot \mathbf{r}' + 2\mathbf{a} \cdot \mathbf{v}' + \mathbf{a} \cdot (\boldsymbol{\omega} \times \mathbf{r}') \right). \end{aligned} \quad (8.21)$$

This agrees with the standard flat-spacetime Fermi-coordinate result (24, Eq. (20)), derived here by paravector multiplication and ordinary differentiation. The last line is the relativistic correction from the time component and the use of the observer's clock. Dropping it would incorrectly treat t' as the test particle's affine time.

If the particle has a nonzero physical four-acceleration, add to the right side of (8.21) the term

$$\left(\frac{ds}{dt'}\right)^2 \left(\text{vec}\left(T \frac{dU}{ds} T^H\right) - \frac{\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}'}{1 + \mathbf{a} \cdot \mathbf{r}'} \text{sc}\left(T \frac{dU}{ds} T^H\right) \right).$$

This follows from the same chain rule before setting $dU/ds = 0$, and is equivalent to the momentum law (8.14).

Which terms are the fictitious forces? The free coordinate acceleration contains the following terms:

Coordinate-acceleration term	Origin
$-(1 + \mathbf{a} \cdot \mathbf{r}')\mathbf{a}$	Translational acceleration of the frame
$-2\boldsymbol{\omega} \times \mathbf{v}'$	Coriolis
$-(d\boldsymbol{\omega}/dt') \times \mathbf{r}'$	Euler
$-\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}')$	Centrifugal

These are apparent accelerations of a particle with zero physical four-force. In the Newtonian limit, multiplying them by m gives the usual fictitious forces. At relativistic speeds the force law is (8.14), with the clock and velocity conversion in (8.18); a physical force is not simply $m d\mathbf{v}'/dt'$. The centrifugal term points outward from the rotation axis. A particle held fixed on that rotating grid needs a real inward centripetal force; centripetal and centrifugal terms are not two additional fictitious forces.

Checks and limiting cases. For a constant frame with an inertial origin, $\Omega = 0$ and hence $\mathbf{a} = \boldsymbol{\omega} = 0$; free motion has constant $\mathbf{v}'$. For a purely rotating frame, $\mathbf{a} = 0$, and the relativistic last line of (8.21) vanishes exactly. The remaining Coriolis, Euler, and centrifugal coordinate accelerations have their classical forms even for relativistic particle speeds. For a time-independent rotation and a particle held at fixed $\mathbf{r}'$, (8.14) requires a real spatial force $E'\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}')$ per unit t' , the usual relativistic centripetal force.

For constant proper acceleration along $\mathbf{u}$ without rotation, put $\mathbf{a} = \dot{v}_0\mathbf{u}$ and $\mathbf{v}' = v'\mathbf{u}$, and use $x' = \mathbf{u} \cdot \mathbf{r}'$. Then

$$\frac{dv'}{dt'} = -\dot{v}_0(1 + \dot{v}_0x') + \frac{2\dot{v}_0v'^2}{1 + \dot{v}_0x'}. \quad (8.22)$$

At $x' = 0$ this is $-\dot{v}_0(1 - 2v'^2)$; the sign change at $|v'| = 1/\sqrt{2}$ is a coordinate effect, not a new physical force. For the null rays $v' = \pm(1 + \dot{v}_0x')$, the same equation gives $dv'/dt' = \dot{v}_0(1 + \dot{v}_0x')$, agreeing with direct differentiation of the light paths in (8.33). For light the derivation uses an affine null parameter in place of s ; it does not define a photon proper-time four-velocity.

Thus the single complex generator Ω packages six real frame rates: three accelerations and three rotations. Its derivative supplies their changes, and its ordered products supply their couplings. All of the inertial terms vanish from the physical free-particle law $dP/ds = 0$; they describe changing coordinates and axes in flat spacetime, rather than additional interactions.

SI units: Relativistic inertial acceleration

Here t' is the origin's proper time in seconds, $\mathbf{r}'$ is in metres, $\mathbf{v}' = d\mathbf{r}'/dt'$ is in metres per second, $\mathbf{a}$ is the origin's proper acceleration in m s^{-2} , and $\boldsymbol{\omega}$ is the axes' angular velocity in s^{-1} . With the proper-length convention $\Omega = (dT/ds)T^{-1}$ along the origin,

$$\mathbf{a} = -2c^2 \text{vec}(\text{re}(\Omega)), \quad \boldsymbol{\omega} = 2c \text{vec}(\text{im}(\Omega)).$$

SI units: Relativistic inertial acceleration (continued)

The exact free coordinate acceleration is

$$\begin{aligned} \frac{d\mathbf{v}'}{dt'} = & - \left(1 + \frac{\mathbf{a} \cdot \mathbf{r}'}{c^2} \right) \mathbf{a} - 2\boldsymbol{\omega} \times \mathbf{v}' - \frac{d\boldsymbol{\omega}}{dt'} \times \mathbf{r}' - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \\ & + \frac{\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}'}{c^2 + \mathbf{a} \cdot \mathbf{r}'} \left(\frac{d\mathbf{a}}{dt'} \cdot \mathbf{r}' + 2\mathbf{a} \cdot \mathbf{v}' + \mathbf{a} \cdot (\boldsymbol{\omega} \times \mathbf{r}') \right). \end{aligned}$$

For fixed dimensional motion as $c \rightarrow \infty$, the last line vanishes and the translational term becomes $-\mathbf{a}$. The four classical inertial accelerations are recovered. For a free particle the momentum law is

$$\frac{dE'}{dt'} = -\mathbf{a} \cdot \mathbf{P}', \quad \frac{d\mathbf{P}'}{dt'} = -\frac{E'}{c^2} \mathbf{a} - \boldsymbol{\omega} \times \mathbf{P}'.$$

The clock and speed condition is

$$\left(\frac{d\tau}{dt'} \right)^2 = \left(1 + \frac{\mathbf{a} \cdot \mathbf{r}'}{c^2} \right)^2 - \frac{\|\mathbf{v}' + \boldsymbol{\omega} \times \mathbf{r}'\|^2}{c^2} > 0$$

for a massive particle; equality gives a null tangent.

8.6 Fixed-axis acceleration

For the orthonormal pair U, N in Section 6.3, the first derivatives exchange the two directions:

$$\dot{U} = \dot{\theta}N, \quad \dot{N} = \dot{\theta}U, \quad \det(\dot{U}) = -\dot{\theta}^2.$$

Differentiating $\beta = \tanh(\theta)$ gives

$$\dot{\beta} = \gamma^{-2} \dot{\theta}.$$

Since $\dot{s} = 1/\gamma$, the chain rule gives

$$\dot{\theta} = \frac{d\theta}{ds} \dot{s}, \quad \frac{d\theta}{ds} = \gamma \dot{\theta} = \gamma^3 \dot{\beta}.$$

These are ordinary derivatives of the same rapidity along the path. The signed proper acceleration is $d\theta/ds$. By scaling the coordinate derivatives,

$$\frac{dU}{ds} = \frac{d\theta}{ds} N = \gamma \dot{\theta} N, \quad \frac{dN}{ds} = \gamma \dot{\theta} U, \quad \det \left(\frac{dU}{ds} \right) = -\gamma^2 \dot{\theta}^2.$$

The proper-acceleration magnitude is therefore $\gamma|\dot{\theta}|$.

8.7 Events in an accelerating frame

The general chart of Section 8.4 can now be specialized to fixed-axis motion. An accelerated observer has an instantaneous rest frame at each event, but assigning coordinates to distant events also requires a rule for simultaneity and spatial axes. Here spacetime is flat, the motion is along the fixed axis $\mathbf{u}$, and the carried axes do not rotate. Ideal clocks are assumed to measure proper time. The construction below uses the observer's instantaneous rest planes for simultaneity, as in Fermi–Walker coordinates (23, Eqs. (4)–(8)). It is a local coordinate choice, not a new gravitational field.

Let $X(s)$ be the reference observer's worldline. Label each rest plane by that observer's clock, $t' = s$, and give its points the spatial coordinates $\mathbf{r}'$. Using the already defined orthonormal directions U, N , the same event has inertial coordinates determined by

$$\boxed{t + \mathbf{r} \cdot \boldsymbol{\sigma} = X(t') + (\mathbf{u} \cdot \mathbf{r}')N(t') + \mathbf{r}'_{\perp} \cdot \boldsymbol{\sigma}.} \quad (8.23)$$

The notation $X(t')$ on the right denotes the reference worldline evaluated at its proper time t' , not the distant event on the left. Its separation from that reference event is orthogonal to $U(t')$; this is precisely the chosen simultaneity condition. At $\mathbf{r}' = 0$ the map reduces to the observer's worldline. Along the origin, $dX/dt' = U$ and $dN/dt' = (d\theta/dt')U$. Differentiating the full map, including its moving spatial direction, gives

$$\begin{aligned} dX &= \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'}\right) U dt' + N \mathbf{u} \cdot d\mathbf{r}' + d\mathbf{r}'_{\perp} \cdot \boldsymbol{\sigma}, \\ dt &= \cosh(\theta) \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'}\right) dt' + \sinh(\theta) \mathbf{u} \cdot d\mathbf{r}', \\ \mathbf{u} \cdot d\mathbf{r} &= \sinh(\theta) \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'}\right) dt' + \cosh(\theta) \mathbf{u} \cdot d\mathbf{r}'. \end{aligned}$$

The inverse comparison of the same neighboring events is therefore

$$\boxed{\begin{aligned} \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'}\right) dt' &= \gamma(dt - \beta \mathbf{u} \cdot d\mathbf{r}), \\ d\mathbf{r}'_{\parallel} &= \gamma(d\mathbf{r}_{\parallel} - \beta \mathbf{u} dt), \\ d\mathbf{r}'_{\perp} &= d\mathbf{r}_{\perp}. \end{aligned}} \quad (8.24)$$

Here $\beta = \tanh(\theta(t'))$ and $\gamma = \cosh(\theta(t'))$ belong to the origin's instantaneous velocity. At $\mathbf{r}' = 0$ the ordinary boost of increments is recovered. Away from the origin the additional factor comes from the changing rest planes. Merely substituting a time-dependent velocity into the finite inertial formula $X' = LXL$ would omit both the moving origin and these derivative terms.

Indeed, $LUL = 1$ and $LNL = \mathbf{u} \cdot \boldsymbol{\sigma}$ give

$$L dXL = \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'}\right) dt' + d\mathbf{r}' \cdot \boldsymbol{\sigma}.$$

These are local orthonormal components of the displacement. They differ from the coordinate paravector $dt' + d\mathbf{r}' \cdot \boldsymbol{\sigma}$ away from the origin. Taking the determinant gives

$$\boxed{ds^2 = \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'}\right)^2 dt'^2 - d\mathbf{r}'^2.} \quad (8.25)$$

In this interval formula ds is the proper time of whichever path is being measured; it equals dt' along the origin, but not generally along other paths. The chart is used on a one-to-one region where $1 + (\mathbf{u} \cdot \mathbf{r}') d\theta/dt' > 0$. At zero this coordinate map becomes singular, rather than creating an observer whose clock has stopped.

Clocks, lengths, and light cones. At a fixed grid point $\mathbf{r}'$, the clock rate relative to the origin is

$$\boxed{\frac{ds}{dt'} = 1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'} \quad (d\mathbf{r}' = 0).}$$

Thus simultaneous rest in the accelerated coordinates does not mean equal clock rates. At an instant when the origin is also at inertial rest, $\theta = 0$, its clock has $dt = dt'$, but distant fixed grid points still have the displayed position-dependent rate. On a slice $dt' = 0$, proper spatial distance is $\|\mathbf{dr}'\|$. Keeping such distances fixed requires different signed proper accelerations at different grid points:

$$\frac{d\theta}{ds} = \frac{d\theta/dt'}{1 + (\mathbf{u} \cdot \mathbf{r}') d\theta/dt'}.$$

Here s belongs to the fixed grid clock. Giving every point the same accelerometer reading would generally change the rest distances.

For light, (8.25) instead gives

$$\left\| \frac{d\mathbf{r}'}{dt'} \right\| = 1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'}.$$

Dividing by the local fixed clock's rate restores measured speed 1. The coordinate light cones can therefore widen or narrow with position and with the acceleration history, while the local light cone and physical light speed are unchanged. For axial rays,

$$\frac{d(\mathbf{u} \cdot \mathbf{r}')}{dt'} = \pm \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'} \right). \quad (8.26)$$

When $d\theta/dt' = 0$ throughout an interval, the construction reduces to an inertial frame on that interval. For finite rapidity and positive clock factor it remains a frame of massive observers, never a light-speed rest frame. Changing directions and rotating axes are treated with the general paravector frame factor in Section 8.4. Figure 6 illustrates the coordinate grid, light paths, and clocks for constant proper acceleration; Figure 8 shows how they change with proper jerk.

SI units: Accelerated event coordinates and clocks

Here t, t' are seconds, $\mathbf{r}, \mathbf{r}'$ are metres, the origin's proper time is t' , and βc is its inertial speed. For an arbitrary clock, $s = c\tau$. The signed proper acceleration of the origin is $c d\theta/dt'$. The final comparison is

$$\begin{aligned} \left(1 + \frac{\mathbf{u} \cdot \mathbf{r}'}{c} \frac{d\theta}{dt'} \right) dt' &= \gamma \left(dt - \frac{\beta}{c} \mathbf{u} \cdot d\mathbf{r} \right), \\ d\mathbf{r}'_{\parallel} &= \gamma (d\mathbf{r}_{\parallel} - \beta c \mathbf{u} dt), \quad d\mathbf{r}'_{\perp} = d\mathbf{r}_{\perp}, \\ d\tau^2 &= \left(1 + \frac{\mathbf{u} \cdot \mathbf{r}'}{c} \frac{d\theta}{dt'} \right)^2 dt'^2 - \frac{d\mathbf{r}'^2}{c^2}. \end{aligned}$$

At fixed $\mathbf{r}'$, $d\tau/dt' = 1 + (\mathbf{u} \cdot \mathbf{r}') d\theta/(c dt')$. Light has coordinate speed c times this factor and locally measured speed c .

8.8 Constant coordinate acceleration

In the following examples, subscript 0 denotes the initial event $t = s = 0$. Constant coordinate acceleration means that $\dot{\mathbf{v}} = \dot{\mathbf{v}}_0$ is a fixed real vector in a chosen inertial frame. With initial position $\mathbf{r}(0) = \mathbf{r}_0$ and velocity $\mathbf{v}(0) = \mathbf{v}_0$, integration gives

$$\mathbf{v}(t) = \mathbf{v}_0 + \dot{\mathbf{v}}_0 t, \quad (8.27)$$

$$X(t) = t + (\mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \dot{\mathbf{v}}_0 t^2) \cdot \boldsymbol{\sigma}. \quad (8.28)$$

These familiar coordinate-time formulas also describe a relativistic massive particle, provided $\|\mathbf{v}_0 + \dot{\mathbf{v}}_0 t\| < 1$ throughout the interval considered. A nonzero constant coordinate acceleration can therefore persist only for a finite coordinate-time interval.

For motion from rest at the origin, the notation $\dot{\mathbf{v}}_0 = \dot{v}_0 \mathbf{u}$ is used, where $\dot{v}_0 > 0$ is the scalar acceleration along the fixed real unit vector $\mathbf{u}$. Then

$$\mathbf{v}(t) = \dot{v}_0 t \mathbf{u}, \quad X(t) = t + \frac{1}{2} \dot{v}_0 t^2 \mathbf{u} \cdot \boldsymbol{\sigma}, \quad \gamma(t) = \frac{1}{\sqrt{1 - \dot{v}_0^2 t^2}}, \quad 0 \leq t < \frac{1}{\dot{v}_0}.$$

The particle's clock runs at $\dot{s} = 1/\gamma(t)$. Taking $s(0) = 0$ in (6.5) gives

$$s(t) = \int_0^t \sqrt{1 - \dot{v}_0^2 t'^2} dt' = \frac{\arcsin(\dot{v}_0 t) + \dot{v}_0 t \sqrt{1 - \dot{v}_0^2 t^2}}{2\dot{v}_0}. \quad (8.29)$$

Four-velocity and four-acceleration follow by differentiating the world line:

$$U(t) = \gamma(t)(1 + \dot{v}_0 t \mathbf{u} \cdot \boldsymbol{\sigma}),$$

$$\frac{dU}{ds} = \gamma \dot{U} = \gamma^4 \dot{v}_0^2 t + \gamma^4 \dot{v}_0 \mathbf{u} \cdot \boldsymbol{\sigma}.$$

Consequently,

$$\det \left(\frac{dU}{ds} \right) = \gamma^8 \dot{v}_0^2 (\dot{v}_0^2 t^2 - 1) = -\gamma^6 \dot{v}_0^2.$$

The proper-acceleration magnitude is therefore

$$\sqrt{-\det \left(\frac{dU}{ds} \right)} = \gamma^3 \dot{v}_0 = \frac{\dot{v}_0}{(1 - \dot{v}_0^2 t^2)^{3/2}}.$$

Although the coordinate acceleration stays fixed, the acceleration measured in the instantaneous rest frame grows without bound as $t \rightarrow 1/\dot{v}_0$. The next example instead holds the proper acceleration fixed.

8.9 Constant proper acceleration

A fixed [real unit axis](#) $\mathbf{u}$ is chosen, and the initial conditions $X(0) = 0$, $U(0) = 1$ are imposed, with initial coordinate acceleration $\dot{\mathbf{v}}_0 = \dot{v}_0 \mathbf{u}$, where $\dot{v}_0 > 0$. At rest $\gamma_0 = 1$ and $\dot{\theta}_0 = \dot{v}_0$. Holding the signed proper acceleration along this axis constant gives $\theta(s) = \dot{v}_0 s$, and hence

$$U(s) = L(\dot{v}_0 s, \mathbf{u})^{-2} = \cosh(\dot{v}_0 s) + \sinh(\dot{v}_0 s) \mathbf{u} \cdot \boldsymbol{\sigma}.$$

The stated acceleration is verified by differentiation:

$$\det \left(\frac{dU}{ds} \right) = -\dot{v}_0^2 (\cosh^2(\dot{v}_0 s) - \sinh^2(\dot{v}_0 s)) = -\dot{v}_0^2.$$

Integrating (6.9) yields (9, Eq. (7.33), pp. 142–143)

$$X(s) = \frac{\sinh(\dot{v}_0 s)}{\dot{v}_0} + \frac{\cosh(\dot{v}_0 s) - 1}{\dot{v}_0} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (8.30)$$

In components,

$$\begin{aligned} \mathbf{v}(t(s)) &= \tanh(\dot{v}_0 s) \mathbf{u}, \\ (1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r})^2 - (\dot{v}_0 t)^2 &= 1. \end{aligned}$$

Thus (8.30) is the usual hyperbolic motion, with speed below light speed at every finite s . Eliminating s using $\dot{v}_0 t = \sinh(\dot{v}_0 s)$ makes the coordinate acceleration explicit:

$$\mathbf{v}(t) = \frac{\dot{v}_0 t}{\sqrt{1 + \dot{v}_0^2 t^2}} \mathbf{u}, \quad \dot{\mathbf{v}}(t) = \frac{\dot{v}_0}{(1 + \dot{v}_0^2 t^2)^{3/2}} \mathbf{u}.$$

Here $\gamma(t) = \sqrt{1 + \dot{v}_0^2 t^2}$, so $\gamma^3 \|\dot{\mathbf{v}}(t)\| = \dot{v}_0$ is constant while the coordinate acceleration decreases as the particle gains speed.

8.10 A uniformly accelerating grid and its horizon

Use the same reference worldline (8.30), with $\dot{v}_0 > 0$ and $\theta = \dot{v}_0 t'$. Substituting it and N in (8.23) gives the explicit coordinates of every event in the accelerated chart:

$$\boxed{\begin{aligned} t &= \left(\frac{1}{\dot{v}_0} + \mathbf{u} \cdot \mathbf{r}' \right) \sinh(\dot{v}_0 t'), \\ \mathbf{u} \cdot \mathbf{r} &= \left(\frac{1}{\dot{v}_0} + \mathbf{u} \cdot \mathbf{r}' \right) \cosh(\dot{v}_0 t') - \frac{1}{\dot{v}_0}, \\ \mathbf{r}_\perp &= \mathbf{r}'_\perp. \end{aligned}} \quad (8.31)$$

The origin $\mathbf{r}' = 0$ is the accelerating reference clock; each other fixed $\mathbf{r}'$ is a different hyperbolic worldline. At $t' = 0$ all are instantaneously at inertial rest, yet their clock rates are $ds/dt' = 1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r}'$ and their signed proper accelerations are $\dot{v}_0/(1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r}')$. For example, the grid clock at $\mathbf{u} \cdot \mathbf{r}' = 1/\dot{v}_0$ advances twice as much as the origin per unit t' and needs half its acceleration.

For a concrete event on the origin, choose $t' = \ln(2)/\dot{v}_0$. Its two coordinate assignments and elapsed proper time from departure are

$$(t, \mathbf{r}) = \left(\frac{3}{4\dot{v}_0}, \frac{\mathbf{u}}{4\dot{v}_0} \right), \quad (t', \mathbf{r}') = \left(\frac{\ln(2)}{\dot{v}_0}, \mathbf{0} \right), \quad \Delta s = \frac{\ln(2)}{\dot{v}_0}.$$

The finite separation has $\sqrt{\det(\Delta X)} = 1/(\sqrt{2}\dot{v}_0) > \Delta s$. This explicitly distinguishes the invariant separation of the two events from the proper time accumulated along the accelerated path between them.

These are Rindler coordinates. The interval and axial light paths are

$$\boxed{ds^2 = (1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r}')^2 dt'^2 - d\mathbf{r}'^2}, \quad (8.32)$$

$$1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r}'(t') = (1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r}'_0) e^{\pm \dot{v}_0 t'}. \quad (8.33)$$

Here $\mathbf{r}'_0$ is the ray's position at $t' = 0$ and its transverse coordinates are fixed. The exponential follows by integrating (8.26); the inward ray approaches zero coordinate speed only as $t' \rightarrow \infty$, always moving at unit speed relative to a nearby grid clock.

The map covers the wedge

$$\frac{1}{\dot{v}_0} + \mathbf{u} \cdot \mathbf{r} > |t|, \quad 1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r}' > 0.$$

Its null boundaries $t = \pm(\mathbf{u} \cdot \mathbf{r} + 1/\dot{v}_0)$ are the past and future horizons for the eternally accelerated observer. The reference trajectory approaches a null direction as $t' \rightarrow \infty$ but is time-like at every finite t' , with $\det(U) = 1$. A finite period of acceleration followed by inertial motion need not produce such a horizon; a horizon is a statement about the full history, not merely the instantaneous clock factor. Taking $\dot{v}_0 \rightarrow 0$ at fixed $t', \mathbf{r}'$ gives $t = t'$ and $\mathbf{r} = \mathbf{r}'$, as follows by expanding the hyperbolic functions.

SI units: Uniformly accelerated grid

Let $\dot{v}_0$ be the origin's constant proper acceleration in m s^{-2} , with t, t', τ in seconds. Then

$$\begin{aligned} ct &= \left(\frac{c^2}{\dot{v}_0} + \mathbf{u} \cdot \mathbf{r}' \right) \sinh(\dot{v}_0 t'/c), \\ \mathbf{u} \cdot \mathbf{r} &= \left(\frac{c^2}{\dot{v}_0} + \mathbf{u} \cdot \mathbf{r}' \right) \cosh(\dot{v}_0 t'/c) - \frac{c^2}{\dot{v}_0}, \quad \mathbf{r}_\perp = \mathbf{r}'_\perp, \\ d\tau^2 &= \left(1 + \frac{\dot{v}_0 \mathbf{u} \cdot \mathbf{r}'}{c^2} \right)^2 dt'^2 - \frac{d\mathbf{r}'^2}{c^2}. \end{aligned}$$

The coordinate boundary is $\mathbf{u} \cdot \mathbf{r}' = -c^2/\dot{v}_0$; the inertial null boundaries are $ct = \pm(\mathbf{u} \cdot \mathbf{r} + c^2/\dot{v}_0)$.

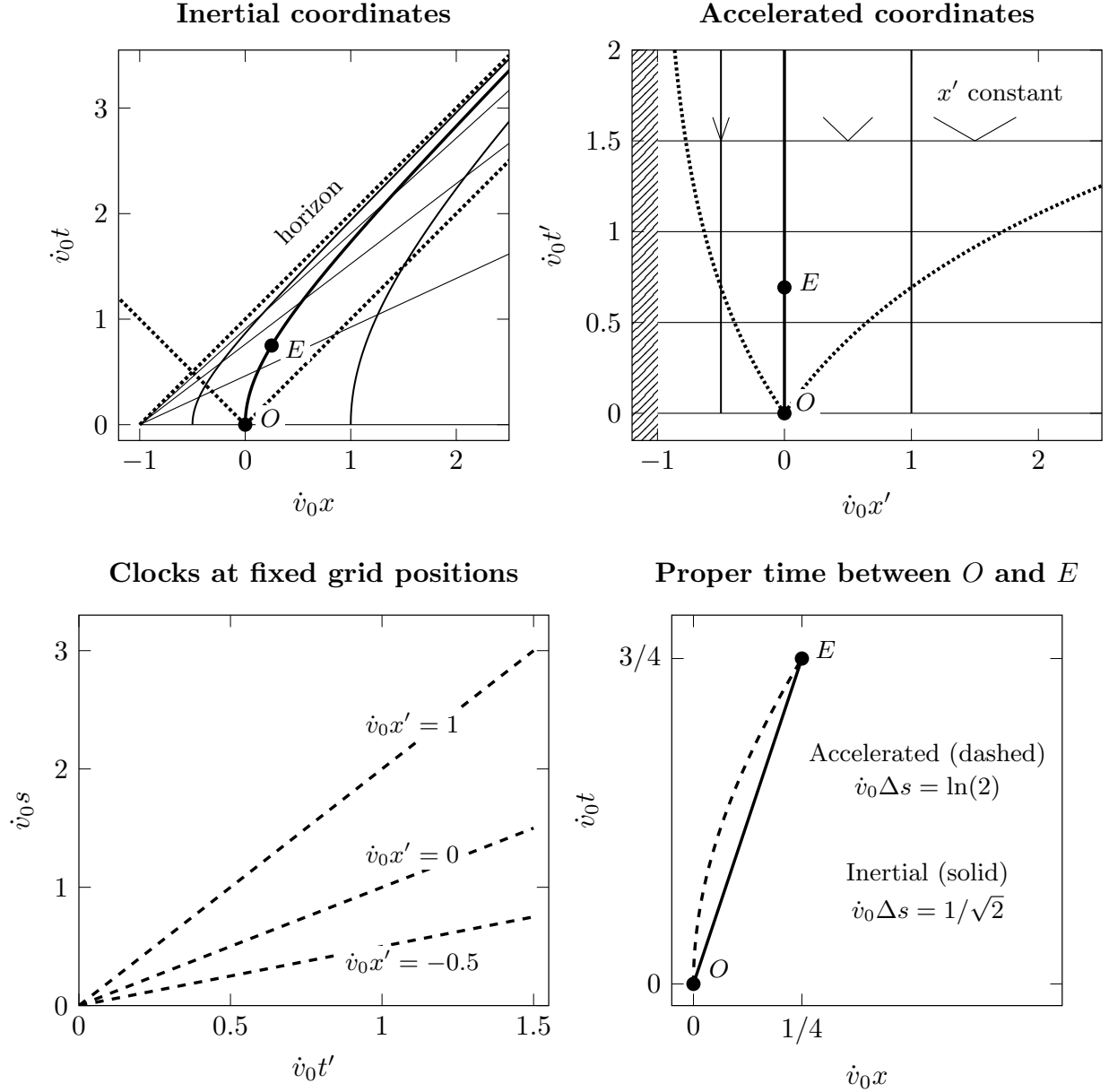


Figure 6: Uniform acceleration, with $x = \mathbf{u} \cdot \mathbf{r}$ and $x' = \mathbf{u} \cdot \mathbf{r}'$. Top: the same grid and light rays in inertial and accelerated coordinates. The three solid worldlines have $\dot{v}_0 x' = -1/2, 0, 1$; thin connecting lines are simultaneous at fixed t' . Dotted light rays leave O ; the additional dotted null line on the left is the future horizon. Hatching excludes $1 + \dot{v}_0 x' \leq 0$. Short V shapes show local coordinate cones: their width varies with position, while locally measured light speed is 1. Bottom left: fixed grid clocks accumulate different proper times. Bottom right: between the same events O and $E : (\dot{v}_0 x, \dot{v}_0 t) = (1/4, 3/4)$, the accelerated clock records $\ln(2)/\dot{v}_0$, less than the inertial clock's $1/(\sqrt{2}\dot{v}_0)$. Thus finite event separation and elapsed time along a curved path differ.

8.11 A trip with midpoint braking

Departure and destination are at rest in one inertial frame, separated by $\ell \in \mathbb{R}_{>0}$ along a fixed real unit vector $\mathbf{u}$. Flat spacetime and initial conditions $t = s = 0$, $\mathbf{r} = 0$, $\mathbf{v} = 0$ are assumed. Two equal burns at constant proper-acceleration magnitude $\dot{v}_0 > 0$ are joined by an instantaneous thrust reversal; arrival is at rest. This is a prescribed trajectory; fuel consumption and the duration of the reversal are not modeled.

Rapidity and derivatives. The reversal times are $s_m, t_m \in \mathbb{R}_{>0}$; arrival occurs at $2s_m$ and $2t_m$. With $d\theta/ds = +\dot{v}_0$ during acceleration and $-\dot{v}_0$ during braking, integration gives

$$\theta(s) = \begin{cases} \dot{v}_0 s, & 0 \leq s \leq s_m, \\ \dot{v}_0(2s_m - s), & s_m \leq s \leq 2s_m. \end{cases} \quad (8.34)$$

Throughout the trip,

$$v = \tanh(\theta), \quad \mathbf{v} = v\mathbf{u}, \quad \gamma = \cosh(\theta), \quad \dot{s} = \gamma^{-1}.$$

On the open burn intervals, the upper sign denotes acceleration and the lower sign braking. The derivatives are obtained directly:

$$\begin{aligned} U &= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \frac{dU}{ds} &= \pm \dot{v}_0 (\sinh(\theta) + \cosh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \dot{U} &= \frac{1}{\gamma} \frac{dU}{ds} = \pm \dot{v}_0 (v + \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \dot{v} &= \pm \frac{\dot{v}_0}{\gamma^3}, \quad \dot{\mathbf{v}} = \dot{v}\mathbf{u}, \quad \dot{\gamma} = \pm \dot{v}_0 v. \end{aligned} \quad (8.35)$$

The acceleration is entirely tangential; the perpendicular component vanishes. Although $\|\dot{\mathbf{v}}\|$ varies, the proper-acceleration magnitude is fixed:

$$\det(U) = 1, \quad \sqrt{-\det\left(\frac{dU}{ds}\right)} = \gamma^3 \|\dot{\mathbf{v}}\| = \dot{v}_0.$$

Position, clocks, and arrival. By integrating $dt/ds = \cosh(\theta)$ and $d\mathbf{r}/ds = \sinh(\theta)\mathbf{u}$, the continuous worldline is obtained:

$$\begin{aligned} t(s) &= \begin{cases} \sinh(\theta(s))/\dot{v}_0, & 0 \leq s \leq s_m, \\ 2t_m - \sinh(\theta(s))/\dot{v}_0, & s_m \leq s \leq 2s_m, \end{cases} \\ \mathbf{r}(s) &= \begin{cases} (\cosh(\theta(s)) - 1)\mathbf{u}/\dot{v}_0, & 0 \leq s \leq s_m, \\ \mathbf{u}(\ell - (\cosh(\theta(s)) - 1)/\dot{v}_0), & s_m \leq s \leq 2s_m. \end{cases} \end{aligned} \quad (8.36)$$

Continuity at the reversal fixes the midpoint and burn durations:

$$\begin{aligned} \mathbf{r}(s_m) &= \frac{1}{2}\mathbf{u}\ell, \quad \gamma_m := \cosh(\dot{v}_0 s_m) = 1 + \frac{\dot{v}_0 \ell}{2}, \\ s_m &= \frac{\text{arcosh}(1 + \dot{v}_0 \ell/2)}{\dot{v}_0}, \quad t_m = \sqrt{\frac{\ell}{\dot{v}_0} + \frac{\ell^2}{4}}, \\ v_{\max} &= \sqrt{1 - \gamma_m^{-2}}. \end{aligned} \quad (8.37)$$

Thus $\mathbf{r}(2s_m) = \mathbf{u}\ell$ and $U(0) = U(2s_m) = 1$. For evaluation in coordinate time, the same expressions are used with

$$\theta(t) = \begin{cases} \operatorname{arsinh}(\dot{v}_0 t), & 0 \leq t \leq t_m, \\ \operatorname{arsinh}(\dot{v}_0(2t_m - t)), & t_m \leq t \leq 2t_m. \end{cases}$$

In particular, $\|\dot{\mathbf{v}}\|$ decreases to $\dot{v}_0/\gamma_m^3$ before the reversal and increases back to $\dot{v}_0$ during braking.

The thrust reversal. Position, both clocks, velocity, and four-velocity are continuous at the midpoint. The one-sided values of $\dot{\mathbf{v}}$, $\dot{U}$, and dU/ds are opposite; these derivatives are undefined at the ideal instantaneous reversal. The thrust is reversed, not the direction of travel. The jerk during each burn and through a finite reversal is discussed in Section 9.5. Figure 7 compares the trajectory, acceleration, and clocks for a trip with $\dot{v}_0\ell = 2$.

SI units: Trip parameters

In the following SI formulas, $\dot{v}_0$ is the initial acceleration in metres per second squared, also equal to the constant proper-acceleration magnitude. The distance ℓ is in metres, t_{trip} is the elapsed departure-frame time in seconds, and τ_{trip} is the elapsed onboard proper time in seconds. Restoring units in (8.37) gives

$$\begin{aligned} t_{\text{trip}} &= \sqrt{\frac{\ell^2}{c^2} + \frac{4\ell}{\dot{v}_0}}, \\ \tau_{\text{trip}} &= \frac{2c}{\dot{v}_0} \operatorname{arcosh}\left(1 + \frac{\dot{v}_0\ell}{2c^2}\right), \\ v_{\text{max}} &= c\sqrt{1 - \left(1 + \frac{\dot{v}_0\ell}{2c^2}\right)^{-2}}. \end{aligned}$$

For a specified onboard duration and peak speed, the required proper acceleration is fixed by the midpoint rapidity:

$$\dot{v}_0 = \frac{2c}{\tau_{\text{trip}}} \operatorname{artanh}\left(\frac{v_{\text{max}}}{c}\right).$$

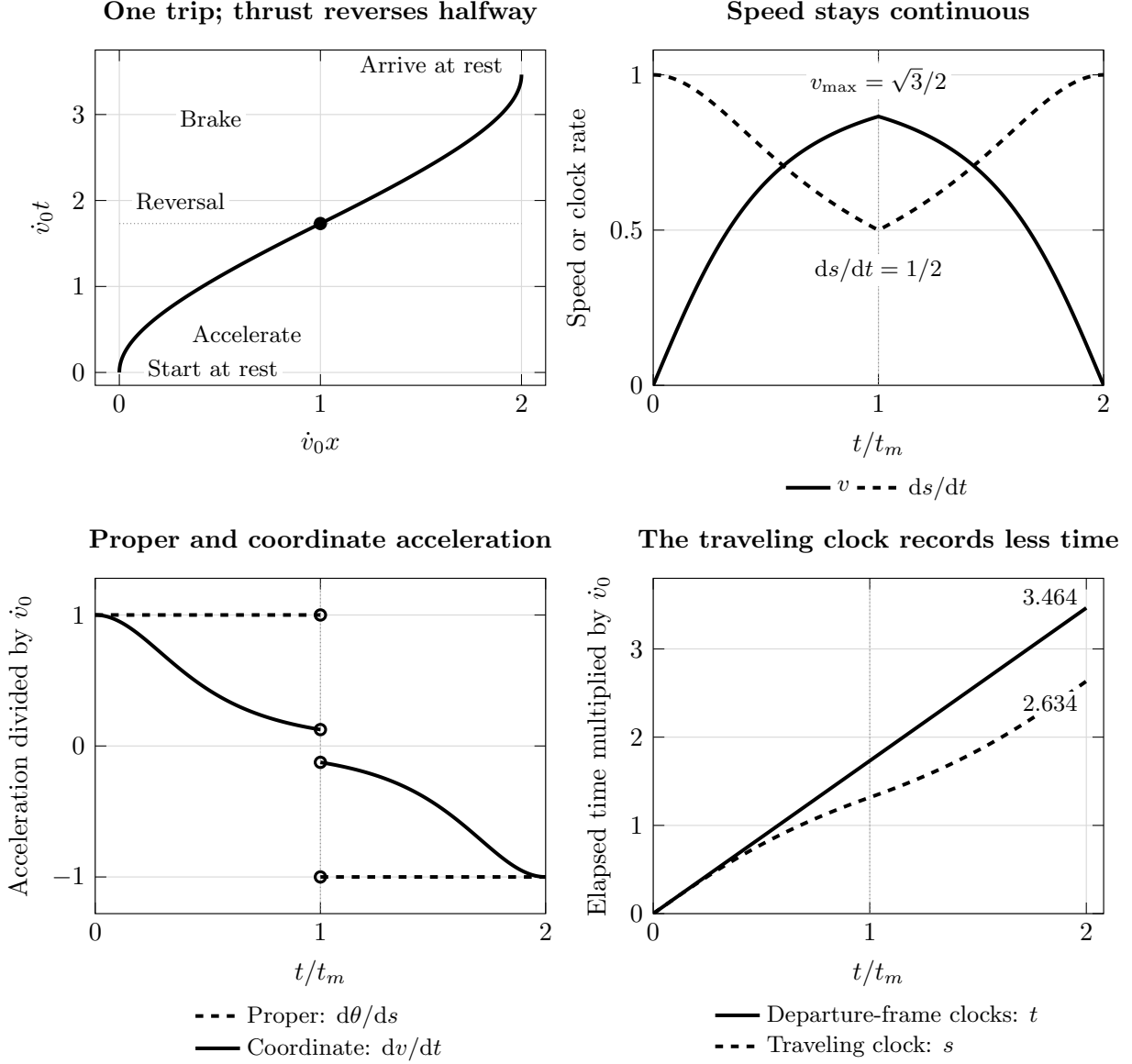


Figure 7: Midpoint braking in flat spacetime, with $x = \mathbf{u} \cdot \mathbf{r} \in \mathbb{R}$ and $\dot{v}_0 \ell = 2$. Dotted guides mark the reversal at $t = t_m$. The trajectory and speed remain continuous while the signed acceleration changes abruptly; open circles show its one-sided values. The proper-acceleration magnitude stays fixed, whereas $|\dot{v}| = \dot{v}_0/\gamma^3$ varies. The reduced clock rate accumulates to a shorter onboard elapsed time. All plotted axes use the natural units of the main text.

9 Jerk

The coordinates t, s , rapidity θ , and their derivatives are real scalars, with $\gamma \in \mathbb{R}_{\geq 1}$ and $\beta \in (-1, 1)$. $\mathbf{r}, \mathbf{v}, \dot{\mathbf{v}}, \ddot{\mathbf{v}}, \mathbf{u} \in \mathbb{R}^3$, with $\mathbf{v}^2 < 1$ and $\mathbf{u}^2 = 1$. The values X, U, N , four-acceleration, unprojected four-jerk, and projected four-jerk lie in $\mathbb{R} \oplus \mathbb{R}^3$. Their determinants are real scalars; their measured magnitudes are in $\mathbb{R}_{\geq 0}$. Initial values inherit these domains. Three quantities are distinguished: the coordinate jerk $\ddot{\mathbf{v}}$, the proper-time derivative of four-acceleration, and its projection into the instantaneous rest space. Proper jerk is the vector rate of change of measured acceleration per unit proper time, referred to nonrotating carried axes. Dots still mean t derivatives.

9.1 Coordinate derivatives, then proper-time derivatives

The expressions for $\dot{\gamma}$ and $\dot{U}$ in Section 8.1 are differentiated once more with respect to t . Using

$$\frac{d(\mathbf{v} \cdot \dot{\mathbf{v}})}{dt} = \dot{\mathbf{v}}^2 + \mathbf{v} \cdot \ddot{\mathbf{v}},$$

the following is obtained by a second differentiation:

$$\begin{aligned} \ddot{\gamma} &= \gamma^3(\dot{\mathbf{v}}^2 + \mathbf{v} \cdot \ddot{\mathbf{v}}) + \frac{3\dot{\gamma}^2}{\gamma}, \\ \ddot{U} &= \ddot{\gamma} + (\gamma\ddot{\mathbf{v}} + 2\dot{\gamma}\dot{\mathbf{v}} + \ddot{\gamma}\mathbf{v}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The conversion $d/ds = \gamma d/dt$ is now applied. The first derivative is $dU/ds = \gamma\dot{U}$, and the second is

$$\frac{d^2U}{ds^2} = \gamma \frac{d(\gamma\dot{U})}{dt} = \gamma^2\ddot{U} + \gamma\dot{\gamma}\dot{U}. \quad (9.1)$$

The same conversion applies to every twice-differentiable scalar, vector, or paravector along the worldline. In particular,

$$\begin{aligned} \frac{d^2\gamma}{ds^2} &= \gamma^2\ddot{\gamma} + \gamma\dot{\gamma}^2, \\ \frac{d^2U}{ds^2} &= \frac{d^2\gamma}{ds^2} + \left(\gamma^3\ddot{\mathbf{v}} + 3\gamma^2\dot{\gamma}\dot{\mathbf{v}} + \frac{d^2\gamma}{ds^2}\mathbf{v} \right) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (9.2)$$

This derivative of four-acceleration is the *unprojected four-jerk*:

$$\frac{d}{ds} \left(\frac{dU}{ds} \right) = \frac{d^2U}{ds^2} = \frac{d^3X}{ds^3}. \quad (9.3)$$

By comparison, $d\dot{\mathbf{v}}/ds = \gamma\ddot{\mathbf{v}}$ still differentiates the coordinate acceleration of the chosen inertial frame.

9.2 Projection into the instantaneous rest space

The coordinate-time normalization identities are used:

$$\text{sc}(U^*\dot{U}) = 0, \quad \text{sc}(U^*\ddot{U}) = -\det(\dot{U}).$$

The second follows by differentiating the first. Using

$$\det\left(\frac{dU}{ds}\right) = \gamma^2 \det(\dot{U}),$$

substitution into (9.1) gives

$$\text{sc}\left(U^* \frac{d^2U}{ds^2}\right) = -\det\left(\frac{dU}{ds}\right).$$

Thus the derivative of four-acceleration need not be orthogonal to four-velocity. Since $\det(U) = 1$, the right-hand side is its coefficient of U in the unprojected four-jerk. Removing that component defines the projected relativistic jerk (22, Eqs. (2.8)–(2.13)). With the signature $(+, -, -, -)$, its orthogonality is

$$\text{sc}\left(U^* \left(\frac{d^2U}{ds^2} + \det\left(\frac{dU}{ds}\right) U\right)\right) = 0. \quad (9.4)$$

At an instant of rest, $U = 1$ and

$$\begin{aligned} \frac{d^2U}{ds^2} &= \dot{\mathbf{v}}^2 + \ddot{\mathbf{v}} \cdot \boldsymbol{\sigma} \\ &= -\det\left(\frac{dU}{ds}\right) U + \ddot{\mathbf{v}} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here the coordinate derivatives are evaluated in the inertial frame in which the particle is instantaneously at rest. The projected expression is therefore the measured proper jerk there; it is space-like whenever nonzero.

Invariant magnitude. Expanding the determinant using the metric pairing and (9.4) gives

$$\det\left(\frac{d^2U}{ds^2} + \det\left(\frac{dU}{ds}\right) U\right) = \det\left(\frac{d^2U}{ds^2}\right) - \det\left(\frac{dU}{ds}\right)^2 \leq 0. \quad (9.5)$$

The negative of this determinant is the squared proper-jerk magnitude. The unprojected four-jerk can instead be time-like, null, or space-like; its determinant includes the positive contribution from the component along U .

When proper jerk vanishes. In (9.2), subtracting the scalar coefficient times $\mathbf{v}$ from the vector coefficient leaves $\gamma^3(\ddot{\mathbf{v}} + 3\dot{\gamma}\dot{\mathbf{v}}/\gamma)$. Its vanishing says that the unprojected four-jerk is parallel to U , so its projection vanishes. Hence (22, Eq. (2.14))

$$\frac{d^2U}{ds^2} + \det\left(\frac{dU}{ds}\right) U = 0 \iff \ddot{\mathbf{v}} = -3\frac{\dot{\gamma}}{\gamma}\dot{\mathbf{v}}. \quad (9.6)$$

Zero proper jerk therefore need not mean zero coordinate jerk. An equivalent first integral is

$$\frac{d(\gamma^3\dot{\mathbf{v}})}{dt} = 0, \quad \gamma^3\dot{\mathbf{v}} = \gamma_0^3\dot{\mathbf{v}}_0.$$

Here subscript 0 denotes any reference event. Any nonzero coordinate acceleration therefore retains a fixed direction, although the velocity need not be parallel to it; the spatial trajectory is confined to a plane. Also, differentiating the acceleration determinant gives

$$\frac{d}{ds} \det\left(\frac{dU}{ds}\right) = 2 \text{sc}\left(\left(\frac{dU}{ds}\right)^* \frac{d^2U}{ds^2}\right).$$

The pairing is unchanged if the second factor is replaced by its projected expression, because dU/ds is orthogonal to U . Consequently, zero projected jerk implies constant proper-acceleration magnitude. The converse does not hold in general: the direction can still change, as in uniform circular motion.

9.3 Fixed-axis jerk

Differentiating the paired first derivatives in Section 8.6 gives

$$\begin{aligned}\ddot{U} &= \dot{\theta}^2 U + \ddot{\theta} N, \\ \ddot{N} &= \dot{\theta}^2 N + \ddot{\theta} U.\end{aligned}$$

The $\dot{\theta}^2$ terms come from differentiating the moving basis; the $\ddot{\theta}$ terms come from differentiating its rate of change. By orthonormality,

$$\det(\ddot{U}) = -\det(\ddot{N}) = \dot{\theta}^4 - \ddot{\theta}^2.$$

Differentiating $\dot{\beta} = \gamma^{-2}\dot{\theta}$ also gives

$$\ddot{\beta} = \gamma^{-2}\ddot{\theta} - 2\gamma^{-3}\dot{\gamma}\dot{\theta}.$$

Conversion to proper time. By (9.1), the signed proper jerk is

$$\frac{d^2\theta}{ds^2} = \gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta}.$$

Applying (9.1) to each member of the pair gives

$$\begin{aligned}\frac{d^2U}{ds^2} &= \gamma^2\dot{\theta}^2 U + (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})N, \\ \frac{d^2N}{ds^2} &= \gamma^2\dot{\theta}^2 N + (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})U.\end{aligned}\tag{9.7}$$

Removing the component along U leaves

$$\frac{d^2U}{ds^2} - \gamma^2\dot{\theta}^2 U = (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})N = \frac{d^2\theta}{ds^2} N.$$

The proper-jerk magnitude is $|\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta}|$. The corresponding determinant identities are

$$\begin{aligned}\det\left(\frac{d^2U}{ds^2} - \gamma^2\dot{\theta}^2 U\right) &= -(\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})^2, \\ \det\left(\frac{d^2U}{ds^2}\right) &= -\det\left(\frac{d^2N}{ds^2}\right) \\ &= \gamma^4\dot{\theta}^4 - (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})^2.\end{aligned}$$

The sign reversal again comes from exchanging the unit time and spatial directions. Using $\dot{\theta} = \gamma^2\dot{\beta}$ and $\dot{\gamma} = \gamma^3\beta\dot{\beta}$ also gives

$$\begin{aligned}\frac{d^2\theta}{ds^2} &= \gamma^4\ddot{\beta} + 3\gamma^3\dot{\gamma}\dot{\beta} = \gamma^4\ddot{\beta} + 3\gamma^6\beta\dot{\beta}^2, \\ \ddot{\beta} &= \gamma^{-4}\frac{d^2\theta}{ds^2} - 3\frac{\dot{\gamma}}{\gamma}\dot{\beta}.\end{aligned}\tag{9.8}$$

For fixed-axis motion, zero proper jerk is exactly $\gamma\ddot{\theta} + \dot{\gamma}\dot{\theta} = 0$, or equivalently $d^2\theta/ds^2 = 0$. Thus $\gamma\dot{\theta}$ is constant and rapidity is affine in proper time. The unprojected four-jerk remains nonzero whenever that constant proper acceleration is nonzero.

Null combinations and constant acceleration. The symmetry of the paired derivatives becomes especially simple on adding or subtracting U and N :

$$U \pm N = e^{\pm\theta}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \det(U \pm N) = 0.$$

These two null directions are fixed; only their scalar factors change. Their coordinate-time derivatives are

$$\begin{aligned} \dot{U} \pm \dot{N} &= \pm\dot{\theta}(U \pm N), \\ \ddot{U} \pm \ddot{N} &= (\dot{\theta}^2 \pm \ddot{\theta})(U \pm N). \end{aligned}$$

The proper-time derivatives follow by the same conversion:

$$\begin{aligned} \frac{d(U \pm N)}{ds} &= \pm\gamma\dot{\theta}(U \pm N), \\ \frac{d^2(U \pm N)}{ds^2} &= (\gamma^2\dot{\theta}^2 \pm (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta}))(U \pm N). \end{aligned}$$

These formulas allow varying acceleration. If the signed proper acceleration $d\theta/ds = \gamma\dot{\theta}$ is constant, then

$$\frac{d^2\theta}{ds^2} = \gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta} = 0.$$

The second-derivative formula therefore reduces to

$$\frac{d^2(U \pm N)}{ds^2} = \gamma^2\dot{\theta}^2(U \pm N) = (\pm\gamma\dot{\theta})^2(U \pm N).$$

Only under this constant-acceleration assumption is each null combination multiplied by the same constant factor at every proper-time differentiation. For integers $n \geq 0$, the $n = 2$ case above is included in

$$\frac{d^n(U \pm N)}{ds^n} = (\pm\gamma\dot{\theta})^n(U \pm N).$$

Adding and subtracting recovers the derivatives of U and N : even orders preserve each direction, while odd orders exchange them. These ordinary derivatives remain nonzero for nonzero constant proper acceleration, even though the projected jerk is zero. The constant-proper-acceleration worldline in Section 8.9 realizes this pattern; Section 9.6 treats nonzero constant proper jerk.

9.4 Events and light cones when acceleration changes

Jerk changes the history of the accelerating coordinates of Section 8.7; it does not introduce a new kind of spacetime interval. Continue to use fixed $\mathbf{u}$, the origin's proper time t' , and its rapidity $\theta(t')$. The signed proper acceleration and proper jerk at the origin are $d\theta/dt'$ and $d^2\theta/dt'^2$. Along the origin, the previously derived four-derivatives become

$$\frac{dU}{dt'} = \frac{d\theta}{dt'}N, \quad \frac{d^2U}{dt'^2} = \left(\frac{d\theta}{dt'}\right)^2 U + \frac{d^2\theta}{dt'^2}N.$$

At the same event, the instantaneous rest boost gives

$$\boxed{\begin{aligned} L \frac{dU}{dt'} L &= \frac{d\theta}{dt'} \mathbf{u} \cdot \boldsymbol{\sigma}, \\ L \frac{d^2U}{dt'^2} L &= \left(\frac{d\theta}{dt'} \right)^2 + \frac{d^2\theta}{dt'^2} \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}} \quad (9.9)$$

The vector term in the second line is the measured proper jerk; its scalar term is removed by the projection in (9.4). Although $LU L = 1$ at every point on this carried frame, the displayed physical derivatives need not vanish. Differentiating the constant rest components alone would lose the derivatives of the changing frame; Section 11.5 gives the general correction.

Changing clock comparisons. For a fixed accelerated grid point, (8.25) gives the clock rate ds/dt' . Its change is

$$\boxed{\frac{d^2s}{dt'^2} = (\mathbf{u} \cdot \mathbf{r}') \frac{d^2\theta}{dt'^2} \quad (\mathbf{r}' \text{ fixed}).} \quad (9.10)$$

Here s is that grid clock's own proper time. Thus acceleration sets the spatial variation of clock rates, and proper jerk changes that variation. Differentiating the grid point's signed proper acceleration with respect to its own time gives

$$\frac{d}{ds} \left(\frac{d\theta/dt'}{1 + (\mathbf{u} \cdot \mathbf{r}') d\theta/dt'} \right) = \frac{d^2\theta/dt'^2}{(1 + (\mathbf{u} \cdot \mathbf{r}') d\theta/dt')^3}. \quad (9.11)$$

The factor follows by applying $d/ds = (1 + (\mathbf{u} \cdot \mathbf{r}') d\theta/dt')^{-1} d/dt'$ at fixed $\mathbf{r}'$. At the origin it reduces to the origin's proper jerk.

Changing light paths. For axial light, differentiate (8.26) along the ray, allowing both t' and $\mathbf{r}'$ to change:

$$\frac{d^2(\mathbf{u} \cdot \mathbf{r}')}{dt'^2} = \pm (\mathbf{u} \cdot \mathbf{r}') \frac{d^2\theta}{dt'^2} + \frac{d\theta}{dt'} \left(1 + (\mathbf{u} \cdot \mathbf{r}') \frac{d\theta}{dt'} \right). \quad (9.12)$$

The first term changes the light-cone slopes through the changing acceleration; the second comes from the ray moving through their spatial variation. Consequently even zero proper jerk can give curved light paths in these coordinates. Both rays still have locally measured speed 1, and a finite proper jerk does not permit a light-speed observer.

At an event where the origin is at rest, $\theta = 0$, the origin still has $dt = dt' = ds$, whether acceleration and jerk vanish or not. If acceleration is zero there but jerk is not, all fixed grid clocks have equal instantaneous rates, yet those rates begin separating according to (9.10). If both acceleration and proper jerk vanish throughout an interval, the chart is inertial there. A zero of the clock factor marks a coordinate failure; its instantaneous location alone does not determine a causal horizon.

For another local view, start at rest at $X(0) = 0$ with $\dot{\mathbf{v}}_0 = \dot{v}_0 \mathbf{u}$ and $\ddot{\mathbf{v}}_0 = \ddot{v}_0 \mathbf{u}$. At that event $d\theta/ds = \dot{v}_0$ and $d^2\theta/ds^2 = \ddot{v}_0$. Taylor expansion of the same worldline in the inertial coordinates gives

$$X(s) = s + \frac{1}{2} \dot{v}_0 s^2 \mathbf{u} \cdot \boldsymbol{\sigma} + \frac{1}{6} (\dot{v}_0^2 + \ddot{v}_0 \mathbf{u} \cdot \boldsymbol{\sigma}) s^3 + O(s^4).$$

Acceleration first changes position at quadratic order, while proper jerk first changes it at cubic order. In the reference observer's own chart these same successive events remain at $\mathbf{r}' = 0$, with $t' = s$.

SI units: Jerk, clock comparisons, and light paths

Here t' is the origin's proper time in seconds, τ belongs to a fixed grid clock, and $\mathbf{r}'$ is measured in metres. The signed proper acceleration and jerk at the origin are $c \, d\theta/dt'$ and $c \, d^2\theta/dt'^2$, respectively. The final formulas are

$$\begin{aligned} \frac{d\tau}{dt'} &= 1 + \frac{\mathbf{u} \cdot \mathbf{r}'}{c} \frac{d\theta}{dt'}, & \frac{d^2\tau}{dt'^2} &= \frac{\mathbf{u} \cdot \mathbf{r}'}{c} \frac{d^2\theta}{dt'^2}, \\ \text{signed proper jerk at fixed } \mathbf{r}' : & \frac{c \, d^2\theta/dt'^2}{(1 + (\mathbf{u} \cdot \mathbf{r}') \, d\theta/(c \, dt'))^3}, \\ \frac{d(\mathbf{u} \cdot \mathbf{r}')}{dt'} &= \pm c \left(1 + \frac{\mathbf{u} \cdot \mathbf{r}'}{c} \frac{d\theta}{dt'} \right) \quad \text{for axial light.} \end{aligned}$$

Proper jerk has units m s^{-3} ; the locally measured light speed remains c .

9.5 Jerk at a thrust reversal

For the trip in Section 8.11, away from the reversal,

$$\frac{d^2U}{ds^2} = \dot{v}_0^2 U, \quad \ddot{\mathbf{v}} = -\frac{3\dot{v}_0^2 v}{\gamma^4} \mathbf{u}.$$

The projected proper jerk is zero during each constant-acceleration burn, although coordinate jerk is generally nonzero. At the ideal instantaneous reversal, acceleration is discontinuous and jerk is undefined. A finite reversal must be described by a transition with nonzero proper jerk.

9.6 Constant jerk

A distinction is made between constant coordinate jerk in a chosen inertial frame, constant signed proper jerk along a fixed axis, and constant components of the unprojected four-jerk. The last condition is preserved by constant Lorentz transformations. Subscript 0 denotes the initial event $t = s = 0$.

9.6.1 Constant coordinate jerk

For $\ddot{\mathbf{v}} = \ddot{\mathbf{v}}_0$ in a chosen inertial frame, integration gives

$$\begin{aligned} \dot{\mathbf{v}}(t) &= \dot{\mathbf{v}}_0 + \ddot{\mathbf{v}}_0 t, \\ \mathbf{v}(t) &= \mathbf{v}_0 + \dot{\mathbf{v}}_0 t + \frac{1}{2} \ddot{\mathbf{v}}_0 t^2, \\ X(t) &= t + (\mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \dot{\mathbf{v}}_0 t^2 + \frac{1}{6} \ddot{\mathbf{v}}_0 t^3) \cdot \boldsymbol{\sigma}. \end{aligned} \tag{9.13}$$

These formulas apply on the interval where $\mathbf{v}(t)^2 < 1$. Equations (6.3), (6.7), and (6.5) give γ , U , and $s(t)$; (9.2) gives the generally varying four-jerk. For example, from rest with zero initial acceleration, the following is chosen: $\ddot{\mathbf{v}}_0 = \ddot{v}_0 \mathbf{u}$, where $\ddot{v}_0 > 0$ is the scalar jerk along the fixed real unit axis $\mathbf{u}$. Then

$$\mathbf{v}(t) = \frac{1}{2} \ddot{v}_0 t^2 \mathbf{u}, \quad \mathbf{r}(t) = \mathbf{r}_0 + \frac{1}{6} \ddot{v}_0 t^3 \mathbf{u}, \quad 0 \leq t < \sqrt{\frac{2}{\ddot{v}_0}}.$$

Here $\dot{\beta} = \ddot{v}_0 t$ and $\beta = \ddot{v}_0 t^2/2$, so (9.8) gives $d^2\theta/ds^2 = \gamma^4 \ddot{v}_0 + 3\gamma^6 \beta (\ddot{v}_0 t)^2$. Thus constant coordinate jerk does not give constant proper jerk.

9.6.2 Constant proper jerk

For fixed-axis motion, the signed quantity $d^2\theta/ds^2$ is held constant. Its value and the initial signed proper acceleration are expressed in coordinate-time data using $\gamma_0 = \cosh(\theta_0)$ and $\dot{\gamma}_0 = \sinh(\theta_0)\dot{\theta}_0$. By integration with respect to proper time,

$$\begin{aligned}\frac{d\theta}{ds} &= \gamma_0 \dot{\theta}_0 + (\gamma_0^2 \ddot{\theta}_0 + \gamma_0 \dot{\gamma}_0 \dot{\theta}_0) s, \\ \theta(s) &= \theta_0 + \gamma_0 \dot{\theta}_0 s + \frac{1}{2} (\gamma_0^2 \ddot{\theta}_0 + \gamma_0 \dot{\gamma}_0 \dot{\theta}_0) s^2.\end{aligned}\tag{9.14}$$

If the particle starts at rest, $\theta_0 = 0$, $\gamma_0 = 1$, and $\dot{\gamma}_0 = 0$, so this reduces to $\theta(s) = \dot{\theta}_0 s + \frac{1}{2} \ddot{\theta}_0 s^2$. The corresponding world line, with $t(0) = 0$, is

$$\begin{aligned}t(s) &= \int_0^s \cosh(\theta(s')) ds', \\ \mathbf{r}(s) &= \mathbf{r}_0 + \mathbf{u} \int_0^s \sinh(\theta(s')) ds', \quad X(s) = t(s) + \mathbf{r}(s) \cdot \boldsymbol{\sigma}.\end{aligned}\tag{9.15}$$

The trajectory is specified exactly by these integrals; a related derivation is given in (22, Eqs. (5.8)–(5.9)). Because $dt/ds = \cosh(\theta) > 0$, they also define a unique coordinate-time trajectory. Its velocity, acceleration, and jerk are

$$\begin{aligned}\mathbf{v}(t(s)) &= \tanh(\theta(s)) \mathbf{u}, \\ \dot{\mathbf{v}}(t(s)) &= \frac{1}{\cosh^3(\theta(s))} \frac{d\theta}{ds} \mathbf{u}, \\ \ddot{\mathbf{v}}(t(s)) &= \frac{1}{\cosh^4(\theta(s))} \left(\frac{d^2\theta}{ds^2} - 3 \left(\frac{d\theta}{ds} \right)^2 \tanh(\theta(s)) \right) \mathbf{u}.\end{aligned}$$

The projected jerk is $(d^2\theta/ds^2)N$, as derived in Section 9.3. Its magnitude is constant, while its inertial-frame components change with N . The unprojected derivative also contains the changing component $(d\theta/ds)^2 U$. Setting $\theta_0 = 0$, $\ddot{\theta}_0 = 0$, and $\dot{\theta}_0 = \dot{v}_0 > 0$ recovers the constant-proper-acceleration example in Section 8.9.

The frame of a clock with constant proper jerk. For a clock starting at rest with zero initial acceleration, take $\dot{\theta}_0 = \ddot{v}_0 > 0$. Then, with $t' = s$ at the origin,

$$\theta(t') = \frac{1}{2} \ddot{v}_0 t'^2, \quad \frac{d\theta}{dt'} = \ddot{v}_0 t', \quad \frac{d^2\theta}{dt'^2} = \ddot{v}_0.$$

Equations (9.15) and (8.23) assign inertial coordinates to each event in this observer's frame. Their local interval is explicitly

$$ds^2 = (1 + \ddot{v}_0 t' \mathbf{u} \cdot \mathbf{r}')^2 dt'^2 - d\mathbf{r}'^2.$$

At $t' = 0$ it agrees with the inertial interval throughout the local grid; thereafter the clock-rate gradient grows linearly with t' . For each fixed event where the factor is positive, the light-cone slopes are $\pm(1 + \ddot{v}_0 t' \mathbf{u} \cdot \mathbf{r}')$. The origin's speed $\tanh(\ddot{v}_0 t'^2/2)$ approaches 1 only as $t' \rightarrow \infty$. This example holds proper jerk constant; constant coordinate jerk has the finite subluminal interval described above.

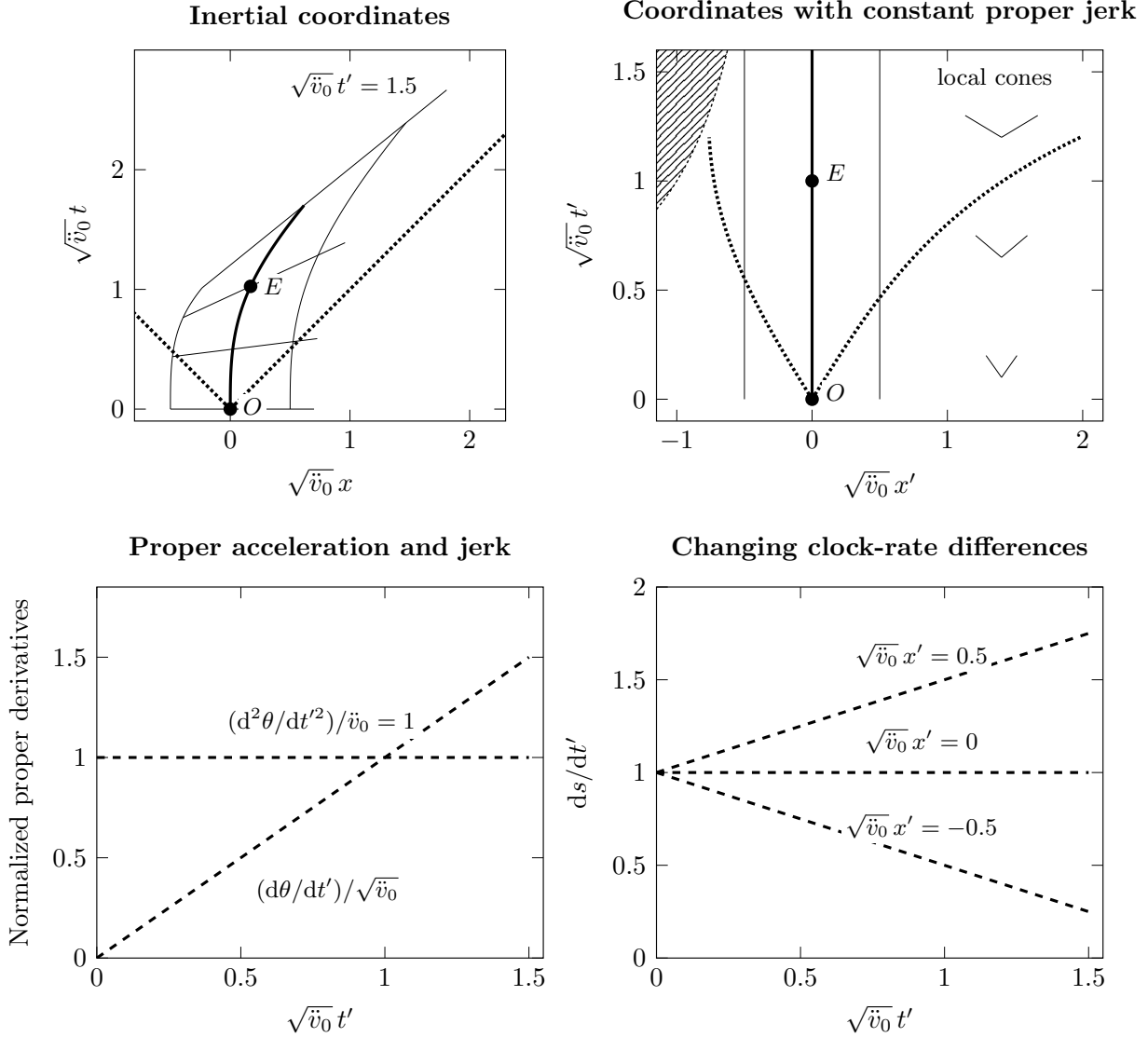


Figure 8: Constant proper jerk from rest and zero initial acceleration, with $x = \mathbf{u} \cdot \mathbf{r}$ and $x' = \mathbf{u} \cdot \mathbf{r}'$. Top: the same grid and dotted light rays in two coordinate charts. The solid grid worldlines have $\sqrt{\ddot{v}_0} x' = -1/2, 0, 1/2$; thin connecting lines on the left have fixed t' . The shared event E is on the origin at $\sqrt{\ddot{v}_0} t' = 1$. The small V shapes on the right show coordinate cones widening with time at a fixed positive x' . Hatching marks failure of $1 + \ddot{v}_0 t' x' > 0$, not by itself a horizon. Bottom: proper acceleration grows linearly, proper jerk stays constant, and the rates of separated grid clocks diverge from their initially common value. All plotted light rays remain null in the inertial chart.

SI units: Constant proper jerk from rest
Let $\ddot{v}_0$ be the constant signed proper jerk in m s^{-3}, and let t' be seconds. Then $\theta(t') = \frac{\ddot{v}_0 t'^2}{2c}, \quad d\tau^2 = \left(1 + \frac{\ddot{v}_0 t' \mathbf{u} \cdot \mathbf{r}'}{c^2}\right)^2 dt'^2 - \frac{d\mathbf{r}'^2}{c^2}.$

9.6.3 Constant unprojected four-jerk

If the components of d^2U/ds^2 are constant in an inertial frame, its value is determined by the initial coordinate-time derivatives:

$$\frac{d^2U}{ds^2} = \gamma_0^2 \ddot{U}_0 + \gamma_0 \dot{\gamma}_0 \dot{U}_0.$$

Here γ_0 and $\dot{\gamma}_0$ are the scalar parts of U_0 and $\dot{U}_0$, respectively. By direct integration,

$$\begin{aligned} U(s) &= U_0 + s\gamma_0 \dot{U}_0 + \frac{s^2}{2}(\gamma_0^2 \ddot{U}_0 + \gamma_0 \dot{\gamma}_0 \dot{U}_0), \\ X(s) &= X_0 + sU_0 + \frac{s^2}{2}\gamma_0 \dot{U}_0 + \frac{s^3}{6}(\gamma_0^2 \ddot{U}_0 + \gamma_0 \dot{\gamma}_0 \dot{U}_0). \end{aligned}$$

With U_0 future-directed, the following coefficient conditions are necessary and sufficient for $\det(U(s)) = 1$. Equating powers of s using the metric pairing in (5.3) gives

$$\begin{aligned} \det(U_0) &= 1, & \text{sc}(U_0^* \dot{U}_0) &= 0, & \text{sc}(U_0^* \ddot{U}_0) &= -\det(\dot{U}_0), \\ \gamma_0 \text{sc}(\dot{U}_0^* \ddot{U}_0) + \dot{\gamma}_0 \det(\dot{U}_0) &= 0, & \det(\gamma_0 \ddot{U}_0 + \dot{\gamma}_0 \dot{U}_0) &= 0. \end{aligned}$$

In particular, a nonzero constant d^2U/ds^2 must be null. If it is zero, the normalization conditions force $\dot{U}_0 = 0$, so the motion is inertial. Thus a constant four-acceleration paravector must vanish; constant proper acceleration does not mean constant inertial-frame components. The same conditions imply constant $\det(dU/ds)$. By (9.5), the proper-jerk magnitude is then the square of the proper-acceleration magnitude for this entire class.

For a concrete example (22, Eq. (3.40)), an initial rest state at the origin is chosen, with nonzero initial coordinate acceleration $\dot{\mathbf{v}}_0$ and jerk $\ddot{\mathbf{v}}_0$ constrained by

$$\dot{\mathbf{v}}_0 \cdot \ddot{\mathbf{v}}_0 = 0, \quad \ddot{\mathbf{v}}_0^2 = (\dot{\mathbf{v}}_0^2)^2.$$

Here $\gamma_0 = 1$ and $\dot{\gamma}_0 = 0$, so the initial paravector data are

$$X_0 = 0, \quad U_0 = 1, \quad \dot{U}_0 = \dot{\mathbf{v}}_0 \cdot \boldsymbol{\sigma}, \quad \ddot{U}_0 = \ddot{\mathbf{v}}_0^2 + \ddot{\mathbf{v}}_0 \cdot \boldsymbol{\sigma}.$$

The resulting world line is

$$X(s) = s + \frac{1}{6}\dot{\mathbf{v}}_0^2 s^3 + \left(\frac{1}{2}\dot{\mathbf{v}}_0 s^2 + \frac{1}{6}\ddot{\mathbf{v}}_0 s^3\right) \cdot \boldsymbol{\sigma}. \quad (9.16)$$

By differentiation, the following is verified:

$$\frac{d^2U}{ds^2} = \dot{\mathbf{v}}_0^2 + \ddot{\mathbf{v}}_0 \cdot \boldsymbol{\sigma}, \quad \det(U) = 1.$$

Its invariants are

$$\det \left(\frac{dU}{ds} \right) = -\dot{\mathbf{v}}_0^2, \quad \det \left(\frac{d^2U}{ds^2} \right) = 0.$$

Thus the proper-acceleration magnitude $\|\dot{\mathbf{v}}_0\|$ and, by (9.5), the proper-jerk magnitude $\dot{\mathbf{v}}_0^2 = \|\dot{\mathbf{v}}_0\|^2$ are both constant. Also $dt/ds = 1 + \dot{\mathbf{v}}_0^2 s^2/2 > 0$, so s is indeed future-directed proper time. This curved spatial path is a different motion from the fixed-axis constant-proper-jerk example.

10 Kinematics in a Frenet frame

Along the path $t, s, \ell \in \mathbb{R}$, $\mathbf{r}, \mathbf{v} \in \mathbb{R}^3$, and $X, U \in \mathbb{R} \oplus \mathbb{R}^3$. On the curved portions used below, $v \in (0, 1)$, $\gamma \in \mathbb{R}_{\geq 1}$, curvature $k \in \mathbb{R}_{>0}$, and torsion $\tau \in \mathbb{R}$ are scalars. The three frame vectors are in $\mathbb{R}^3$; their primes label directions rather than derivatives.

Velocity, acceleration, and jerk are resolved along the tangent and two normals of the spatial path. Turning and twisting are described by curvature and torsion.

10.1 Spatial frame and curvature

In the chosen inertial frame, the speed is $v := \|\mathbf{v}\| \in (0, 1)$ and spatial arc length is ℓ , with $\dot{\ell} = v$. Dots denote t derivatives. On a curved portion of the spatial path, the real unit tangent, principal normal, and binormal are defined by

$$\mathbf{u} := \frac{\mathbf{v}}{v}, \quad k := \left\| \frac{d\mathbf{u}}{d\ell} \right\| = \frac{\|\dot{\mathbf{u}}\|}{v} > 0, \quad \mathbf{u}' := \frac{\dot{\mathbf{u}}}{vk}, \quad \mathbf{u}'' := \mathbf{u} \times \mathbf{u}'.$$

Here $\mathbf{u}, \mathbf{u}', \mathbf{u}'' \in \mathbb{R}^3$ form a right-handed orthonormal frame. The primes are frame labels; derivatives are indicated by dots or fractions. The instantaneous plane of curvature is spanned by $\mathbf{u}, \mathbf{u}'$. Spatial curvature k is the turning angle per unit distance along the path; the radius of its local osculating circle is $1/k$. The signed torsion $\tau \in \mathbb{R}$ is defined by

$$\tau := -\frac{d\mathbf{u}''}{d\ell} \cdot \mathbf{u}' = -\frac{\dot{\mathbf{u}}'' \cdot \mathbf{u}'}{v}.$$

Here τ denotes spatial torsion; proper time remains s . The turning of the curvature plane per unit distance is described by τ ; for a planar curved path, $\tau = 0$.

By differentiating the orthonormality relations and using $d/dt = v d/d\ell$, the Frenet equations are obtained (25, Secs. 2.4–2.6):

$$\dot{\mathbf{u}} = vk\mathbf{u}', \quad \dot{\mathbf{u}}' = -vk\mathbf{u} + v\tau\mathbf{u}'', \quad \dot{\mathbf{u}}'' = -v\tau\mathbf{u}'. \quad (10.1)$$

At $v = 0$ the tangent is undefined; at $k = 0$ the principal normal and binormal are not fixed by the path. The unsplit kinematic formulas remain valid there. On a straight segment, fixed transverse axes may be chosen and the turning terms set to zero.

10.2 Coordinate velocity, acceleration, and jerk

Differentiating $\mathbf{v} = v\mathbf{u}$ and applying (10.1) gives

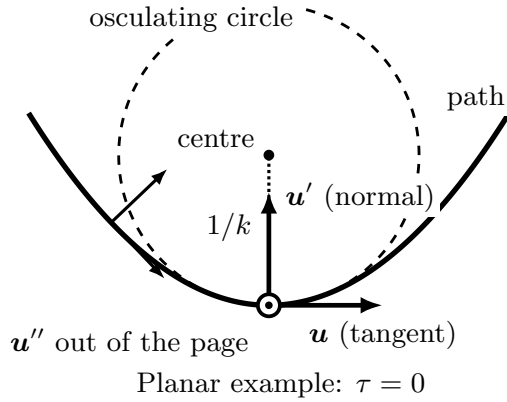
$$\begin{aligned}\mathbf{v} &= v\mathbf{u}, \\ \dot{\mathbf{v}} &= \underbrace{\dot{v}\mathbf{u}}_{\dot{\mathbf{v}}_{\parallel}: \text{tangential}} + \underbrace{v^2 k \mathbf{u}'}_{\dot{\mathbf{v}}_{\perp}: \text{perpendicular}}, \\ \ddot{\mathbf{v}} &= (\ddot{v} - v^3 k^2)\mathbf{u} + (3v\dot{v}k + v^2 \dot{k})\mathbf{u}' + v^3 k \tau \mathbf{u}''.\end{aligned}\tag{10.2}$$

Speed is changed by tangential acceleration; direction is changed by perpendicular acceleration. Acceleration is confined to the tangent–normal plane; a binormal component is introduced in jerk through torsion. Curvature is therefore the perpendicular-acceleration magnitude divided by speed squared. Both geometric quantities are recovered from coordinate derivatives as

$$\begin{aligned}k &= \frac{\|\dot{\mathbf{v}}_{\perp}\|}{v^2} = \frac{\|\mathbf{v} \times \dot{\mathbf{v}}\|}{v^3}, \\ \tau &= \frac{(\mathbf{v} \times \dot{\mathbf{v}}) \cdot \ddot{\mathbf{v}}}{\|\mathbf{v} \times \dot{\mathbf{v}}\|^2}.\end{aligned}\tag{10.3}$$

Figure 9 shows the moving axes and the acceleration decomposition for a planar example.

The path sets the local axes



Acceleration: speed and turning

changes direction

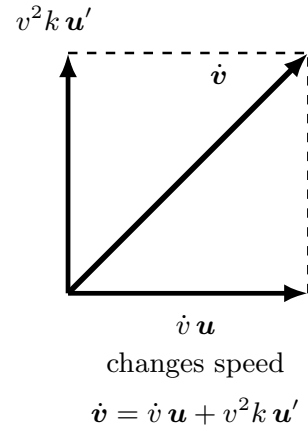


Figure 9: The Frenet frame follows the spatial path. At the marked point, the tangent $\mathbf{u}$ and principal normal $\mathbf{u}'$ lie in the osculating plane; $\mathbf{u}''$ points out of the page. The dashed osculating circle has radius $1/k$. Smaller arrows show how the tangent and normal change along the path. Right: tangential acceleration changes speed, while normal acceleration changes direction. The example is planar, so $\tau = 0$; primes label frame vectors, not derivatives.

10.3 Spacetime derivatives in the moving frame

The earlier longitudinal rest direction N is extended to the varying tangent $\mathbf{u}(t)$:

$$U = \gamma(1 + v\mathbf{u} \cdot \boldsymbol{\sigma}), \quad N := \gamma(v + \mathbf{u} \cdot \boldsymbol{\sigma}), \quad U, N \in \mathbb{R} \oplus \mathbb{R}^3.$$

The four directions $U, N, \mathbf{u}' \cdot \boldsymbol{\sigma}, \mathbf{u}'' \cdot \boldsymbol{\sigma}$ are Minkowski-orthogonal, with determinants $1, -1, -1, -1$. Thus a spatial Frenet frame is used to resolve spacetime quantities; U is the spacetime tangent, while $\mathbf{u}$ is the spatial tangent. The three rest directions generally rotate relative to nonrotating carried axes.

With $\dot{\gamma} = \gamma^3 v \dot{v}$ and $\mathbf{u} \cdot \boldsymbol{\sigma} = \gamma(N - vU)$, the complete frame evolution is

$$\begin{aligned} \dot{U} &= \gamma^2 \dot{v} N + \gamma v^2 k \mathbf{u}' \cdot \boldsymbol{\sigma}, \\ \dot{N} &= \gamma^2 \dot{v} U + \gamma v k \mathbf{u}' \cdot \boldsymbol{\sigma}, \\ \dot{\mathbf{u}}' \cdot \boldsymbol{\sigma} &= \gamma v^2 k U - \gamma v k N + v \tau \mathbf{u}'' \cdot \boldsymbol{\sigma}, \\ \dot{\mathbf{u}}'' \cdot \boldsymbol{\sigma} &= -v \tau \mathbf{u}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

Coefficients are paired with the same sign between time and spatial directions, and with opposite signs between spatial directions. Thus Minkowski orthonormality is preserved. Differentiating the first row gives

$$\begin{aligned} \ddot{U} &= (\gamma^4 \dot{v}^2 + \gamma^2 v^4 k^2) U \\ &+ (\gamma^2 (\ddot{v} - v^3 k^2) + 2\gamma \dot{\gamma} \dot{v}) N \\ &+ (3\gamma v \dot{v} k + 2\dot{\gamma} v^2 k + \gamma v^2 \dot{k}) \mathbf{u}' \cdot \boldsymbol{\sigma} + \gamma v^3 k \tau \mathbf{u}'' \cdot \boldsymbol{\sigma}. \end{aligned}$$

The proper-time derivatives follow by $d/ds = \gamma d/dt$:

$$\begin{aligned} \frac{dU}{ds} &= \gamma \dot{U} = \gamma^3 \dot{v} N + \gamma^2 v^2 k \mathbf{u}' \cdot \boldsymbol{\sigma}, \\ \frac{d^2 U}{ds^2} &= \gamma^2 \ddot{U} + \gamma \dot{\gamma} \dot{U}. \end{aligned} \tag{10.4}$$

The first is four-acceleration and the second is unprojected four-jerk. In the instantaneous rest frame, the tangential and perpendicular acceleration components are scaled by γ^3 and γ^2 , respectively, relative to their coordinate values. By orthogonality, the projection of four-jerk as in (9.4) is

$$\begin{aligned} \frac{d^2 U}{ds^2} + \det \left(\frac{dU}{ds} \right) U \\ = (\gamma^4 (\ddot{v} - v^3 k^2) + 3\gamma^6 v \dot{v}^2) N \\ + (3\gamma^5 v \dot{v} k + \gamma^3 v^2 \dot{k}) \mathbf{u}' \cdot \boldsymbol{\sigma} + \gamma^3 v^3 k \tau \mathbf{u}'' \cdot \boldsymbol{\sigma}. \end{aligned} \tag{10.5}$$

The squared proper-jerk magnitude is the sum of the squares of these three rest-space coefficients. In particular, differentiating only the acceleration coefficients would omit the rotation of the frame.

Spatial curvature versus worldline curvature. The invariant curvature of the spacetime worldline is the magnitude of its proper-time tangent's derivative:

$$\sqrt{-\det \left(\frac{dU}{ds} \right)} = \sqrt{\gamma^6 \dot{v}^2 + \gamma^4 v^4 k^2}. \tag{10.6}$$

In the natural units used here, this is the proper-acceleration magnitude. Both changing speed and changing spatial direction contribute. Spatial curvature k depends on the inertial frame; worldline curvature is Lorentz invariant. A spatially straight accelerated path has $k = 0$ but nonzero worldline curvature. For the trip in Section 8.11, these are $k = 0$ and $\dot{v}_0$ on each burn, respectively. In both cases the curvature is that of a path in flat spacetime; no curvature of spacetime itself is implied.

For uniform circular motion, v and k are constant and $\tau = 0$. The coordinate derivatives reduce to

$$\dot{\mathbf{v}} = v^2 k \mathbf{u}', \quad \ddot{\mathbf{v}} = -v^3 k^2 \mathbf{u}.$$

The proper-acceleration magnitude is the constant $\gamma^2 v^2 k$, while (10.5) reduces to

$$\frac{d^2 U}{ds^2} + \det \left(\frac{dU}{ds} \right) U = -\gamma^4 v^3 k^2 N \neq 0.$$

Thus constant acceleration magnitude is compatible with nonzero proper jerk: the acceleration direction is changing.

11 Coordinates, frames, and differentiation

Fields have domain $\mathbb{R}^4$ (or an open coordinate region). The symbols ∂_t , $\partial_{\mathbf{r}}$, ∂ , ∂^* , and $\square$ are differential operators. On a scalar field $f : \mathbb{R}^4 \rightarrow \mathbb{C}$, $\partial_t f, \square f \in \mathbb{C}$, $\partial_{\mathbf{r}} f \in \mathbb{C}^3$, and $\partial f, \partial^* f \in \mathbb{C} \oplus \mathbb{C}^3$ pointwise. Equivalently, on smooth scalar fields ∂_t and $\square$ map $C^\infty(\mathbb{R}^4, \mathbb{C})$ to itself, $\partial_{\mathbf{r}}$ maps it to $C^\infty(\mathbb{R}^4, \mathbb{C}^3)$, and ∂, ∂^* map it to $C^\infty(\mathbb{R}^4, \mathbb{C} \oplus \mathbb{C}^3)$. For real f , these outputs lie in the corresponding real spaces. On a paravector field, ∂ and ∂^* use algebra multiplication, so even a real input may have a complex output.

11.1 Paravector differential operators

Cartesian inertial coordinates are denoted by $t \in \mathbb{R}$ and $\mathbf{r} = \{r_1, r_2, r_3\} \in \mathbb{R}^3$. The scalar and vector derivatives are defined by

$$\partial_t := \frac{\partial}{\partial t}, \quad \partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}.$$

The partial paravector and its sigma conjugate are defined by (8, Eq. (176))

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sum_{n=1}^3 \sigma_n \frac{\partial}{\partial r_n}, \quad (11.1)$$

$$\partial^* = \text{adj}(\partial) = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}.$$

The sigmas are constant. Derivatives act from the left on everything to their right unless parentheses specify their scope. For fields with commuting second derivatives, anticommutation cancels all mixed spatial terms:

$$\begin{aligned} (\boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})^2 &= \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} + \sum_{n < m} (\sigma_n \sigma_m + \sigma_m \sigma_n) \frac{\partial^2}{\partial r_n \partial r_m} \\ &= \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

The mixed time–space terms likewise cancel. Both orders therefore give the scalar d’Alembertian (8, Eq. (178))

$$\partial\partial^* = \partial^*\partial = \partial_t^2 - \boldsymbol{\partial}_{\mathbf{r}} \cdot \boldsymbol{\partial}_{\mathbf{r}} =: \square. \quad (11.2)$$

These are operator identities: $\partial\partial^*Z$ means $\partial(\partial^*Z)$.

11.2 Components and the product rule

Real scalar functions $a, b : \mathbb{R}^4 \rightarrow \mathbb{R}$ and real vector functions $\mathbf{X}, \mathbf{Y} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the complex paravector field

$$Z(t, \mathbf{r}) = (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}.$$

Ordered vector multiplication gives the real and imaginary scalar and vector parts of ∂Z :

$$\begin{aligned} \partial Z &= (\partial_t a + \boldsymbol{\partial}_{\mathbf{r}} \cdot \mathbf{X}) + i(\partial_t b + \boldsymbol{\partial}_{\mathbf{r}} \cdot \mathbf{Y}) \\ &\quad + (\boldsymbol{\partial}_{\mathbf{r}} a + \partial_t \mathbf{X} - \boldsymbol{\partial}_{\mathbf{r}} \times \mathbf{Y}) \cdot \boldsymbol{\sigma} \\ &\quad + i(\boldsymbol{\partial}_{\mathbf{r}} b + \partial_t \mathbf{Y} + \boldsymbol{\partial}_{\mathbf{r}} \times \mathbf{X}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Multiplication by i exchanges real and imaginary parts, with the former imaginary parts acquiring a minus sign.

Here $Z : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$. For a second field $Z' : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$, the prime labels the field rather than a derivative. The scalar product rule is applied before any factors are reordered:

$$\begin{aligned} \partial(ZZ') &= (\partial_t Z)Z' + Z(\partial_t Z') \\ &\quad + \sum_{n=1}^3 \sigma_n \left(\frac{\partial Z}{\partial r_n} Z' + Z \frac{\partial Z'}{\partial r_n} \right) \\ &= (\partial Z)Z' + Z(\partial Z') + \sum_{n=1}^3 [\sigma_n, Z] \frac{\partial Z'}{\partial r_n}. \end{aligned}$$

For a complex scalar first factor, the commutators vanish and the usual product rule is recovered. Partial derivatives are taken at fixed remaining coordinates. Along a path $\mathbf{r} : I \rightarrow \mathbb{R}^3$, $I \subseteq \mathbb{R}$, with $\mathbf{v} = \dot{\mathbf{r}} \in \mathbb{R}^3$, the total derivative is instead

$$\dot{Z} = \partial_t Z + \sum_{n=1}^3 \dot{r}_n \frac{\partial Z}{\partial r_n} = (\partial_t + \mathbf{v} \cdot \boldsymbol{\partial}_{\mathbf{r}})Z.$$

All fields and partial derivatives are evaluated at $(t, \mathbf{r}(t))$.

11.3 Generalized coordinates and clock rates

Here $(t, \mathbf{r})$ are real chart coordinates. Smooth functions $a : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{X} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, restricted to an open region with nonsingular Jacobian, specify inertial coordinates:

$$X = a(t, \mathbf{r}) + \mathbf{X}(t, \mathbf{r}) \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3.$$

Here a is scalar-valued inertial time and $\mathbf{X}$ is vector-valued inertial position. Chart time t and chart position $\mathbf{r}$ need not be clock readings and Cartesian distances.

Interval. By differentiation,

$$\begin{aligned} dX &= (\partial_t a + \partial_t \mathbf{X} \cdot \boldsymbol{\sigma}) dt + \sum_{k=1}^3 (\partial_{r_k} a + \partial_{r_k} \mathbf{X} \cdot \boldsymbol{\sigma}) dr_k, \\ \det(dX) &= da^2 - d\mathbf{X}^2. \end{aligned}$$

This is the flat Minkowski interval in changed coordinates; varying coefficients alone do not imply spacetime curvature.

Motion and clock rates. Along a future-directed time-like path, s is proper time, $ds^2 = \det(dX) > 0$, and $da > 0$. On a segment parametrized by increasing chart time t , overdots denote total t derivatives. The inertial velocity and particle Lorentz factor are

$$\mathbf{v} := \frac{d\mathbf{X}}{da} = \frac{\dot{\mathbf{X}}}{\dot{a}} \in \mathbb{R}^3, \quad \mathbf{v}^2 < 1, \quad \gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} = \frac{da}{ds}.$$

Consequently,

$$\dot{s} = \sqrt{\dot{a}^2 - \dot{\mathbf{X}}^2} = \frac{\dot{a}}{\gamma}, \quad \frac{d}{ds} = \frac{\gamma}{\dot{a}} \frac{d}{dt}, \quad U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (11.3)$$

Here $\dot{a} > 0$. The earlier rule $d/ds = \gamma d/dt$ is recovered when $t = a$. The two acceleration derivatives are related by

$$\dot{\mathbf{v}} = \frac{\ddot{\mathbf{X}}\dot{a} - \dot{\mathbf{X}}\ddot{a}}{\dot{a}^2}, \quad \frac{d\mathbf{v}}{da} = \frac{\dot{\mathbf{v}}}{\dot{a}}.$$

Equation (8.2) is applied with inertial time a and acceleration $d\mathbf{v}/da$. For inertial motion, $dU/ds = 0$ implies $d\mathbf{v}/da = 0$, hence

$$\mathbf{X} = \mathbf{X}_0 + \mathbf{v}_0 a, \quad \mathbf{X}_0, \mathbf{v}_0 \in \mathbb{R}^3, \quad \mathbf{v}_0^2 < 1.$$

The reference inertial time is $a = 0$; the same path may be nonlinear in chart coordinates.

11.4 Varying frame transformations

A coordinate chart is held fixed while the local frame is varied along a curve. Here s is any real curve parameter; it is specialized to proper time in Section 11.5. A smooth complex paravector $T(s)$ with $\det(T) = 1$ is chosen. Here $T(s), Z(s) \in \mathbb{C} \oplus \mathbb{C}^3$. Use the scalar-free generator $\Omega = (dT/ds)T^{-1}$ from Section 8.3. That section derives the second derivative of a congruence; here the same differentiation rule is also applied to fields transformed by similarity. By the inverse rule,

$$\frac{dT^{-1}}{ds} = -T^{-1}\Omega.$$

For a complex paravector $Z(s)$ transformed by similarity, $Z' = TZT^{-1}$, product differentiation yields

$$\frac{dZ'}{ds} = T \frac{dZ}{ds} T^{-1} + [\Omega, Z']. \quad (11.4)$$

Differentiating again and replacing $T(dZ/ds)T^{-1}$ by its first-derivative expression gives

$$\begin{aligned} \frac{d^2 Z'}{ds^2} &= T \frac{d^2 Z}{ds^2} T^{-1} + 2 \left[\Omega, \frac{dZ'}{ds} \right] \\ &\quad + \left[\frac{d\Omega}{ds}, Z' \right] - [\Omega, [\Omega, Z']]. \end{aligned}$$

For a real paravector transformed instead by congruence, $Z' = TZT^H$, the same rule gives

$$\frac{dZ'}{ds} = T \frac{dZ}{ds} T^H + \Omega Z' + Z' \Omega^H. \quad (11.5)$$

Equations (11.4) and (11.5) include the frame terms omitted by a constant-factor transformation. For a varying rotation $T = R = \exp(-i\theta(s)\mathbf{u}(s) \cdot \boldsymbol{\sigma}/2)$, with real $\theta(s)$ and **real unit vector** $\mathbf{u}(s)$, unitarity gives $\Omega^H = -\Omega$ and $\Omega \in \{0\} \oplus i\mathbb{R}^3$, so both formulas agree.

Position-dependent sigma bases and metric constructions are developed in Section 19.

11.5 Physical derivatives in a carried frame

Here s is proper time along a future-directed time-like worldline, $U = dX/ds \in \mathbb{R} \oplus \mathbb{R}^3$, and $\det(U) = 1$. From (11.5), the physical four-acceleration expressed in the local frame is

$$T \frac{dU}{ds} T^H = \frac{dU'}{ds} - \Omega U' - U' \Omega^H. \quad (11.6)$$

The derivative of the transformed components includes motion of the frame; its contribution is removed by the two Ω terms. Similarly, dX'/ds includes frame terms and generally differs from the transformed tangent U' .

If the frame is kept instantaneously at rest with the particle, then $U' = TUT^H = 1$ along the curve, so

$$T \frac{dU}{ds} T^H = -(\Omega + \Omega^H) = -2\operatorname{re}(\Omega).$$

A constant rest-frame four-velocity therefore does not imply zero physical acceleration. Its components are encoded in the Hermitian part of Ω .

Differentiating this expression, removing the frame terms, and applying the projection in (9.4) gives

$$T \left(\frac{d^2U}{ds^2} + \det \left(\frac{dU}{ds} \right) U \right) T^H = -\frac{d(\Omega + \Omega^H)}{ds} + [\Omega, \Omega^H]. \quad (11.7)$$

Rotation of the carried axes is accounted for by the commutator. For axes kept aligned with ideal torque-free gyroscopes, $\Omega = \Omega^H$, so the proper jerk is $-2 d\Omega/ds$: the proper-time rate of the measured acceleration. The fixed-axis results of Section 9.3 are recovered with $T = \exp(-\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and fixed $\mathbf{u}$.

12 Electromagnetism

All coefficient fields depend on real inertial coordinates $(t, \mathbf{r}) \in \mathbb{R} \times \mathbb{R}^3$. The electric and magnetic fields $\mathbf{E}, \mathbf{B} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the complex field vector $\mathbf{F} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$ and its scalar-free paravector $F \in \{0\} \oplus \mathbb{C}^3$ (5, Eq. (23)):

$$\mathbf{F} := \mathbf{E} + i\mathbf{B}, \quad F := \mathbf{F} \cdot \boldsymbol{\sigma}. \quad (12.1)$$

The charge density $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}$ and current density $\mathbf{J} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the real four-current paravector:

$$J := \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \quad J^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.$$

The paravector derivative of (11.1) is used, with spatial gradient $\partial_{\mathbf{r}}$ and d'Alembertian $\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2$. The symbols ∂ , ∂^* , and $\square$ denote differential operators.

The dot/cross product (2.8) combines metric and orientation. With a spatial derivative it gives divergence and curl, packing the six real electromagnetic components into one complex field F . The product explains the compact form; Maxwell dynamics remains physical input, derived from an action in Section 12.3.

12.1 Maxwell's equations and wave propagation

Maxwell's four equations are the single field law (8, Eq. (177)):

$$\boxed{\partial F = J^*} \quad (12.2)$$

The product rule expands its left side into

$$\begin{aligned} \partial F = & \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i \partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \\ & + \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \\ & + \underbrace{i(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \end{aligned}$$

Equating the four parts gives (7, Eqs. (1.4)–(1.7)):

$$\text{Real scalar: } \partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (12.3)$$

$$\text{Imaginary scalar: } \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (12.4)$$

$$\text{Real vector: } \partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} = -\mathbf{J}, \quad (12.5)$$

$$\text{Imaginary vector: } \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E} = 0. \quad (12.6)$$

These are Gauss' electric and magnetic laws, Ampère–Maxwell's law, and Faraday's law.

SI units: Maxwell's equation

With t in seconds, $\mathbf{r}$ in metres, electric field in volts per metre, magnetic field in teslas, charge density in coulombs per cubic metre, and current density in amperes per square metre,

$$\begin{aligned} F &= (\mathbf{E} + ic\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad \partial = \frac{1}{c} \partial_t + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}, \\ \partial F &= \frac{1}{\epsilon_0} \left(\rho - \frac{\mathbf{J}}{c} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

The electric and magnetic terms now have the same units. By separating the scalar and vector parts, the four SI Maxwell equations are recovered; the vacuum constants satisfy $\epsilon_0 \mu_0 c^2 = 1$.

In natural units, applying ∂^* gives the sourced wave equation:

$$\partial^*(\partial F) = \square F = \partial^* J^*.$$

Taking its star yields (8, Eq. (152)):

$$\boxed{\square F^* = \partial J.} \quad (12.7)$$

By expansion of the source derivative,

$$\partial J = (\partial_t \rho + \boldsymbol{\partial}_r \cdot \mathbf{J}) + (\partial_t \mathbf{J} + \boldsymbol{\partial}_r \rho + i \boldsymbol{\partial}_r \times \mathbf{J}) \cdot \boldsymbol{\sigma}.$$

Because F is scalar-free, its real and imaginary scalar parts vanish. Matching all four parts gives (7, Eq. (1.1)):

$$\text{Real scalar: } 0 = \partial_t \rho + \boldsymbol{\partial}_r \cdot \mathbf{J}, \quad (12.8)$$

$$\text{Imaginary scalar: } 0 = 0,$$

$$\text{Real vector: } \square \mathbf{E} = -\partial_t \mathbf{J} - \boldsymbol{\partial}_r \rho, \quad (12.9)$$

$$\text{Imaginary vector: } \square \mathbf{B} = \boldsymbol{\partial}_r \times \mathbf{J}. \quad (12.10)$$

Equation (12.8) is charge conservation; (12.9) and (12.10) are the sourced field waves. In source-free vacuum, $J = 0$ and $\square F = 0$: both fields obey the free wave equation.

12.2 Potentials and gauge freedom

The scalar potential $V : \mathbb{R}^4 \rightarrow \mathbb{R}$ and vector potential $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the real four-potential; its four-divergence is also defined:

$$A := V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \quad S := \text{sc}(\partial A) \in \mathbb{R}.$$

By the product rule,

$$\partial A = (\partial_t V + \boldsymbol{\partial}_r \cdot \mathbf{A}) + (\partial_t \mathbf{A} + \boldsymbol{\partial}_r V) \cdot \boldsymbol{\sigma} + i(\boldsymbol{\partial}_r \times \mathbf{A}) \cdot \boldsymbol{\sigma}.$$

Subtracting its scalar part gives the field (7, p. 105):

$$\boxed{F^* = \partial A - S.} \quad (12.11)$$

Thus $\partial A = S + F^* \in \mathbb{R} \oplus \mathbb{C}^3$, with components:

$$\text{Real scalar: } S = \partial_t V + \boldsymbol{\partial}_r \cdot \mathbf{A},$$

$$\text{Imaginary scalar: } 0 = 0,$$

$$\text{Real vector: } \mathbf{E} = -\partial_t \mathbf{A} - \boldsymbol{\partial}_r V, \quad (12.12)$$

$$\text{Imaginary vector: } \mathbf{B} = \boldsymbol{\partial}_r \times \mathbf{A}. \quad (12.13)$$

Equation (12.11) separates the four-divergence from the field; (12.12) and (12.13) satisfy (12.4) and (12.6) identically.

Applying ∂^* gives the source equations in any gauge:

$$\square A = \partial^*(\partial A) = \partial^*(S + F^*) = \partial^* S + (\partial F)^*.$$

Using Maxwell's equation (12.2) gives (7, Eq. (6.1)):

$$\boxed{\square A = \partial^* S + J.}$$

With scalar and vector parts separated,

$$\text{Real scalar: } \square V - \partial_t S = \rho,$$

$$\text{Imaginary scalar: } 0 = 0,$$

$$\text{Real vector: } \square \mathbf{A} + \boldsymbol{\partial}_r S = \mathbf{J},$$

$$\text{Imaginary vector: } \mathbf{0} = \mathbf{0}.$$

These source equations hold before gauge fixing. For a real gauge function $\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}$, the transformation is defined by (7, Eq. (6.2)):

$$A' := A + \partial^* \lambda.$$

By differentiation, $\partial A' = \partial A + \square \lambda$. Only the scalar changes, so the field is gauge invariant:

$$S' = S + \square \lambda, \quad F' = F.$$

In the Lorenz gauge $S = 0$, the potential wave equations are (7, Eq. (6.4)):

$$\text{Real scalar: } \square V = \rho,$$

$$\text{Real vector: } \square \mathbf{A} = \mathbf{J}.$$

12.3 The electromagnetic field action

For real scalar potential $V(t, \mathbf{r})$ and vector potential $\mathbf{A}(t, \mathbf{r})$, the notation $A = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ is used. With prescribed conserved current $J = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$, Maxwell dynamics follows from the scalar Lagrangian density $L : \mathbb{R}^4 \rightarrow \mathbb{R}$ (7, Eq. (5.42)) (7, Eq. (5.43))

$$L = \frac{1}{2} \text{re}(F^2) - \text{sc}(A^* J) = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) - \rho V + \mathbf{J} \cdot \mathbf{A}. \quad (12.14)$$

For boundary-vanishing variations $\delta V \in \mathbb{R}$ and $\delta \mathbf{A} \in \mathbb{R}^3$, the induced field variations $\delta \mathbf{E}, \delta \mathbf{B} \in \mathbb{R}^3$ satisfy

$$\delta \mathbf{E} = -\partial_{\mathbf{r}} \delta V - \partial_t \delta \mathbf{A},$$

$$\delta \mathbf{B} = \partial_{\mathbf{r}} \times \delta \mathbf{A}.$$

By integration by parts,

$$\delta \int L \, dt \, d^3 \mathbf{r} = \int ((\partial_{\mathbf{r}} \cdot \mathbf{E} - \rho) \delta V + (\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} + \mathbf{J}) \cdot \delta \mathbf{A}) \, dt \, d^3 \mathbf{r}.$$

From independent variations, (12.3) and (12.5); the potential identities supply (12.4) and (12.6). Together they recover (12.2).

12.4 Field invariants

Since $F \in \{0\} \oplus \mathbb{C}^3$, its adjugate is $-F$. Its square is a complex scalar, $F^2 \in \mathbb{C}$ (3, p. 106)

$$F^2 = -\det(F) = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B}. \quad (12.15)$$

For a complex [invertible paravector](#) T , the field similarity $F' = T F T^{-1}$ preserves this scalar by (5.1). Physical rotations and boosts use $R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, respectively, with [real unit vector](#) $\mathbf{u}$ and real angle or rapidity θ ; their field laws are derived in Section 18. The real and imaginary parts of (12.15) are therefore observer-independent. At a point with a nonzero field, they give the classification

Condition	Consequence
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 > \mathbf{B}^2$	Some frame has $\mathbf{B}' = 0$.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 < \mathbf{B}^2$	Some frame has $\mathbf{E}' = 0$.
$\mathbf{E} \cdot \mathbf{B} \neq 0$	Neither field can be eliminated.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 = \mathbf{B}^2$	Null field: equal, perpendicular fields.

For the first two rows, the following factors are chosen, respectively:

$$\tanh(\theta) \mathbf{u} = \frac{\mathbf{E} \times \mathbf{B}}{\mathbf{E}^2}, \quad \tanh(\theta) \mathbf{u} = \frac{\mathbf{E} \times \mathbf{B}}{\mathbf{B}^2}.$$

Both speeds are less than one. Substitution in (18.7) and (18.8) gives $\mathbf{B}' = 0$ and $\mathbf{E}' = 0$, respectively.

12.5 Lorentz force from the action

A dot is used for a total derivative along the particle's path with respect to coordinate time t , for example $\dot{\mathbf{P}} = d\mathbf{P}/dt$. The field derivatives ∂_t are partial derivatives at fixed position.

Notation: the real four-velocity $U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$, an [invertible paravector](#) with $UU^{-1} = 1$. Charge and constant mass are denoted by $q \in \mathbb{R}$ and $m > 0$, respectively. The real momentum is $P = mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, with $E = m\gamma$ and $\mathbf{P} = m\gamma\mathbf{v}$. The particle variables are $\mathbf{v}, \mathbf{P} \in \mathbb{R}^3$ and $U, P \in \mathbb{R} \oplus \mathbb{R}^3$, with $\gamma \in \mathbb{R}_{\geq 1}$ and $E \in \mathbb{R}$. Here $L : \mathbb{R} \times \mathbb{R}^3 \times \{\mathbf{v} \in \mathbb{R}^3 : \mathbf{v}^2 < 1\} \rightarrow \mathbb{R}$ is a particle Lagrangian, distinct from the field Lagrangian density above. It is fixed by the action (7, Eq. (5.41))

$$\int L dt = -m \int ds - q \int \text{sc}(A^*U) ds. \quad (12.16)$$

Since $\text{sc}(A^*U) = \gamma(V - \mathbf{v} \cdot \mathbf{A})$, the Lagrangian is

$$L = -m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}.$$

Its Euler–Lagrange terms are

$$\begin{aligned} \partial_{\mathbf{v}} L &= \mathbf{P} + q\mathbf{A}, \\ \frac{d(\partial_{\mathbf{v}} L)}{dt} &= \dot{\mathbf{P}} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \\ \partial_{\mathbf{r}} L &= -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}). \end{aligned}$$

Here spatial differentiation holds the independent velocity argument fixed. Using $\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A} = \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})$ gives (7, Eq. (5.32))

$$\dot{\mathbf{P}} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (12.17)$$

Here (12.12) and (12.13) turn the potential derivatives into the fields. The Hamiltonian is

$$H = \mathbf{v} \cdot (\mathbf{P} + q\mathbf{A}) - L = E + qV. \quad (12.18)$$

Using (7.18) in (12.18),

$$\dot{E} + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V = -\partial_t L = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}.$$

Canceling $q\partial_t V$ and using (12.12) gives (7, p. 116)

$$\dot{E} = q\mathbf{v} \cdot \mathbf{E}. \quad (12.19)$$

This is the mechanical work rate. Thus $\mathbf{P} + q\mathbf{A}$ and $H = E + qV$ are canonical momentum and energy. The momentum $\mathbf{P}$ is kinetic; the particle energy E includes rest energy and excludes the potential energy qV . Canonical energy is conserved for static potentials in the chosen gauge. A time-dependent gauge transformation can change H while leaving E and the force unchanged.

The complex product of the field and four-velocity is expanded:

$$\begin{aligned} \frac{FU}{\gamma} &= \mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &\quad + i(\mathbf{v} \cdot \mathbf{B} + (\mathbf{B} - \mathbf{v} \times \mathbf{E}) \cdot \boldsymbol{\sigma}). \end{aligned} \quad (12.20)$$

The real scalar and vector parts reproduce (12.19) and (12.17). Since P is real and $d/ds = \gamma d/dt$, they combine into (10, Eq. (2))

$$\boxed{\frac{dP}{ds} = q \operatorname{re}(FU) = \frac{q}{2}(FU + UF^H).} \quad (12.21)$$

Hermitian extraction enforces real kinetic four-momentum and reproduces the action-derived force. The imaginary part need not vanish for an electric charge.

12.6 Acceleration and the mass shell

Notation: the real four-velocity $U = dX/ds$, an [invertible paravector](#) with $UU^{-1} = 1$ and $U^{-1} = U^*$, real momentum $P = mU$, and coordinate acceleration $\dot{\mathbf{v}} = d\mathbf{v}/dt$. Scalar-part cyclicity proves tangency to the mass shell:

$$\begin{aligned} \operatorname{sc}(U^* dP/ds) &= \frac{q}{2} \operatorname{sc}(U^* FU + F^H) \\ &= \frac{q}{2} (\operatorname{sc}(F) + \operatorname{sc}(F^H)) = 0. \end{aligned}$$

This agrees with $\operatorname{sc}(U^* dU/ds) = 0$ in Section 8.1. Combining (8.2) with the force components in (12.20) gives

$$\begin{aligned} m\gamma^3 \mathbf{v} \cdot \dot{\mathbf{v}} &= q\mathbf{v} \cdot \mathbf{E}, \\ m\gamma(\dot{\mathbf{v}} + \gamma^2(\mathbf{v} \cdot \dot{\mathbf{v}})\mathbf{v}) &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \end{aligned}$$

The coordinate acceleration is obtained by eliminating $\mathbf{v} \cdot \dot{\mathbf{v}}$

$$\boxed{\dot{\mathbf{v}} = \frac{q}{m\gamma}(\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E})\mathbf{v}).}$$

The magnetic term changes momentum without doing work.

Parallel and perpendicular components. For $\mathbf{v} \neq 0$, all projections here are relative to the instantaneous velocity, as in Section 8.1:

$$\mathbf{E}_{\parallel} = \frac{(\mathbf{v} \cdot \mathbf{E})\mathbf{v}}{\mathbf{v}^2}, \quad \mathbf{E}_{\perp} = \mathbf{E} - \mathbf{E}_{\parallel}.$$

Since $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$, projecting (12.17) and using (8.3) gives

$$\begin{aligned} \dot{\mathbf{P}}_{\parallel} &= m\gamma^3 \dot{\mathbf{v}}_{\parallel} = q\mathbf{E}_{\parallel}, \\ \dot{\mathbf{P}}_{\perp} &= m\gamma \dot{\mathbf{v}}_{\perp} = q(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}). \end{aligned}$$

The subscripts denote projections of the force along the instantaneous velocity axis. Thus

$$\begin{aligned}\dot{\mathbf{v}}_{\parallel} &= \frac{q}{m\gamma^3} \mathbf{E}_{\parallel}, \\ \dot{\mathbf{v}}_{\perp} &= \frac{q}{m\gamma} (\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}).\end{aligned}\tag{12.22}$$

The parallel component changes speed; the perpendicular component changes direction. The magnetic force contributes only to the latter. At $\mathbf{v} = 0$, the velocity axis is undefined and the unsplit equation gives $\dot{\mathbf{v}} = q\mathbf{E}/m$.

Proper acceleration and the rest-frame field. Taking the determinant of the real force gives

$$\begin{aligned}-\det\left(\frac{dP}{ds}\right) &= -m^2 \det\left(\frac{dU}{ds}\right) \\ &= q^2 \gamma^2 ((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2).\end{aligned}$$

The invariant squared proper acceleration is obtained by division by m^2 . For $\mathbf{v} \neq 0$, orthogonality gives

$$(\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2 = \gamma^{-2} \mathbf{E}_{\parallel}^2 + \|\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}\|^2.$$

Equivalently, substituting (12.22) into the acceleration determinant from Section 8.1 gives

$$\begin{aligned}-\det\left(\frac{dU}{ds}\right) &= \gamma^6 \dot{\mathbf{v}}_{\parallel}^2 + \gamma^4 \dot{\mathbf{v}}_{\perp}^2 \\ &= \frac{q^2}{m^2} (\mathbf{E}_{\parallel}^2 + \gamma^2 \|\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}\|^2).\end{aligned}\tag{12.23}$$

The boost into the instantaneous rest frame gives $\mathbf{E}' = \mathbf{E}_{\parallel} + \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B})$, as derived in (18.7). Thus the squared proper acceleration is $q^2 \|\mathbf{E}'\|^2 / m^2$: the particle experiences only electric force at that instant in its rest frame.

SI units: Lorentz force and acceleration

Here t is in seconds, $\mathbf{v}$ in metres per second, m in kilograms, q in coulombs, $\mathbf{E}$ in volts per metre, and $\mathbf{B}$ in teslas. Dots denote derivatives with respect to this t , and $\gamma = (1 - \mathbf{v}^2/c^2)^{-1/2}$. For $\mathbf{v} \neq 0$, the same projections give

$$\begin{aligned}\dot{\mathbf{P}}_{\parallel} &= q\mathbf{E}_{\parallel}, & \dot{\mathbf{P}}_{\perp} &= q(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}), \\ \dot{\mathbf{v}}_{\parallel} &= \frac{q}{m\gamma^3} \mathbf{E}_{\parallel}, & \dot{\mathbf{v}}_{\perp} &= \frac{q}{m\gamma} (\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}).\end{aligned}$$

The force is in newtons and acceleration in ms^{-2} . The unsplit form, also valid at rest, is

$$\dot{\mathbf{v}} = \frac{q}{m\gamma} \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} - \frac{\mathbf{v} \cdot \mathbf{E}}{c^2} \mathbf{v} \right).$$

SI units: Lorentz force and acceleration (continued)

With τ in seconds, $\mathbf{r}$ in metres, $s = c\tau$, $X = ct + \mathbf{r} \cdot \boldsymbol{\sigma}$, and dimensionless $U = dX/ds$, the squared proper-acceleration magnitude is

$$\begin{aligned} -c^4 \det \left(\frac{dU}{ds} \right) &= \gamma^6 \dot{\mathbf{v}}_{\parallel}^2 + \gamma^4 \dot{\mathbf{v}}_{\perp}^2 \\ &= \frac{q^2}{m^2} (\mathbf{E}_{\parallel}^2 + \gamma^2 \|\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}\|^2), \end{aligned}$$

in $\text{m}^2 \text{s}^{-4}$. At rest it reduces to $q^2 \|\mathbf{E}\|^2 / m^2$ without choosing a projection axis.

12.7 Energy and momentum flux

Notation: $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Its Hermitian product is (8, Eq. (180))

$$\frac{1}{2} F F^{\text{H}} = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (12.24)$$

The energy density $u = (\mathbf{E}^2 + \mathbf{B}^2)/2 \in \mathbb{R}_{\geq 0}$ and Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B} \in \mathbb{R}^3$ are defined. These coefficients are frame-dependent, whereas F^2 contains the Lorentz invariants.

SI units: Field energy and flux

Using the SI field $F = (\mathbf{E} + ic\mathbf{B}) \cdot \boldsymbol{\sigma}$,

$$\begin{aligned} u &= \frac{\epsilon_0}{2} (\mathbf{E}^2 + c^2 \mathbf{B}^2), \quad \mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}, \\ \frac{\epsilon_0}{2} F F^{\text{H}} &= u + \frac{\mathbf{S}}{c} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here u is energy per unit volume in J m^{-3} , and $\mathbf{S}$ is energy flux in W m^{-2} .

In natural units, $\frac{1}{2} F F^{\text{H}} = u + \mathbf{S} \cdot \boldsymbol{\sigma}$. By (2.15) and scalar scaling,

$$u^2 - \mathbf{S}^2 = \frac{1}{4} \|\det(F)\|^2 \geq 0.$$

Since $u \geq 0$, this gives $\|\mathbf{S}\| \leq u$, with equality for null fields.

SI units: Field determinant and flux bound

In SI units, the determinant and flux bound are

$$\begin{aligned} u^2 - \frac{\mathbf{S}^2}{c^2} &= \frac{\epsilon_0^2}{4} \|\det(F)\|^2 \geq 0, \\ \|\mathbf{S}\| &\leq cu. \end{aligned}$$

Equality holds for null fields.

Returning to natural units, reversal of the order gives $F^{\text{H}} F = (F F^{\text{H}})^*$: the same energy density and opposite flux. The transformation of these energy and flux components is derived in Section 18.

Using Maxwell's vector equations,

$$\begin{aligned}\partial_t u &= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J}) - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \\ &= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E},\end{aligned}$$

where $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$. Thus Poynting's energy-balance law is (7, Eq. (5.45))

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (12.25)$$

Equivalently, the left side is $\frac{1}{2}\partial_t \text{sc}(FF^H) + \frac{1}{2}\partial_{\mathbf{r}} \cdot \text{vec}(FF^H)$. The term $\mathbf{J} \cdot \mathbf{E}$ is the power delivered to matter per unit volume.

13 Quantum amplitudes and differential operators

Here $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ has real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$, $m > 0$ is mass, $q \in \mathbb{R}$ is charge, and $\hbar = c = 1$. Amplitudes are complex functions of these real coordinates; their scalar or paravector type is specified below.

13.1 Spinor amplitudes and probability density

All products and norms here act on paravector values at one real event X . They sum algebra components only; no spatial integration is implied. For $\varphi, \psi \in \mathbb{C} \oplus \mathbb{C}^3$, their algebraic inner product is $\langle \varphi | \psi \rangle := \text{sc}(\varphi^H \psi) \in \mathbb{C}$. Two entries separated by a vertical bar denote this inner product; one entry in angle brackets denotes a normalized expectation below.

For arbitrary constant perpendicular **real unit vectors** $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$, a constant $\psi_+ \in \mathbb{C} \oplus \mathbb{C}^3$ of unit norm with $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_+ = \psi_+$ is chosen, and $\psi_- = (\mathbf{u}' \cdot \boldsymbol{\sigma})\psi_+$ is defined. Then

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_{\pm} = \pm\psi_{\pm}. \quad (13.1)$$

The pair is orthonormal:

$$\begin{aligned}\text{sc}(\psi_+^H \psi_+) &= 1, \\ \text{sc}(\psi_-^H \psi_-) &= 1, \\ \text{sc}(\psi_+^H \psi_-) &= \text{sc}(\psi_-^H \psi_+) = 0.\end{aligned}$$

The two cross terms are complex conjugates. A Pauli amplitude or either Dirac entry is (11, Eq. (69a))

$$\psi(t, \mathbf{r}) = c_+(t, \mathbf{r})\psi_+ + c_-(t, \mathbf{r})\psi_-. \quad (13.2)$$

Here $c_+, c_- : \mathbb{R}^4 \rightarrow \mathbb{C}$ and $\psi : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$. The fixed basis in (13.2) restricts the amplitude to two independent complex coefficients, with column $(c_+, c_-)^T \in \mathbb{C}^2$. Left multiplication by any paravector preserves this two-component form. Here $\psi_{\pm}$ denote constant basis elements; in the Dirac sections they denote fields taking values in this same subspace. Left multiplication by $\mathbf{u} \cdot \boldsymbol{\sigma}$ or $\mathbf{u}' \cdot \boldsymbol{\sigma}$ preserves this two-component form; these generate all paravector left actions. The matrix verification is in Appendix A.5.

The local squared norm $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}_{\geq 0}$ is (12, Eq. (1.1a))

$$\rho := \|\psi\|^2 = \text{sc}(\psi^H \psi) \geq 0. \quad (13.3)$$

This ρ denotes probability density, distinct from charge density in the electromagnetic sections. Orthonormality gives $\rho = \|c_+\|^2 + \|c_-\|^2$. For a Pauli wavefunction normalized to unit total probability, this is its position density; Dirac uses the sum over its two entries. Total spatial normalization is separate from the pointwise norm. This probability interpretation does not apply to an arbitrary scalar KG field.

13.2 Weighted densities and local expectations

$Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is taken to be real ($Z^H = Z$), with $c \in \mathbb{R}$ and $\mathbf{Z} \in \mathbb{R}^3$. It acts by multiplication, not differentiation. For $\mathbf{Z} \neq 0$, the real unit vector $\mathbf{u} = \mathbf{Z}/\|\mathbf{Z}\|$ is used with the projectors $\Pi_\pm = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ from Section 3.2. Their spectral decomposition gives

$$\text{sc}(\psi^H Z \psi) = (c + \|\mathbf{Z}\|)\|\Pi_+ \psi\|^2 + (c - \|\mathbf{Z}\|)\|\Pi_- \psi\|^2. \quad (13.4)$$

For $\mathbf{Z} = 0$, this reduces to $c\|\psi\|^2$. Thus the product is nonnegative for every ψ exactly when $c \geq \|\mathbf{Z}\|$. For such a *nonnegative weight*, the following is defined:

$$\|\psi\|_Z^2 := \text{sc}(\psi^H Z \psi) = \|\sqrt{Z} \psi\|^2 \geq 0. \quad (13.5)$$

The square root has nonnegative eigenvalues, as in Section 3. For $Z = 1$, this is ρ . It is a true squared norm when $c > \|\mathbf{Z}\|$; at equality it can vanish for nonzero ψ , so the notation then denotes a weighted density. Projectors are an important example. Reality alone is insufficient: $Z = -1$ gives $\text{sc}(\psi^H Z \psi) = -\rho$.

Local expectations. For $\rho > 0$ and $Z \in \mathbb{C} \oplus \mathbb{C}^3$ acting by multiplication, the scalar expectation $\langle Z \rangle_\psi \in \mathbb{C}$ is defined by (13, Eq. (6.1))

$$\langle Z \rangle_\psi := \frac{\text{sc}(\psi^H Z \psi)}{\rho}. \quad (13.6)$$

For real Z , this is real and may have either sign. Dividing (13.4) by ρ shows its meaning: it averages the two eigenvalues with their projected probabilities. For fields this mean is conditional on position; it is not a spatial average. A whole-wavefunction expectation additionally requires spatial weighting. The same formula extends linearly to complex Z , including commutators. For a nonnegative weight, $\langle Z \rangle_\psi = \|\psi\|_Z^2/\rho$. For a projector, this ratio is a Born probability.

Transforming the amplitude. For a constant complex *invertible paravector* T , the transformation $\psi' = T\psi$ is applied. For a nonnegative weight Z , the following is obtained by substitution:

$$\|\psi'\|_Z^2 = \text{sc}(\psi^H T^H Z T \psi) = \|\psi\|_{T^H Z T}^2. \quad (13.7)$$

This also proves that the transformed weight is nonnegative. The same substitution in (13.6) applies to any complex Z . With $\rho' = \|\psi'\|^2 > 0$, it gives

$$\langle Z \rangle_{\psi'} = \frac{\rho}{\rho'} \langle T^H Z T \rangle_\psi. \quad (13.8)$$

For $T = R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real angle θ and real unit $\mathbf{u}$, unitarity preserves both $\rho' = \rho$ and $\langle \varphi' | \psi' \rangle = \langle \varphi | \psi \rangle$. For $T = L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real rapidity θ , this one-sided law instead gives $\rho' = \|\psi\|_{L^2}^2$; the norm and inner product are generally not invariant. The two Dirac entries use opposite boost factors, as derived in Section 18.

13.3 Energy and momentum as differential operators

Notation: $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$.

SI units: Wave energy and momentum

In the following wave relations, SI units are restored and Planck's constant $h > 0$ and $\hbar = h/(2\pi)$ are retained. For a free wave, angular frequency in radians per second is denoted by $\omega \in \mathbb{R}$, and the wavenumber vector in cycles per metre by $\mathbf{k} \in \mathbb{R}^3$. For $\mathbf{k} \neq 0$, the wavelength $\lambda > 0$ satisfies $\|\mathbf{k}\| = 1/\lambda$.

The physical input is the Planck–Einstein energy relation and de Broglie momentum relation (14, p. 248) (14, p. 249):

$$E = \hbar\omega, \quad (13.9)$$

$$\mathbf{P} = \hbar\mathbf{k}. \quad (13.10)$$

Here $E \in \mathbb{R}$ is total energy in joules, including rest energy, and $\mathbf{P} \in \mathbb{R}^3$ is momentum in kg ms^{-1} . Photoelectric measurements test the energy relation (15, p. 62); electron diffraction tests the momentum relation (14, p. 255).

For constant E and $\mathbf{P}$, with time t in seconds and position $\mathbf{r}$ in metres, the plane wave $f : \mathbb{R}^4 \rightarrow \mathbb{C}$ is

$$\begin{aligned} f(t, \mathbf{r}) &:= \exp(-i(\omega t - 2\pi\mathbf{k} \cdot \mathbf{r})), \\ f(t, \mathbf{r}) &= \exp\left(-\frac{i}{\hbar}(Et - \mathbf{P} \cdot \mathbf{r})\right). \end{aligned}$$

The second line substitutes (13.9) and (13.10). By differentiation,

$$\begin{aligned} +i\hbar\partial_t f &= Ef, \\ -i\hbar\partial_{\mathbf{r}} f &= \mathbf{P}f. \end{aligned}$$

These operators represent energy and momentum on each mode and extend to superpositions by linearity.

Returning to natural units $\hbar = c = 1$ gives $h = 2\pi$. With $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ and $P = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, the following is obtained (16, Eq. (4.104))

$$f(X, P) = \exp(-i(Et - \mathbf{P} \cdot \mathbf{r})). \quad (13.11)$$

The two eigenvalue equations combine into

$$i\partial^* f = Pf. \quad (13.12)$$

The canonical energy and momentum operators act on smooth complex fields. On scalar inputs their value spaces are $\mathbb{C}$ and $\mathbb{C}^3$, respectively; the combined four-momentum output lies in $\mathbb{C} \oplus \mathbb{C}^3$. They are defined by

$$\begin{aligned} \hat{E} &:= +i\partial_t, \\ \hat{\mathbf{P}} &:= -i\partial_{\mathbf{r}}, \end{aligned} \quad (13.13)$$

which combine to form the canonical four-momentum operator:

$$\hat{P} := i\partial^* = \hat{E} + \hat{\mathbf{P}} \cdot \boldsymbol{\sigma}. \quad (13.14)$$

Here $\hat{E}$ is a scalar differential operator, $\hat{\mathbf{P}}$ is a triple of scalar differential operators, and $\hat{P}$ is a paravector differential operator. These act on fields, not elements of the numerical coefficient spaces. On a smooth scalar field ψ , the pointwise outputs satisfy $\hat{E}\psi \in \mathbb{C}$, $\hat{\mathbf{P}}\psi \in \mathbb{C}^3$, and $\hat{P}\psi \in \mathbb{C} \oplus \mathbb{C}^3$. On spinor fields, $\hat{P}$ preserves the fixed subspace of (13.2). The coupled operators have the same types. In a field, the following notation is retained: the real potential paravector $A(t, \mathbf{r}) = V(t, \mathbf{r}) + \mathbf{A}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$; $V : \mathbb{R}^4 \rightarrow \mathbb{R}$, $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, and $A : \mathbb{R}^4 \rightarrow \mathbb{R} \oplus \mathbb{R}^3$. Within the coupling formulas, subscript 0 labels the uncoupled operators of (13.13):

$$\begin{aligned}\hat{E}_0 &:= +i\partial_t, \\ \hat{\mathbf{P}}_0 &:= -i\partial_{\mathbf{r}}, \\ \hat{P}_0 &:= i\partial^*.\end{aligned}$$

The charge is $q \in \mathbb{R}$. The particle action gives canonical energy $E + qV$ and momentum $\mathbf{P} + q\mathbf{A}$; identifying them with translation operators defines the gauge-covariant energy and kinetic momentum operators (12, Eqs. (2.11)–(2.12)):

$$\hat{E}_q := \hat{E}_0 - qV, \quad (13.15)$$

$$\hat{\mathbf{P}}_q := \hat{\mathbf{P}}_0 - q\mathbf{A}, \quad (13.16)$$

which combine to form the kinetic four-momentum operator:

$$\hat{P}_q := \hat{P}_0 - qA = \hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}. \quad (13.17)$$

Here $\hat{E}_q$ includes rest energy and excludes the potential energy qV . Hats distinguish named differential operators from physical values and mode eigenvalues $E, \mathbf{P}$. The derivative symbols ∂ , $\partial_{\mathbf{r}}$, and $\square$ remain unhatted. Fields obtained by differentiating potentials are coefficient fields, not differential operators. The operators act to the right. The adjugate of $\hat{P}_q$ reverses the vector part without conjugating i . Operator order matters in a field; an ordinary paravector determinant cannot replace their product. The canonical operators commute, giving

$$\hat{P} \text{ adj}(\hat{P}) = \det(\hat{P}) = \hat{E}^2 - \hat{\mathbf{P}}^2 = -\square. \quad (13.18)$$

The same result is obtained by reversal of the order.

13.4 Field strength from noncommuting derivatives

Notation: the real electric and magnetic fields from (12.12) and (12.13), and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real scalar in (12.11). The real potential is $A = V + \mathbf{A} \cdot \boldsymbol{\sigma}$; $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is a complex vector paravector.

The operators are applied to a differentiable complex amplitude. The product rule cancels all derivatives of the amplitude in the commutators; writing $\hat{P}_{q,j} := (\hat{\mathbf{P}}_q)_j$ for Cartesian components $j, k = 1, 2, 3$,

$$\begin{aligned}[\hat{E}_q, \hat{P}_{q,j}] &= -iq(\partial_t A_j + \partial_j V) = iqE_j, \\ [\hat{P}_{q,j}, \hat{P}_{q,k}] &= iq(\partial_j A_k - \partial_k A_j), \\ \hat{\mathbf{P}}_q \times \hat{\mathbf{P}}_q &= iq\mathbf{B}.\end{aligned}$$

Sigma multiplication and the energy commutator consequently give (12, Eq. (2.13)) (12, p. 6)

$$(\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 = \hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}, \quad (13.19)$$

$$[\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}. \quad (13.20)$$

Here E_j is an electric-field component. Reversing the commutator order and combining the electric and magnetic terms also gives

$$\begin{aligned} [\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}, \hat{E}_q] &= -iq\mathbf{E} \cdot \boldsymbol{\sigma}, \\ \hat{\mathbf{P}}_q^2 + iqF &= (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 - [\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}, \hat{E}_q]. \end{aligned}$$

Using (13.19) and (13.20) in the ordered products gives

$$\begin{aligned} (\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})(\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) &= \hat{E}_q^2 - (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 - [\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] \\ &= \hat{E}_q^2 - \hat{\mathbf{P}}_q^2 - iqF, \end{aligned} \quad (13.21)$$

$$\begin{aligned} (\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})(\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) &= \hat{E}_q^2 - (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 + [\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] \\ &= \hat{E}_q^2 - \hat{\mathbf{P}}_q^2 + iqF^*. \end{aligned} \quad (13.22)$$

Their common scalar part gives KG; the vector terms give the Dirac spin coupling.

14 The Klein–Gordon equation

Here $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, and $m \in \mathbb{R}_{>0}$. Except during the explicit quantization step below, the amplitudes $\psi, \psi_{\pm}$ are scalar fields $\mathbb{R}^4 \rightarrow \mathbb{C}$. $\hat{E}$, $\square$, and the spectral operator $\hat{\Omega}$ act on such scalar fields; $\hat{\mathbf{P}}$ is a triple of scalar operators, and $\hat{P}$ is a paravector operator. In the charge-conjugation and coupled subsections, $q \in \mathbb{R}$, $V : \mathbb{R}^4 \rightarrow \mathbb{R}$, $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, and $A : \mathbb{R}^4 \rightarrow \mathbb{R} \oplus \mathbb{R}^3$.

14.1 The mass shell as a wave equation

A scalar field $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ of constant mass $m \in \mathbb{R}_{>0}$ is considered. The derivation in Section 7.6 gives the real momentum $P = mU \in \mathbb{R} \oplus \mathbb{R}^3$ and its mass shell $\det(P) = E^2 - \mathbf{P}^2 = m^2$. Represent $E \in \mathbb{R}$ and $\mathbf{P} \in \mathbb{R}^3$ by the translation operators $\hat{E} = i\partial_t$ and $\hat{\mathbf{P}} = -i\partial_{\mathbf{r}}$ from (13.13). Imposing the mass shell on the scalar wavefunction gives

$$\begin{aligned} 0 &= (\det(\hat{P}) - m^2)\psi \\ &= (\hat{E}^2 - \hat{\mathbf{P}}^2 - m^2)\psi \\ &= (-\partial_t^2 + \partial_{\mathbf{r}}^2 - m^2)\psi \\ &= -(\square + m^2)\psi. \end{aligned}$$

Here the free coefficients commute, so the operator determinant is the scalar operator in (13.18). This yields the Klein–Gordon equation (16, Eq. (1.13)):

$$\boxed{(\square + m^2)\psi = 0.} \quad (14.1)$$

14.2 Positive and negative frequency

Notation: the complex scalar field $\psi(t, \mathbf{r})$ and real mass $m > 0$. Here the branch fields $\psi_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}$ are scalars, not Dirac entries. The positive scalar operator is defined by

$$\hat{\Omega} := \sqrt{m^2 - \partial_{\mathbf{r}}^2}, \quad \hat{\Omega}^2 = \det(m + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}).$$

Since $\hat{\Omega}$ commutes with ∂_t , KG factors as

$$\square + m^2 = \partial_t^2 + \hat{\Omega}^2 = (\partial_t - i\hat{\Omega})(\partial_t + i\hat{\Omega}).$$

The reverse product is the same. Its two frequency branches obey

$$(\partial_t \pm i\hat{\Omega})\psi_{\pm} = 0, \quad \psi_{\pm}(t, \mathbf{r}) = e^{\mp i\hat{\Omega}t} \psi_{\pm}(0, \mathbf{r}).$$

Their sum gives the general solution:

$$\boxed{\psi(t, \mathbf{r}) = e^{-i\hat{\Omega}t} \psi_+(0, \mathbf{r}) + e^{i\hat{\Omega}t} \psi_-(0, \mathbf{r}).} \quad (14.2)$$

The two initial data determine its complex branch fields by

$$\psi_{\pm}(0, \mathbf{r}) = \frac{1}{2}(\psi(0, \mathbf{r}) \pm i\hat{\Omega}^{-1} \partial_t \psi(0, \mathbf{r})).$$

14.3 Square roots from Fourier series

For spatially periodic free KG solutions, the following discrete Fourier series is used:

$$\psi = \sum_n c_n f_n, \quad f_n(t, \mathbf{r}) = \exp(i(\omega_n t + 2\pi \mathbf{k}_n \cdot \mathbf{r})).$$

Here $n \in \mathbb{N}$ enumerates modes, $f_n : \mathbb{R}^4 \rightarrow \mathbb{C}$, $c_n \in \mathbb{C}$ are constant coefficients, $\omega_n \in \mathbb{R}$ are signed angular phase frequencies, and $\mathbf{k}_n \in \mathbb{R}^3$ are wavenumber vectors in cycles per unit length. For $\mathbf{k}_n \neq 0$, its real wavelength $\lambda_n > 0$ satisfies $\|\mathbf{k}_n\| = 1/\lambda_n$. By (13.10), the mode momentum is $2\pi \mathbf{k}_n$ in these units. The $+\omega_n t$ convention gives energy eigenvalue $-\omega_n$. By differentiation,

$$\begin{aligned} \square \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 - \omega_n^2) c_n f_n, \\ (\square + m^2) \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 + m^2 - \omega_n^2) c_n f_n. \end{aligned}$$

Thus each nonzero KG coefficient requires

$$\omega_n^2 = 4\pi^2 \mathbf{k}_n^2 + m^2, \quad \omega_n = \pm \sqrt{4\pi^2 \mathbf{k}_n^2 + m^2}.$$

Both signs are independent modes (16, pp. 33–34); no common time period is assumed. A square-root operator acts on the same coefficients by

$$\sqrt{\square} \psi = \sum_n \sqrt{4\pi^2 \mathbf{k}_n^2 - \omega_n^2} c_n f_n.$$

One square-root branch is chosen consistently. Squaring restores $\square$, giving

$$\square + m^2 = (m - i\sqrt{\square})(m + i\sqrt{\square}).$$

For the positive spatial root, the same series gives

$$\begin{aligned}\hat{\Omega}^2\psi &= \sum_n (m^2 + 4\pi^2\mathbf{k}_n^2) c_n f_n, \\ \hat{\Omega}\psi &= \sum_n \sqrt{m^2 + 4\pi^2\mathbf{k}_n^2} c_n f_n.\end{aligned}$$

Thus $e^{\mp i\hat{\Omega}t}$ multiplies each spatial mode by $e^{\mp i\sqrt{m^2 + 4\pi^2\mathbf{k}_n^2}t}$. On every KG mode, $4\pi^2\mathbf{k}_n^2 - \omega_n^2 = -m^2$. Thus a fixed scalar square-root branch gives

$$\sqrt{\square}\psi = im\psi \quad \text{or} \quad \sqrt{\square}\psi = -im\psi,$$

with the same choice for both signs of ω_n . Unlike $\hat{\Omega}$, this root has imaginary eigenvalues on KG modes; its branch does not select the frequency sign. That separation comes from the factors $(\partial_t - i\hat{\Omega})(\partial_t + i\hat{\Omega})$, whose solutions evolve as $e^{+i\hat{\Omega}t}$ and $e^{-i\hat{\Omega}t}$, respectively.

14.4 Particles, antiparticles, and charge conjugation

KG supplies two frequency branches. Charge conjugation relates them; field quantization identifies their positive-energy particle excitations. We retain the scalar amplitudes $\psi, \psi_{\pm}$, mass $m > 0$, charge $q \in \mathbb{R}$, and Fourier modes defined above. On scalars, $*$ is complex conjugation.

Opposite charge, equal mass. The free spatial operator $\hat{\Omega}$ is real, so

$$i\partial_t\psi_- = -\hat{\Omega}\psi_- \implies i\partial_t\psi_-^* = \hat{\Omega}\psi_-^*.$$

Conjugation turns the negative-energy amplitude into a positive-energy one and reverses its momentum. In the same real background $V, \mathbf{A}$,

$$(\hat{E}_q\psi)^* = -(i\partial_t + qV)\psi^*, \quad (\hat{\mathbf{P}}_q\psi)^* = -(-i\partial_{\mathbf{r}} + q\mathbf{A})\psi^*.$$

Applying these identities twice gives

$$(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi = 0 \implies (-(i\partial_t + qV)^2 + (-i\partial_{\mathbf{r}} + q\mathbf{A})^2 + m^2)\psi^* = 0. \quad (14.3)$$

Thus ψ^* obeys KG with charge $-q$ and the same mass. This relation holds for varying potentials, although a fixed-frequency split need not survive a time-dependent background.

Charge density is signed. The free scalar Lagrangian density $L = (\partial_t\psi)^*\partial_t\psi - (\partial_{\mathbf{r}}\psi)^* \cdot \partial_{\mathbf{r}}\psi - m^2\psi^*\psi$ gives KG and the conserved current, with charge sign fixed by $\hat{E}_q = i\partial_t - qV$ (16, Eqs. (2.67), (2.74)):

$$\begin{aligned}\rho &= iq(\psi^*\partial_t\psi - (\partial_t\psi)^*\psi), \\ \mathbf{J} &= -iq(\psi^*\partial_{\mathbf{r}}\psi - (\partial_{\mathbf{r}}\psi)^*\psi), \quad J = \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ \partial_t\rho + \partial_{\mathbf{r}} \cdot \mathbf{J} &= iq(\psi^*\square\psi - (\square\psi)^*\psi) = 0.\end{aligned}$$

For $\psi = c_n f_n$, $\rho = -2q\omega_n|c_n|^2$ and $\mathbf{J} = 4\pi q\mathbf{k}_n|c_n|^2$. The two frequency signs therefore carry opposite charge densities, not opposite probability densities. Conjugating both ψ and q preserves this physical current.

Positive-energy particles and antiparticles. Canonical bosonic quantization is an additional physical assumption. For this step only, reuse c_n as quantum mode operators and ψ as an operator-valued scalar field; H is the operator adjoint. For this step, absorb the spatial normalization into f_n : its squared magnitude integrates to one over the periodic spatial region, while its phase is unchanged. Including both frequency signs gives

$$\psi = \sum_n \frac{c_n f_n}{\sqrt{2|\omega_n|}}.$$

Quantization of L gives the oscillator commutators

$$[c_n, c_k^H] = \begin{cases} +\delta_{nk}, & \omega_n < 0, \\ -\delta_{nk}, & \omega_n > 0, \end{cases} \quad [c_n, c_k] = 0. \quad (14.4)$$

For $\omega_n < 0$, c_n annihilates and c_n^H creates; for $\omega_n > 0$, their roles reverse. The second sign reflects this reversed order, not a negative state norm. All annihilation operators kill the vacuum (16, Eqs. (2.69)–(2.73)).

The classical energy integrates $(\partial_t \psi)^* \partial_t \psi + (\partial_{\mathbf{r}} \psi)^* \cdot \partial_{\mathbf{r}} \psi + m^2 \psi^* \psi$; charge integrates ρ . Replacing conjugation by the operator adjoint and substituting the modes eliminates cross terms by spatial orthogonality and the mass shell. Placing creation operators first and subtracting vacuum constants gives

$$\text{energy above vacuum} = \sum_{\omega_n < 0} (-\omega_n) c_n^H c_n + \sum_{\omega_n > 0} (+\omega_n) c_n c_n^H, \quad (14.5)$$

$$\text{charge above vacuum} = \sum_{\omega_n < 0} (+q) c_n^H c_n + \sum_{\omega_n > 0} (-q) c_n c_n^H. \quad (14.6)$$

Both products are nonnegative occupation operators; the commutators show that each creation raises its occupation by one. The energy, momentum, and charge added by one creation are

	creation	E	$\mathbf{P}$	charge
particle ($\omega_n < 0$)	c_n^H	$-\omega_n > 0$	$+2\pi \mathbf{k}_n$	$+q$
antiparticle ($\omega_n > 0$)	c_n	$+\omega_n > 0$	$-2\pi \mathbf{k}_n$	$-q$

Both satisfy $E^2 - \mathbf{P}^2 = m^2$: equal mass, opposite charge, and positive excitation energy. The negative-frequency field sector creates antiparticles, not negative-energy particles (16, pp. 33–34).

For a real field, $\psi^* = \psi$ relates the two branches: quantization gives one neutral species, its own antiparticle. A neutral complex field can still have distinct conjugate species; $q = 0$ alone does not impose reality. This derivation concerns spin-zero particles and does not fix their relative abundance. Spin one-half follows from the spinor representation in the Dirac sections. Subsequent wave equations again use ordinary amplitudes rather than quantum field operators.

14.5 Electromagnetic coupling

Notation: the real potentials $V, \mathbf{A}$, charge q , and coupled operators $\hat{E}_q = i\partial_t - qV$, $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$. The real potential paravector is $A = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, with real scalar $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$. The scalar equation in a field is (17, Eq. (299))

$$\boxed{(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi = 0.} \quad (14.7)$$

The mass-shell equation uses the scalar part $\hat{E}_q^2 - \hat{\mathbf{P}}_q^2$ of (13.21) or (13.22). By the product rule,

$$\hat{E}_q^2 = -\partial_t^2 - iq(\partial_t V) - 2iqV\partial_t + q^2V^2, \quad (14.8)$$

$$\hat{\mathbf{P}}_q^2 = -\partial_{\mathbf{r}}^2 + iq(\partial_{\mathbf{r}} \cdot \mathbf{A}) + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2\mathbf{A}^2. \quad (14.9)$$

Substituting (14.8) and (14.9) into (14.7) gives

$$\boxed{(\square + iqS - q^2 \det(A) + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}))\psi = 0.}$$

All potential derivatives are retained.

15 The Dirac equation

The independent variables are $(t, \mathbf{r}) \in \mathbb{R} \times \mathbb{R}^3$, with mass $m \in \mathbb{R}_{>0}$. The entry fields are $\psi_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$, each restricted to the fixed two-component form of (13.2). Their two-entry column ψ therefore has four independent complex coordinates. The free and coupled Dirac operators act on smooth fields of this form. When coupling is introduced, $q \in \mathbb{R}$, $V : \mathbb{R}^4 \rightarrow \mathbb{R}$, $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, and $A : \mathbb{R}^4 \rightarrow \mathbb{R} \oplus \mathbb{R}^3$.

15.1 Block form and free factorization

Notation: $m > 0$ and real space–time coordinates $(t, \mathbf{r})$. In the Weyl representation, a Dirac field is the two-entry block

$$\psi(t, \mathbf{r}) = \begin{Bmatrix} \psi_+(t, \mathbf{r}) \\ \psi_-(t, \mathbf{r}) \end{Bmatrix}.$$

Each entry lies in $\mathbb{C} \oplus \mathbb{C}^3$ and has the fixed two-component form of (13.2). Each has coordinates in $\mathbb{C}^2$, so the block has coordinates in $\mathbb{C}^4$. On blocks, Hermitian conjugation also transposes entries; sigma conjugation acts entrywise. Summing the two local squared norms gives the nonnegative real density (16, Eqs. (4.97)–(4.99))

$$\|\psi\|^2 = \text{sc}(\psi_+^H \psi_+) + \text{sc}(\psi_-^H \psi_-). \quad (15.1)$$

For $Z \in \mathbb{C} \oplus \mathbb{C}^3$, the complex-linear block map is defined by

$$\mathbf{W}(Z) := \begin{Bmatrix} 0 & \text{adj}(Z) \\ Z & 0 \end{Bmatrix}. \quad (15.2)$$

Intrinsically $\mathbf{W}(Z)$ is a 2×2 array with paravector entries acting on the two Dirac entries by left multiplication. In spinor coordinates, $\mathbf{W}(Z)$ and $\mathbf{I}$ are matrices in $\mathbb{C}^{4 \times 4}$. For differential-operator arguments, $\mathbf{W}(\hat{P})$, $\hat{\mathbf{D}}$, and $\hat{\mathbf{D}}_q$ are instead matrix differential operators on four-component fields. Here $\mathbf{I}$ is the two-entry block identity, represented by $\mathbf{I}_4$. Applying $\mathbf{W}(Z)$ twice multiplies the first entry by $\text{adj}(Z)Z$ and the second by $Z \text{adj}(Z)$. Both products are the paravector determinant, so

$$\mathbf{W}(Z)^2 = \det(Z)\mathbf{I}.$$

Sigma conjugation on each block entry gives $\mathbf{W}(Z)^* = \mathbf{W}(Z^*)$; Hermitian conjugation also transposes the block:

$$\mathbf{W}(Z)^H = \mathbf{W}(Z^*).$$

Thus $\mathbf{W}(Z)$ is not generally Hermitian. For operator arguments, the adjugate reverses the vector part without reordering derivatives. For the free field ($q = 0$), the canonical operators in (13.13) are used. The block map and Dirac operator are

$$\begin{aligned}\mathbf{W}(\hat{P}) &= \begin{Bmatrix} 0 & \text{adj}(\hat{P}) \\ \hat{P} & 0 \end{Bmatrix}, \\ \hat{\mathbf{D}}(+m) &:= \mathbf{W}(\hat{P}) - m\mathbf{I}, \\ \hat{\mathbf{D}}(-m) &= \mathbf{W}(\hat{P}) + m\mathbf{I}.\end{aligned}\tag{15.3}$$

The Dirac equation is (16, Eq. (4.51))

$$\boxed{\hat{\mathbf{D}}(m)\psi = 0.}\tag{15.4}$$

Using $\mathbf{W}(\hat{P})^2 = (\det(\hat{P}))\mathbf{I} = -\square\mathbf{I}$ from (13.18), multiplication by the opposite-mass operator gives (16, Eqs. (4.53)–(4.54))

$$\begin{aligned}\hat{\mathbf{D}}(+m)\hat{\mathbf{D}}(-m) &= \hat{\mathbf{D}}(-m)\hat{\mathbf{D}}(+m) \\ &= \mathbf{W}(\hat{P})^2 - m^2\mathbf{I} = -(\square + m^2)\mathbf{I}.\end{aligned}\tag{15.5}$$

Thus every free Dirac solution satisfies the wave form of the mass shell $\det(P) = E^2 - \mathbf{P}^2 = m^2$ in (7.12):

$$(\det(\hat{P}) - m^2)\psi = 0 \iff (\square + m^2)\psi = 0.$$

The original first-order constraint is the coupled pair (16, Eq. (4.108))

$$i\partial\psi_- = (\hat{E} - \hat{\mathbf{P}} \cdot \boldsymbol{\sigma})\psi_- = m\psi_+, \tag{15.6}$$

$$i\partial^*\psi_+ = (\hat{E} + \hat{\mathbf{P}} \cdot \boldsymbol{\sigma})\psi_+ = m\psi_-. \tag{15.7}$$

On one Fourier mode, the eigenvalue $s \in \mathbb{C}$ is defined by $\hat{\mathbf{D}}(m)\psi = s\psi$ for a nonzero amplitude. The spectral parameter s here is distinct from proper time. Write $\square\psi = \lambda\psi$, with $\lambda = -E^2 + \mathbf{P}^2 \in \mathbb{R}$ for real mode energy E and momentum $\mathbf{P}$. Thus $\square$ remains a differential operator. Since $\hat{\mathbf{D}}(m) + m\mathbf{I} = \mathbf{W}(\hat{P})$, (15.5) gives

$$(\hat{\mathbf{D}}(m) + m\mathbf{I})^2\psi = -\square\psi \implies ((s + m)^2 + \lambda)\psi = 0.$$

Thus the eigenvalues are

$$s = -m \pm i\sqrt{\lambda}.$$

When distinct, each allows two independent complex coefficients in ψ_+ , with $\psi_- = i\partial^*\psi_+/(s + m)$. Each root therefore occurs twice, giving the scalar characteristic equation

$$\boxed{((s + m)^2 + \lambda)^2 = 0.}\tag{15.8}$$

At $\lambda = 0$ the roots coalesce with total algebraic multiplicity four. For KG modes, $\lambda = -m^2$, so $s = 0, -2m$; the Dirac equation selects $s = 0$. The matrix verification is (A.14).

15.2 Reconstructing free solutions from Klein–Gordon

A KG solution $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$ is combined with the constant unit basis amplitude $\eta_+ \in \mathbb{C} \oplus \mathbb{C}^3$ from the fixed basis of (13.2). For constants $a, b \in \mathbb{C}$, a two-constant construction is

$$\psi = \frac{1}{m} \begin{Bmatrix} ma + ib\partial \\ ia\partial^* + mb \end{Bmatrix} \psi' \eta_+. \quad (15.9)$$

Substituting (15.9) into (15.6) and (15.7) gives

$$\begin{aligned} i\partial\psi_- - m\psi_+ &= -\frac{a}{m}(\square + m^2)\psi' \eta_+ = 0, \\ i\partial^*\psi_+ - m\psi_- &= -\frac{b}{m}(\square + m^2)\psi' \eta_+ = 0. \end{aligned}$$

Taking $a = 1, b = 0$ gives one scalar-seeded solution. To span every Dirac field, ψ is instead expressed in the form (13.2), with both scalar coefficients solving KG. Then

$$\boxed{\psi = \begin{Bmatrix} \psi \\ i\partial^*\psi/m \end{Bmatrix}.} \quad (15.10)$$

The second Dirac equation holds by construction; the first is KG. Conversely, every massive free solution has this form with $\psi = \psi_+$. Equation (14.2) evolves its two coefficients.

15.3 Potential and paravector components

Notation: the real $A = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, real charge q , and $\hat{E}_q = i\partial_t - qV$, $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$. The kinetic four-momentum is $\hat{P}_q = \hat{P}_0 - qA$ from (13.17). For the fixed potential A and charge q , the following interacting Dirac operator is used:

$$\begin{aligned} \hat{\mathbf{D}}_q(m) &:= \mathbf{W}(\hat{P}_q) - m\mathbf{I} \\ &= \begin{Bmatrix} -m & \text{adj}(\hat{P}_q) \\ \hat{P}_q & -m \end{Bmatrix}. \end{aligned} \quad (15.11)$$

Minimal coupling gives (8, Eq. (191))

$$\boxed{\hat{\mathbf{D}}_q(m)\psi = (\mathbf{W}(\hat{P}_q) - m\mathbf{I})\psi = 0.} \quad (15.12)$$

Its entries are

$$(\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\psi_- = m\psi_+, \quad (15.13)$$

$$(\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\psi_+ = m\psi_-. \quad (15.14)$$

With the time derivative isolated, $i\partial_t\psi = \hat{\mathbf{H}}\psi$, where the Hamiltonian is a matrix differential operator on the two-entry Dirac field: (17, Eq. (412))

$$\hat{\mathbf{H}} := \begin{Bmatrix} qV - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma} & m \\ m & qV + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma} \end{Bmatrix}. \quad (15.15)$$

For real potentials, the Dirac equation conserves total probability when the spatial boundary flux vanishes. The pointwise density (15.1) can still vary through probability flux.

Each coefficient is exposed by the decomposition $\psi_{\pm} = a_{\pm} + \mathbf{b}_{\pm} \cdot \boldsymbol{\sigma}$, with complex scalar functions $a_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}$ and vector functions $\mathbf{b}_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$. Their pairs $(a_{\pm}, \mathbf{b}_{\pm})$ are restricted to the fixed spinor space, so their four complex coefficients per entry are not independent. By sigma multiplication,

$$\begin{aligned}\hat{E}_q a_- - \hat{\mathbf{P}}_q \cdot \mathbf{b}_- &= m a_+, \\ \hat{E}_q \mathbf{b}_- - \hat{\mathbf{P}}_q a_- - i \hat{\mathbf{P}}_q \times \mathbf{b}_- &= m \mathbf{b}_+, \\ \hat{E}_q a_+ + \hat{\mathbf{P}}_q \cdot \mathbf{b}_+ &= m a_-, \\ \hat{E}_q \mathbf{b}_+ + \hat{\mathbf{P}}_q a_+ + i \hat{\mathbf{P}}_q \times \mathbf{b}_+ &= m \mathbf{b}_-.\end{aligned}$$

Every operator acts on the following coefficient. Each entry still has the two independent complex components of (13.2). Appendix A.7 converts the system to the conventional four-component Dirac equation.

15.4 Interacting factorization and reconstruction

Notation: the real potential $A = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, the complex field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, and $S = \partial_t V + \boldsymbol{\partial}_r \cdot \mathbf{A}$. For any field $\psi : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$, the product rule gives

$$\partial^*(A^*\psi) + A\partial\psi = (S + F)\psi + 2(V\partial_t + \mathbf{A} \cdot \boldsymbol{\partial}_r)\psi, \quad (15.16)$$

$$\partial(A\psi) + A^*\partial^*\psi = (S + F^*)\psi + 2(V\partial_t + \mathbf{A} \cdot \boldsymbol{\partial}_r)\psi. \quad (15.17)$$

The difference of (15.16) and (15.17) is $(F - F^*)\psi = 2(\mathbf{E} \cdot \boldsymbol{\sigma})\psi$.

Applying the block map $\mathbf{W}$ twice gives

$$\begin{aligned}\mathbf{W}(\partial^*)\mathbf{W}(A)\psi &= \left\{ \begin{array}{c} \partial(A\psi_+) \\ \partial^*(A^*\psi_-) \end{array} \right\}, \\ \mathbf{W}(A)\mathbf{W}(\partial^*)\psi &= \left\{ \begin{array}{c} A^*\partial^*\psi_+ \\ A\partial\psi_- \end{array} \right\}, \\ \mathbf{W}(A)^2 &= \det(A) \mathbf{I}.\end{aligned}$$

Differentiating a parenthesized product includes the potential derivative. The argument $-m$ changes only the mass sign, keeping q and A fixed. Multiplying and using the preceding mixed-derivative identities gives

$$\hat{\mathbf{D}}_q(-m)\hat{\mathbf{D}}_q(+m) = (\hat{E}_q^2 - \hat{\mathbf{P}}_q^2 - m^2)\mathbf{I} - iq \begin{Bmatrix} F^* & 0 \\ 0 & F \end{Bmatrix}. \quad (15.18)$$

The derivatives act on potentials as well as amplitudes. For every Dirac solution,

$$\hat{\mathbf{D}}_q(-m)\hat{\mathbf{D}}_q(+m)\psi = 0.$$

At $q = 0$, this recovers (15.5) and hence the free Klein–Gordon equation. With coupling, its entries obey

$$\begin{aligned}((\partial + iqA^*)(\partial^* + iqA) + m^2)\psi_+ &= 0, \\ ((\partial^* + iqA)(\partial + iqA^*) + m^2)\psi_- &= 0.\end{aligned}$$

The exact second-order equations are obtained by expansion (18, Eq. (6))

$$(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2 + iqF^*)\psi_+ = 0, \quad (15.19)$$

$$(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2 + iqF)\psi_- = 0. \quad (15.20)$$

For the spin-term components, the notation $\mathbf{Z} = \mathbf{E} + i\mathbf{B}$ is used, with $\psi = a + \mathbf{b} \cdot \boldsymbol{\sigma}$, with complex functions $a : \mathbb{R}^4 \rightarrow \mathbb{C}$ and $\mathbf{b}, \mathbf{Z} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$. Here $\mathbf{Z}^*$ conjugates its scalar coefficients. By sigma multiplication,

$$\begin{aligned} iqF\psi &= iq\mathbf{Z} \cdot \mathbf{b} + (iqa\mathbf{Z} - q\mathbf{Z} \times \mathbf{b}) \cdot \boldsymbol{\sigma}, \\ iqF^*\psi &= -iq\mathbf{Z}^* \cdot \mathbf{b} + (-iqa\mathbf{Z}^* + q\mathbf{Z}^* \times \mathbf{b}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Solutions are constructed from a complex paravector amplitude ψ of (13.2) satisfying (15.20) with ψ in place of ψ_- . The Dirac field is defined by

$$\boxed{\psi = \left\{ \begin{array}{c} (\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\psi/m \\ \psi \end{array} \right\}.} \quad (15.21)$$

The first Dirac equation holds by construction; the second is the seed equation. Every massive solution has this form with $\psi = \psi_-$. That seed equation must be solved for the chosen potential: a scalar KG seed lacks the spin term, and independently solved squared entries need not satisfy the first-order coupling.

15.5 Gauge phase

Notation: the real scalar $\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}$ and gauge transformation $A' = A + \partial^*\lambda$, or $V' = V + \partial_t\lambda$, $\mathbf{A}' = \mathbf{A} - \partial_{\mathbf{r}}\lambda$. For $\hat{E}'_q := i\partial_t - qV'$, $\hat{\mathbf{P}}'_q := -i\partial_{\mathbf{r}} - q\mathbf{A}'$, and $\hat{P}'_q := \hat{E}'_q + \hat{\mathbf{P}}'_q \cdot \boldsymbol{\sigma}$, the product rule gives

$$\begin{aligned} (\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})(e^{iq\lambda}\psi'_{\pm}) &= e^{iq\lambda}(\hat{E}_q - q\partial_t\lambda + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma} + q(\partial_{\mathbf{r}}\lambda) \cdot \boldsymbol{\sigma})\psi'_{\pm} \\ &= e^{iq\lambda}(\hat{E}'_q + \hat{\mathbf{P}}'_q \cdot \boldsymbol{\sigma})\psi'_{\pm}. \end{aligned}$$

The adjugate block follows with the vector signs reversed. Thus the new amplitude is

$$\psi' = e^{-iq\lambda}\psi.$$

Equivalently,

$$\begin{aligned} \hat{E}'_q\psi' &= e^{-iq\lambda}\hat{E}_q\psi, \\ \hat{\mathbf{P}}'_q\psi' &= e^{-iq\lambda}\hat{\mathbf{P}}_q\psi. \end{aligned}$$

Consequently (16, Eqs. (6.68)–(6.69))

$$\begin{aligned} (\mathbf{W}(\hat{P}'_q) - m\mathbf{I})\psi' \\ = e^{-iq\lambda}(\mathbf{W}(\hat{P}_q) - m\mathbf{I})\psi. \end{aligned} \quad (15.22)$$

Equation (15.22) preserves the Dirac equation; the phase sign is fixed by the potential shift. The factor $e^{-iq\lambda} \in \mathbb{C}$ has modulus one and preserves the spinor space, so the norm is unchanged. The same product rules preserve the KG and Pauli equations.

16 Low-energy limits

Notation: $m \in \mathbb{R}_{>0}$, $q \in \mathbb{R}$, and potentials $V : \mathbb{R}^4 \rightarrow \mathbb{R}$, $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. Scalar KG envelopes have domain $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$; Dirac and Pauli entry fields map $\mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$ and have the fixed two-component form of (13.2), unless an alternative encoding is explicitly introduced. The electric and magnetic fields take values in $\mathbb{R}^3$. The coupled operators are $\hat{E}_q = i\partial_t - qV$ and $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$; the fields are $\mathbf{E} = -\partial_{\mathbf{r}}V - \partial_t\mathbf{A}$ and $\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}$. The rest oscillation is removed and the leading nonrelativistic terms are retained. Schrödinger is obtained from KG, and Pauli from Dirac.

16.1 Klein–Gordon to Schrödinger

Schrödinger dynamics is obtained from the scalar Klein–Gordon equation after removal of the rest oscillation. The coupled operators $\hat{E}_q, \hat{\mathbf{P}}_q$ defined above are retained. For the full scalar KG field $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$, its scalar envelope $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ is defined by (16, Eqs. (2.103)–(2.104))

$$\psi'(t, \mathbf{r}) = e^{-imt}\psi(t, \mathbf{r}). \quad (16.1)$$

The product rule makes the rest-energy shift explicit:

$$\begin{aligned} \hat{E}_q\psi' &= (i\partial_t - qV)(e^{-imt}\psi) \\ &= e^{-imt}(m\psi + i\partial_t\psi - qV\psi) \\ &= e^{-imt}(m + \hat{E}_q)\psi, \\ \hat{\mathbf{P}}_q\psi' &= e^{-imt}\hat{\mathbf{P}}_q\psi. \end{aligned}$$

Since m is constant, applying $\hat{E}_q$ again gives $\hat{E}_q^2\psi' = e^{-imt}(\hat{E}_q + m)^2\psi$; spatial derivatives leave the phase untouched. Substitution into (14.7) therefore gives

$$\begin{aligned} 0 &= (-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi' \\ &= e^{-imt}(-(\hat{E}_q + m)^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi \\ &= e^{-imt}(-\hat{E}_q^2 - 2m\hat{E}_q + \hat{\mathbf{P}}_q^2)\psi. \end{aligned}$$

On the envelope, $\hat{E}_q$ measures energy above rest, excluding qV . The rest terms cancel. By division by the nonzero phase, the following is obtained:

$$\hat{E}_q\psi = \frac{\hat{\mathbf{P}}_q^2 - \hat{E}_q^2}{2m}\psi. \quad (16.2)$$

For weak, slowly varying fields and a slow positive-energy envelope, a real expansion parameter $0 < v \ll 1$ is chosen with $\hat{E}_q/m = O(v^2)$. Then $\hat{E}_q^2\psi$ is smaller than $2m\hat{E}_q\psi$ by $O(v^2)$. Dropping it at leading order gives (19, Eq. (21.3))

$$\boxed{i\partial_t\psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi.} \quad (16.3)$$

With $\mathbf{A} = 0$, hence $\mathbf{B} = 0$, the kinetic-energy operator is $-\partial_{\mathbf{r}}^2/(2m)$, and the conventional scalar form is

$$i\partial_t\psi = \left(-\frac{\partial_{\mathbf{r}}^2}{2m} + qV \right) \psi.$$

Schrödinger and Pauli occur at the same leading order $1/m$. The following Dirac limit retains the magnetic spin term; when $\mathbf{B} = 0$, each complex coefficient of the Pauli amplitude obeys this same scalar Schrödinger equation.

16.2 Dirac to Pauli

The fields $\psi_{\pm}, \psi'_{\pm}, \phi_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$ all have the fixed two-component form of (13.2), with coordinate columns in $\mathbb{C}^2$. The phase and change of blocks below preserve this domain.

The Dirac envelopes are defined by the same rest-energy phase shift (12, Eq. (2.1)):

$$\psi'_{\pm} := e^{imt} \psi_{\pm}. \quad (16.4)$$

By the product rule,

$$\hat{E}_q \psi_{\pm} = e^{-imt} (m + \hat{E}_q) \psi'_{\pm}.$$

On the envelopes, $\hat{E}_q$ measures energy above rest, excluding qV . The two rows of the Dirac equation, (15.13) and (15.14), become

$$\begin{aligned} (m + \hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \psi'_- &= m \psi'_+, \\ (m + \hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \psi'_+ &= m \psi'_-. \end{aligned}$$

At free positive-energy rest, $\psi'_+ = \psi'_-$. The large and small components are defined by

$$\phi_+ := \frac{\psi'_+ + \psi'_-}{\sqrt{2}}, \quad (16.5)$$

$$\phi_- := \frac{\psi'_- - \psi'_+}{\sqrt{2}}. \quad (16.6)$$

The exact pair is obtained by addition and subtraction (12, Eqs. (2.15)–(2.16)):

$$\hat{E}_q \phi_+ = (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \phi_-, \quad (16.7)$$

$$(2m + \hat{E}_q) \phi_- = (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \phi_+. \quad (16.8)$$

For nonrelativistic positive-energy states in weak, slowly varying fields, $\hat{E}_q \phi_-$ is neglected relative to $2m \phi_-$. The small component is substituted into the first equation, and (13.19) is applied:

$$\begin{aligned} \phi_- &\simeq \frac{\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}}{2m} \phi_+, \\ \hat{E}_q \phi_+ &\simeq \frac{(\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2}{2m} \phi_+ = \frac{\hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m} \phi_+. \end{aligned}$$

Restoring $\hat{E}_q = i\partial_t - qV$ gives Pauli's equation through order $1/m$ (12, Eq. (2.13)):

$$\boxed{i\partial_t \phi_+ = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \phi_+}. \quad (16.9)$$

SI units: Pauli's equation

With m in kilograms, q in coulombs, V in volts, $\mathbf{A}$ in T m, and $\mathbf{B}$ in teslas, restoring $\hbar$ gives

$$i\hbar\partial_t\phi_+ = \left(\frac{(-i\hbar\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q\hbar}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right) \phi_+.$$

Each term in parentheses has energy units. The same nonrelativistic approximation as in (16.9) is retained.

Returning to natural units, the change of entries preserves the density exactly:

$$\|\boldsymbol{\psi}\|^2 = \|\phi_+\|^2 + \|\phi_-\|^2.$$

The leading probability density is $\|\phi_+\|^2$. Its two complex components are the Pauli amplitudes, as in Appendix A.5. This is the amplitude used in the spin discussion. For a real expansion parameter $0 < v \ll 1$, the scaling $\hat{\mathbf{P}}_q/m = O(v)$, $\hat{E}_q/m = O(v^2)$, $qV/m = O(v^2)$, $q\mathbf{B}/m^2 = O(v^2)$, $q\mathbf{E}/m^2 = O(v^3)$, with fields varying on spatial scales $1/(mv)$ and time scales $1/(mv^2)$ or longer, gives $\phi_- = O(v\phi_+)$ and omitted energy terms of order mv^4 .

16.3 Spin components in a uniform magnetic field

For a constant field $\mathbf{B} \in \mathbb{R}^3 \setminus \{0\}$, its [unit direction](#) and [spin projectors](#) are defined by:

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|} \in \mathbb{R}^3, \quad \Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) \in \mathbb{R} \oplus \mathbb{R}^3.$$

[Spectral decomposition](#) gives

$$\mathbf{B} \cdot \boldsymbol{\sigma} = \|\mathbf{B}\|(\Pi_+ - \Pi_-), \quad \Pi_{\pm}(\mathbf{B} \cdot \boldsymbol{\sigma}) = \pm\|\mathbf{B}\|\Pi_{\pm}.$$

The two spin amplitudes $\phi_{++}, \phi_{+-} : \mathbb{R}^4 \rightarrow \mathbb{C}$ are encoded in an aligned paravector $\phi_+ : \mathbb{R}^4 \rightarrow \text{span}_{\mathbb{C}}(1, \mathbf{u} \cdot \boldsymbol{\sigma}) \subset \mathbb{C} \oplus \mathbb{C}^3$ by spectral decomposition. This subsection temporarily uses this aligned encoding of two scalars in place of the form in (13.2); the two subspaces are different:

$$\begin{aligned} \phi_+ &= \phi_{++}\Pi_+ + \phi_{+-}\Pi_-, \\ \Pi_{\pm}\phi_+ &= \phi_{\pm}\Pi_{\pm} = \phi_{\pm\pm}\Pi_{\pm}. \end{aligned}$$

Constant $\Pi_{\pm}$ commute with $\hat{E}_q$ and $\hat{\mathbf{P}}_q^2$, so the Pauli equation gives

$$\begin{aligned} 0 &= \Pi_{\pm}(2m\hat{E}_q - \hat{\mathbf{P}}_q^2 + q\mathbf{B} \cdot \boldsymbol{\sigma})\phi_+ \\ &= ((2m\hat{E}_q - \hat{\mathbf{P}}_q^2 \pm q\|\mathbf{B}\|)\phi_{\pm\pm})\Pi_{\pm}. \end{aligned}$$

Since $\Pi_{\pm} \neq 0$, the scalar coefficients vanish. Restoring $\hat{E}_q = i\partial_t - qV$ gives

$$\boxed{i\partial_t\phi_{\pm\pm} = \left(\frac{\hat{\mathbf{P}}_q^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \phi_{\pm\pm}.}$$

States with [spin projections](#) $\pm\frac{1}{2}$ along $\mathbf{u}$ have opposite [Zeeman shifts](#). The [kinetic momentum](#) $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$ retains the orbital coupling.

The separation $\Delta E \in \mathbb{R}_{\geq 0}$ of the two spin-energy contributions, excluding orbital energy, is

$$\Delta E = \frac{|q|}{m} \|\mathbf{B}\|.$$

SI units: Zeeman splitting	
Restoring SI units gives	$\Delta E = \frac{ q \hbar}{m} \ \mathbf{B}\ .$
This is the splitting from the derived Dirac magnetic coupling; anomalous magnetic-moment corrections are not included.	

This representation applies to uniform-field Pauli evolution; the full Dirac equation and spin measurements use the fixed spinor space.

The four scalar fields $\phi_{ab} : \mathbb{R}^4 \rightarrow \mathbb{C}$, $a, b \in \{+, -\}$, of the Dirac spinor after removal of the rest-energy phase are arranged as:

$$\phi := \begin{pmatrix} \phi_{++} & \phi_{+-} \\ \phi_{-+} & \phi_{--} \end{pmatrix} \in \mathbb{C}^{2 \times 2}.$$

Dirac blocks are indexed by rows and spin projections by columns; the upper row is governed by the Pauli limit.

In a basis diagonalizing $\mathbf{u} \cdot \boldsymbol{\sigma}$, the aligned paravector is $\text{diag}(\phi_{++}, \phi_{+-})$. The corresponding spinor in the fixed subspace of (13.2) is $\phi_{++}Z_+ + \phi_{+-}Z_-$, where constant orthonormal $Z_{\pm} \in \mathbb{C} \oplus \mathbb{C}^3$ lie in that same subspace and satisfy $\Pi_{\pm}Z_{\pm} = Z_{\pm}$ and $\Pi_{\pm}Z_{\mp} = 0$. Both representations give the same scalar Pauli equations. The full Dirac coupling $\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}$ need not preserve alignment.

For the original spinor blocks, the lower block still satisfies $\phi_- \simeq (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\phi_+/(2m)$. For a nonzero field with varying direction, its projectors depend on $(t, \mathbf{r})$. Here $\hat{H}$ is the paravector differential operator in (16.9), acting on smooth fields of the form (13.2) (a 2×2 matrix differential operator in spinor coordinates). With this Pauli Hamiltonian, the product rule for the original spinor gives

$$\begin{aligned} i\partial_t(\Pi_{\pm}\phi_{\pm}) &= i(\partial_t\Pi_{\pm})\phi_{\pm} + \Pi_{\pm}\hat{H}\phi_{\pm} \\ &= \hat{H}(\Pi_{\pm}\phi_{\pm}) + \left(i\partial_t\Pi_{\pm} + \frac{[\Pi_{\pm}, \hat{\mathbf{P}}_q^2]}{2m} \right) \phi_{\pm}. \end{aligned}$$

Only the projector derivatives and kinetic commutator couple the sectors; the magnetic term commutes with its own projectors.

16.4 Evolution about an alternate spin axis

A constant unit vector $\mathbf{u}' \in \mathbb{R}^3$ and $0 \leq \theta \leq \pi$ are chosen. Initial scalar wavefunctions $\phi_{++}, \phi_{+-} : \mathbb{R}^3 \rightarrow \mathbb{C}$ are chosen along this axis:

$$\begin{aligned} \mathbf{u}' \cdot \mathbf{u} &= \cos(\theta), \quad \Pi'_{\pm} := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}) \in \mathbb{R} \oplus \mathbb{R}^3, \\ \phi_+(0) &= \phi_{++}\Pi'_+ + \phi_{+-}\Pi'_- \in \mathbb{C} \oplus \mathbb{C}^3. \end{aligned}$$

The magnetic term has both diagonal and off-diagonal parts in these projectors:

$$\begin{aligned}\Pi'_\pm(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi'_\pm &= \pm \cos(\theta) \Pi'_\pm, \\ \Pi'_\mp(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi'_\pm &= ((\mathbf{u} - \cos(\theta) \mathbf{u}') \cdot \boldsymbol{\sigma})\Pi'_\pm.\end{aligned}$$

By projection of the Pauli equation,

$$\begin{aligned}2m\hat{E}_q\Pi'_\pm\phi_+ &= (\hat{\mathbf{P}}_q^2 \mp q\|\mathbf{B}\|\cos(\theta))\Pi'_\pm\phi_+ \\ &\quad - q\|\mathbf{B}\|\Pi'_\pm(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi'_\mp\phi_+.\end{aligned}$$

Initially $\Pi'_\mp\phi_+(0)\Pi'_\pm = 0$, but

$$\mathrm{i}\partial_t(\Pi'_\mp\phi_+\Pi'_\pm)|_{t=0} = -\frac{q\|\mathbf{B}\|}{2m}\phi_{+\pm}((\mathbf{u} - \cos(\theta) \mathbf{u}') \cdot \boldsymbol{\sigma})\Pi'_\pm.$$

For $q \neq 0$ and $\sin(\theta) \neq 0$, evolution generates off-diagonal components: two scalar multiples of the fixed $\Pi'_\pm$ no longer suffice. At $\theta = 0$ the earlier equations return; at $\theta = \pi$ the signs exchange. For perpendicular axes, the magnetic term is entirely off-diagonal.

For time-independent $V : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $\mathbf{A} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, the orbital operator commutes with the constant magnetic term. The full Pauli evolution of this initial paravector is

$$\begin{aligned}\phi_+(t) &= e^{-it(\hat{\mathbf{P}}_q^2/(2m)+qV)}e^{\mathrm{i}q\|\mathbf{B}\|t\mathbf{u}\cdot\boldsymbol{\sigma}/(2m)}\phi_+(0) \\ &= e^{-it(\hat{\mathbf{P}}_q^2/(2m)+qV)}\left(\cos\left(\frac{q\|\mathbf{B}\|t}{2m}\right) + \mathrm{i}\mathbf{u} \cdot \boldsymbol{\sigma} \sin\left(\frac{q\|\mathbf{B}\|t}{2m}\right)\right)\phi_+(0).\end{aligned}$$

The exponential follows from $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$. Its transverse contribution produces the off-diagonal terms above.

16.5 The next free order

For the free complex scalar envelope $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$, the conditions $qV = 0$ and $q\mathbf{A} = 0$ are imposed, so $\hat{E}_q = \hat{E}$, $\hat{\mathbf{P}}_q = \hat{\mathbf{P}}$, and the operators commute. On the positive-frequency KG branch, solving (16.2) gives

$$\hat{E}\psi = \left(\sqrt{m^2 + \hat{\mathbf{P}}^2} - m\right)\psi.$$

The same energy acts on either normalized positive-energy Dirac spin component. Iteration starts with $\hat{E}\psi \simeq \hat{\mathbf{P}}^2\psi/(2m)$; its square supplies the next term (12, Eq. (2.19)):

$$\hat{E}\psi = \left(\frac{\hat{\mathbf{P}}^2}{2m} - \frac{\hat{\mathbf{P}}^4}{8m^3} + O\left(\frac{\hat{\mathbf{P}}^6}{m^5}\right)\right)\psi.$$

For the charged scalar equation, the same iteration applies whenever $\hat{E}_q$ commutes with $\hat{\mathbf{P}}_q^2$. On the positive-energy branch,

$$\begin{aligned}\hat{E}_q\psi &= \left(\sqrt{m^2 + \hat{\mathbf{P}}_q^2} - m\right)\psi \\ &= \left(\frac{\hat{\mathbf{P}}_q^2}{2m} - \frac{\hat{\mathbf{P}}_q^4}{8m^3} + O\left(\frac{\hat{\mathbf{P}}_q^6}{m^5}\right)\right)\psi.\end{aligned}$$

This expansion applies when the kinetic momenta in the state are small compared with m .

17 Spin and probability

Here $\psi : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$ takes values in the fixed spinor space of (13.2), with coordinates in $\mathbb{C}^2$. Axes and polarizations $\mathbf{u}, \mathbf{u}', \mathbf{a} \in \mathbb{R}^3$ are unit vectors; angles $\theta, \varphi \in \mathbb{R}$, projectors $\Pi_{\pm}, \Pi'_{\pm} \in \mathbb{R} \oplus \mathbb{R}^3$, and probabilities lie between 0 and 1, inclusive. The density $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}_{\geq 0}$ is a probability density. Multiplying observables $Z \in \mathbb{R} \oplus \mathbb{R}^3$ have real scalar expectations. Rotation factors $R \in \mathbb{R} \oplus i\mathbb{R}^3$ preserve this two-component form; probabilities are real-valued functions of the state and axis.

Spin is specified by the intrinsic rotation of an amplitude. Here the state is rotated at fixed coordinates. A scalar KG amplitude stays unchanged under this intrinsic action and has spin zero; a two-component amplitude has the one-sided law below. The wave equation alone does not fix spin: free Dirac components also satisfy KG.

17.1 Spin one-half and rotations

Notation: the complex paravector amplitude $\psi(t, \mathbf{r})$ of (13.2). For a **real unit axis** $\mathbf{u}$ and real angle θ , the following unitary rotation factor is used: $R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ of (4.5). The spinor law is (11, p. 4)

$$\boxed{\psi' = R\psi.} \quad (17.1)$$

Writing its infinitesimal action as $\psi' = \psi - i\theta S(\mathbf{u})\psi + O(\theta^2)$ defines the spin operator along $\mathbf{u}$, given by left multiplication by the real pure-vector paravector $S(\mathbf{u}) \in \{0\} \oplus \mathbb{R}^3$ (13, p. 62):

$$S(\mathbf{u})\psi = \frac{1}{2}(\mathbf{u} \cdot \boldsymbol{\sigma})\psi. \quad (17.2)$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, its two spin projections are $\pm 1/2$ in units $\hbar = 1$. A 2π rotation gives $R = -1$, reversing the amplitude's sign; a 4π rotation restores it. Probabilities are unchanged by the overall sign, while a relative sign can be detected by interference (20, Eq. (6.18)). Each Dirac entry obeys this same active rotation law.

For a real paravector observable Z , the state law (17.1) gives, by (13.8),

$$\langle Z \rangle_{\psi'} = \langle R^H Z R \rangle_{\psi}.$$

The observable inside the original state's pairing rotates through $-\theta$. If the observable rotates with the state, $Z' = R Z R^H$, then $\langle Z' \rangle_{\psi'} = \langle Z \rangle_{\psi}$.

17.2 Projectors and probabilities

Notation: a nonzero Pauli amplitude ψ of (13.2). For a real unit measurement axis $\mathbf{u}$, the real projectors are

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).$$

Their algebra in Section 3.2 gives

$$\begin{aligned} \psi &= \Pi_+ \psi + \Pi_- \psi, \\ S(\mathbf{u})(\Pi_{\pm} \psi) &= \pm \frac{1}{2} \Pi_{\pm} \psi, \\ \text{sc}((\Pi_+ \psi)^H (\Pi_- \psi)) &= 0, \\ \|\Pi_{\pm} \psi\|^2 &= \text{sc}(\psi^H \Pi_{\pm} \psi) = \|\psi\|_{\Pi_{\pm}}^2. \end{aligned}$$

By (13.5), these are nonnegative weighted densities. The Born probability is this density divided by $\rho = \|\psi\|^2$. In the fixed orthonormal basis (13.1), $\psi = c_+\psi_+ + c_-\psi_-$ with complex scalars $c_\pm$, so $\rho = \|c_+\|^2 + \|c_-\|^2$. When the measurement axis is that basis axis, the probabilities are $\|c_\pm\|^2/\rho$. The same-side and cross-side projector products are already derived in Section 3.2.

The real spin-polarization vector $\mathbf{u}' \in \mathbb{R}^3$ is defined by its projection along every real unit measurement axis $\mathbf{u}$:

$$\mathbf{u} \cdot \mathbf{u}' := \frac{\text{sc}(\psi^H(\mathbf{u} \cdot \boldsymbol{\sigma})\psi)}{\rho}. \quad (17.3)$$

The right side is real and linear in $\mathbf{u}$, so it uniquely specifies $\mathbf{u}'$. For the arbitrary perpendicular basis axes $\mathbf{u}, \mathbf{a}$ in (13.2), expanding the same pairing gives

$$\begin{aligned} \rho \mathbf{u}' &= (\|c_+\|^2 - \|c_-\|^2)\mathbf{u} \\ &\quad + 2\text{re}(c_+^*c_-)\mathbf{a} + 2\text{im}(c_+^*c_-)(\mathbf{u} \times \mathbf{a}). \end{aligned}$$

These three directions are orthonormal, hence

$$\rho^2(\mathbf{u}')^2 = (\|c_+\|^2 - \|c_-\|^2)^2 + 4\|c_+^*c_-\|^2 = \rho^2.$$

Thus $\mathbf{u}'$ is a real unit vector, with mean spin $\mathbf{u}'/2$. For any real unit measurement axis, inserting $\Pi_\pm = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ directly into the pairing gives

$$\|\Pi_\pm \psi\|^2 = \text{sc}(\psi^H \Pi_\pm \psi) = \frac{1}{2}\rho(1 \pm \mathbf{u} \cdot \mathbf{u}').$$

Therefore (11, Eqs. (66)–(67))

$$\boxed{\text{prob}(\pm | \mathbf{u}) = \frac{\|\Pi_\pm \psi\|^2}{\rho} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{u}').} \quad (17.4)$$

The mean spin along $\mathbf{u}$ is $(\text{prob}(+) - \text{prob}(-))/2 = \mathbf{u} \cdot \mathbf{u}'/2$. For a spatial wavefunction, (17.4) is the spin probability conditional on position, not a position density.

17.3 Measurement along a second axis

The prepared spin direction and measurement axis are denoted by the real unit vectors $\mathbf{u}'$ and $\mathbf{u}$, respectively. A nonzero spin-up amplitude $\psi = \Pi'_+ \psi$ is prepared, where $\Pi'_+ = (1 + \mathbf{u}' \cdot \boldsymbol{\sigma})/2$. The angle $0 \leq \theta \leq \pi$ is fixed by $\mathbf{u} \cdot \mathbf{u}' = \cos(\theta)$. The Pauli product and the unit lengths give

$$(\mathbf{u}' \cdot \boldsymbol{\sigma})^2 = 1, \quad (\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) + (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = 2\cos(\theta).$$

Multiplying the second identity on the right by $\mathbf{u}' \cdot \boldsymbol{\sigma}$ yields

$$(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = 2\cos(\theta)(\mathbf{u}' \cdot \boldsymbol{\sigma}) - \mathbf{u} \cdot \boldsymbol{\sigma}.$$

Thus the prepared-state projector gives

$$\begin{aligned} \Pi'_+(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi'_+ &= \frac{1}{4}(1 + \mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma})(1 + \mathbf{u}' \cdot \boldsymbol{\sigma}) \\ &= \frac{1}{4}\left[\mathbf{u} \cdot \boldsymbol{\sigma} + 2\cos(\theta) + (\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})\right] \\ &= \frac{1}{4}\left[2\cos(\theta) + 2\cos(\theta)(\mathbf{u}' \cdot \boldsymbol{\sigma})\right] = \cos(\theta)\Pi'_+. \end{aligned}$$

The measurement projectors $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ therefore satisfy

$$\Pi'_+ \Pi_{\pm} \Pi'_+ = \frac{1}{2}(\Pi'_+ \pm \Pi'_+(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi'_+) = \frac{1 \pm \cos(\theta)}{2} \Pi'_+. \quad (17.5)$$

Using $\|Z\|^2 = \text{sc}(ZZ^H)$ for $Z \in \mathbb{C} \oplus \mathbb{C}^3$, the denominator is

$$\|\psi\|^2 = \text{sc}(\psi\psi^H) > 0.$$

For the prepared state, the projected squared norm is

$$\begin{aligned} \|\Pi_{\pm}\psi\|^2 &= \text{sc}((\Pi_{\pm}\psi)(\Pi_{\pm}\psi)^H) \\ &= \text{sc}(\Pi_{\pm}\psi\psi^H) \\ &= \text{sc}(\Pi'_+ \Pi_{\pm} \Pi'_+ \psi\psi^H) \\ &= \frac{1 \pm \cos(\theta)}{2} \text{sc}(\psi\psi^H) = \frac{1 \pm \cos(\theta)}{2} \|\psi\|^2. \end{aligned}$$

The Born rule divides by the nonzero squared norm, so no prior normalization is needed:

$$\text{prob}(\pm) := \frac{\|\Pi_{\pm}\psi\|^2}{\|\psi\|^2} = \frac{1 \pm \cos(\theta)}{2}, \quad (17.6)$$

$$\boxed{\text{prob}(+) = \cos^2\left(\frac{\theta}{2}\right), \quad \text{prob}(-) = \sin^2\left(\frac{\theta}{2}\right).}$$

Here $+$ and $-$ label measured spin projections $+1/2$ and $-1/2$ along $\mathbf{u}$. The probabilities sum to one. Aligned axes give spin up with certainty, perpendicular axes give equal probabilities, and opposite axes give spin down with certainty. Preparing $\psi = \Pi'_-\psi$ interchanges the probabilities. For a spatial wavefunction with the same spin preparation at every position, integrating both norm densities gives the same result. Figure 10 connects the angle between the axes with these probabilities.

17.4 Rotating the polarization

For $R = \exp(-i\varphi \mathbf{a} \cdot \boldsymbol{\sigma}/2)$, with real unit rotation axis $\mathbf{a}$ and real angle φ , the polarization of the rotated amplitude $\psi' = R\psi$ is denoted by $\mathbf{u}''$. The amplitude law (17.1) gives

$$\mathbf{u} \cdot \mathbf{u}'' = \langle R^H(\mathbf{u} \cdot \boldsymbol{\sigma})R \rangle_{\psi}.$$

This is the inverse of Rodrigues' rotation (5.5): the measurement axis rotates through $-\varphi$, so $\mathbf{u}''$ rotates through $+\varphi$ about $\mathbf{a}$. Thus the polarization rotates as a vector although its amplitude has spin one-half. Its overlap with the original polarization is

$$\mathbf{u}' \cdot \mathbf{u}'' = (\mathbf{a} \cdot \mathbf{u}')^2 + (1 - (\mathbf{a} \cdot \mathbf{u}')^2) \cos(\varphi).$$

For $\mathbf{a} \cdot \mathbf{u}' = 0$, the probability to retain the initial spin is $\cos^2(\varphi/2)$.

Preparation and measurement

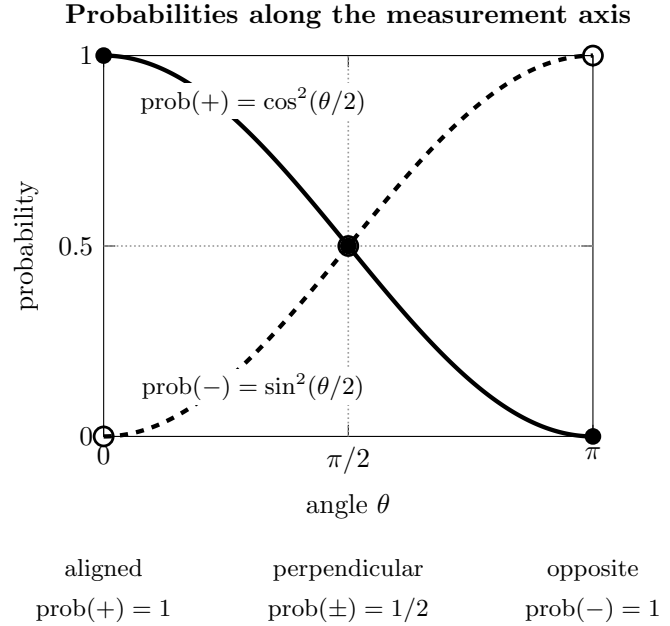
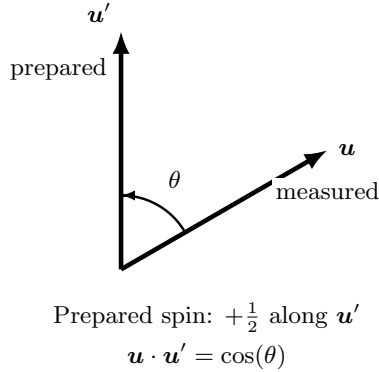


Figure 10: Spin prepared along $\mathbf{u}'$ and measured along $\mathbf{u}$. Only the angle θ between the axes enters the probabilities. The solid curve gives the $+1/2$ outcome; the dashed curve gives the $-1/2$ outcome. Aligned axes give certainty, perpendicular axes give equal probabilities, and opposite axes reverse the certain outcome. The two curves sum to one at every angle.

18 Lorentz covariance and electromagnetic transformations

The algebraic actions of Section 5 are applied here to fields and equations at corresponding events. Primed and unprimed quantities describe the same event in different coordinates and retain their types. We compare inertial frames with a common coordinate origin. The frame factors are constant; varying factors are treated in Section 11.4. Appendix C compares exponent signs and sandwich orders. Natural units $c = \hbar = 1$ are used. These transformations preserve spatial and time orientation; fields are assumed twice continuously differentiable where second derivatives occur. Mass $m > 0$ and charge $q \in \mathbb{R}$ are Lorentz scalars. The particle coupling, Maxwell equation, and spin-one-half representation are the physical inputs already specified in the preceding sections. The algebra then determines how their factors must combine. Being a paravector alone does not determine a transformation law: the definition and physical role of each quantity do.

18.1 Events, proper time, and momentum

Coordinates satisfy $(t, \mathbf{r}) \in \mathbb{R} \times \mathbb{R}^3$, with $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$. The constant frame factor $T \in \mathbb{C} \oplus \mathbb{C}^3$ has $\det(T) = 1$ and may be any ordered product of rotations and boosts,

$$R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) \in \mathbb{R} \oplus i\mathbb{R}^3, \quad L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) \in \mathbb{R} \oplus \mathbb{R}^3,$$

where $\theta \in \mathbb{R}$ is an angle or rapidity and $\mathbf{u} \in \mathbb{R}^3$ is a unit axis. The event coordinates transform as (5, Eq. (24))

$$\boxed{X' = T X T^H.} \tag{18.1}$$

This is the Lorentz representation established in Section 5. It preserves reality and the interval:

$$\begin{aligned} X'^{\mathbf{H}} &= TX^{\mathbf{H}}T^{\mathbf{H}} = X', \\ \det(\mathrm{d}X') &= \det(T) \det(\mathrm{d}X) \det(T^{\mathbf{H}}) = \det(\mathrm{d}X). \end{aligned}$$

The continuous rotation and boost factors preserve orientation and the future cone, as shown there. On a future-directed time-like path, take the positive proper-time increment $\mathrm{d}s = \sqrt{\det(\mathrm{d}X)}$. Then $\mathrm{d}s' = \mathrm{d}s$, and differentiating the event law gives

$$\begin{aligned} U' &= \frac{\mathrm{d}X'}{\mathrm{d}s'} = T \frac{\mathrm{d}X}{\mathrm{d}s} T^{\mathbf{H}} = TUT^{\mathbf{H}}, \\ P' &= mU' = T(mU)T^{\mathbf{H}} = TPT^{\mathbf{H}}. \end{aligned}$$

Thus $U, P \in \mathbb{R} \oplus \mathbb{R}^3$ inherit the event transformation from $U = \mathrm{d}X/\mathrm{d}s$ and $P = mU$; their determinants remain 1 and m^2 . For the same reason, proper-time differentiation preserves this law:

$$\frac{\mathrm{d}U'}{\mathrm{d}s'} = T \frac{\mathrm{d}U}{\mathrm{d}s} T^{\mathbf{H}}, \quad \frac{\mathrm{d}P'}{\mathrm{d}s'} = T \frac{\mathrm{d}P}{\mathrm{d}s} T^{\mathbf{H}}.$$

Further proper-time derivatives follow in the same way. Coordinate-time derivatives require the additional factor $\mathrm{d}t/\mathrm{d}t'$.

18.2 Coordinate derivatives

To derive the transformation of $\partial = \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\partial}_{\mathbf{r}}$, express the same scalar differential in both coordinate systems. Let $f : \mathbb{R}^4 \rightarrow \mathbb{C}$ be twice continuously differentiable, with $f' = f$ at corresponding events; then $\mathrm{d}f' = \mathrm{d}f$. The constant frame factor gives $\mathrm{d}X = T^{-1} \mathrm{d}X' T^{-\mathbf{H}}$, and $\det(T) = 1$ implies $T^* = T^{-\mathbf{H}}$. Scalar-part cyclicity therefore gives

$$\begin{aligned} \mathrm{d}f &= (\partial_t f) \mathrm{d}t + (\boldsymbol{\partial}_{\mathbf{r}} f) \cdot \mathrm{d}\mathbf{r} \\ &= \text{sc}((\partial f) \mathrm{d}X) \\ &= \text{sc}\left((T^*(\partial f) T^{-1}) \mathrm{d}X'\right). \end{aligned}$$

In the primed coordinates, the same differential is

$$\mathrm{d}f' = (\partial'_t f') \mathrm{d}t' + (\boldsymbol{\partial}'_{\mathbf{r}'} f') \cdot \mathrm{d}\mathbf{r}' = \text{sc}((\partial' f') \mathrm{d}X').$$

The increments $\mathrm{d}t'$ and the three components of $\mathrm{d}\mathbf{r}'$ are independent. Comparing their complex coefficients in $\mathrm{d}f' = \mathrm{d}f$ gives

$$\partial' f' = T^*(\partial f) T^{-1}.$$

Since this holds for every such scalar field, the coordinate derivative operators obey the chain rule (7, p. 101)

$$\boxed{\partial' = T^* \partial T^{-1}}. \tag{18.2}$$

Here derivatives act on the field to their right; the frame factors are constant. The partial derivatives commute on these fields, so determinant multiplicativity applies to the operator coefficients. Since $\det(T^*) = \det(T^{-1}) = 1$,

$$\square' = \det(\partial') = \det(T^*) \det(\partial) \det(T^{-1}) = \square.$$

Equivalently, $\square' f' = \square f$ at corresponding events. Sigma conjugation of the operator coefficients also gives

$$\partial'^* = T \partial^* T^{\mathbf{H}}.$$

Here the star acts on the algebra coefficients, not on the field to which the operator will be applied.

18.3 Potential and current

Write $A = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $J = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$, where $V, \rho \in \mathbb{R}$, $\mathbf{A}, \mathbf{J} \in \mathbb{R}^3$, and ρ is charge density. Their transformation laws require their physical definitions.

Potential from the particle coupling. The interaction in (12.16) is $-q \int \text{sc}(A^*U) \, ds$. We require the same interaction in every inertial frame. First compare the same gauge representative, so the local coupling agrees, without adding a total derivative to the action. For a nonzero test charge, invariance of q and ds therefore requires, for every possible time-like particle tangent (7, Eq. (5.41)),

$$\text{sc}(A'^*U') = \text{sc}(A^*U).$$

Substitute the derived law for U' and use scalar-part cyclicity:

$$\text{sc}(A'^*TUT^H) = \text{sc}((T^H A'^*T)U) = \text{sc}(A^*U).$$

Future-directed unit time-like tangents span the real paravectors: a rest tangent determines the scalar coefficient, and opposite velocities along each axis determine the three vector coefficients. Consequently

$$T^H A'^*T = A^*, \quad A'^* = T^* A^* T^{-1}.$$

Applying sigma conjugation yields the potential law:

$$\boxed{A' = T A T^H}. \quad (18.3)$$

Thus A transforms like U and P because of its coupling to charge motion.

Gauge freedom. For the real gauge scalar λ already introduced, let $\lambda' = \lambda$ at corresponding events. The derivative law gives

$$\partial'^* \lambda' = T(\partial^* \lambda) T^H, \quad A' + \partial'^* \lambda' = T(A + \partial^* \lambda) T^H.$$

The potential law therefore respects gauge freedom. Choosing an additional gauge change only in the primed frame adds $\partial'^* \lambda'$ to (18.3). Its contribution to the particle coupling is a boundary term, because

$$\text{sc}((\partial^* \lambda)^* U) \, ds = \text{sc}((\partial \lambda) \, dX) = d\lambda.$$

In the frame transformations below no additional gauge change is made.

Current from charge transport. For one local stream of charges with velocity $\mathbf{v}$, the definitions give $\mathbf{J} = \rho \mathbf{v}$ and

$$J = \rho(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{\rho}{\text{sc}(U)} U.$$

Charge is invariant, while Lorentz contraction divides a small comoving volume by $\text{sc}(U) = dt/ds$. Hence $\rho/\text{sc}(U)$ is the charge density measured in that stream's rest frame, a Lorentz scalar (7, Sec. 5.2). It follows that

$$J' = \frac{\rho'}{\text{sc}(U')} U' = \frac{\rho}{\text{sc}(U)} TUT^H = T J T^H.$$

$$\boxed{J' = T J T^H}. \quad (18.4)$$

A general current is a sum of contributions from streams with their own velocities and signed charge densities. Linearity gives the same law for the total, even when it has no rest frame. Charge conservation is preserved:

$$\text{sc}(\partial' J') = \text{sc}(T^*(\partial J)T^H) = \text{sc}(\partial J) = 0.$$

This is $\partial_{t'}\rho' + \partial_{\mathbf{r}'} \cdot \mathbf{J}' = 0$ whenever the unprimed continuity equation holds.

18.4 Electromagnetic field and Maxwell covariance

The electric and magnetic fields $\mathbf{E}, \mathbf{B} \in \mathbb{R}^3$ form $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \in \{0\} \oplus \mathbb{C}^3$. The field transformation follows from the potential law. With $S = \text{sc}(\partial A) \in \mathbb{R}$, (12.11) gives

$$\begin{aligned} S' + F'^* &= \partial' A' = (T^* \partial T^{-1})(T A T^H) \\ &= T^*(\partial A)T^H \\ &= T^*(S + F^*)T^H = S + T^* F^* T^H. \end{aligned}$$

The derivatives act on A ; the constant adjacent factors $T^{-1}T$ cancel. Since $T^H T^* = 1$, scalar-part cyclicity gives $\text{sc}(T^* F^* T^H) = 0$. The scalar and vector parts therefore give, separately,

$$S' = S, \quad F'^* = T^* F^* T^H.$$

Sigma conjugation preserves product order and satisfies $(T^H)^* = T^{-1}$. Applying it to the second equality gives (5, Eq. (33))

$$\boxed{F' = T F T^{-1}}. \quad (18.5)$$

Thus the field transforms by similarity. Maxwell's equation then follows in the primed frame by cancellation:

$$\begin{aligned} \partial' F' &= (T^* \partial T^{-1})(T F T^{-1}) \\ &= T^*(\partial F)T^{-1} = T^* J^* T^{-1} \\ &= (T J T^H)^* = J'^*. \end{aligned}$$

The transformed potentials and fields therefore satisfy the same Maxwell equation (12.2). An additional gauge change would give $S' \mapsto S' + \square' \lambda'$ but leave F' unchanged: the added derivative $\partial' \partial'^* \lambda' = \square' \lambda'$ is purely scalar. Thus $S' = S$ refers to the frame change alone, whereas the field law is independent of the gauge representative.

18.5 Boost components

For a frame moving with velocity $+\beta \mathbf{u}$, use the boost factor L above, with $\beta = \tanh(\theta) \in (-1, 1)$ and $\gamma = \cosh(\theta) \in \mathbb{R}_{\geq 1}$. Parallel and perpendicular components are taken relative to $\mathbf{u}$. The expansion $L = \cosh(\theta/2) - \sinh(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}$ and the vector product give

$$\begin{aligned} L^2 &= \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma}), \\ L(\mathbf{u} \cdot \boldsymbol{\sigma})L &= \gamma(\mathbf{u} \cdot \boldsymbol{\sigma} - \beta), \\ L(\mathbf{r}_\perp \cdot \boldsymbol{\sigma})L &= \mathbf{r}_\perp \cdot \boldsymbol{\sigma}. \end{aligned}$$

In the last line $\mathbf{u} \cdot \boldsymbol{\sigma}$ anticommutes with $\mathbf{r}_\perp \cdot \boldsymbol{\sigma}$, so the two boost factors cancel. Applying these identities to $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ gives

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad \mathbf{r}'_\parallel = \gamma(\mathbf{r}_\parallel - \beta t \mathbf{u}), \quad \mathbf{r}'_\perp = \mathbf{r}_\perp.$$

Differentiation with respect to invariant proper time gives the component law for U , and multiplication by m gives that for P . The identical congruence laws for the potential and current give

$$\begin{aligned} V' &= \gamma(V - \beta \mathbf{u} \cdot \mathbf{A}), & \rho' &= \gamma(\rho - \beta \mathbf{u} \cdot \mathbf{J}), \\ \mathbf{A}'_\parallel &= \gamma(\mathbf{A}_\parallel - \beta V \mathbf{u}), & \mathbf{J}'_\parallel &= \gamma(\mathbf{J}_\parallel - \beta \rho \mathbf{u}), \\ \mathbf{A}'_\perp &= \mathbf{A}_\perp, & \mathbf{J}'_\perp &= \mathbf{J}_\perp. \end{aligned}$$

Since $L^H = L$ and $L^* = L^{-1}$, (18.2) becomes $\partial' = L^{-1} \partial L^{-1}$. Resolving the spatial derivatives parallel and perpendicular to $\mathbf{u}$ gives

$$\begin{aligned} \partial_{t'} &= \gamma(\partial_t + \beta \mathbf{u} \cdot \boldsymbol{\partial}_\mathbf{r}), \\ \partial_{\mathbf{r}'_\parallel} &= \gamma(\partial_{\mathbf{r}_\parallel} + \beta \mathbf{u} \partial_t), \\ \partial_{\mathbf{r}'_\perp} &= \partial_{\mathbf{r}_\perp}. \end{aligned}$$

Here $\partial_{\mathbf{r}_\parallel} = \mathbf{u}(\mathbf{u} \cdot \boldsymbol{\partial}_\mathbf{r})$ and $\partial_{\mathbf{r}_\perp} = \boldsymbol{\partial}_\mathbf{r} - \partial_{\mathbf{r}_\parallel}$, with the same projections in the primed frame. Only the parallel derivative mixes with the time derivative; the plus signs express the inverse coordinate transformation in the chain rule.

At corresponding events, with $X' = LXL$, the field law reduces to

$$F' = LFL^{-1}. \quad (18.6)$$

Similarity preserves the field's zero scalar part and mixes its real and imaginary vector parts. A nonzero boost congruence would instead introduce a scalar part for a nonzero field parallel to $\mathbf{u}$; it is the law for events and the real four-potential, not for F . Expanding this product by (5.14) gives

$$F' = (\mathbf{E}_\parallel + i\mathbf{B}_\parallel + \gamma(\mathbf{E}_\perp + i\mathbf{B}_\perp) - i\gamma\beta \mathbf{u} \times (\mathbf{E} + i\mathbf{B})) \cdot \boldsymbol{\sigma}.$$

Separating real and imaginary parts gives (7, Eqs. (5.17)–(5.18))

$$\boxed{\mathbf{E}' = \mathbf{E}_\parallel + \gamma(\mathbf{E}_\perp + \beta \mathbf{u} \times \mathbf{B}),} \quad (18.7)$$

$$\boxed{\mathbf{B}' = \mathbf{B}_\parallel + \gamma(\mathbf{B}_\perp - \beta \mathbf{u} \times \mathbf{E}).} \quad (18.8)$$

The parallel components are unchanged, and reversing β gives the inverse field law. For a rotation, $R^H = R^{-1}$, so congruence and similarity coincide. Boosts distinguish the two laws because $L^H = L \neq L^{-1}$ unless the boost is zero.

18.6 Field invariants, energy flux, and force

Field invariants. Since F^2 is scalar, similarity preserves $F'^2 = TF^2T^{-1} = F^2$. Equation (12.15) therefore gives

$$\boxed{\begin{aligned} \mathbf{E}'^2 - \mathbf{B}'^2 &= \mathbf{E}^2 - \mathbf{B}^2, \\ \mathbf{E}' \cdot \mathbf{B}' &= \mathbf{E} \cdot \mathbf{B}. \end{aligned}}$$

Energy density and flux. The energy density $u = (\mathbf{E}^2 + \mathbf{B}^2)/2 \in \mathbb{R}_{\geq 0}$ and Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B} \in \mathbb{R}^3$ depend on the observer. Their primed values follow directly from (12.24):

$$u' + \mathbf{S}' \cdot \boldsymbol{\sigma} = \frac{1}{2} T F T^{-1} T^{-\mathbf{H}} F^{\mathbf{H}} T^{\mathbf{H}}.$$

For a boost this becomes, step by step,

$$\begin{aligned} u' + \mathbf{S}' \cdot \boldsymbol{\sigma} &= \frac{1}{2} F' F'^{\mathbf{H}} \\ &= \frac{1}{2} (L F L^{-1}) (L^{-1} F^{\mathbf{H}} L) \\ &= \frac{1}{2} L F L^{-2} F^{\mathbf{H}} L. \end{aligned}$$

Explicitly, $u' = (\mathbf{E}'^2 + \mathbf{B}'^2)/2$ and $\mathbf{S}' = \mathbf{E}' \times \mathbf{B}'$. The middle factor L^{-2} shows why the real paravector $u + \mathbf{S} \cdot \boldsymbol{\sigma}$ does not itself obey the four-momentum law: it combines field energy density and flux for a specified observer. For example, if $\mathbf{B} = 0$ and $\mathbf{u} \cdot \mathbf{E} = 0$, the component laws give

$$2u' = \|F'\|^2 = \gamma^2(1 + \beta^2)\mathbf{E}^2, \quad \mathbf{E}'^2 - \mathbf{B}'^2 = \mathbf{E}^2.$$

The energy density changes while the field invariant remains fixed.

Lorentz force. The field and velocity laws give

$$\begin{aligned} F' U' &= (T F T^{-1})(T U T^{\mathbf{H}}) = T F U T^{\mathbf{H}}, \\ U' F'^{\mathbf{H}} &= (T U T^{\mathbf{H}})(T^{-\mathbf{H}} F^{\mathbf{H}} T^{\mathbf{H}}) = T U F^{\mathbf{H}} T^{\mathbf{H}}. \end{aligned}$$

Their sum therefore transforms like dP/ds , proving covariance of the force law (12.21) for every frame factor T . For a boost, the identity insertions are explicit:

$$\begin{aligned} \frac{dP'}{ds} &= L \frac{q}{2} (F U + U F^{\mathbf{H}}) L \\ &= \frac{q}{2} (L F (L^{-1} L) U L + L U (L L^{-1}) F^{\mathbf{H}} L) \\ &= \frac{q}{2} ((L F L^{-1})(L U L) + (L U L)(L^{-1} F^{\mathbf{H}} L)). \end{aligned}$$

Thus the transformed fields give the same form of the real four-force in the moving frame:

$$\frac{dP'}{ds} = \frac{q}{2} (F' U' + U' F'^{\mathbf{H}}).$$

18.7 Momentum operators, scalar waves, and Dirac covariance

Momentum operators. The definition $\hat{P} = i\partial^*$ and the coefficient transformation of ∂^* give

$$\begin{aligned} \hat{P}' &= i\partial'^* = T(i\partial^*)T^{\mathbf{H}} = T\hat{P}T^{\mathbf{H}}, \\ \hat{P}'_q &= \hat{P}' - qA' = T(\hat{P} - qA)T^{\mathbf{H}} = T\hat{P}_qT^{\mathbf{H}}. \end{aligned}$$

These are operator equalities under the coordinate change, with all derivatives acting to the right. Their factors are constant except for the potentials. The adjugate is complex-linear and reverses product order, so

$$\text{adj}(\hat{P}'_q) = \text{adj}(T^{\mathbf{H}}) \text{adj}(\hat{P}_q) \text{adj}(T) = T^* \text{adj}(\hat{P}_q) T^{-1}.$$

Thus the kinetic operator and its adjugate have the two complementary laws needed in the Dirac equations. Sigma conjugation cannot replace this adjugate: it would also conjugate the factor i .

Wave phase and Klein–Gordon. The momentum $P \in \mathbb{R} \oplus \mathbb{R}^3$ transforms as $P' = TPT^H$. Adjunction and cyclicity give the invariant metric pairing with the event X (5.3):

$$\text{sc}(P'^* X') = \text{sc}(T^* P^* X T^H) = \text{sc}(P^* X).$$

Consequently $E't' - \mathbf{P}' \cdot \mathbf{r}' = Et - \mathbf{P} \cdot \mathbf{r}$, so the complex plane wave (13.11) has $f' = f$ at corresponding events and momenta. A scalar Klein–Gordon field $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ likewise uses $\psi' = \psi$ at corresponding events. With $\hat{E}_q = i\partial_t - qV$ and $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$, scalar extraction gives

$$\text{sc}(\text{adj}(\hat{P}_q)\hat{P}_q) = \text{sc}((\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})(\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})) = \hat{E}_q^2 - \hat{\mathbf{P}}_q^2.$$

The mixed terms are vectors; the scalar part of the spatial product is the sum of its three component squares. This extraction does not require the kinetic components to commute. Under the transformation just derived,

$$\begin{aligned} \text{sc}(\text{adj}(\hat{P}'_q)\hat{P}'_q) &= \text{sc}(T^* \text{adj}(\hat{P}_q)T^{-1}T\hat{P}_qT^H) \\ &= \text{sc}(\text{adj}(\hat{P}_q)\hat{P}_q). \end{aligned}$$

Here cyclicity moves only constant frame factors past the derivatives. Consequently

$$\boxed{(-(\hat{E}'_q)^2 + (\hat{\mathbf{P}}'_q)^2 + m^2)\psi' = (-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi.}$$

Derivatives act on the potentials as well as the amplitude. Thus a zero right-hand side remains zero in the primed frame; $q = 0$ gives the free case.

Dirac entries. The fields $\psi_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$ have the fixed two-component form of (13.2). Choose the spin-one-half representation $\psi'_- = T\psi_-$: consecutive frame factors act by their ordered product, and a rotation gives the established law (17.1). Left multiplication preserves that fixed two-component form. This representation choice is the spin input. For $m > 0$, the first interacting equation (15.13) then determines the partner's transformation:

$$\begin{aligned} m\psi'_+ &= \text{adj}(\hat{P}'_q)\psi'_- = (T^* \text{adj}(\hat{P}_q)T^{-1})(T\psi_-) \\ &= T^* \text{adj}(\hat{P}_q)\psi_- = mT^*\psi_+. \end{aligned}$$

Hence

$$\boxed{\psi'_+ = T^*\psi_+, \quad \psi'_- = T\psi_-}.$$

The second equation (15.14) now follows by the complementary cancellation:

$$\hat{P}'_q\psi'_+ = (T\hat{P}_qT^H)(T^*\psi_+) = T\hat{P}_q\psi_+ = mT\psi_- = m\psi'_-.$$

Here $\text{adj}(\hat{P}_q) = \hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}$ and $\hat{P}_q = \hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}$ are exactly the operators in those two equations. Both equations retain their form; $q = 0$ gives the free case. For $T = R$, both entries transform by R , consistently with (17.1); the spatial coordinates also rotate. A rotation of the axes through $+\theta$ about $\mathbf{u}$ uses the inverse factor $T = R^{-1}$. For $T = L$, the entries instead use L^{-1} and L , respectively (16, Eq. (4.31)). The boost factor already contains half the rapidity.

Spinor pairing and density. The cross-pairing is invariant because $(T^*)^H = T^{-1}$:

$$\text{sc}(\psi_+^H \psi'_-) = \text{sc}(\psi_+^H T^{-1} T \psi_-) = \text{sc}(\psi_+^H \psi_-).$$

The summed positive density (15.1) instead becomes

$$\|\psi'_+\|^2 + \|\psi'_-\|^2 = \text{sc}(\psi_+^H T^{-1} T^* \psi_+) + \text{sc}(\psi_-^H T^H T \psi_-).$$

For rotations both middle products are 1; for a boost they are L^{-2} and L^2 . Thus the density is preserved by rotations and generally changes under boosts. The full Dirac pair is used for boosts before the Pauli low-energy limit is taken.

Transformation laws collected. For the same constant frame factor and gauge representative,

$$\boxed{\begin{aligned} X' &= T X T^H, & U' &= T U T^H, & P' &= T P T^H, \\ A' &= T A T^H, & J' &= T J T^H, & \hat{P}'_q &= T \hat{P}_q T^H, \\ \partial' &= T^* \partial T^{-1}, & F' &= T F T^{-1}, \\ \psi'_+ &= T^* \psi_+, & \psi'_- &= T \psi_-. \end{aligned}}$$

These are different representations within the same algebra. The field law follows from differentiating the potential; it does not replace the potential's four-vector law.

19 Position-dependent paravector bases

Coordinates satisfy $(t, \mathbf{r}) \in \mathbb{R} \times \mathbb{R}^3$ on an open chart. The coordinate-map functions are $a : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{X} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ locally. The frame and metric fields $\mathbf{M}, \mathbf{g}$ take values in $\mathbb{R}^{4 \times 4}$, with $\mathbf{M}$ invertible and $\mathbf{g}$ symmetric of signature $(+, -, -, -)$. A local basis entry $\sigma'_t \in \mathbb{R} \oplus \mathbb{R}^3$ is a paravector; $\sigma'_r \in (\mathbb{R} \oplus \mathbb{R}^3)^3$ is a column of three paravectors. Local displacement and four-velocity values lie in $\mathbb{R} \oplus \mathbb{R}^3$; $dt', s \in \mathbb{R}$, $d\mathbf{r}', \mathbf{v} \in \mathbb{R}^3$, and $\gamma \in \mathbb{R}_{\geq 1}$ for time-like motion. Coordinate velocity $\dot{\mathbf{r}} \in \mathbb{R}^3$ is distinguished from measured velocity $\mathbf{v}$.

Section 11.3 starts with an inertial coordinate map $X = a(t, \mathbf{r}) + \mathbf{X}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$. Its exact differential gives

$$dX = da + d\mathbf{X} \cdot \boldsymbol{\sigma}, \quad ds^2 = da^2 - d\mathbf{X}^2.$$

This is flat spacetime in generalized coordinates. In the block notation used below,

$$\begin{Bmatrix} da \\ d\mathbf{X} \end{Bmatrix} = \mathbf{M}^T \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix};$$

so $\mathbf{M}^T$ is the Jacobian of the same functions $a, \mathbf{X}$.

Here we assign a local displacement directly:

$$dX = dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r, \quad ds^2 = \det(dX).$$

A smooth real invertible $\mathbf{M}(t, \mathbf{r})$ specifies the basis elements, without requiring $\mathbf{M}^T$ to be a coordinate Jacobian. The notation dX now denotes that local assignment; it need not be an exact differential, even locally. For fixed coordinate increments $dt, d\mathbf{r}$, the basis specifies their metric. This is the

usual local-frame (tetrad) construction, expressed in paravector form (26, Eq. (3.119)). The product, conjugations, and determinant retain their earlier definitions.

Generalized coordinates are a flat special case. A general basis can describe curvature, although position dependence or failure of exactness alone does not establish it: a varying Lorentz basis can still give a flat metric. We use the metric's unique torsion-free, metric-compatible parallel transport to determine free fall and curvature. This is the standard geometric input; Einstein's field equation is additional physical input selecting metrics from matter and boundary conditions.

19.1 The basis and its interval

Let $\mathbf{M}(t, \mathbf{r})$ be a smooth real invertible 4×4 matrix, partitioned as

$$\mathbf{M} = \begin{Bmatrix} M_{tt} & \mathbf{M}_{tr} \\ \mathbf{M}_{rt} & \mathbf{M}_{rr} \end{Bmatrix}.$$

Here $M_{tt} \in \mathbb{R}$, $\mathbf{M}_{tr} \in \mathbb{R}^{1 \times 3}$, $\mathbf{M}_{rt} \in \mathbb{R}^{3 \times 1}$, and $\mathbf{M}_{rr} \in \mathbb{R}^{3 \times 3}$. The 3×1 columns are identified with numerical vectors in $\mathbb{R}^3$ when dot products are used. The blocks are independent: no symmetry of $\mathbf{M}$ is assumed, and $\mathbf{M}_{rt} = \mathbf{M}_{tr}^\top$ is not required. Let $\mathbf{I} \in \mathbb{R}^{3 \times 3}$ be the identity and $\mathbf{0} \in \mathbb{R}^3$ the zero three-vector. The local basis is

$$\begin{Bmatrix} \sigma'_t \\ \boldsymbol{\sigma}'_r \end{Bmatrix} = \mathbf{M} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}. \quad (19.1)$$

Here σ'_t is one real paravector and $\boldsymbol{\sigma}'_r$ is a triple of real paravectors, all Hermitian. Their scalar and vector parts are

$$\sigma'_t = M_{tt} + \mathbf{M}_{tr}^\top \cdot \boldsymbol{\sigma}, \quad \boldsymbol{\sigma}'_r = \mathbf{M}_{rt} + \mathbf{M}_{rr} \boldsymbol{\sigma}.$$

In the second expression, $\mathbf{M}_{rt}$ collects the three scalar parts and each row of $\mathbf{M}_{rr}$ gives the corresponding vector part. The labels t, r identify basis entries, not algebraic scalar and vector grades. Sigma conjugation acts componentwise and leaves the real entries of $\mathbf{M}$ unchanged:

$$\begin{aligned} \begin{Bmatrix} \sigma'_t \\ \boldsymbol{\sigma}'_r \end{Bmatrix}^* &= \mathbf{M} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}^* = \mathbf{M} \begin{Bmatrix} 1 \\ -\boldsymbol{\sigma} \end{Bmatrix} \\ &= \mathbf{M} \begin{Bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & -\mathbf{I} \end{Bmatrix} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}. \end{aligned} \quad (19.2)$$

Write the local displacement as $dX = dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r$. Its interval is

$$\boxed{ds^2 = \det(dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r)}. \quad (19.3)$$

Its scalar–vector coefficient column in the fixed basis is $\mathbf{M}^\top \{dt, d\mathbf{r}\}^\top$, as derived below. Taking the scalar part squared minus the vector part squared gives the metric matrix

$$\begin{aligned} \mathbf{g} &= \mathbf{M} \begin{Bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & -\mathbf{I} \end{Bmatrix} \mathbf{M}^\top = \begin{Bmatrix} g_{tt} & \mathbf{g}_{tr} \\ \mathbf{g}_{tr}^\top & \mathbf{g}_{rr} \end{Bmatrix}, \\ ds^2 &= \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}^\top \mathbf{g} \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix} \\ &= g_{tt} dt^2 + 2 dt \mathbf{g}_{tr} d\mathbf{r} + d\mathbf{r}^\top \mathbf{g}_{rr} d\mathbf{r}. \end{aligned} \quad (19.4)$$

Here $g_{tt} \in \mathbb{R}$, $\mathbf{g}_{tr} \in \mathbb{R}^{1 \times 3}$, and $\mathbf{g}_{rr} \in \mathbb{R}^{3 \times 3}$ is symmetric. Explicitly,

$$\begin{aligned} g_{tt} &= M_{tt}^2 - \mathbf{M}_{tr} \mathbf{M}_{tr}^\top, \\ \mathbf{g}_{tr} &= M_{tt} \mathbf{M}_{rt}^\top - \mathbf{M}_{tr} \mathbf{M}_{rr}^\top, \\ \mathbf{g}_{rr} &= \mathbf{M}_{rt} \mathbf{M}_{rt}^\top - \mathbf{M}_{rr} \mathbf{M}_{rr}^\top. \end{aligned}$$

Invertibility of $\mathbf{M}$ ensures signature $(+, -, -, -)$ and $\det(\mathbf{g}) = -(\det(\mathbf{M}))^2 < 0$. Every smooth Lorentzian metric has such a representation locally (26, Eq. (3.119)). Along a time-like path with $dt/ds > 0$,

$$\frac{ds}{dt} = \sqrt{g_{tt} + 2\mathbf{g}_{tr} \dot{\mathbf{r}} + \dot{\mathbf{r}}^\top \mathbf{g}_{rr} \dot{\mathbf{r}}}. \quad (19.5)$$

For comparison, a similarity basis $\boldsymbol{\sigma}'_r = T \boldsymbol{\sigma} T^{-1}$, with invertible $T \in \mathbb{C} \oplus \mathbb{C}^3$, is generally a triple in $(\{0\} \oplus \mathbb{C}^3)^3$, outside the real-basis construction above. It preserves

$$\det(dt + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r) = dt^2 - d\mathbf{r}^2,$$

so it retains the flat interval. The general real matrix $\mathbf{M}$ allows the metric blocks above to vary independently of that similarity law.

Local measurements and velocity. The primed basis represents coordinate directions; it is generally not an observer's orthonormal time and space basis. The matrix $\mathbf{M}$ converts coordinate increments to components in an orthonormal local frame (26, Eqs. (3.119)–(3.123)):

$$\begin{aligned} \begin{Bmatrix} dt' \\ d\mathbf{r}' \end{Bmatrix} &= \mathbf{M}^\top \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}, \\ dX &= dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r = dt' + d\mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned} \quad (19.6)$$

Here dt' , $d\mathbf{r}'$ denote local measurement increments. They need not be exact differentials of coordinate functions t' , $\mathbf{r}'$ on a region. The fixed algebraic symbols $1, \boldsymbol{\sigma}$ describe orthonormal axes at each point, not a single global inertial frame. Their interval is

$$ds^2 = dt'^2 - d\mathbf{r}'^2.$$

The observer at rest in this frame has $d\mathbf{r}' = \mathbf{0}$ and $dt' = ds$. Choose the time orientation so that future-directed time-like displacements have $dt' > 0$. For a massive particle with proper time s , the earlier four-velocity relation becomes

$$\begin{aligned} U &= \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ \boxed{\mathbf{v} = \frac{d\mathbf{r}'}{dt'}, \quad \gamma = \frac{dt'}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}.} \end{aligned} \quad (19.7)$$

These are measurements relative to that observer. Writing $\dot{\mathbf{r}} = d\mathbf{r}/dt$ and $\dot{s} = ds/dt > 0$, with $\dot{s}$ given by (19.5), gives

$$\boxed{\begin{aligned} \gamma &= \frac{M_{tt} + \mathbf{M}_{rt} \cdot \dot{\mathbf{r}}}{\dot{s}}, \\ \mathbf{v} &= \frac{\mathbf{M}_{tr}^\top + \mathbf{M}_{rr}^\top \dot{\mathbf{r}}}{M_{tt} + \mathbf{M}_{rt} \cdot \dot{\mathbf{r}}}. \end{aligned}} \quad (19.8)$$

Thus γ is generally not dt/ds , and $\mathbf{v}$ is generally not $\dot{\mathbf{r}}$. A local Lorentz change of observer preserves $\mathbf{g}$ but changes these measurements; the metric alone does not select an observer. Nor does choosing a frame imply that its observers are in free fall.

19.2 Algebra and differentiation

For the real local displacement $dX = dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r$, the coefficient row multiplies the basis column. Substituting (19.1) gives

$$\begin{aligned} dX &= \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}^\top \begin{Bmatrix} \sigma'_t \\ \boldsymbol{\sigma}'_r \end{Bmatrix} \\ &= \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}^\top \mathbf{M} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix} = \left(\mathbf{M}^\top \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix} \right)^\top \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}. \end{aligned}$$

Thus expansion in the fixed basis uses $\mathbf{M}^\top$ on the coefficient column. The transpose introduces no complex conjugation. Transposition swaps and transposes the off-diagonal blocks; it does not equate them:

$$\begin{Bmatrix} dt' \\ d\mathbf{r}' \end{Bmatrix} = \mathbf{M}^\top \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix} = \begin{Bmatrix} M_{tt} & \mathbf{M}_{rt}^\top \\ \mathbf{M}_{tr}^\top & \mathbf{M}_{rr}^\top \end{Bmatrix} \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}. \quad (19.9)$$

Equivalently, the paravector is

$$dX = (M_{tt} dt + \mathbf{M}_{rt} \cdot d\mathbf{r}) + (dt \mathbf{M}_{tr}^\top + \mathbf{M}_{rr}^\top d\mathbf{r}) \cdot \boldsymbol{\sigma}.$$

The first term is dt' ; the coefficient of $\boldsymbol{\sigma}$ is $d\mathbf{r}'$. In general these are not dt and $d\mathbf{r}$. To multiply paravectors, apply the existing rule (2.9) to their scalar–vector parts, then apply $\mathbf{M}^{-\top}$ to the resulting coefficient column to return to the local basis. The product may have complex coefficients even when both factors are real.

Differentiation. Holding the coordinate increments dt , $d\mathbf{r}$ fixed while varying the base point gives

$$\begin{aligned} \partial_t(dX) &= ((\partial_t M_{tt}) dt + (\partial_t \mathbf{M}_{rt}) \cdot d\mathbf{r}) \\ &\quad + (dt \partial_t \mathbf{M}_{tr}^\top + (\partial_t \mathbf{M}_{rr}^\top) d\mathbf{r}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Refactoring into the local basis gives

$$\partial_t(dX) = \begin{Bmatrix} \sigma'_t & \boldsymbol{\sigma}'_r{}^\top \end{Bmatrix} \mathbf{M}^{-\top} (\partial_t \mathbf{M}^\top) \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}. \quad (19.10)$$

The same rule holds for any coordinate or fixed-direction spatial derivative. If the increments also vary, their derivatives contribute the usual product-rule terms. Refactoring an ordinary derivative changes its representation; it does not supply the metric’s torsion-free parallel transport.

19.3 Expansion, free fall, and curvature

For a smooth scale factor $\lambda : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ on the time interval considered, choose

$$\mathbf{M} = \begin{Bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \lambda \mathbf{I} \end{Bmatrix}, \quad \sigma'_t = 1, \quad \boldsymbol{\sigma}'_r = \lambda \boldsymbol{\sigma}. \quad (19.11)$$

The metric blocks are

$$g_{tt} = 1, \quad \mathbf{g}_{tr} = \mathbf{0}^\top, \quad \mathbf{g}_{rr} = -\lambda(t)^2 \mathbf{I}. \quad (19.12)$$

For real vector increments $d\mathbf{X}, d\mathbf{Y} \in \mathbb{R}^3$, the algebra gives

$$(d\mathbf{X} \cdot \boldsymbol{\sigma}'_r)(d\mathbf{Y} \cdot \boldsymbol{\sigma}'_r) = \lambda^2 d\mathbf{X} \cdot d\mathbf{Y} + i\lambda(d\mathbf{X} \times d\mathbf{Y}) \cdot \boldsymbol{\sigma}'_r.$$

For $dX = dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r$, the interval is

$$\boxed{ds^2 = \det(dX) = dt^2 - \lambda(t)^2 d\mathbf{r}^2,} \quad (19.13)$$

the spatially flat Robertson–Walker metric (26, Eqs. (8.7)–(8.9)). Its constant-time spatial slices are flat; its spacetime need not be.

Free fall in an expanding universe. Observers at fixed $\mathbf{r}$ are comoving: their clocks measure t , and the scale factor $\lambda(t)$ converts coordinate separations into physical distances. Here (19.6) reduces to $dt' = dt$ and $d\mathbf{r}' = \lambda d\mathbf{r}$, so the local frame's rest observers are exactly the comoving observers. For this metric, the quadratic geodesic action reduces to

$$\frac{1}{2} \int \left(\left(\frac{dt}{ds} \right)^2 - \lambda(t)^2 \left(\frac{d\mathbf{r}}{ds} \right)^2 \right) ds.$$

Varying t and $\mathbf{r}$ with fixed endpoints gives the affinely parameterized geodesic equations (26, pp. 68–70)

$$\frac{d^2 t}{ds^2} = -\lambda \dot{\lambda} \left(\frac{d\mathbf{r}}{ds} \right)^2, \quad \frac{d}{ds} \left(\lambda^2 \frac{d\mathbf{r}}{ds} \right) = \mathbf{0}. \quad (19.14)$$

Here $\dot{\lambda} = d\lambda/dt$. For a massive particle, choose s to be proper time. Its velocity relative to a comoving observer is $\mathbf{v} = \lambda d\mathbf{r}/dt$, so $\gamma = dt/ds = (1 - \mathbf{v}^2)^{-1/2}$. For constant mass $m > 0$, the earlier momentum law $P = mU$ gives

$$E = m\gamma, \quad \mathbf{P} = m\lambda \frac{d\mathbf{r}}{ds} = m\gamma \mathbf{v}.$$

The spatial geodesic equation therefore has the direct physical meaning

$$\boxed{\frac{d}{dt}(\lambda \mathbf{P}) = \mathbf{0}, \quad \frac{d\mathbf{P}}{dt} = -\frac{\dot{\lambda}}{\lambda} \mathbf{P}.} \quad (19.15)$$

The momentum measured by comoving observers decreases as $1/\lambda$ during expansion. This is the standard cosmological redshift of particle momentum (26, Eq. (8.64)). At low speeds, $\mathbf{P} \simeq m\mathbf{v}$, so the velocity relative to those observers also decreases as $1/\lambda$. A particle initially at fixed $\mathbf{r}$ remains there.

Parallel transport and redshift. Parallel transport carries a four-vector along a chosen path using the metric's torsion-free, metric-compatible connection. Write the real paravector in terms of its measured scalar and vector parts:

$$Y = b\sigma'_t + \frac{\mathbf{Y}}{\lambda} \cdot \sigma'_r = b + \mathbf{Y} \cdot \sigma, \quad b \in \mathbb{R}, \quad \mathbf{Y} \in \mathbb{R}^3.$$

The factor $1/\lambda$ converts the measured vector $\mathbf{Y}$ into coefficients of the scaled basis. The torsion-free transport law follows from the quadratic velocity terms in (19.14): replace one tangent by the carried vector and average the two placements. Converting to the measured components above and using t as the path parameter, with $\mathbf{v} = \lambda \, d\mathbf{r}/dt$, gives

$$\frac{db}{dt} = -\frac{\dot{\lambda}}{\lambda} \mathbf{v} \cdot \mathbf{Y}, \quad \frac{d\mathbf{Y}}{dt} = -\frac{\dot{\lambda}}{\lambda} b \mathbf{v}. \quad (19.16)$$

These equations preserve $\det(Y) = b^2 - \mathbf{Y}^2$. Choosing $Y = U$, with $b = \gamma$ and $\mathbf{Y} = \gamma \mathbf{v}$, recovers free fall: a geodesic parallel-transport its own tangent.

For a freely propagating photon, take $Y = P = E + \mathbf{P} \cdot \sigma$, with $\mathbf{P} = E\mathbf{v}$ and $\mathbf{v}^2 = 1$. The same transport rule gives

$$\boxed{\frac{dE}{dt} = -\frac{\dot{\lambda}}{\lambda} E, \quad \lambda E = \text{constant}.} \quad (19.17)$$

Photon energy and frequency scale as $1/\lambda$: expansion produces the familiar cosmological redshift (26, Eq. (8.66)). Both results follow from the assigned metric for any smooth positive $\lambda(t)$; determining $\lambda(t)$ from matter requires Einstein's field equation.

Tidal acceleration and curvature. Two nearby comoving observers are both in free fall, with proper time t and fixed coordinate separation $\Delta \mathbf{r}$. Their physical separation ℓ therefore obeys

$$\ell = \lambda \|\Delta \mathbf{r}\|, \quad \dot{\ell} = \frac{\dot{\lambda}}{\lambda} \ell, \quad \boxed{\ddot{\ell} = \frac{\ddot{\lambda}}{\lambda} \ell.}$$

The middle relation is Hubble's law: recession speed is proportional to distance. The last relation is geodesic deviation, the tidal acceleration between neighboring freely falling observers (26, Eq. (3.113)). Each observer has zero proper acceleration, yet their separation can accelerate. The ratio $\ddot{\lambda}/\lambda$ measures this tidal effect.

Transport around infinitesimal loops gives curvature magnitudes $|\ddot{\lambda}/\lambda|$ in each of the three orthonormal time-space planes and $(\dot{\lambda}/\lambda)^2$ in each of the three spatial planes. These are spacetime transport effects; each fixed-time spatial slice has zero intrinsic curvature. Contracting the Riemann tensor with itself counts each plane four times. The resulting coordinate-independent *Kretschmann scalar* is

$$\boxed{12 \left(\left(\frac{\ddot{\lambda}}{\lambda} \right)^2 + \left(\frac{\dot{\lambda}}{\lambda} \right)^4 \right)}. \quad (19.18)$$

A nonzero value establishes spacetime curvature that no coordinate change can remove. In this spatially flat family it is nonzero wherever $\dot{\lambda}$ or $\ddot{\lambda}$ is nonzero; the spacetime is flat on a time interval precisely when λ is constant there. In particular, $\ddot{\lambda} = 0$ alone does not imply flat spacetime.

The role of the assigned displacement. The curved metric above uses $dX = dt + \lambda d\mathbf{r} \cdot \boldsymbol{\sigma}$. An ordinary rescaling of inertial spatial coordinates instead gives

$$\begin{aligned} d(t + \lambda(t)\mathbf{r} \cdot \boldsymbol{\sigma}) &= dt + (\lambda d\mathbf{r} + \dot{\lambda}\mathbf{r} dt) \cdot \boldsymbol{\sigma}, \\ \det(d(t + \lambda(t)\mathbf{r} \cdot \boldsymbol{\sigma})) &= dt^2 - (\lambda d\mathbf{r} + \dot{\lambda}\mathbf{r} dt)^2. \end{aligned}$$

Since $\lambda > 0$, this is an invertible coordinate change of flat Minkowski spacetime. The product-rule term $\dot{\lambda}\mathbf{r} dt$ distinguishes this flat interval from (19.13): the two displacements assign different metrics to the same $dt, d\mathbf{r}$.

19.4 Newtonian motion and black-hole metrics

Schwarzschild and Kerr belong to a common family: their intervals can be written as the Minkowski interval plus a term proportional to the square of a null form. This is the Kerr–Schild construction (27, Eq. (1)). We now express it in our paravector basis. It is a special choice of $\mathbf{M}$, within the general metric construction of Section 19.1; the expansion example above uses a different choice.

A common basis. On an open coordinate region in $\mathbb{R} \times \mathbb{R}^3$, let $\lambda(t, \mathbf{r}) \in \mathbb{R}$ be a smooth dimensionless scalar field and $\mathbf{u}(t, \mathbf{r}) \in \mathbb{R}^3$ a smooth unit vector field, $\mathbf{u}^2 = 1$. Here λ denotes the Kerr–Schild scalar, distinct from the scale factor $\lambda(t)$ in the preceding example. We omit the field arguments $(t, \mathbf{r})$ below. The paravector $1 - \mathbf{u} \cdot \boldsymbol{\sigma}$ is null:

$$\det(1 - \mathbf{u} \cdot \boldsymbol{\sigma}) = 1 - \mathbf{u}^2 = 0.$$

To change the displacement along this null direction, choose

$$\begin{aligned} \sigma'_t &= (1 - \lambda) + (\lambda\mathbf{u}) \cdot \boldsymbol{\sigma}, \\ \sigma'_r &= (-\lambda\mathbf{u}) + (\mathbf{I} + \lambda\mathbf{u}\mathbf{u}^\top) \cdot \boldsymbol{\sigma}. \end{aligned} \tag{19.19}$$

In the second line, the dot contracts each row of the matrix with $\boldsymbol{\sigma}$: $-\lambda\mathbf{u}$ lists the three scalar parts, and the matrix rows give their vector parts. The assigned displacement is therefore

$$\begin{aligned} dX &= dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r \\ &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma} - \lambda(dt + \mathbf{u} \cdot d\mathbf{r})(1 - \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Its scalar and vector parts give the local measurement increments:

$$\begin{aligned} dt' &= dt - \lambda(dt + \mathbf{u} \cdot d\mathbf{r}), \\ d\mathbf{r}' &= d\mathbf{r} + \lambda(dt + \mathbf{u} \cdot d\mathbf{r})\mathbf{u}. \end{aligned} \tag{19.20}$$

Taking $\det(dX) = dt'^2 - d\mathbf{r}'^2$, the terms quadratic in λ cancel because $\mathbf{u}^2 = 1$. The resulting metric blocks are

$$\begin{aligned} g_{tt} &= 1 - 2\lambda, & \mathbf{g}_{tr} &= -2\lambda\mathbf{u}^\top, \\ \mathbf{g}_{rr} &= -\mathbf{I} - 2\lambda\mathbf{u}\mathbf{u}^\top. \end{aligned} \tag{19.21}$$

Combining the three blocks gives

$$\boxed{ds^2 = dt^2 - d\mathbf{r}^2 - 2\lambda(dt + \mathbf{u} \cdot d\mathbf{r})^2.} \tag{19.22}$$

This is exactly the Kerr–Schild interval, with no weak-field approximation. Setting $\lambda = 0$ recovers Minkowski spacetime. The radial choice $\mathbf{u} = \mathbf{r}/\|\mathbf{r}\|$ with $\lambda = Gm/\|\mathbf{r}\|$ gives Schwarzschild; the spin-dependent fields below give Kerr. Newtonian motion follows in the weak-field, slow-motion limit. The basis is invertible for every finite λ : its inverse change is obtained by replacing λ with $-\lambda$, and $\det(\mathbf{M}) = 1$, so $\det(\mathbf{g}) = -1$. The algebra converts the chosen fields into a metric. Einstein’s equation is the additional physical law that constrains those fields for a given source; the basis identities alone do not determine the Schwarzschild or Kerr solutions.

Local velocity and observer. For a future-directed time-like path parametrized by t , the metric gives

$$\dot{s} = \sqrt{1 - \dot{\mathbf{r}}^2 - 2\lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})^2}. \quad (19.23)$$

Equations (19.20) and (19.7) then give

$$\boxed{\begin{aligned} \gamma &= \frac{1 - \lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})}{\dot{s}}, \\ \mathbf{v} &= \frac{\dot{\mathbf{r}} + \lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})\mathbf{u}}{1 - \lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})}. \end{aligned}} \quad (19.24)$$

The observer at rest in this frame has $d\mathbf{r}' = \mathbf{0}$. For the black-hole examples below, $\lambda \geq 0$, and this observer follows

$$\frac{d\mathbf{r}_{\text{obs}}}{dt} = -\frac{\lambda}{1 + \lambda}\mathbf{u}, \quad \frac{dt}{ds} = 1 + \lambda.$$

Substitution gives $\mathbf{v} = \mathbf{0}$ and $\gamma = 1$, as required. This observer moves in the coordinate chart and need not be freely falling. Substituting the Schwarzschild or Kerr fields below into the basis therefore supplies both their exact metric and this choice of local measurement frame wherever the fields are regular. The increments dt' , $d\mathbf{r}'$ do not generally integrate to a coordinate system on a region. Their Minkowski-form interval expresses local measurements; the varying metric still determines curved-spacetime free fall and curvature.

Newtonian limit. For stationary $\lambda, \mathbf{u}$ with $|\lambda| \ll 1$ and $\|\dot{\mathbf{r}}\| \ll 1$, expand (19.23), retaining terms linear in λ and quadratic in velocity, dropping their products. The free-particle action per unit mass becomes

$$-\int ds \simeq \int (-1 + \frac{1}{2}\dot{\mathbf{r}}^2 + \lambda) dt.$$

The Euler–Lagrange equation gives

$$\boxed{\ddot{\mathbf{r}} \simeq -\frac{1}{2}\partial_{\mathbf{r}}g_{tt} = \partial_{\mathbf{r}}\lambda.} \quad (19.25)$$

Thus $-\lambda$ is the Newtonian gravitational potential (26, Eqs. (4.20)–(4.22)).

Schwarzschild. For stationary spherical symmetry, take $r = \|\mathbf{r}\| > 0$ and $\mathbf{u} = \mathbf{r}/r$, with λ depending on position only through r . For this radial ansatz, the vacuum Einstein equation with zero cosmological constant gives the angular condition

$$\frac{d}{dr}(r\lambda) = 0 \quad \implies \quad \lambda = \frac{Gm}{r}. \quad (19.26)$$

The remaining radial condition follows by differentiating this equation. Matching the Newtonian potential fixes the integration constant to Gm (26, Eqs. (7.23)–(7.29)). Here $G \in \mathbb{R}_{>0}$ is Newton's gravitational constant and $m \in \mathbb{R}_{>0}$ is the source mass. Substitution into (19.21) gives

$$\begin{aligned} g_{tt} &= 1 - \frac{2Gm}{r}, & \mathbf{g}_{tr} &= -\frac{2Gm}{r^2} \mathbf{r}^\top, \\ \mathbf{g}_{rr} &= -\mathbf{I} - \frac{2Gm}{r^3} \mathbf{r} \mathbf{r}^\top. \end{aligned} \quad (19.27)$$

Since $\mathbf{u} \cdot d\mathbf{r} = dr$, these blocks give

$$\boxed{ds^2 = dt^2 - d\mathbf{r}^2 - \frac{2Gm}{r} (dt + d\mathbf{r})^2.} \quad (19.28)$$

This is the Schwarzschild metric in coordinates regular at the future horizon $r = 2Gm$ (28, Table 3). The weak-field acceleration is

$$\ddot{\mathbf{r}} \simeq -\frac{Gm}{r^3} \mathbf{r}.$$

For radial light, setting $ds^2 = 0$ gives the two coordinate velocities

$$\frac{dr}{dt} = -1, \quad \frac{dr}{dt} = \frac{r - 2Gm}{r + 2Gm}.$$

The outward branch has zero radial velocity at the horizon; inside, both future-directed radial light directions point toward smaller r . Figure 11 shows these directions alongside the scalar-field contours.

Kerr. Let the constant vector $\boldsymbol{\omega} \in \mathbb{R}^3$ be the outer horizon's angular velocity relative to a nonrotating observer at infinity. For $2Gm\|\boldsymbol{\omega}\| \leq 1$, define the length $\ell \in \mathbb{R}_{>0}$ by

$$\ell^2 = \frac{4(Gm)^2}{1 + 4(Gm)^2 \boldsymbol{\omega}^2}.$$

The angular momentum per unit mass is $\ell^2 \boldsymbol{\omega}$. This scaling follows from the Kerr horizon angular-velocity relation (28, Eq. (145)). Use the ellipsoidal-radius branch $\epsilon \in \mathbb{R}_{>0}$, on the coordinate region where it is smooth. It is determined by $\mathbf{r}$ and the spin, and generally differs from $\|\mathbf{r}\|$:

$$\epsilon^4 - \epsilon^2(\mathbf{r}^2 - \ell^4 \boldsymbol{\omega}^2) - \ell^4(\boldsymbol{\omega} \cdot \mathbf{r})^2 = 0. \quad (19.29)$$

The standard Kerr black-hole solution is represented by

$$\begin{aligned} \mathbf{u} &= \frac{\epsilon \mathbf{r} - \ell^2 \boldsymbol{\omega} \times \mathbf{r} + \ell^4 \boldsymbol{\omega} (\boldsymbol{\omega} \cdot \mathbf{r}) / \epsilon}{\epsilon^2 + \ell^4 \boldsymbol{\omega}^2}, \\ \lambda &= \frac{Gm \epsilon^3}{\epsilon^4 + \ell^4 (\boldsymbol{\omega} \cdot \mathbf{r})^2}. \end{aligned} \quad (19.30)$$

The radius equation verifies the unit direction:

$$\mathbf{u}^2 = \frac{\mathbf{r}^2 + \ell^4 (\boldsymbol{\omega} \cdot \mathbf{r})^2 / \epsilon^2}{\epsilon^2 + \ell^4 \boldsymbol{\omega}^2} = 1.$$

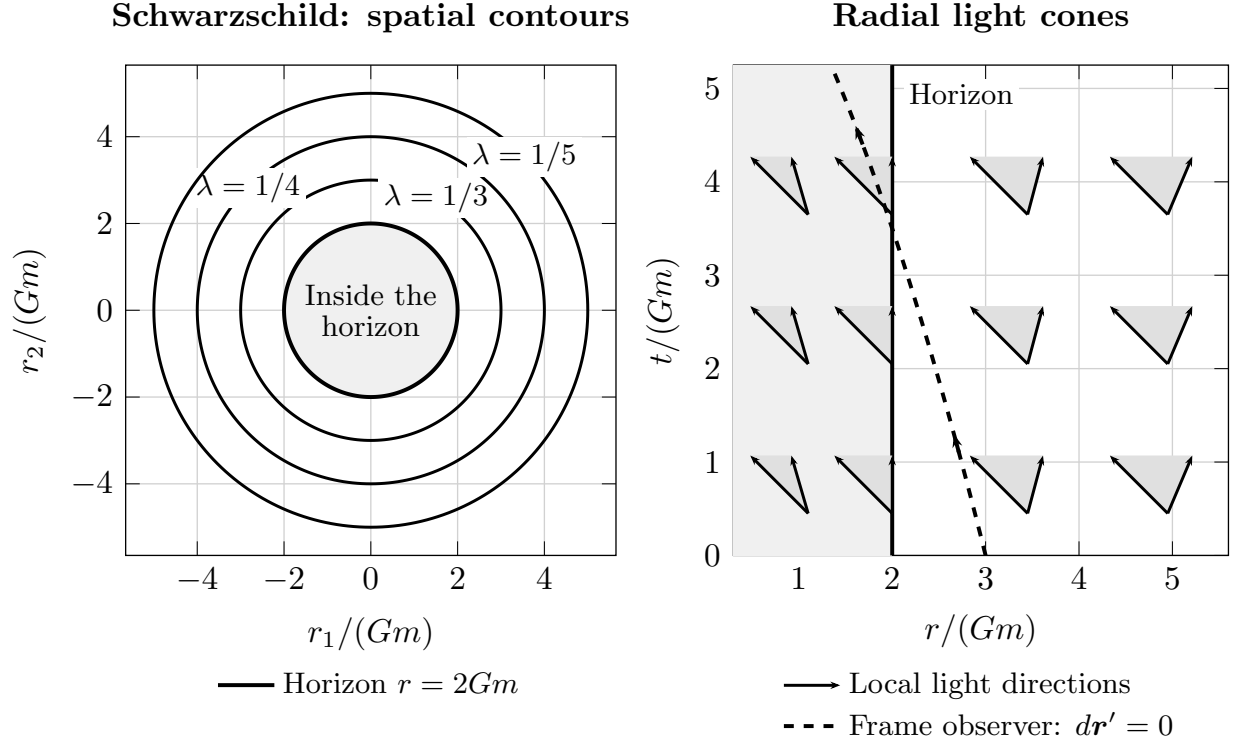


Figure 11: Schwarzschild in the coordinates of (19.28), with lengths and times in units of Gm . Left: constant- λ circles in the $r_3 = 0$ plane; the solid black circle is the horizon, with its interior shaded pale gray. Right: future radial light cones, drawn as local tangent directions, with time increasing upward. The dashed black curve follows the chosen frame observer, $dr/dt = -Gm/(r + Gm)$, starting at $r = 3Gm$ when $t = 0$; it is not a free-fall trajectory. The spatial panel shows coordinate contours, not proper distances or an isometric embedding.

Substituting into (19.21) gives the explicit metric blocks

$$\begin{aligned} g_{tt} &= 1 - \frac{2Gm\epsilon^3}{\epsilon^4 + \ell^4(\boldsymbol{\omega} \cdot \mathbf{r})^2}, \\ \mathbf{g}_{tr} &= -\frac{2Gm\epsilon^3}{\epsilon^4 + \ell^4(\boldsymbol{\omega} \cdot \mathbf{r})^2} \mathbf{u}^\top, \\ \mathbf{g}_{rr} &= -\mathbf{I} - \frac{2Gm\epsilon^3}{\epsilon^4 + \ell^4(\boldsymbol{\omega} \cdot \mathbf{r})^2} \mathbf{u} \mathbf{u}^\top, \end{aligned} \tag{19.31}$$

where $\mathbf{u}$ is given explicitly in (19.30). Thus the scalar–vector determinant gives

$$\begin{aligned} ds^2 &= \det(dX) = g_{tt} dt^2 + 2 dt \mathbf{g}_{tr} d\mathbf{r} + d\mathbf{r}^\top \mathbf{g}_{rr} d\mathbf{r} \\ &= dt^2 - d\mathbf{r}^2 - \frac{2Gm\epsilon^3}{\epsilon^4 + \ell^4(\boldsymbol{\omega} \cdot \mathbf{r})^2} (dt + \mathbf{u} \cdot d\mathbf{r})^2. \end{aligned} \tag{19.32}$$

This is the Cartesian Kerr–Schild vacuum solution with signature $(+, -, -, -)$ (27, Sec. 1, Eqs. (5)–(6)). Setting $\boldsymbol{\omega} = \mathbf{0}$ gives $\epsilon = \|\mathbf{r}\|$ and recovers Schwarzschild. The outer horizon is the larger root of $\epsilon^2 - 2Gm\epsilon + \ell^4\boldsymbol{\omega}^2 = 0$:

$$\epsilon = \frac{\ell^2}{2Gm} = \frac{2Gm}{1 + 4(Gm)^2\boldsymbol{\omega}^2}.$$

The metric remains regular there, with $\det(\mathbf{g}) = -1$. At this radius, the tangent displacement $d\mathbf{r} = (\boldsymbol{\omega} \times \mathbf{r}) dt$ gives $ds^2 = 0$, verifying that $\boldsymbol{\omega}$ is the horizon’s angular velocity. The outer stationary limit is instead the outer surface $g_{tt} = 0$, or $\lambda = 1/2$. Between it and the horizon lies the exterior ergoregion: observers cannot remain at fixed $\mathbf{r}$, since that direction has $ds^2 = g_{tt} dt^2 < 0$ (28, Sec. 3.5). Figure 12 distinguishes these surfaces and shows the directions of our chosen frame observers.

These substitutions reproduce the known Kerr solution. A general paravector construction of transport, curvature, and Einstein’s field equation remains to be developed.

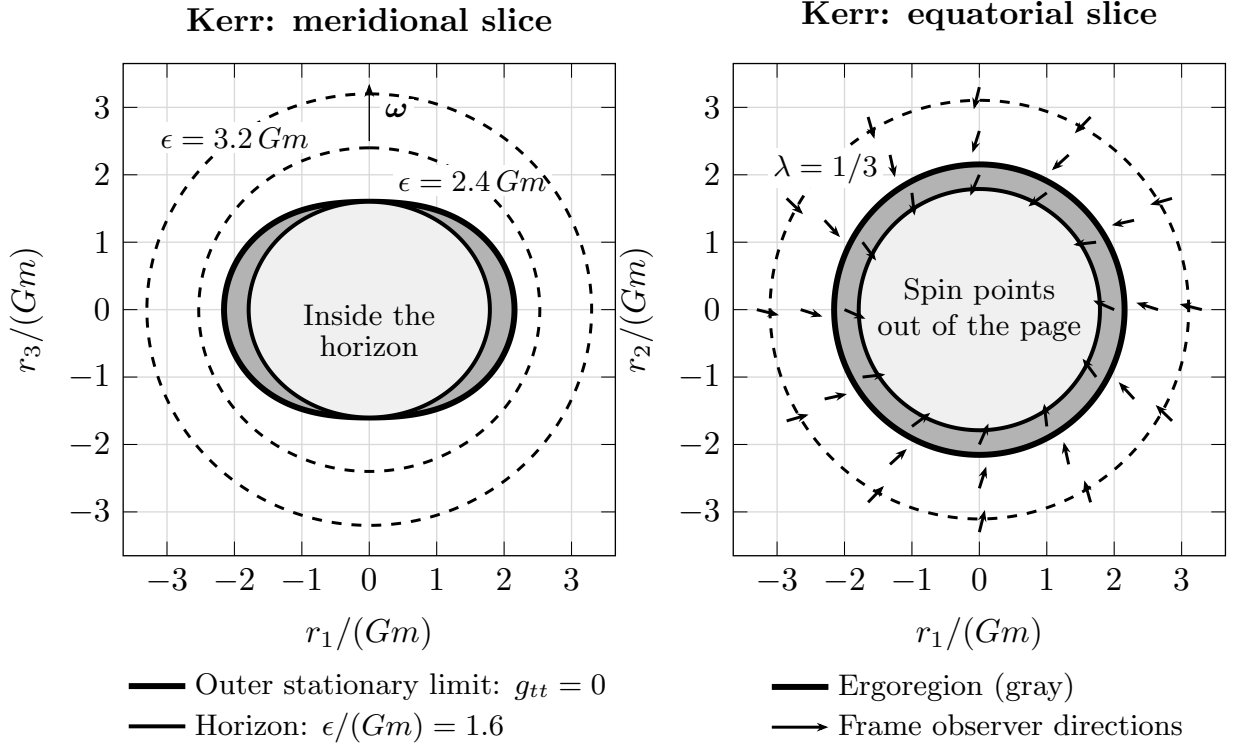


Figure 12: Kerr coordinate slices with spin along $+r_3$ and $Gm\|\boldsymbol{\omega}\| = 1/4$, so $\ell^2\|\boldsymbol{\omega}\| = 0.8Gm$ and the horizon is $\epsilon = 1.6Gm$. Left: the $r_2 = 0$ slice, with dashed black contours of the ellipsoidal radius ϵ . Right: the $r_3 = 0$ slice, with a dashed constant- λ contour and equal-length arrows showing the direction of $d\mathbf{r}/dt = -\lambda\mathbf{u}/(1+\lambda)$. Solid black curves mark the horizon and outer stationary limit, with the exterior ergoregion shaded gray. Pale gray marks the horizon's interior. The arrows describe the chosen frame observers, not general free fall; the coordinate shapes are not an isometric embedding.

A Pauli-matrix representation

The paravector formulas are verified in matrix form in this appendix. The decomposition $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is used, with $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$. The matrix map takes $Z \in \mathbb{C} \oplus \mathbb{C}^3$ to $\mathbf{M} = \text{mat}(Z) \in \mathbb{C}^{2 \times 2}$. Real paravectors map to Hermitian matrices, not necessarily real matrices. The identity matrices are $\mathbf{I}_2 \in \mathbb{R}^{2 \times 2}$ and $\mathbf{I}_4 \in \mathbb{R}^{4 \times 4}$. This 2×2 complex $\mathbf{M}$ replaces the 4×4 real frame matrix of the preceding section; it is a different use of the letter. Numerical columns introduced below have their own $\mathbb{C}^2$ or $\mathbb{C}^4$ domains.

A.1 Sigmas and products

The sigmas are represented by the Pauli matrices (6, p. 11)

$$\text{mat}(\sigma_1) = \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_2) = \begin{Bmatrix} 0 & -i \\ i & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_3) = \begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}.$$

Consequently,

$$\text{mat}(c + \mathbf{Z} \cdot \boldsymbol{\sigma}) = \begin{Bmatrix} c + Z_3 & Z_1 - iZ_2 \\ Z_1 + iZ_2 & c - Z_3 \end{Bmatrix}. \quad (\text{A.1})$$

The sigma product rule is satisfied by these matrices, which are linearly independent over $\mathbb{C}$ together with $\mathbf{I}_2$. Thus every complex 2×2 matrix corresponds to exactly one paravector. The map preserves addition, sends 1 to $\mathbf{I}_2$, and satisfies

$$\text{mat}(Z'Z) = \text{mat}(Z') \text{mat}(Z).$$

Here Z' is any second complex paravector. Thus algebra multiplication is ordinary matrix multiplication.

A.2 Conjugations and product order

For $\mathbf{M}, \mathbf{N} \in \mathbb{C}^{2 \times 2}$, transpose T exchanges row and column indices; entrywise complex conjugation C sends each $a + ib$ to $a - ib$ for real a, b . Their composition is $\mathbf{M}^H = (\mathbf{M}^C)^T$. Introduce the fixed matrix $\mathbf{S} \in \mathbb{R}^{2 \times 2}$:

$$\mathbf{S} = \begin{Bmatrix} 0 & 1 \\ -1 & 0 \end{Bmatrix}, \quad \mathbf{S}^{-1} = -\mathbf{S}.$$

The paravector sigma conjugation has the matrix image

$$\text{mat}(Z^*) = \mathbf{S}^{-1} \mathbf{M}^C \mathbf{S} =: \mathbf{M}^*. \quad (\text{A.2})$$

Substituting the basis in (A.2) gives

Algebra basis	1	i	σ_k	$i\sigma_k$
Hermitian image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$\text{mat}(\sigma_k)$	$-i \text{mat}(\sigma_k)$
Sigma-conjugate image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$-\text{mat}(\sigma_k)$	$i \text{mat}(\sigma_k)$

Here $k = 1, 2, 3$. These are exactly the operations of Section 1.3, so $\text{mat}(Z^H) = \mathbf{M}^H$. Their product-order rules are

$$\begin{aligned}(\mathbf{MN})^T &= \mathbf{N}^T \mathbf{M}^T, \\ (\mathbf{MN})^C &= \mathbf{M}^C \mathbf{N}^C, \\ (\mathbf{MN})^H &= \mathbf{N}^H \mathbf{M}^H.\end{aligned}$$

For sigma conjugation the middle factors cancel:

$$(\mathbf{MN})^* = \mathbf{S}^{-1} \mathbf{M}^C \mathbf{S} \mathbf{S}^{-1} \mathbf{N}^C \mathbf{S}.$$

Thus sigma conjugation preserves product order:

$$(\mathbf{MN})^* = \mathbf{M}^* \mathbf{N}^*.$$

This verifies (1.5) in matrices. Hermitian and sigma conjugation commute, and applying either twice restores the original matrix. The matrix-only operations can be recovered as

$$\begin{aligned}\mathbf{M}^C &= \mathbf{S} \mathbf{M}^* \mathbf{S}^{-1}, \\ \mathbf{M}^T &= \mathbf{S} (\mathbf{M}^*)^H \mathbf{S}^{-1}.\end{aligned}$$

A.3 Scalar part, determinant, and norm

For $\mathbf{M} = (m_{jk}) \in \mathbb{C}^{2 \times 2}$, with $m_{jk} \in \mathbb{C}$, its trace is $m_{11} + m_{22}$ and its determinant is $m_{11}m_{22} - m_{12}m_{21}$. Substituting (A.1) gives

$$\text{sc}(Z) = \frac{1}{2} \text{tr}(\mathbf{M}) = c, \tag{A.3}$$

$$(\text{vec}(Z))_k = \frac{1}{2} \text{tr}(\text{mat}(\sigma_k) \mathbf{M}) = Z_k,$$

$$\det(Z) = \det(\mathbf{M}) = c^2 - \mathbf{Z}^2. \tag{A.4}$$

By (A.3), matrix-trace cyclicity is precisely the scalar-part cyclicity used in the main text. For $\mathbf{M}' = \text{mat}(Z')$,

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}' \mathbf{M}) = \text{sc}(ZZ').$$

The remaining extraction and norm formulas are

$$\text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) = \mathbf{M} - \text{sc}(Z) \mathbf{I}_2,$$

$$\text{mat}(\text{re}(Z)) = \frac{1}{2} (\mathbf{M} + \mathbf{M}^H), \tag{A.5}$$

$$\text{mat}(\text{im}(Z)) = (\mathbf{M} - \mathbf{M}^H)/(2i), \tag{A.6}$$

$$\|Z\|^2 = \frac{1}{2} \text{tr}(\mathbf{M}^H \mathbf{M}) = \|c\|^2 + \|\mathbf{Z}\|^2. \tag{A.7}$$

Equations (A.5)–(A.6) extract Hermitian matrices, so a real paravector has a Hermitian image. The squared norm in (A.7) is half the sum of the squared magnitudes of the matrix entries. The determinant in (A.4) may be complex.

The matrix adjugate is

$$\text{adj}(\mathbf{M}) = (\text{tr}(\mathbf{M})) \mathbf{I}_2 - \mathbf{M} = \text{mat}(\text{adj}(Z)) = (\mathbf{M}^*)^H.$$

It satisfies

$$\text{mat}(\text{adj}(Z) Z) = \det(\mathbf{M}) \mathbf{I}_2.$$

Consequently $\mathbf{M}^{-1} = \text{adj}(\mathbf{M})/\det(\mathbf{M})$ when the determinant is nonzero. The determinant product and adjugate order rules are the same in both representations. The quadratic identity (3.1) is precisely Cayley–Hamilton for a 2×2 matrix.

A.4 Boost and rotation matrices

Notation: the real unit axis $\mathbf{u} = \{u_1, u_2, u_3\}$ and real rapidity θ . The matrix image of (4.6) is

$$\text{mat}(L) = \cosh(\theta/2)\mathbf{I}_2 - \sinh(\theta/2) \begin{Bmatrix} u_3 & u_1 - iu_2 \\ u_1 + iu_2 & -u_3 \end{Bmatrix}. \quad (\text{A.8})$$

The displayed axis matrix is Hermitian, has square $\mathbf{I}_2$, and has eigenvalues ± 1 . Thus (A.8) has eigenvalues $e^{\mp\theta/2}$, determinant one, and squared norm $\frac{1}{2}\text{tr}(\text{mat}(L)^H \text{mat}(L)) = \cosh(\theta)$. For a real rotation angle θ , replacing the hyperbolic coefficients by $\cos(\theta/2)$ and $i\sin(\theta/2)$ gives $\text{mat}(R)$, which is unitary.

Multiplication preserves the distinct spacetime and field laws:

$$\begin{aligned} \text{mat}(X') &= \text{mat}(L) \text{mat}(X) \text{mat}(L), \\ \text{mat}(F') &= \text{mat}(L) \text{mat}(F) \text{mat}(L)^{-1}. \end{aligned}$$

Here $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ is real and $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is complex, with $t \in \mathbb{R}$ and $\mathbf{r}, \mathbf{E}, \mathbf{B} \in \mathbb{R}^3$. Extracting scalar and vector coefficients by (A.3) and the preceding component rule gives (7.1)–(7.2) and (18.7)–(18.8) for the same arbitrary axis. The first matrix stays Hermitian; the second stays traceless.

A.5 Matrix amplitudes and column spinors

The matrix image of $\psi \in \mathbb{C} \oplus \mathbb{C}^3$ is denoted by $\mathbf{M}_\psi \in \mathbb{C}^{2 \times 2}$ at one event $X \in \mathbb{R} \oplus \mathbb{R}^3$. For a nonnegative weight $Z \in \mathbb{R} \oplus \mathbb{R}^3$, the local densities (13.3) and (13.5) become

$$\begin{aligned} \|\psi\|^2 &= \frac{1}{2} \text{tr}(\mathbf{M}_\psi^H \mathbf{M}_\psi), \\ \|\psi\|_Z^2 &= \frac{1}{2} \text{tr}(\mathbf{M}_\psi^H \text{mat}(Z) \mathbf{M}_\psi). \end{aligned}$$

The first is one half the sum of the squared magnitudes of the matrix entries: the squared Frobenius norm with the factor $1/2$ fixed by $\text{sc}(Z) = \text{tr}(\text{mat}(Z))/2$. The second is its weighted form. Both are pointwise; neither contains a spatial integral. Matrix multiplication gives exactly the congruence identity (13.7).

An arbitrary spin basis. Notation: the arbitrary constant perpendicular **real unit vectors** $\mathbf{u}, \mathbf{a} \in \mathbb{R}^3$ of (13.1). The projector $\Pi_+ = (1 + \mathbf{u} \cdot \boldsymbol{\sigma})/2$ is defined, and a constant complex paravector ψ' with $\Pi_+ \psi' \neq 0$ is chosen. Then

$$\begin{Bmatrix} \psi_+ \\ \psi_- \end{Bmatrix} := \frac{1}{\|\Pi_+ \psi'\|} \begin{Bmatrix} \Pi_+ \psi' \\ (\mathbf{a} \cdot \boldsymbol{\sigma}) \Pi_+ \psi' \end{Bmatrix}. \quad (\text{A.9})$$

This constructs (13.1) with $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_\pm = \pm\psi_\pm$ and $\text{sc}(\psi_s^H \psi_{s'}) = \delta_{ss'}$ for $s, s' = +, -$. The two complex scalar components give

$$\psi = c_+ \psi_+ + c_- \psi_- = (c_+ + c_- \mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+. \quad (\text{A.10})$$

For fields, $c_\pm : \mathbb{R}^4 \rightarrow \mathbb{C}$, and (13.3) is $\rho = \|c_+\|^2 + \|c_-\|^2$.

The two-component column. Every column of $\mathbf{M}_{\psi_+}$ has eigenvalue $+1$ under $\text{mat}(\mathbf{u} \cdot \boldsymbol{\sigma})$, whose $+1$ eigenspace is one-dimensional. Since $\|\psi_+\| = 1$, its matrix therefore factors as $\mathbf{M}_{\psi_+} = \sqrt{2} \boldsymbol{\psi}_+ \mathbf{v}^H$ for constant columns $\boldsymbol{\psi}_+, \mathbf{v} \in \mathbb{C}^2$ (here $\mathbf{v}$ is a reference column, not velocity) satisfying $\boldsymbol{\psi}_+^H \boldsymbol{\psi}_+ = \mathbf{v}^H \mathbf{v} = 1$.

By left multiplication, $\mathbf{M}_{\psi_-} = \sqrt{2} \psi_- \mathbf{v}^H$, where $\psi_- = \text{mat}(\mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+$. The columns ψ_+, ψ_- are orthonormal. Thus every amplitude in (A.10) has the conventional Pauli column

$$\psi := \frac{1}{\sqrt{2}} \mathbf{M}_{\psi} \mathbf{v} = c_+ \psi_+ + c_- \psi_- \in \mathbb{C}^2.$$

Conversely, $\mathbf{M}_{\psi} = \sqrt{2} \psi \mathbf{v}^H$, so this correspondence is one-to-one. For two amplitudes in this fixed basis and any complex multiplying paravector Z , by substitution,

$$\begin{aligned} \text{sc}(\psi'^H \psi) &= \psi'^H \psi, \\ \text{sc}(\psi^H Z \psi) &= \psi^H \text{mat}(Z) \psi. \end{aligned}$$

For nonnegative Z , the second line is $\|\psi\|_Z^2$. Dividing by $\rho = \psi^H \psi > 0$ gives $\langle Z \rangle_{\psi}$, the ordinary finite-component expectation. Left action $T\psi$ maps to $\text{mat}(T)\psi$ for any complex paravector T .

$\mathbf{C} \in \mathbb{C}^{2 \times 2}$ is defined as the matrix with columns ψ_+, ψ_- . Then $\mathbf{C}^H \mathbf{C} = \mathbf{I}_2$ and $\psi = \mathbf{C} \{c_+, c_-\}^T$. For a complex paravector $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$, its matrix in this basis is

$$\text{mat}'(Z) := \mathbf{C}^H \text{mat}(Z) \mathbf{C} = \begin{Bmatrix} c + \mathbf{u} \cdot \mathbf{Z} & (\mathbf{a} - i\mathbf{u} \times \mathbf{a}) \cdot \mathbf{Z} \\ (\mathbf{a} + i\mathbf{u} \times \mathbf{a}) \cdot \mathbf{Z} & c - \mathbf{u} \cdot \mathbf{Z} \end{Bmatrix}. \quad (\text{A.11})$$

Here the prime denotes only this change of matrix basis. For nonzero squared norm $\rho = \psi^H \psi$, the polarization (17.3) and probabilities for $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ become

$$\mathbf{u} \cdot \mathbf{u}' = \psi^H \text{mat}(\mathbf{u} \cdot \boldsymbol{\sigma}) \psi / \rho, \quad (\text{A.12})$$

$$\text{prob}(\pm | \mathbf{u}) = \psi^H \text{mat}(\Pi_{\pm}) \psi / \rho. \quad (\text{A.13})$$

Equations (A.12) and (A.13) are the ordinary spin expectation and Born rule.

A.6 Dirac columns

In this subsection $\psi = \{\psi_+, \psi_-\}^T$ again denotes a Dirac block. Each entry $\psi_{\pm} \in \mathbb{C} \oplus \mathbb{C}^3$ has the fixed two-component form of (13.2); the corresponding columns are $\mathbf{c}_+, \mathbf{c}_- \in \mathbb{C}^2$. Then

$$\mathbf{W}(Z) \psi \longleftrightarrow \begin{Bmatrix} 0 & \text{mat}(\text{adj}(Z)) \\ \text{mat}(Z) & 0 \end{Bmatrix} \begin{Bmatrix} \mathbf{c}_+ \\ \mathbf{c}_- \end{Bmatrix}.$$

The block squared norm in (15.1) becomes the squared norm of a four-component column:

$$\rho = \mathbf{c}_+^H \mathbf{c}_+ + \mathbf{c}_-^H \mathbf{c}_-.$$

Characteristic polynomial. On one Fourier mode, $s \in \mathbb{C}$ is the spectral parameter and $\lambda = -E^2 + \mathbf{P}^2 \in \mathbb{R}$ is the wave-operator eigenvalue for $E \in \mathbb{R}$, $\mathbf{P} \in \mathbb{R}^3$; s is not proper time here. Each differential-operator block in the determinant is evaluated on that mode. The two off-diagonal blocks multiply to $-\lambda \mathbf{I}_2$, so block elimination gives the ordinary matrix determinant

$$\begin{aligned} \det \left(\begin{Bmatrix} -(s+m)\mathbf{I}_2 & \text{mat}(i\partial) \\ \text{mat}(i\partial^*) & -(s+m)\mathbf{I}_2 \end{Bmatrix} \right) &= \det ((s+m)^2 + \lambda) \mathbf{I}_2 \\ &= ((s+m)^2 + \lambda)^2. \end{aligned} \quad (\text{A.14})$$

This verifies (15.8), including multiplicities. The polynomial identity also holds at $s = -m$.

A.7 Four components and the standard Dirac basis

The fixed basis elements of (A.9) are denoted by $\eta_{\pm} \in \mathbb{C} \oplus \mathbb{C}^3$. Both entries are expressed in this complex span: $\psi_+ = a\eta_+ + b\eta_-$ and $\psi_- = c'\eta_+ + d\eta_-$, where $a, b, c', d : \mathbb{R}^4 \rightarrow \mathbb{C}$ are the four scalar components, forming a column in $\mathbb{C}^4$ at each event. Their columns are $\mathbf{c}'_+ = \{a, b\}^\top$ and $\mathbf{c}'_- = \{c', d\}^\top$; the primes on the columns denote the basis change $\mathbf{c}'_{\pm} = \mathbf{C}^H \mathbf{c}_{\pm}$. Notation: real mass $m > 0$, real charge q , and real potentials $V : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. The gauge-covariant energy and kinetic momentum operators are $\hat{E}_q = i\partial_t - qV$ and $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$. Their momentum components in this frame are

$$\hat{P}_{q,1} = \mathbf{a} \cdot \hat{\mathbf{P}}_q, \quad \hat{P}_{q,2} = (\mathbf{u} \times \mathbf{a}) \cdot \hat{\mathbf{P}}_q, \quad \hat{P}_{q,3} = \mathbf{u} \cdot \hat{\mathbf{P}}_q.$$

Using (A.11), the coupled paravector equations give

$$(\hat{E}_q - \hat{P}_{q,3})c' - (\hat{P}_{q,1} - i\hat{P}_{q,2})d = ma, \quad (\text{A.15})$$

$$-(\hat{P}_{q,1} + i\hat{P}_{q,2})c' + (\hat{E}_q + \hat{P}_{q,3})d = mb, \quad (\text{A.16})$$

$$(\hat{E}_q + \hat{P}_{q,3})a + (\hat{P}_{q,1} - i\hat{P}_{q,2})b = mc', \quad (\text{A.17})$$

$$(\hat{P}_{q,1} + i\hat{P}_{q,2})a + (\hat{E}_q - \hat{P}_{q,3})b = md. \quad (\text{A.18})$$

Operators in (A.15)–(A.18) act on the scalar immediately to their right. Their Hamiltonian on $\{a, b, c', d\}^\top$ is

$$\hat{\mathbf{H}} = \begin{Bmatrix} qV - \hat{P}_{q,3} & -\hat{P}_{q,1} + i\hat{P}_{q,2} & m & 0 \\ -\hat{P}_{q,1} - i\hat{P}_{q,2} & qV + \hat{P}_{q,3} & 0 & m \\ m & 0 & qV + \hat{P}_{q,3} & \hat{P}_{q,1} - i\hat{P}_{q,2} \\ 0 & m & \hat{P}_{q,1} + i\hat{P}_{q,2} & qV - \hat{P}_{q,3} \end{Bmatrix}. \quad (\text{A.19})$$

This is the component image of (15.15).

Conventional gamma matrices are identified through $\sigma_k = \text{mat}(\sigma_k)$ for $k = 1, 2, 3$. Here $\sigma_k \in \mathbb{C}^{2 \times 2}$ and $\gamma^\mu \in \mathbb{C}^{4 \times 4}$, $\mu \in \{0, 1, 2, 3\}$. The σ_k also equal $\text{mat}'(\mathbf{a} \cdot \sigma)$, $\text{mat}'((\mathbf{u} \times \mathbf{a}) \cdot \sigma)$, and $\text{mat}'(\mathbf{u} \cdot \sigma)$, respectively. Their numerical entries apply to every choice of $\mathbf{u}, \mathbf{a}$. The following matrices are defined (16, Eqs. (4.16)–(4.17))

$$\gamma^0 = \begin{Bmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{Bmatrix}, \quad (\text{A.20})$$

$$\gamma^k = \begin{Bmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{Bmatrix}. \quad (\text{A.21})$$

Equations (A.20)–(A.21) obey $(\gamma^0)^2 = \mathbf{I}_4$, $(\gamma^k)^2 = -\mathbf{I}_4$, and $\gamma^j \gamma^k + \gamma^k \gamma^j = 0$ for distinct $j, k = 0, 1, 2, 3$. The standard chiral Dirac operator in this basis, with signature $(+, -, -, -)$, is

$$\mathbf{W}(\hat{P}_q) \longleftrightarrow \gamma^0 \hat{E}_q - \sum_{k=1}^3 \gamma^k \hat{P}_{q,k}.$$

Here $A = V + \mathbf{A} \cdot \sigma$ is the real potential paravector.

The unitary change to the standard Dirac basis commutes with the scalar rest-phase factor explained in (16.4). Together they define the envelope columns

$$\begin{Bmatrix} \phi_+ \\ \phi_- \end{Bmatrix} = \frac{e^{imt}}{\sqrt{2}} \begin{Bmatrix} \mathbf{I}_2 & \mathbf{I}_2 \\ -\mathbf{I}_2 & \mathbf{I}_2 \end{Bmatrix} \begin{Bmatrix} \mathbf{c}'_+ \\ \mathbf{c}'_- \end{Bmatrix}. \quad (\text{A.22})$$

Here $\phi_+, \phi_- \in \mathbb{C}^2$ are the coefficient columns in the same spin basis of the paravector fields $\phi_+ = e^{imt}(\psi_+ + \psi_-)/\sqrt{2}$ and $\phi_- = e^{imt}(\psi_- - \psi_+)/\sqrt{2}$. Before removing the rest phase, the basis change in (A.22) sends (A.19) to

$$\hat{\mathbf{H}}' = \begin{Bmatrix} (qV + m)\mathbf{I}_2 & \sum_k \hat{P}_{q,k}\boldsymbol{\sigma}_k \\ \sum_k \hat{P}_{q,k}\boldsymbol{\sigma}_k & (qV - m)\mathbf{I}_2 \end{Bmatrix}. \quad (\text{A.23})$$

Multiplying the full column by e^{imt} subtracts m from its Hamiltonian: $i\partial_t(e^{imt}\psi) = e^{imt}(i\partial_t - m)\psi$. Thus (A.23) becomes $\hat{\mathbf{H}}' - m\mathbf{I}_4$: its diagonal entries are qV and $qV - 2m$. Moving qV into $\hat{E}_q = i\partial_t - qV$ gives the exact envelope equations

$$\begin{aligned} \hat{E}_q\phi_+ &= \sum_k \hat{P}_{q,k}\boldsymbol{\sigma}_k\phi_-, \\ (\hat{E}_q + 2m)\phi_- &= \sum_k \hat{P}_{q,k}\boldsymbol{\sigma}_k\phi_+. \end{aligned}$$

These identify the large and small amplitudes used in the Pauli limit.

For real electric and magnetic components E_k, B_k in the same frame, the squared Dirac field coupling has the explicit image

$$\begin{aligned} \text{mat}'(iqF) &= q \begin{Bmatrix} iE_3 - B_3 & iE_1 + E_2 - B_1 + iB_2 \\ iE_1 - E_2 - B_1 - iB_2 & -iE_3 + B_3 \end{Bmatrix}, \\ \text{mat}'(iqF^*) &= \text{mat}'(iqF)^{\mathbf{H}}. \end{aligned}$$

The indexed E_k are field components; whereas $\hat{E}_q$ is the gauge-covariant energy operator. The two couplings are $-q\mathbf{B} \cdot \boldsymbol{\sigma} \pm iq\mathbf{E} \cdot \boldsymbol{\sigma}$.

A.8 Quaternion matrix image

For the quaternion $Q = a + \mathbf{Q} \cdot (-i\boldsymbol{\sigma}) = a - i\mathbf{Q} \cdot \boldsymbol{\sigma}$ of Section 5.4, with $a \in \mathbb{R}$, $\mathbf{Q} \in \mathbb{R}^3$, and $Q \in \mathbb{R} \oplus i\mathbb{R}^3$, the explicit matrix image is (6, p. 10)

$$\text{mat}(Q) = \begin{Bmatrix} a - iQ_3 & -Q_2 - iQ_1 \\ Q_2 - iQ_1 & a + iQ_3 \end{Bmatrix}. \quad (\text{A.24})$$

Here $\text{mat}((-i\sigma_k)) = -i\text{mat}(\sigma_k)$. The conjugation and norm identities become

$$\begin{aligned} \text{mat}(Q^{\mathbf{H}}) &= \text{mat}(Q)^{\mathbf{H}}, \\ \text{mat}(Q)^{\mathbf{H}}\text{mat}(Q) &= (a^2 + \mathbf{Q}^2)\mathbf{I}_2, \\ \det(\text{mat}(Q)) &= a^2 + \mathbf{Q}^2. \end{aligned}$$

For a unit quaternion, $a^2 + \mathbf{Q}^2 = 1$, these identities give a unitary matrix of determinant one.

B Equation reference

Each displayed formula repeats a numbered result; its right-hand number links to the definition or derivation. Natural units are $\hbar = c = 1$. The linked text supplies domains, component expansions, and approximation conditions.

Algebra. Notation: complex paravectors $Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ and $Z' = c' + \mathbf{Z}' \cdot \boldsymbol{\sigma}$, with $c, c' \in \mathbb{C}$, $\mathbf{Z}, \mathbf{Z}' \in \mathbb{C}^3$, and $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$. The prime here labels a second element. Their product is

$$\begin{aligned} ZZ' &= (cc' + \mathbf{Z} \cdot \mathbf{Z}') \\ &\quad + (c\mathbf{Z}' + c'\mathbf{Z} + i\mathbf{Z} \times \mathbf{Z}') \cdot \boldsymbol{\sigma}. \end{aligned} \quad (2.9)$$

The basic operations are

$$Z^H = c^* + \mathbf{Z}^* \cdot \boldsymbol{\sigma}, \quad (1.7)$$

$$Z^* = c^* - \mathbf{Z}^* \cdot \boldsymbol{\sigma}, \quad (1.6)$$

$$\text{adj}(Z) = Z^{*H} = c - \mathbf{Z} \cdot \boldsymbol{\sigma}, \quad (1.8)$$

$$\text{tr}(Z) = 2 \text{sc}(Z), \quad (1.9)$$

$$\text{re}(Z) = \frac{1}{2}(Z + Z^H), \quad (1.16)$$

$$\text{im}(Z) = (Z - Z^H)/(2i), \quad (1.17)$$

$$\det(Z) = \text{adj}(Z) Z = c^2 - \mathbf{Z}^2, \quad (1.10)$$

$$Z^{-1} = \text{adj}(Z)/\det(Z) \quad (\det(Z) \neq 0), \quad (1.12)$$

$$\|Z\|^2 = \text{sc}(Z^H Z). \quad (2.14)$$

The coefficient form is (2.1); product-order and cyclicity rules are (1.4), (1.5), and (1.13). Projectors and the spectral rule are (3.2) and (3.3). The quaternion component map is (5.6); matrix images are (A.1) and (A.24).

Rotations and boosts. Here $\mathbf{u} \in \mathbb{R}^3$ is a [unit axis](#), $\theta \in \mathbb{R}$ an angle or rapidity, $R \in \mathbb{R} \oplus i\mathbb{R}^3$, and $L \in \mathbb{R} \oplus \mathbb{R}^3$; $\beta = \tanh(\theta)$ and $\gamma = \cosh(\theta)$ in the boost formulas. The factors and their actions are

$$R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cos(\theta/2) - i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (4.5)$$

$$L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cosh(\theta/2) - \sinh(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (4.6)$$

$$L = \sqrt{\frac{\gamma+1}{2}} \left(1 - \frac{\gamma\beta}{\gamma+1} \mathbf{u} \cdot \boldsymbol{\sigma} \right), \quad (4.15)$$

$$L^2 = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (4.16)$$

$$Z' = RZR^H, \quad (5.4)$$

$$Z' = LZL, \quad (5.10)$$

$$F' = LFL^{-1}. \quad (18.6)$$

Both Z transformations use $Z' = TZT^H$, with $R^H = R^{-1}$ and $L^H = L$. The Z boost acts on a real four-vector; the complex electromagnetic field F uses similarity. Both refer to the same frame. Rodrigues' rotation is (5.5); boost components are (5.11)–(5.12). The general determinant-one congruence is (5.2); interval preservation is (7.4). Physical field components are derived in Section 18, with matrix verification in Appendix A.4.

Spacetime and particles. Notation: $X, U, P \in \mathbb{R} \oplus \mathbb{R}^3$, $s, t, E \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and constant $m \in \mathbb{R}_{>0}$. The real four-velocity $U = dX/ds$ is an [invertible paravector](#) with $UU^{-1} = 1$. The real kinetic four-momentum P has particle energy E (including rest energy, excluding potential energy) and spatial kinetic momentum $\mathbf{P}$. The kinematic identities are

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad (11.1)$$

$$\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2, \quad (11.2)$$

$$ds^2 = \det(dX) = dt^2 - d\mathbf{r}^2, \quad (6.1)$$

$$U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad (6.7)$$

$$U^*U = 1, \quad (6.8)$$

$$P = mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad (7.11)$$

$$E^2 - \mathbf{P}^2 = m^2. \quad (7.12)$$

Here t is coordinate time, s is proper time along the time-like path, and $\gamma = dt/ds = \frac{1}{\sqrt{1-\mathbf{v}^2}}$. Four-velocity is real, with [determinant one](#); Hermitian unitarity is the separate condition (2.4). For real proper times $s_0 < s_1$ and coordinate times $t_0 < t$,

$$X(s_1) - X(s_0) = \int_{s_0}^{s_1} U(s) ds, \quad (6.9)$$

$$s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} dt'. \quad (6.5)$$

Clock comparison is (6.6); constant proper acceleration gives (8.30); the future cone is (6.2). Proper acceleration, inertia, and particle action are (8.2), (8.3), and (7.16). Moving-frame velocity and its causal bound are (7.7)–(7.8); clock dilation and measured contraction are (7.9) and (7.10); energy and momentum transform by (7.13)–(7.14).

Electromagnetism. Notation: the real four-velocity $U = dX/ds$, an [invertible paravector](#) with $UU^{-1} = 1$. Notation: $V, \rho, S, q \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $A, J \in \mathbb{R} \oplus \mathbb{R}^3$, and $F \in \{0\} \oplus \mathbb{C}^3$. The coefficient fields depend on real spacetime; q is constant charge. Their boost law at corresponding events is (18.6), with components (18.7)–(18.8). The field and particle equations are

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (12.1)$$

$$\partial F = J^*, \quad (12.2)$$

$$\square F^* = \partial J, \quad \square F = \partial^* J^*, \quad (12.7)$$

$$F^* = \partial A - S, \quad (12.11)$$

$$F^2 = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B}, \quad (12.15)$$

$$\frac{dP}{ds} = q \operatorname{re}(FU), \quad (12.21)$$

$$u + \mathbf{S} \cdot \boldsymbol{\sigma} = \frac{1}{2} F F^H, \quad (12.24)$$

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (12.25)$$

Here $u \in \mathbb{R}_{\geq 0}$ is field energy density and $\mathbf{S} \in \mathbb{R}^3$ is energy flux. Potential components are (12.12)–(12.13); force components are (12.17) and (12.19). The action derivation uses (12.14) and (12.16).

Relativistic waves. Notation: $m \in \mathbb{R}_{>0}$, $q, V \in \mathbb{R}$, $\mathbf{A} \in \mathbb{R}^3$, and $A = V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$. Here $\hat{E}_q$ is a scalar differential operator and $\hat{\mathbf{P}}_q$ a triple of them; $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ is a KG field. The Dirac field ψ has two paravector entries $\psi_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$, each of the form (13.2), giving four independent complex coordinates. Here, subscript 0 labels the uncoupled operators of (13.13). The operators and wave equations are

$$\hat{E}_q = \hat{E}_0 - qV = i\partial_t - qV, \quad (13.15)$$

$$\hat{\mathbf{P}}_q = \hat{\mathbf{P}}_0 - q\mathbf{A} = -i\partial_{\mathbf{r}} - q\mathbf{A}, \quad (13.16)$$

$$\hat{P}_q = \hat{P}_0 - qA = \hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}, \quad (13.17)$$

$$(\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 = \hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}, \quad (13.19)$$

$$[\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}, \quad (13.20)$$

$$(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi = 0, \quad (14.7)$$

$$\mathbf{W}(Z) = \begin{Bmatrix} 0 & \text{adj}(Z) \\ Z & 0 \end{Bmatrix}. \quad (15.2)$$

Here $Z \in \mathbb{C} \oplus \mathbb{C}^3$; $\mathbf{W}(Z)$ and $\mathbf{I}$ are represented by matrices in $\mathbb{C}^{4 \times 4}$, whereas $\hat{\mathbf{D}}$ and $\hat{\mathbf{D}}_q$ are matrix differential operators. For the free field ($q = 0$),

$$\hat{P} \text{adj}(\hat{P}) = \det(\hat{P}) = \hat{E}^2 - \hat{\mathbf{P}}^2 = -\square, \quad (13.18)$$

$$\hat{\mathbf{D}}(m) = \mathbf{W}(\hat{P}) - m\mathbf{I}, \quad (15.3)$$

$$\hat{\mathbf{D}}(m)\psi = 0, \quad (15.4)$$

$$\hat{\mathbf{D}}(-m)\hat{\mathbf{D}}(m) = -(\square + m^2)\mathbf{I}. \quad (15.5)$$

For fixed q and A , the interacting definition and equation are

$$\hat{\mathbf{D}}_q(m) = \mathbf{W}(\hat{P}_q) - m\mathbf{I}, \quad (15.11)$$

$$\hat{\mathbf{D}}_q(m)\psi = (\mathbf{W}(\hat{P}_q) - m\mathbf{I})\psi = 0. \quad (15.12)$$

Both $\hat{\mathbf{D}}(-m)$ and $\hat{\mathbf{D}}_q(-m)$ change only the sign of the mass term. The free equations are (14.1) and (15.4); the two-branch solution is (14.2). Massive Dirac reconstruction is given by (15.10) and (15.21); the squared interacting entries are (15.19) and (15.20).

Low-energy limits. After removing rest energy by (16.4) and (16.1), the complex scalar envelope ψ obeys Schrödinger's equation to leading order:

$$i\partial_t\psi = \left(qV + \frac{\hat{\mathbf{P}}_q^2}{2m} \right) \psi. \quad (16.3)$$

The envelope $\phi_+ : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$, restricted to the form (13.2), obeys Pauli's equation:

$$i\partial_t\phi_+ = \left(qV + \frac{\hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m} \right) \phi_+. \quad (16.9)$$

The derivations and approximation conditions are in Section 16; these are the same leading order for scalar and spinor amplitudes respectively.

Spin and probabilities. Notation: a nonzero $\psi \in \mathbb{C} \oplus \mathbb{C}^3$ of the form (13.2), $\rho \in \mathbb{R}_{>0}$, and polarization and measurement axes $\mathbf{u}', \mathbf{u} \in \mathbb{R}^3$ of unit norm. The real observable $Z \in \mathbb{R} \oplus \mathbb{R}^3$ acts from the left. A rotation uses

$$\psi' = R\psi. \quad (17.1)$$

It preserves the norm and inner product; the expectation transforms as $\langle Z \rangle_{\psi'} = \langle R^H Z R \rangle_{\psi}$ by (13.8). All formulas are pointwise. For a nonnegative real weight Z from (13.5), the densities and measurements are

$$\rho = \|\psi\|^2 = \text{sc}(\psi^H \psi), \quad (13.3)$$

$$\|\psi\|_Z^2 = \text{sc}(\psi^H Z \psi), \quad (13.5)$$

$$\mathbf{u} \cdot \mathbf{u}' = \text{sc}(\psi^H (\mathbf{u} \cdot \boldsymbol{\sigma}) \psi) / \rho, \quad (17.3)$$

$$\text{prob}(\pm \mid \mathbf{u}) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{u}'), \quad (17.4)$$

$$\langle Z \rangle_{\psi} = \text{sc}(\psi^H Z \psi) / \rho. \quad (13.6)$$

The expectation formula also applies to signed real observables. The matrix verification is (A.12)–(A.13).

C Conventions

Unadorned source symbols below retain each source's meaning. In the coefficient notation, $Z, G \in \mathbb{C} \oplus \mathbb{C}^3$, $X, Y, A, P, J \in \mathbb{R} \oplus \mathbb{R}^3$, and $F \in \{0\} \oplus \mathbb{C}^3$. Physical factors G have determinant one; spinor amplitudes have the fixed two-component form of (13.2). Here $t, E, V, \rho, q, \theta \in \mathbb{R}$, $m \in \mathbb{R}_{>0}$, $\mathbf{r}, \mathbf{P}, \mathbf{v}, \mathbf{u}, \mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\beta \in (-1, 1)$, and $\gamma \in \mathbb{R}_{\geq 1}$, with $\mathbf{u}^2 = 1$. The factors $T \in \mathbb{C} \oplus \mathbb{C}^3$, $R, Q \in \mathbb{R} \oplus i\mathbb{R}^3$, and $L, B \in \mathbb{R} \oplus \mathbb{R}^3$ have the roles stated locally. The amplitudes $\psi, \varphi, \psi_{\pm}$ take values in $\mathbb{C} \oplus \mathbb{C}^3$ with that restriction; $\psi_{\text{KG}} : \mathbb{R}^4 \rightarrow \mathbb{C}$ is scalar. In expectation formulas $\rho = \|\psi\|^2 \in \mathbb{R}_{\geq 0}$ is probability density; in J it is charge density. The algebra coefficient $c \in \mathbb{C}$ and vector coefficient $\mathbf{Z} \in \mathbb{C}^3$ are unrelated to the physical constant $c = 1$.

Names and algebra classifications. The name *space-time algebra* is used in this paper for the formulation in complex paravectors defined by the sigma multiplication rules. In Clifford-algebra terminology, this eight-real-dimensional algebra is classified as $\text{Cl}_{3,0}$ and is commonly called the *algebra of physical space* (APS). It is isomorphic to the even subalgebra $\text{Cl}_{1,3}^+$ of the sixteen-real-dimensional algebra $\text{Cl}_{1,3}$, for which the name *spacetime algebra* (STA) is used by Hestenes (1, pp. 1–2). In $\text{Cl}_{p,q}$, p positive and q negative basis-vector squares are specified; the signature $(+, -, -, -)$ is denoted by $(1, 3)$, and the opposite signature by $(3, 1)$.

In this terminology, the sigmas are called *generators*. The basis elements $1, \sigma_k, i\sigma_k$, and i are assigned grades 0, 1, 2, and 3, respectively, and are called scalars, vectors, bivectors, and pseudoscalars. Thus scalar and pseudoscalar grades are combined in the complex scalar part, and vector and bivector grades in the complex vector term. Hermitian conjugation is called *reversion* in APS; sigma conjugation is called *grade involution*; and adjugation is called *Clifford conjugation* (1, pp. 4–6). These correspondences depend on the algebra: the four-dimensional comparison is given below.

A rotation factor is also called a *rotor*, a projector an *idempotent*, and an invertible element an *algebraic unit*; neither unit norm nor determinant one is implied by the last name. The fixed two-dimensional complex spinor space of (13.2) is called a *minimal left ideal*: it is preserved by left multiplication by every paravector. In particle dynamics, *mechanical momentum* is another name for kinetic momentum. The quantity γm in Section 8.2 is sometimes called *relativistic mass*; invariant mass is used throughout this paper.

Mapping to the literature. The conventions of Baylis and Sobczyk are compared below (1, pp. 4–6). The table translates the notation of the *Paravector* article; A denotes a general element and x a real paravector.

Object or operation	This paper	APS notation
Spatial basis	σ_k	$\mathbf{e}_k$
Central imaginary unit	$\mathbf{i} = \sigma_1 \sigma_2 \sigma_3$	$i = \mathbf{e}_{123}$
Real paravector	$X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$	$x = x^0 + \mathbf{x}$
General algebra element	$Z = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$	A (complex paravector)
Reversion	$Z^{\mathbf{H}}$	$A^\dagger$
Clifford conjugation	$\text{adj}(Z)$	$\bar{A}$
Grade involution	Z^*	$\bar{A}^\dagger$
Complex scalar part	$\text{sc}(Z)$	$\langle A \rangle_S$
Complex vector part	$\text{vec}(Z) \cdot \boldsymbol{\sigma}$	$\langle A \rangle_V$
Real part	$\text{re}(Z)$	$\langle A \rangle_R$
Imaginary part	$\mathbf{i} \text{im}(Z)$	$\langle A \rangle_I$
Spacetime quadratic form	$\det(X) = t^2 - \mathbf{r}^2$	$x\bar{x}$
Projectors	Π_+, Π_-	$P_{\mathbf{u}}, \bar{P}_{\mathbf{u}}$

Real paravectors have four real components; complex Z packages all eight APS components. Here $\text{sc}(Z)$ includes the pseudoscalar, and $\text{im}(Z)$ removes the imaginary part's factor $\mathbf{i}$.

In particular, the bar is represented here by the adjugate, not by sigma conjugation:

$$\bar{A} \longleftrightarrow c - \mathbf{Z} \cdot \boldsymbol{\sigma}, \quad \bar{A}^\dagger \longleftrightarrow c^* - \mathbf{Z}^* \cdot \boldsymbol{\sigma}.$$

They coincide only for real paravectors. Complex coefficients are conjugated by sigma conjugation, which is identified with grade involution on the full algebra. Wikipedia's [paragradiant](#) is represented here by the starred derivative:

$$\partial_{\text{article}} = \partial_t - \nabla \longleftrightarrow \partial^*, \quad \bar{\partial}_{\text{article}} \longleftrightarrow \partial, \quad \square = \partial \partial^*.$$

Baylis's rotor action $x' = GxG^\dagger$ is written here as $X' = GXG^{\mathbf{H}}$; Baylis's field $\mathbf{E} + i\mathbf{B}$ and law $F' = GF\bar{G}$ are written here as $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ and $F' = GFG^{-1}$ for $\det(G) = 1$ ($c = 1$) (5, Eq. (23)) (5, Eq. (24)) (5, Eq. (33)). Baylis's positively oriented spatial-plane generator is $\mathbf{e}_2\mathbf{e}_1 = -i\mathbf{e}_3$, giving $R = e^{-i\theta\sigma_3/2}$ about $+\sigma_3$ (5, Eqs. (17)–(19)). Baylis's positive-rapidity boost factor, denoted here by B , satisfies

$$B = e^{+\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2} = L^{-1}, \quad B^2 = \gamma(1 + \beta\mathbf{u} \cdot \boldsymbol{\sigma}), \quad X' = B^{-1}XB^{-\mathbf{H}} = LXL^{\mathbf{H}}.$$

Thus the passive boost is obtained by inversion of Baylis's positive active boost (5, p. 23); the active/passive distinction is explicit in (1, Eq. (38)). The algebra is standard; exponent signs must be read with the transformation direction and sandwich order.

For translation to Hestenes's four-dimensional STA, an observer axis γ_0 is chosen, with $\gamma_0^2 = 1$ and $\gamma_k^2 = -1$. The identification with its even subalgebra is (1, Eq. (40)) (1, Eq. (41))

$$\sigma_k \longleftrightarrow \gamma_k \gamma_0, \quad \text{adj}(Z) \longleftrightarrow \tilde{Z}_{\text{STA}}, \quad Z^{\mathbf{H}} \longleftrightarrow \gamma_0 \tilde{Z}_{\text{STA}} \gamma_0.$$

Here $\gamma_\mu \in \text{Cl}_{1,3}$ are spacetime basis vectors, not the numerical gamma matrices of Appendix A.7. The image Z_{STA} lies in the even subalgebra $\text{Cl}_{1,3}^+$, and the tilde reverses STA products: reversion in APS and reversion in STA are different operations.

Source comparisons for the cited equations. Body citations point to matching definitions or physical component results. The source locators below identify those comparisons; intervening algebra and the block-map packaging are derived in this paper.

Result here	Direct source comparison
Sigma product, conjugations, and extractions (1.1)–(1.17)	(1, pp. 4–6); (2, Eq. (3)).
Squared norm (2.14); inverse (1.12)	(2, Eqs. (46)–(47)).
Quaternions (5.6); projectors (3.2)	(3, p. 92); (3, p. 121).
Spectral rule (3.3); coalescence (3.5)	(4, pp. 3–4).
Interval and four-velocity (6.1)–(6.7)	(5, Eqs. (7)–(8)).
Momentum, acceleration, and world lines (7.11), (8.2), (8.30)	(9, Eq. (7.24)); (9, p. 141); (9, Eq. (7.33), p. 143).
Maxwell, field energy, and energy flux (12.2), (12.24)	(8, Eq. (177)); (8, Eq. (167)); (8, Eq. (180)).
Potential, gauge, and field action (12.12)–(12.13), (12.14)	(7, p. 105); (7, Eq. (5.42)); (7, Eq. (5.43)); (7, Eqs. (6.1)–(6.3)); (7, Eq. (6.4)).
Lorentz force (12.21); components (12.17), (12.19)	(10, Eq. (2)); (7, Eqs. (5.32)–(5.33)).
Boosted components (7.2), (18.8)	(7, Eq. (5.1)); (7, Eqs. (5.17)–(5.18)).
KG and minimally coupled Dirac (14.1), (14.7), (15.12)	(16, Eq. (1.13)); (17, Eq. (299)); (8, Eqs. (190)–(191)).
Chiral squared Dirac (15.19)	(18, Eq. (6)).
Kinetic operators, small amplitude, and Pauli (13.16), (16.8), (16.9)	(12, Eqs. (2.11)–(2.16)).
KG to Schrödinger (16.3)	(19, Eq. (21.3)); (16, Eqs. (2.103)–(2.104)); (16, Eq. (2.105)).
Spin amplitudes and probabilities (13.2), (17.4)	(11, Eqs. (66)–(67)); (11, Eq. (69a)); (13, Eq. (6.1)); (13, p. 62).

Translating the physical comparisons.

SI units: Comparison with Tong

SI units are used in Tong's component electrodynamics; $c = \epsilon_0 = \mu_0 = 1$ is used outside these boxes.

Tong's gauge parameter in Eq. (5.13) is $\chi = -\lambda$, whereas Eq. (6.2) is written with the same plus-sign convention (7, Eq. (5.13)) (7, Eq. (6.2)). Tong's positive-exponent spatial rotation is a passive axis rotation; the negative-exponent boost is identified with the passive boost used here (9, Eq. (7.38)) (9, Eq. (7.39)).

Chappell et al.'s scalar Z is represented here by c , and the vector–bivector F by $\mathbf{Z} \cdot \boldsymbol{\sigma}$, with $|F|^2 = -\mathbf{Z}^2$. Thus the trigonometric exponential is represented here by the hyperbolic form (2, Eq. (52)).

For the quaternion basis, the Hamilton units are identified by Baylis with $\mathbf{e}_{32}, \mathbf{e}_{13}, \mathbf{e}_{21}$, respectively (3, p. 92). With $\mathbf{e}_k \leftrightarrow \sigma_k$, the units are represented here by $(-i\sigma_1), (-i\sigma_2), (-i\sigma_3)$. Higham's spectral and repeated-root formulas apply through the faithful matrix image in Appendix A; no change of the intrinsic product is required (4, pp. 3–4).

Lenz's scalar potentials ϕ, Φ are represented here by V , with $\hbar = c = 1$ and $e = q$. The Dirac Hamiltonian is represented by the matrix form (A.23) (17, Eq. (412)).

For Hestenes's Maxwell equation, the spacetime derivative and current are denoted by ∇_{STA} and J_{STA} , respectively. Using the STA embedding above,

$$\nabla_{\text{STA}} = \gamma_0 \partial, \quad J_{\text{STA}} = J \gamma_0 = \gamma_0 J^*, \quad \partial F = J^* \iff \nabla_{\text{STA}} F = J_{\text{STA}}.$$

Hestenes's explicit spacetime split is represented here by the component equation (8, Eqs. (175)–(177)). With $F^{\text{H}} = \gamma_0 \tilde{F} \gamma_0$, Hestenes's energy and flux components become

$$\frac{1}{2} F \gamma_0 \tilde{F} \gamma_0 = \frac{1}{2} F F^{\text{H}} = u + \mathbf{S} \cdot \boldsymbol{\sigma},$$

as in (12.24) (8, Eq. (180)).

For Dirac and Pauli, the complex unit is represented by Hestenes and Gurtler–Hestenes through right multiplication of the real spinor by the spatial bivector $i\sigma_3$; it acts as ordinary i in the corresponding column spinor (8, Eqs. (187)–(188)) (8, Eqs. (192)–(193)). With $e = q$ and $\hbar = c = 1$, Gurtler–Hestenes's potential energy V is represented here by qV ; after removal of rest energy, the operators K and p are identified with $\hat{E}_q$ and $\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}$. The same components are then obtained for the large/small equations and Pauli limit (12, Eqs. (2.11)–(2.19)).

Charge is absorbed into the potential by Feynman–Gell-Mann. With charge restored, Eq. (6) is identified with the first squared Dirac block because

$$-iqF^* = q(\mathbf{B} + i\mathbf{E}) \cdot \boldsymbol{\sigma}, \quad -iqF = q(\mathbf{B} - i\mathbf{E}) \cdot \boldsymbol{\sigma}.$$

The two blocks therefore have the same magnetic sign and opposite electric signs (18, Eq. (6)).

Baylis et al.'s RP_3 is a two-component minimal-ideal spinor; its rotations are represented by the left amplitude action. From the density-operator formula is obtained the same $(1 + \mathbf{u} \cdot \mathbf{u}')/2$ probability after basis normalization, extending to mixed states with $\|\mathbf{u}'\| \leq 1$ (11, Eqs. (66)–(67)). Passive basis rotations are used in Feynman's spin chapter; its 2π sign and probability statements agree, while an active rotor uses the inverse rotation (20, Eq. (6.19)).

The right-handed sigma product is fixed, and the following notation is used: $N = \mathbf{u} \cdot \boldsymbol{\sigma} \in \{0\} \oplus \mathbb{R}^3$, with $\mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, and $\theta \in \mathbb{R}$. The alternatives below change notation or transformation direction, not the underlying physics. The main text uses $R = R_-$ and $L = L_-$, with

$$R_{\pm} = e^{\pm i\theta N/2}, \quad L_{\pm} = e^{\pm \theta N/2}, \quad R_{\pm}^{\mathbf{H}} = R_{\pm}^{-1} = R_{\mp}, \quad L_{\pm}^{\mathbf{H}} = L_{\pm}, \quad L_{\pm}^{-1} = L_{\mp}.$$

Here $R_{\pm} \in \mathbb{R} \oplus i\mathbb{R}^3$, $L_{\pm} \in \mathbb{R} \oplus \mathbb{R}^3$, and θ is a rotation angle or rapidity respectively.

Exponent sign and sandwich order. For $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, the four choices for each kind of factor are summarized by the two action columns:

Factor T	$X' = TXT^{\mathbf{H}}$	$X' = T^{\mathbf{H}}XT$
R_-	Active rotation $+\theta$	Active rotation $-\theta$
R_+	Active rotation $-\theta$	Active rotation $+\theta$
L_-	Frame velocity $+\beta\mathbf{u}$	Frame velocity $+\beta\mathbf{u}$
L_+	Frame velocity $-\beta\mathbf{u}$	Frame velocity $-\beta\mathbf{u}$

For rotations $t' = t$. Both $+\theta$ entries give

$$\mathbf{r}' = \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta) + (1 - \cos(\theta))(\mathbf{u} \cdot \mathbf{r})\mathbf{u}.$$

This is the active right-handed Rodrigues formula (5.5); a passive rotation of axes uses the inverse map.

For boosts, $\beta = \tanh(\theta)$ and $\gamma = \cosh(\theta)$. Writing $\mathbf{r}_{\parallel} = (\mathbf{u} \cdot \mathbf{r})\mathbf{u}$ and $\mathbf{r}_{\perp} = \mathbf{r} - \mathbf{r}_{\parallel}$, both sandwiches with $L_{\pm}$ give

$$t' = \gamma(t \pm \beta\mathbf{u} \cdot \mathbf{r}), \quad \mathbf{r}' = \mathbf{r}_{\perp} + \gamma(\mathbf{r}_{\parallel} \pm \beta t\mathbf{u}).$$

The minus choice sends $\mathbf{r} = \beta t\mathbf{u}$ to $\mathbf{r}' = 0$, fixing its frame velocity. Moving the adjoint reverses a rotation but has no effect on a boost; reversing the boost exponent gives the inverse frame transformation. The same L_+ map can instead describe an active boost to velocity $+\beta\mathbf{u}$ in fixed coordinates.

Hamilton quaternions fix the basis orientation. With the sigma orientation held fixed, the two uniform sign choices obey

$$\begin{aligned} (-i\sigma_j)(-i\sigma_k) &= -\delta_{jk} + \epsilon_{jkl}(-i\sigma_l), \\ (+i\sigma_j)(+i\sigma_k) &= -\delta_{jk} - \epsilon_{jkl}(+i\sigma_l). \end{aligned}$$

Repeated l is summed. Hamilton's orientation is obtained with $(-i\sigma_k)$, and the opposite orientation with $(+i\sigma_k)$. For the Hamilton basis, Section 5.4 gives

$$Q = e^{\theta\mathbf{u} \cdot (-i\boldsymbol{\sigma})/2} = R_-, \quad Q^{\mathbf{H}} = Q^{-1}, \quad \mathbf{r}' \cdot (-i\boldsymbol{\sigma}) = Q(\mathbf{r} \cdot (-i\boldsymbol{\sigma}))Q^{\mathbf{H}}.$$

Since $\mathbf{r} \cdot (-i\boldsymbol{\sigma}) = -i\mathbf{r} \cdot \boldsymbol{\sigma}$ and i commutes with Q , the same sandwich rotates $\mathbf{r} \cdot \boldsymbol{\sigma}$, with no extra sign. For example, $Q = (1 - i\sigma_3)/\sqrt{2}$ sends σ_1 to σ_2 .

Using $(+i\sigma_k)$ with $\tilde{Q} = e^{\theta\mathbf{u} \cdot (+i\boldsymbol{\sigma})/2} = R_+$ and the right-adjoint sandwich instead gives $-\theta$. $+\theta$ is recovered by reversing either the exponent or the sandwich. Reflecting the basis also restores Hamilton multiplication, but changes its identification with the physical axes. Replacing a factor by its negative leaves both sandwiches unchanged; the spinor acquires an overall minus sign. Consequently rotations are 2π -periodic on vectors and 4π -periodic on amplitudes.

Rotating a state versus changing the measured observable. With the Hamilton basis, the two consistent descriptions of the same active $+\theta$ rotation are

Choice	Quaternion relation	Rotated observable	Rotated amplitude
$R = R_-$	$R = Q$	$Z' = RZR^H$	$\psi' = R\psi$
$R = R_+$	$R = Q^H$	$Z' = R^H Z R$	$\psi' = R^H \psi$

Both have physical amplitude factor $e^{-i\theta N/2}$ and spin generator $N/2$. Changing only the vector sandwich would reverse its rotation relative to the spin polarization.

For $G \in \mathbb{C} \oplus \mathbb{C}^3$ with $\det(G) \neq 0$ acting as $\psi' = G\psi$, the definitions (13.3) and (13.6) give

$$\begin{aligned}\rho &= \|\psi\|^2, & \rho' &= \rho \langle G^H G \rangle_\psi, \\ \langle \varphi' | \psi' \rangle &= \text{sc}(\varphi^H G^H G \psi), \\ \langle Z \rangle_{\psi'} &= \frac{\langle G^H Z G \rangle_\psi}{\langle G^H G \rangle_\psi}.\end{aligned}$$

Here $\varphi' = G\varphi$, and $\rho > 0$. For a rotation $G^H G = 1$:

$$\boxed{\langle Z \rangle_{\psi'} = \langle G^H Z G \rangle_\psi, \quad \langle G Z G^H \rangle_{\psi'} = \langle Z \rangle_\psi.} \quad (\text{C.1})$$

The first identity measures fixed Z in the rotated state; the second rotates the observable with the state. Thus the opposite sandwich orders in (C.1) are required, not competing conventions. For $G = L_\pm$, the denominator is $\langle L_\pm^2 \rangle_\psi$: boosts generally change the local positive density and inner product.

Fields and wave amplitudes under either notation. G is taken to be the constant physical factor of determinant one, so $X' = G X G^H$. For notation $T X T^H$, $G = T$ is used; for $T^H X T$, $G = T^H$ is used. At corresponding events the laws are

$$\begin{aligned}Z' &= G Z G^H, & Z &= X, P, A, J, \\ F' &= G F G^{-1}, & \partial' &= G^* \partial G^{-1}, \\ \psi'_+ &= G^* \psi_+, & \psi'_- &= G \psi_-, \\ \psi'_{\text{KG}}(X') &= \psi_{\text{KG}}(X).\end{aligned} \quad (\text{C.2})$$

These are the covariance laws of Section 18; $G^* = G^{-H}$. The field F uses similarity, not the four-vector sandwich. Equation (C.2) preserves the Maxwell, Klein–Gordon, and Dirac equations.

For either rotation description, both Dirac entries use R_- . For $G = L_\pm$, they use $L_\pm^{-1}$ and $L_\pm$, respectively; the factor already contains half the rapidity. With $X' = T^H X T$, the same laws read $\psi'_+ = T^{-1} \psi_+$ and $\psi'_- = T^H \psi_-$. The scalar paired quantity $\text{sc}(\psi_+^H \psi_-)$ is invariant; the sum of the two positive densities generally changes under boosts. The conserved Dirac current transforms as a four-vector; its time component alone is not invariant. For products, the physical factors compose as $G_2 G_1$. Hence the right-adjoint parameters compose as $T_2 T_1$, but the left-adjoint parameters compose as $T_1 T_2$.

Preserved quantities. The sign of the angle or rapidity changes the direction, not these invariance statements. Here Z is a complex paravector and $\|Z\|^2 = \text{sc}(Z^H Z)$; “no” means not generally preserved.

Action	$\det(Z)$	$\text{tr}(Z), \text{sc}(Z)$	$\ Z\ ^2$
Rotation RZR^H	Yes	Yes	Yes
Boost LZL^H	Yes	No	No
Field similarity GZG^{-1}	Yes	Yes	If G is unitary

For real four-vectors transformed together, both rotations and boosts preserve

$$\text{sc}(X'^* Y') = \text{sc}(X^* Y).$$

Similarity also preserves $\text{sc}(ZY)$ when both factors transform by similarity. The positive amplitude inner product is rotation invariant; boost invariance instead applies to the spacetime metric and the paired Dirac quantity above. Normalized expectations remain equal for a state and observable rotated together, not for an arbitrary fixed observable.

The main convention $R = Q = R_-$, $X' = RXR^H$, $\psi' = R\psi$ uses the Hamilton quaternion directly. The alternative $R = Q^H = R_+$, $X' = R^H X R$, $\psi' = R^H \psi$ is equally consistent. Either can use $L^H X L = L X L^H$ with the boost sign fixed by the chosen frame velocity.

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